Core Representation Theory

***EYNTKA* Series**

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Part I

Representation Theory

There are many domains of mathematics which have made huge amounts of progress, having very-well understood objects and morphisms, theorems, results, and properties. One of the most well-understood area's of mathematics has become linear algebra, where vector-spaces and linear transformations. One of the goals of this chapter will be do be able to study the following 3 objects where the operations becomes matrix addition and multiplication

1. (Finite) groups as a subgroup of $\mathrm{GL}(V)$, where the group operation becomes the matrix multiplication
2. Associative algebras
3. Lie Algebras

The further advantage of bringing these objects into linear algebra is that many field intersect with it: If the vector-spaces are infinite dimensional, we can make them Hilbert spaces and use tools from analysis. Vector-spaces are at the heart of differential geometry allowing us to use such theories to study groups!

For the reader who is really interested in number theory, section 1.3 shows that matrix algebras have many parallels with some number theory results. The reader who goes onto learn class field theory[1] will to their delight find that these parallels continue. It was observed by Langland[2] that the study of simple k-algebras (algebras with trivial two-sided ideals) can be a higher-dimensional parallel to field extensions and the central results of class field mathematics known as the *Langland's Program*.

(The reader may want to go back to section 12.3 to review about direct sums of modules)

(Also, this entire link is really cool: from wiki. Also has some cool examples of duality!!!)

[1] See [12] for the continuation of this textbook into class field theory
[2] Or at least formalized by Langland, as these ideas were floating around during his time

1

Semisimple Rings and Modules

As mentioned in the introduction, representation theory is all about decomposing structures. In part I, we showed in chapter 3, section 3.7, that all finite groups can uniquely be represented as a series of extensions with simple factors (i.e. G_i/G_{i+1} is simple). The simplest type of composition series would be when G is a direct product of simple groups, which would be in the terminology of exact sequences a series of trivial extensions with simple factors: such a group is sometimes *semi-simple* group[1]. When it came to modules, we found that we may decompose it in different ways depending on what rings acts on it, in particular we saw in chapter 13 that if $R = k$ is a field k, then any k-module is free, and in chapter 14 we saw that if R is a Principal Ideal Domain, where we got that any R-module decompsoed into a free part R-isomorphic to R^k, and a torsion part $R/(a_1) \times \cdots \times R/(a_n)$. It is evident that the ring that acts on the abelian group can have a large impact on it's structure. In this section, we define a new type of ring, which will come to be called a *semi-simple* ring, that will force the module M to *always* be a direct product of *simple modules* (so that we are analysing the "building blocks" of modules).

To define this ring, we first shift focus and start by work with the module itself to find the "simplest" structure it may be[2], and explore in a manner which at the beginning parallels groups: we will look for *simple R-modules*, and direct product of simple modules (since they are the simplest extension of simple module).[3]. After defining such a strucutre, we will define which rings acting on moduels will *always* make the module semi-simple

[1] the term "semi-simple group" is not a completely standard terminology, as direct product of simple groups is not generally very heavily studied, see this

[2] "simple" here means no non-trivial sub-structures that can be used to build an extension, just like for simple groups or simple rings

[3] It might be tempting to say that this means we are exploring projective modules. However, we have yet to establish whether simple modules are free-modules. Are they?

1.1 Decomposing Modules

Definition 1.1.1: Decomposition Terminology

Let M be a nonzero R-module.

1. M is said to be *simple* (or *irreducible*) if the only submodules of M are 0 and M. Otherwise, M is *reducible*
2. M is said to be *indecomposable* if there do not exist submodules $M_1, M_2 \subseteq M$ such that $M = M_1 \oplus M_2$. Otherwise, M is said to be *decomposable*.
3. The module M is said to be *semisimple* (or *completely reducible*) if it is the direct summand of simple modules
4. The module M is said to be semi-indecomposable (or completely indecomposable) if it is a direct summand of indecomposable modules.
5. If M is semisimple, any direct summand of M is called a *constituent* of M (i.e. N is a constituent of M if there exists a submodulde N' such that $M = N \oplus N'$)

A moment should be taken to find the relations between these definitions:

1. Simple modules are semi-simple and indecomposable. The prototypical example of a simple module is a field k over itself.
2. Semi-Simple $\not\Rightarrow$ Simple: take k^2 over k (the 2 dimensional vector-space over k). The cannonical example of a semi-simple module would be vector-spaces, since they are all direct sums of the underlying field, which is a simple k-module.
3. Indecomposable $\not\Rightarrow$ Simple: this statement is equivalent to saying that not all short exact sequence spilt (i.e., not all modules are *projective*, see section 17.2). For a concrete example, let $\mathbb{Z}^2$ be a module over $\mathbb{Z}$ with the action:

$$r \cdot (a, b) = (a + rb, b)$$

   Then it should be checked that it is not simple since $\mathbb{Z} \times \{0\}$ is a submodule, but it is indecomposable[4].
4. Every semi-simple module is a semi-indecomposable module, however the converse is not true: $\mathbb{Z}$ is a semi-indecomposable $\mathbb{Z}$-module (more genreally any module module over a PID is semi-indecomposable due to the Decomposition Theorem from chapter 14), however $\mathbb{Z}$ is not semi-simple a semi-simple $\mathbb{Z}$-module as $p\mathbb{Z}$ is a submodule (the general case will be immediate after lemma 1.1.7)[5]

As stated in the introduction, we want to determine a semi-simple module or semi-simple modules in terms of the ring that is acting on it. The following defintion was made for this purpose:

[4] Hint: Is it possible to write $\mathbb{Z} = n\mathbb{Z} \oplus m\mathbb{Z}$?

[5] This is a version of Krull-Schmidt's theorem for the decomposition of Noetherian or Artinian modules into the direct sum of indecomposable

Definition 1.1.2: Semi-Simple Ring

Let R be a ring. Then R is *semi-simple* if every R-module is semi-simple. Recall that R is *simple* if the only two-sided ideals are (0) and R.

Notice how a semi-simple ring was *not* defined as the direct product of simple rings. In the following proofs, we will show that the definition of semi-simple ring implies that R is the direct product of simple rings, however the converse is not true: there exists simple rings that are *not* semi-simple modules over themselves (and hence not simple R-modules). The problem is that of dimension; we will soon explore the relation between being semi-simple and Noetherian/Artinian. If we impose the condition that R be Noetherian and Artinian, then we are able to say that R is a semi-simple ring if and only if R is the direct product of simple rings.

We next establish many properties of simple and semi-simple modules, and show how closely related they are to the ring acting on them:

Lemma 1.1.3: Simple Module Characterization

Let M be an R-module. Then M is a simple R-module if and only if there exists a maximal left ideal $\mathfrak{m} \subsetneq R$ such that

$$M \cong_R R/\mathfrak{m}$$

***Proof*:**

($\Rightarrow$) Let M be a simple R-module, so $M \neq 0$ and its only submodules are $\{0\}$ and M. Fix $0 \neq x \in M$ and consider the R-module homomorphism

$$\varphi : R \to M \qquad r \mapsto r \cdot x$$

Its image is a nonzero submodule, so $\varphi(R) = M$, and $R/\ker(\varphi) \cong M$, where $\ker(\varphi)$ is a left ideal. The submodules of $R/\ker(\varphi)$ correspond to the left ideals J with $\ker(\varphi) \subseteq J \subseteq R$, so simplicity of M leaves only $J = \ker(\varphi)$ and $J = R$, and $\ker(\varphi) \neq R$ since $M \neq 0$. Thus $\ker(\varphi)$ is a maximal left ideal.

($\Leftarrow$) Let $\mathfrak{m}$ be a maximal left ideal and $M \cong R/\mathfrak{m}$. The submodules of $R/\mathfrak{m}$ correspond to the left ideals J with $\mathfrak{m} \subseteq J \subseteq R$, which are only $\mathfrak{m}$ and R, and $R/\mathfrak{m} \neq 0$. Hence M is simple, as we sought to show.

If R is commutative, a maximal left ideal is a maximal ideal and $R/\mathfrak{m}$ is a field. For noncommutative R, a maximal left ideal need not be two-sided, and $R/\mathfrak{m}$ is then only a left R-module. For instance, the only two-sided ideals of $R = M_n(k)$ are 0 and R (box 1.2.6), while the simple module k^n is $R/\mathfrak{m}$ for the maximal left ideal $\mathfrak{m}$ of matrices with vanishing first column, the kernel of $A \mapsto Ae_1$. Do not confuse the isomorphism in the above statement with a ring isomorphism: the structure of M is isomorphic to the R-module structure of left-multiplication on $R/\mathfrak{m}$. The proof also shows that a simple R-module is cyclic, generated by any of its nonzero elements. If $R = k$ is a field, then M is isomorphic to a 1-dimensional vector-space over k. There do exist noncommutative simple rings that are *not* of the form $M_n(k)$ for a field k, for instance the Hamilton quaternions $\mathbb{H}$, a noncommutative division ring that is a central simple $\mathbb{R}$-algebra of dimension 4 (example 1.3.3).

Using this result, we see that any semi-simple R-module is of the form:

$$M \cong_R \bigoplus_{i \in I} R/\mathfrak{m}_i$$

for maximal left ideals $\mathfrak{m}_i \subseteq R$. In fact, we can further characterize $\mathfrak{m}_i$:

Corollary 1.1.4: Simple Module's And Annihilators

Let M be a simple left R-module. Then $M \cong R/\text{Ann}(m)$ for any nonzero $m \in M$, where $\text{Ann}(m) = \{r \in R \mid rm = 0\}$, and $\text{Ann}(m)$ is a maximal left ideal.

***Proof*:**
For $0 \neq m \in M$, the submodule Rm is nonzero, so $Rm = M$ because M is simple. The map $R \to M$, $r \mapsto rm$, is therefore a surjective homomorphism of left R-modules with kernel $\text{Ann}(m)$, a left ideal, so

$$M \cong_R R/\text{Ann}(m)$$

The submodules of $R/\text{Ann}(m)$ correspond to the left ideals containing $\text{Ann}(m)$, and M is simple, so $\text{Ann}(m)$ is a maximal left ideal, completing the proof.

Since we have seen the limitations of the structure of simple modules, it is also natural to ask if there are any limitations on the structure of the hom-set given two simple modules:

Lemma 1.1.5: Schur's Lemma

Let M and N be simple R-modules, $\varphi : M \to N$ an R-module homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. As a consequence, $\text{End}_R(M)$ is a division ring.

***Proof*:**
Since N is simple, either $\varphi(M) = N$ or $\varphi(M) = 0$. If $\varphi(M) = N$, it has to be that $\ker(\varphi) = 0$, since M is simple, and hence φ would be an isomorphism.

As a consequence, every element in $\text{End}_R(M)$ is either 0 or is a unit, i.e. has a multiplicative inverse, which by definition makes $\text{End}_R(M)$ a divising ring under $(+, \circ)$.

Since $M \cong_R R/\mathfrak{m}$ for some maximal left ideal $\mathfrak{m} \subsetneq R$, we have that $\text{End}_R(R/\mathfrak{m})$ is always a division ring, whether R is commutative or not. In a sense, this can be thought of as "generalization" to proposition 9.4.19, where it was shown that $R/\mathfrak{m}$ is a field if R is commutative.

Notice that if $M = V$ was an 1-dimensional vector-space over k, then $\text{End}_R(V)$ is a known division ring, in particular it's ring-isomorphic to k. If V is a finite dimensional semi-simple module over k, i.e. it is n-dimensional, then $\text{End}_R(V)$ is ring-isomorphic to $M_n(k)$. We might hope for something similar for the structure of $\text{End}_R(M)$ is M is semi-simple. Take a moment to ponder this before reading the next lemma (hint: recall proposition 12.3.36)

Lemma 1.1.6: Hom's of Semisimple Modules

Let M be a semisimple R-module, so that:

$$M = S_1^{\oplus n_1} \oplus S_2^{\oplus n_2} \oplus \cdots \oplus S_k^{\oplus n_k}$$

where each S_i are simple R-modules, $S_i \ncong S_j$. Then as a ring:

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \mathrm{End}_R(S_1^{\oplus n_1}) \oplus \cdots \oplus \mathrm{End}_R(S_k^{\oplus n_k})$$

Furthermore, for each $S_i^{\oplus n_i}$, we have that direct sums "pull out" to become matrices:

$$\mathrm{End}_R(S_i^{\oplus n_i}) \cong_{\mathbf{Ring}} M_{n_i}(\mathrm{End}_R(S)) = M_{n_i}(D_i)$$

where $D = \mathrm{End}_R(S_i)$ and $M_{n_i}(D_i)$ is the matrix ring over the division ring $\mathrm{End}_R(S_i)$.

***Proof*:**
We invoke proposition 12.3.36 to get the result in terms of groups, and then use Schur's lemma and check that the ring operation is compatible:

$$\begin{aligned}
\mathrm{End}_R(M) &= \mathrm{Hom}_R\left(\bigoplus_i S_i^{\bigoplus n_i}, \bigoplus_j S_j^{\bigoplus n_j}\right) \\
&\cong_{\mathbf{Grp}} \bigoplus_{i,j} \mathrm{Hom}_R\left(S_i^{\bigoplus n_i}, S_j^{\bigoplus n_j}\right) \\
&\cong_{\mathbf{Grp}} \bigoplus_{i,j} \mathrm{Hom}_R(S_i, S_j)^{\bigoplus n_i n_j}
\end{aligned}$$

where when $S_i \neq S_j$, by Shur's lemma we must have:

$$\mathrm{Hom}_R(S_i, S_j)^{n_i n_j} = 0$$

Hence we get that $\mathrm{End}_R(M) \cong_{\mathbf{Grp}} \oplus_i \mathrm{Hom}_R(S_i^{n_i}, S_i^{n_i}) = \oplus_i \mathrm{End}_R(S_i^{n_i})$. Next, this map is compatible with the ring structure, since the map:

$$\varphi \to (\varphi|_{S^{n_i}} : S_i^{n_i} \to S_i^{n_i} \subseteq M)$$

is clearly clearly compatible with composition, and hence

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \bigoplus_i \mathrm{End}_R(S_i^{n_i})$$

For part 2, we follow the same steps as for part (1): re-indexing so that $S^n = S_1 \oplus S_2 \oplus \cdots \oplus S_n$ so that $S_i = S_j = S$, we get

$$\mathrm{Hom}_R\left(\bigoplus_i S_i, \bigoplus_j S_j\right) \cong_{\mathbf{Grp}} \bigoplus_{i,j} \mathrm{Hom}_R(S_i, S_j) \overset{!}{=} \bigoplus_{i,j} D \cong_{\mathbf{Grp}} M_n(D)$$

where $\overset{!}{=}$ comes from Shur's lemma with $D = \text{End}_R(S)$ since $S_i = S_j = S$. The last equality comes from the fact that the matrix $M_n(D)$ when treated like a group is simply re-arranging where the addition is happening in $\bigoplus_{i,j} D$. To show that the ring structure is compatible, observe that

$$\varphi \mapsto \varphi_{ji}$$

is the desired map, where

$$\varphi \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2n} \\ \vdots & \vdots & \ddots & \cdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}$$

which is simply a formalization of what the group-isomorphism give us, showing us that composition is compatible, completing the proof. (see this for a more detailed explanation)

Hence, for any semi-simple R-module M, we get that:

$$\text{End}_R(M) \cong_{\textbf{Ring}} \prod_i M_{n_i}(D_i) \cong_{\textbf{Ring}} \prod_i \text{End}_R(S_i^{\oplus n_i})$$

and since each S_i is a simple R-module, $S_i \cong_R R/\mathfrak{m}_i$ for an appropriate maximal left ideal $\mathfrak{m}_i \subseteq R$, $\text{End}_R(S_i) \cong \text{End}_R(R/\mathfrak{m}_i)$. Thus:

$$\text{End}_R(M) \cong_{\textbf{Ring}} \prod_i M_{n_i}(\text{End}_R(R/\mathfrak{m}_i)) = \prod_i M_{n_i}(D_i)$$

If $R = k$ is a field making $M = V$ a vector-sapce over k (say n-dimensonal), then the only simple k-submodule up to isomorphism is k, and so this result reduces to:

$$\text{End}_R(V) \cong_{\textbf{Ring}} M_n(\text{End}_k(k/(0))) = M_n(k)$$

where $\text{End}_k(k) \cong_{\textbf{Ring}} k$, aligning with the results we found in secion 13.2. This shows that when generalizing from linear algebra, due to the fact that a ring is non-commutative, we may have several non-isomorphic simple summands, giving us products of several matrix rings. With a better understanding of simple modules and their endomorphisms, we continue by characterizing semi-simple modules:

Lemma 1.1.7: Semi-Simple Module Equivalence

Let M be an R-module. Then the following are equivalent, and M is called *semi-simple* when they hold:

1. M is an external direct sum of simple modules:
$$M \cong \bigoplus_{i \in I} S_i$$
2. M is the internal direct sum of simple submodules:
$$M = \bigoplus_{i \in I} S_i$$
3. If $N \subseteq M$, there exists a complementary submodule, that is there exists another submodule $N' \subseteq M$ such that
$$M = N \oplus N'$$

In particular, if M is semi-simple, every short exact sequence $0 \to N \to M \to M/N \to 0$ splits. Note that semi-simplicity is not assumed: each of the three conditions implies the other two for an arbitrary module, which is what makes (3) a usable criterion for *establishing* semi-simplicity (as in theorem 3.1.9).

***Proof*:**
For (1) $\Rightarrow$ (2), let $\varphi\colon \bigoplus_{i\in I} S_i \to M$ be an isomorphism. The submodules $\varphi(S_i)$ are simple, and every element of M is uniquely a finite sum of elements of the $\varphi(S_i)$, so M is their internal direct sum. For (2) $\Rightarrow$ (1), if $M = \bigoplus_{i\in I} S_i$ internally, then $(s_i)_{i\in I} \mapsto \sum_i s_i$ is an isomorphism from the external direct sum onto M.

For (2) $\Rightarrow$ (3), let $M = \bigoplus_{i\in I} S_i$ and $N \subseteq M$. The submodules $N' \subseteq M$ with $N \cap N' = 0$ form a nonempty set (it contains 0) in which the union of a chain is again such a submodule, so Zorn's lemma gives a maximal one, N'. Suppose $N + N' \neq M$. Since $M = \sum_i S_i$, some S_i is not contained in $N + N'$, so $S_i \cap (N + N')$ is a proper submodule of the simple module S_i, hence
$$S_i \cap (N + N') = 0$$
Then $(S_i + N') \cap N = 0$: if $s + n' \in N$ with $s \in S_i$ and $n' \in N'$, then $s \in S_i \cap (N + N') = 0$, and so $n' \in N \cap N' = 0$. By maximality of N', $S_i + N' = N'$, so $S_i \subseteq N' \subseteq N + N'$, a contradiction. Hence $M = N \oplus N'$.

For (3) $\Rightarrow$ (2), assume every submodule of M has a complement.

Step 1: Every submodule K of M has the same property. Let $L \subseteq K$ and write $M = L \oplus L'$. For $x \in K$ write $x = l + l'$ with $l \in L$ and $l' \in L'$. Then $l' = x - l \in K$, so $K = L \oplus (L' \cap K)$.

Step 2: Every nonzero submodule $N \subseteq M$ contains a simple submodule. Take $x \in N$ nonzero.

The kernel I of $R \to Rx$, $r \mapsto rx$, is a left ideal not containing 1, and Zorn's lemma, applied to the left ideals containing I but not 1 (a union of a chain of them still omits 1), gives a maximal left ideal $\mathfrak{m} \supseteq I$. Under $Rx \cong R/I$, the submodule $\mathfrak{m}x$ of Rx corresponds to $\mathfrak{m}/I$. By Step 1, $Rx = \mathfrak{m}x \oplus L$ for some submodule L, and then

$$L \cong Rx/\mathfrak{m}x \cong (R/I)/(\mathfrak{m}/I) \cong R/\mathfrak{m}$$

which is simple because its submodules correspond to the left ideals between $\mathfrak{m}$ and R. So $L \subseteq Rx \subseteq N$ is a simple submodule.

Step 3: M is an internal direct sum of simple submodules. Call a family of simple submodules of M *independent* if every element of its sum is uniquely a finite sum of elements of its members. Independence is decided by finite subfamilies, so the union of a chain of independent families is independent, and Zorn's lemma gives a maximal independent family $(S_i)_{i\in I}$ (the empty family is independent). Put $N = \sum_{i\in I} S_i = \bigoplus_{i\in I} S_i$. By hypothesis $M = N \oplus N'$ for some N'. If $N' \neq 0$, Step 2 gives a simple submodule $S \subseteq N'$, and $S \cap N = 0$ makes the family $(S_i)_{i\in I}$ together with S independent, against maximality. Hence $N' = 0$ and $M = \bigoplus_{i\in I} S_i$.

Finally, if M is semi-simple and $N \subseteq M$, write $M = N \oplus N'$ by (3). The projection of M onto N along N' is a left inverse of the inclusion $N \to M$, so the sequence $0 \to N \to M \to M/N \to 0$ splits, completing the proof.

This theorem shows that every vector space over k is a semisimple k-module. On the other hand, $\mathbb{Z}$ is *not* a semi-simple $\mathbb{Z}$-module, since $2\mathbb{Z}$ is a submodule with no complement: if $\mathbb{Z} = 2\mathbb{Z} \oplus N'$, then $N' \neq 0$, and for $0 \neq n \in N'$ the element $2n \neq 0$ lies in $2\mathbb{Z} \cap N' = 0$, a contradiction. So not all free modules are semi-simple. We can characterize semi-simple rings by saying that R is semi-simple if and only if every R-module is projective. This also shows us that $\mathbb{Z}$ is not a semi-simple ring: $\mathbb{Z}$ is projective over itself, but any nonzero finite abelian group is a non-projective $\mathbb{Z}$-module. Thus every module over a semi-simple ring is projective. A semi-simple module over an arbitrary ring need not be projective, since $\mathbb{Z}/2\mathbb{Z}$ is a simple $\mathbb{Z}$-module that is not projective, and a module over a semi-simple ring need not be free: even though many examples of semi-simple modules are free modules, we would require the underlying ring to be semi-simple over itself. In fact, it is enough that the underlying ring be a semi-simple module over itself, giving us an easier condition for defining a semi-simple ring.

Corollary 1.1.8: Closure Properties Of Semisimple Modules

Let M be an R-module. Then all submodules and quotient modules of M are semisimple if and only if M is semi-simple. Furthermore, if $(M_i)_{i\in I}$ is a collection of semi-simple modules, then $\bigoplus_{i\in I} M_i$ is a semi-simple module.

***Proof*:**
Every module is a submodule of itself, so if all submodules of M are semi-simple, so is M. Conversely, let M be semi-simple, so that every submodule of M has a complement by lemma 1.1.7.

Submodules. Let $N \subseteq M$ and $L \subseteq N$, and write $M = L \oplus L'$. For $x \in N$ write $x = l + l'$ with $l \in L$ and $l' \in L'$. Then $l' = x - l \in N$, so $N = L \oplus (L' \cap N)$. Thus every submodule of N has a

complement in N, and N is semi-simple by lemma 1.1.7.

Quotients. Let $K \subseteq M$. A submodule of M/K has the form L/K with $K \subseteq L \subseteq M$. Write $M = L \oplus L'$, so that $M/K = L/K + (L' + K)/K$. If $x \in L \cap (L' + K)$, say $x = l' + k$ with $l' \in L'$ and $k \in K$, then $l' = x - k \in L \cap L' = 0$ because $K \subseteq L$, so $x \in K$. Hence $M/K = L/K \oplus (L' + K)/K$. Thus every submodule of M/K has a complement, and M/K is semi-simple by lemma 1.1.7.

Direct sums. If each M_i is the internal direct sum of simple submodules S_{ij}, $j \in J_i$, then $\bigoplus_{i \in I} M_i$ is the internal direct sum of the simple submodules S_{ij} over all i and j, so it is semi-simple by lemma 1.1.7, completing the proof.

These properties of semi-simple modules allow us to characterize semi-simple rings much more easily:

Corollary 1.1.9: Semi-Simple Rings Condition

Let R be a ring. Then R is a semi-simple ring if and only if R is a semi-simple R-module, that is we only need to check that R is a semi-simple R-module for one R-module structure (namely R itself)

***Proof*:**
If R is a semi-simple ring, then every R-module is semi-simple, including R over itself.

Conversely, suppose that R is a semi-simple R-module. Then so is $\bigoplus_{i \in I} R$ for every index set I, by corollary 1.1.8. Let M be an R-module and let $(m_i)_{i \in I}$ be a family of generators of M (for example, all elements of M). Then

$$\bigoplus_{i \in I} R \to M \qquad (r_i)_{i \in I} \mapsto \sum_i r_i m_i$$

is a surjective homomorphism, so M is a quotient of a semi-simple module and is semi-simple by corollary 1.1.8. Thus every R-module is semi-simple, and R is a semi-simple ring by definition, as we sought to show.

Due to this, we get a nice description of semi-simple rings which is even stronger than semi-simple modules:

Lemma 1.1.10: Decomposing Semi-Simple Rings

Let R be a semi-simple ring. Then:

$$R = I_1 \oplus \cdots \oplus I_n$$

for a finite collection of I_k, where I_k are simple left-submodules of R, i.e. *minimal* left ideals[a] of R

[a] These are *left ideals* as we are working with *left* submodules

***Proof*:**
Since R is a semi-simple ring, it is a semi-simple R-module and so by definition:

$$R = \bigoplus_{i \in \mathcal{C}} I_i$$

for simple submodules I_i (i.e. minimal left ideals I_i). To make this a finite sum, since $1 \in R$, we get for an appropraite finite subset $A \subseteq \mathcal{C}$

$$1 \in I_{\alpha_1} + \cdots + I_{\alpha_n}$$

But then, since these are all ideals,

$$R = R \cdot 1 \subseteq I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n} \subseteq R$$

and hence $R = I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n}$, completing the proof.

1.1.11 Remark As a consequence of this, any semi-simple ring R is a left-Noetherian/Artinian ring. Interestingly, we will soon show that this also implies that R is a *right*-Noetherian/Artinian ring (see rightNoethArtRemark)

Furthermore, this shows how it is necessary for the condition that R be left-Noetherian/Artinian so that the notion of being semi-simple coincide between the two definitions.

Since we characterized simple R-modules for any ring R to be R-isomorphic to $R/\mathfrak{m}$, and we have given a structure of R if R is a semisimple ring, we can make lemma 1.1.3 more precise:

Corollary 1.1.12: Structure Of Simple Modules Of Semi-Simple Ring

Let R be a semi-simple ring so that $R = I_1 \oplus I_2 \oplus \cdots \oplus I_n$ for minimal left ideal I_k (equivalently, simple left R-modules). Then if M is a simple left R-module,

$$M \cong_{R\textbf{-Mod}} I_i$$

for some $1 \leq i \leq n$.

***Proof*:**
Let M be a simple nonzero left R-module, and consider the R-module homomorphism:

$$\varphi : R \to M \qquad r \mapsto r \cdot x$$

where $x \neq 0$. Since M is simple, it is nonzero, so φ is surjective. Since $R = \bigoplus_i^n I_i$, for some i's

$$\varphi|_{I_i} \neq 0$$

and since $\varphi|_{I_i} = \pi_i \circ \varphi$ we have the R-module homomorphism:

$$\varphi|_{I_i} I_i \to M$$

and since both modules are simple and this is a nonzero homomorphism, by Schur's lemma this map must be an isomorphism, as we sought to show.

Note that there can be multiple I_i's that are isomrophic to M, which we will show in a following theorem.

This shows that any simple R-module over a semi-simple ring R is in fact isomorphic to a simple R-submodule of R (i.e. R-isomorphic to a minimal left ideal of R). Thus, studying simple R-modules over semi-simple rings comes down to studying the ring R itself, and that there are only *finitely many* simple R-modules. Furthermore, we have that $R/(I_1 \oplus \cdots \oplus I_{n-1}) \cong_R I_n$. Here is a good example of where the ring structure of the left hand side cannot exist in the right hand side. For example, if R is commutative, then $R/\mathfrak{m}$ is a field that is R-module isomorphic to a minimal ideal, but ideals cannot be fields (over the same identity[6]). Thus, in the case where R is semi-simple, for a maximal left ideal $\mathfrak{m}$, $R/\mathfrak{m}$ is R-isomorphic to a minimal left ideal of R.

Exercise 1.1.1

1. Let S be a simple module. Show that any nonzero submodule of the injective enveloppe $E(S)$ is indecomposable, but not simple.
2. Show that if M is a simple R-module, then it necessarily has to be cyclic (the converse, however, is not true).
3. Show that $A = \mathbb{Q}[x, y]/(xy - yx - 1)$ is a simple ring, but no a semi-simple ring (this is a special example of an algebra called a *Weyl* Algebra)
4. Let R be a ring, not necessarily commutative, and let $[R, R] = \{rs - sr \mid r, s \in R\}$. Show that the trace map induces an isomorphism

$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$

   From this perspective, matrix rings are the natural "nonabelian" extension of a ring!

1.2 Artin-Wederburn

We just showed how studying simple and semi-simple R-modules comes down to studying the ring R, especially if the ring R is semi-simple. Thus, it is fruitful to classify semi-simple rings, which leads us to the main classification result for semi-simple rings, namely we build on lemma 1.1.10 and show that we can find more precisely how to represent each minimal left ideal I_i, and furthermore show that this classification is exhaustive, and that any semi-simple ring uniquely breaks down into the result we will show. We will then finish by showing that if we upgrade our ring to k-algebras (that is, we can put a vector-space structure on our ring), we get some stronger results due to the fact we can count dimensions.

[6] $k \times \{0\} \subseteq k \times k$ is an ideal and a field with identity $(1, 0)$

Definition 1.2.1: Opposite Ring

Let R be a ring. Then we define a new ring labelled R^{op} that has the same underlying set and addition operation, whith the multiplication operation $*$ being:

$$x * y = yx$$

where the multiplication yx happens in R

If R is commutative, then $R = R^{\text{op}}$, so the opposite ring carries new information only when R is not commutative.

Proposition 1.2.2: Properties Of Opposite Ring

Let R be a ring and R^{op} its oppposite ring. Then:

1. $(R^{\text{op}})^{\text{op}} = R$
2. $(R \times S)^{\text{op}} = R^{\text{op}} \times S^{\text{op}}$
3. $M_n(R)^{\text{op}} \cong M_n(R^{\text{op}})$ where the isomorphism is $A \mapsto A^t$ (since transposing converts BA into the product whose entries multiply in R^{op})

***Proof*:**
In all three claims the two rings share their underlying set and their addition, so only the multiplications need comparison. Throughout, $*$ denotes an opposite multiplication as in definition 1.2.1.

For (1), write $\star$ for the multiplication of $(R^{\text{op}})^{\text{op}}$, the opposite of the multiplication $*$ of R^{op}. For any $x, y \in R$:

$$x \star y = y * x = xy$$

so $\star$ is the original multiplication of R, and $(R^{\text{op}})^{\text{op}} = R$ as rings.

For (2), both rings have underlying set $R \times S$ with componentwise addition. Multiplying in $(R \times S)^{\text{op}}$ gives

$$(r, s) * (r', s') = (r', s')(r, s) = (r'r, s's)$$

while multiplying in $R^{\text{op}} \times S^{\text{op}}$ gives $(r * r', s * s') = (r'r, s's)$ as well. The two multiplications agree, so the rings are equal.

For (3), let $\tau : M_n(R)^{\text{op}} \to M_n(R^{\text{op}})$ be the transpose map $\tau(A) = A^t$, where $(A^t)_{ij} = A_{ji}$. The map τ is additive, fixes the identity matrix, and is a bijection because it is its own inverse on the underlying set. For multiplicativity, let $A, B \in M_n(R)^{\text{op}}$ and let $*$ denote the multiplication of $M_n(R)^{\text{op}}$, so that $A * B$ is the matrix BA computed in $M_n(R)$. For each pair (i, j):

$$\tau(A * B)_{ij} = (BA)_{ji} = \sum_k B_{jk} A_{ki}$$

Computing $\tau(A)\tau(B)$ in $M_n(R^{\text{op}})$ multiplies the entries in R^{op}:

$$(\tau(A)\tau(B))_{ij} = \sum_k (A^t)_{ik} * (B^t)_{kj} = \sum_k A_{ki} * B_{jk} = \sum_k B_{jk} A_{ki}$$

The two entries agree, so $\tau(A*B) = \tau(A)\tau(B)$ and τ is a ring isomorphism. For noncommutative R the identity $(BA)^t = A^tB^t$ fails inside $M_n(R)$ itself. The computation above succeeds because the entries of $\tau(A)\tau(B)$ are multiplied in R^{op}, which restores the order of each product.

With this, we can make a link between R and R^{op}:

Lemma 1.2.3: Yonneda's Lemma for opposite Rings

Let R be a ring and consider the ring $\text{End}_R(R)$ to be the set of all R-module homomorphisms from R (as an R-module) to itself. Then $\varphi : R^{\text{op}} \to \text{End}_R(R)$, $r \mapsto (x \mapsto xr)$ is a ring isomorphism:

$$R^{\text{op}} \cong_{\textbf{Ring}} \text{End}_R(R)$$

***Proof*:**
Let φ_r be the given map. Let's first check that $\varphi_r \in \text{End}_R(R)$. Then we see that:

$$\begin{aligned}\varphi_r(\lambda x + \lambda y) &= (\lambda x + \lambda y)r \\ &= \lambda(xr) + \lambda(yr) \\ &= \lambda\varphi_r(x) + \lambda\varphi_r(y)\end{aligned}$$

showing that $\varphi_r \in \text{End}_R(R)$. Next, we check that φ is ineed a ring-homomorphism. By the module axioms:

1. $\varphi_1 = \text{id}$, and hence φ sends the identity to the identity.
2. $\varphi_{r+r'} = \varphi_r + \varphi_{r'}$
3. $\varphi_{rr'} = \varphi_{r'} \circ \varphi_r$ (do not forget we are working with R^{op})

Finally, showing bijectivity, for injectivity we see that if $\varphi_r = 0$, then $0 = \varphi_r(1) = r$, and hence $\varphi_r = \varphi_0$. For surjectivity, let $\varphi \in \text{End}_R(R)$. Let's say $\varphi(1) = r$. Then $\varphi(x) = x\varphi(1) = xr$, so $\varphi = \varphi_r$, completing the proof.

In the special case where R is commutative, then $R \cong R^{\text{op}}$, showing that in this case $R \cong \text{End}_R(R)$. Using this, we will use lemma 1.2.3 along with lemma 1.1.6 to upgrade lemma 1.1.10:

Theorem 1.2.4: Artin-Wedderburn, one Direction

Let R be a semisimple ring. Then:

$$R \cong_{\textbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_i is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i

The key realization about this theorem is that while the multiplication on the left hand side can "send you" all over the place, the multiplication on the right hand side only sends you back into your particular component. To see this more clearly, let's say R was a k-algebra for a field k so that it has a basis. If R was a semi-simple ring, then that means that when multiplying elements of R (so

figuring out where the basis elements go while multiplying), we can isolate the problem down to the components on the right hand side, and figure out where they go through matrix multiplication!

***Proof*:**
First, lemma 1.1.10 decomposes R into simple R-submodules, that is, minimal left ideals of R,

$$R = I_1 \oplus \cdots \oplus I_n$$

Now, we can always re-order the sums (i.e. $I_1 \oplus I_2 \cong I_2 \oplus I_1$, since $R \cong I_1 \oplus \cdots \oplus I_n$ by [4, Equivalent Conditions for Linear Combinations]). Hence, we will group all the I_i's that belong to the same isomrophism class (as R-modules), letting us re-write R as:

$$R \cong_{\mathbf{Ring}} I_1^{n_1} \oplus \cdots \oplus I_k^{n_k}$$

where $I_i \ncong I_j$ for $i \neq j$. Hence by lemma 1.2.3 and lemma 1.1.6:

$$\begin{aligned} R^{\text{op}} &\cong_{\mathbf{Ring}} \text{End}_R(R) \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k) \end{aligned}$$

where $D_i = \text{End}_R(I_i)$ is a divison ring. Finally, Taking the opposite ring on both sides, we get:

$$\begin{aligned} R &\cong_{\mathbf{Ring}} (R^{\text{op}})^{\text{op}} \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1)^{\text{op}} \times \cdots \times M_{n_k}(D_k)^{\text{op}} \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1^{\text{op}}) \times \cdots \times M_{n_k}(D_k^{\text{op}}) \end{aligned}$$

and it is clear the D_i^{op} is still a division ring, hence completing the proof.

Thus, if we have a simple-ring then the ring breaks down into the product of matrix rings over division rings. In the special case where $R = k$ is a field, then k has only one minimal ideal, namely itself, and $\text{End}_k(k) \cong M_1(k) \cong k$, and hence we reduce to $k \cong k$. If $R = k^n$ is a product of the same field, then the minimal ideals of R are $R = k \oplus \cdots \oplus k$, so that we get back $k^n \cong_{\mathbf{Ring}} k^n$.

We will now show that the converse of this theorem is true, that the product of matrix rings over division rings is semi simple. We start by showing that matrix ring are simple modules[7].

Proposition 1.2.5: Matrix Ring over Division Rings are semi-simple

Let $R = M_n(D)$ be a matrix ring over a divison ring. Let $C = D^{\oplus n}$ be the left $M_n(D)$-module where the action is left matrix multiplication (imagine C as a column). Then C is a simple $M_n(D)$-module, and

$$M_n(D) \cong_{M_n(D)\text{-}\mathbf{Mod}} C^{\oplus n} = (D^{\oplus n})^{\oplus n}$$

showing that $M_n(D)$ is semisimple ring and $M_n(D) \cong_{\mathbf{Ring}} C^{\oplus n}$

***Proof*:**
Let $R = M_n(D)$ and $C = D^{\oplus n}$ (C for *column*). For notational simplicity, let $E_{ij} \in R$ be the elementary matrix which is 1 in position ij and zero everywehre else. Let $e_i \in C$ be the column

[7] This will be an example of R being both a simple R-module and a simple ring

where it is 1 at position i and zero everywehre else:

$$E_{ij} = \begin{pmatrix} 0 & & & & 0 \\ & \ddots & & \iddots & \\ & & 1 & & \\ & \iddots & & \ddots & \\ 0 & & & & 0 \end{pmatrix} \qquad e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

Then $E_{ij}e_k = \delta_{jk}e_i$ where δ_{jk} is the Kronecker Delta:

$$\delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

Now, we will start by showing that C is a simple $M_n(D)$-module. Since $C = D^{\oplus n}$,

$$C = \bigoplus_i De_i$$

for some basis $e_1, e_2, ..., e_n$. To show that C is simlpe, it is enough to show that for any $0 \neq x \in C$, $C = Rx$ since then any nonzero submodule would in fact be C. Given our above notation, we have that:

$$x = \sum_k d_k e_k$$

for appropriate $d_k \in D$, where at least one of these is nonzero, let's say $d_j \neq 0$. Then consider $E_{ij}x \in Rx$:

$$E_{ij}x = \sum_k \delta_{jk} d_k e_k = d_j e_i$$

since $d_j \neq 0$. Since D is a division ring, we get that $d_j^{-1}e_i \in Rx$, for all $1 \leq i \leq n$. But then, we get:

$$C = \sum De_i \subseteq Rx \subseteq C$$

showing that $C = Rx$ for any $0 \neq x \in C$, showing that C is indeed simple.

Now, we can re-write this ring as the internal direct sum of matrices I_i which are zero outisde the ith column:

$$R = M_n(D) = I_1 \oplus \cdots \oplus I_n$$

$$I_k = \begin{pmatrix} 0 & \cdots & a_{1,k} & \cdots & 0 \\ 0 & & a_{2,k} & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & & a_{2,n-1} & & 0 \\ 0 & \cdots & a_{n,k} & \cdots & 0 \end{pmatrix}$$

This decomposition works since each I_i is stable under *left* multiplication by elements of $M_n(D)$ (since left multiplication is a *column* transformation, right multiplication would be *row* transformation). Finally, $C \cong_{R\textbf{-mod}} I_i$ by simply moving the column to the right place, and so each I_i is simple. Hence, we have decomposed $M_n(D)$ as a dierct product of simple left $M_n(D)$-modules, and hence $M_n(D)$ is a semisimple $M_n(D)$-module, which by corollary 1.1.9 makes $M_n(D)$ a semi-simple ring. Finally, since $C^{\oplus n}$ represents the n rows of n-length columns, we see that the action of $M_n(D)$ on $C^{\oplus n}$ is identical to that of matrix multiplying elements of $C^{\oplus n}$, which gives us that $M_n(D) \cong_{\textbf{Ring}} C^{\oplus n}$, as we sought to show.

1.2.6 Remark In fact, $M_n(D)$ is a simple ring, meaning the only *two-sided* ideals of R are 0 and R; the above proof can be modified to show this. This gives us a nice class of non-commutative simple rings (since all simple commutative rings are fields).

Notice that it is quite sepcial that the ring opration and the module operation coincided, this because of the special way in which we decompoed the $M_n(D)$-module $M_n(D)$ and by the fact that it is a module over itself. In the previous proposition, we showed that $D^{\oplus n}$ is a simple $M_n(D)$-module. In fact, all simple modules of $M_n(D)$ are isomorphic to $D^{\oplus n}$:

Corollary 1.2.7: Structure of Simple $M_n(D)$-Module

Let $R = M_n(D)$ be a matrix ring over a division ring. Then any simple $M_n(D)$-module is isomorphic to $D^{\oplus n}$.

***Proof*:**
Let M be a simple $M_n(D)$-module. By proposition 1.2.5

$$M_n(D) \cong_{\mathbf{Ring}} D^{\oplus n} \oplus \cdots_{\mathbf{Ring}} \oplus D^{\oplus n}$$

And by corollary 1.1.12,

$$M \cong D^{\oplus n}$$

Corollary 1.2.8: Endomorphism Of Simple $M_n(D)$-Module

Let S be a simple $M_n(D)$-module. Then

$$\mathrm{End}_{M_n(D)}(S) \cong_{\mathbf{Ring}} D^{\mathrm{op}}$$

***Proof*:**
Since S is simple, $S \cong_R D^{\oplus n}$ for some n. Let $\varphi_d : S \to S$, $\varphi_d(x) = x \cdot d$ where $x \cdot d$ is x times dI. Then:

$$\varphi_d(Ax) = (Ax) \cdot d = A(x \cdot d) = A\varphi_d(x)$$

and

$$\varphi_d(x+y) = (x+y) \cdot d = (x+y)dI = xdI + ydI = x \cdot d + y \cdot d = \varphi_d(x) + \varphi_d(y)$$

hence $\varphi_d \in \mathrm{End}_R(S)$. Conversely, if $\varphi \in \mathrm{End}_R(S)$, Since $S \cong D^{\oplus n}$, and

$$D^{\oplus n} \cong \left(D^{\oplus n}\ 0^{\oplus n}\ \cdots\ 0^{\oplus n}\right)$$

we will treat S as a $M_n(D)$ submodule of $M_n(D)$. Now, since φ is a group homomorphism, we know that:

$$\varphi \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \cdots & \varphi_{n1} \\ \vdots & \ddots & \vdots \\ \varphi_{n1} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11}(1) + \cdots + \varphi_{n1}(1) \\ \vdots \\ 0 \end{pmatrix}$$

hence let $d = \sum_i^n \varphi_{i1}(1)$, and let the right-action on $\mathrm{End}_{M_n(D)}(S)$ be right multiplication by the identity matrix dI. Then: Then for all $x \in S$, we have:

$$\begin{aligned}\varphi\begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} &= \varphi\left(\begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix}\begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}\right) \\ &\overset{!}{=} \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix}\varphi\begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \\ &= \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \cdot d\end{aligned}$$

where $\overset{!}{=}$ comes from the fact that we can treat that matrix as an element of $M_n(D)$, and since φ is an $M_n(D)$-module homomorphism, we can pull it out. But then, the action is completely dependent on on d, an so we have a bijection between right multiplication by $d \in D$ an $M_n(D)$-endomrophisms. It is an easy taks to check that:

$$\varphi_{dd'} = \varphi_{d'} \circ \varphi_d$$

showing that

$$D^{\mathrm{op}} \cong_{\mathbf{Ring}} \mathrm{End}_{M_n(D)}(S)$$

as we sought to show.

The next thing we will need is to show that if R, S are semisimple rings, then so is $R \times S$. Since R and S are semi-simple R-modules and S-modules, if we can show that $R \times S$ is a semi-simple $(R \times S)$-module, we are done. For this to be the case, we must be able to decompose $R \times S$ into simple $(R \times S)$-submodules. Forunately, many universal propeties play in our favour to be able to easily identify simple $(R \times S)$-modules of $R \times S$ given we know the simple R-modules and S-modules.

To see the key idea of the following proof, we define some terms

Definition 1.2.9: Idempotent And Orthogonal Elements

Let R be a ring. Then $e \in R$ is said to be *idempotent* if

$$e^2 = e$$

Furthermore, if e_1, e_2 are idempotent elements in R, if

$$e_1 e_2 = e_2 e_1 = 0$$

then they are said to be *orthogonal*. If e is a nonzero idempotent and cannot be written as the sum of two nonzero orthogonal idempotent elements, then e is said to be a *primitive* idempotent element.

Example 1.2.10: Idempotent Elements

Naturally, for any ring R, $1 \in R$ is an idempotent element.

For a more interesting example the product ring $R \times S$, and take $e = (1,0) \in R \times S$, $f = (0,1) \in R \times S$. Then:

1. $e^2 = e$ and $f^2 = f$ (i.e. they are both *idempotent*)
2. $ef = fe = 0$ (implying also that $ef \in Z(R \times S)$)
3. $e + f = 1$ (i.e. e and f are *not* primitive)

What happens if the number of terms in the product increases?

Next, if M is an R-module an N is an S-module, we can M and N to be $(R \times S)$-module by ignoring what happens to S (resp. R), i.e. think of the action being defined by the following function

$$R \times S \twoheadrightarrow R \to \mathrm{End}_{\mathbf{ab}}(M)$$

and similarly for a S-module N. Hence, we can define $M \oplus N$ as an $(R \times S)$-module. Conversely, if M is an $(R \times S)$-module, then using what we have shown in example 1.2.10, given $e = (1,0)$ and $f = (0,1)$:

$$M = eM + fM$$

Notice that eM is killed by Rf, so it's a $A \cong A \times B/(\{0\} \times B)$-module, and similarly for fM. In other words, idempoent elements characterize the splitting of modules! This gives us the following:

Proposition 1.2.11: Direct Product Decomposition

Let A and B be rings, let $e = (1,0)$ and $f = (0,1)$ be the central idempotents of $A \times B$, and let A-**Mod** $\times$ B-**Mod** be the product category, whose objects are pairs (M, N) of an A-module M and a B-module N, and whose morphisms are pairs of module homomorphisms. For an $(A \times B)$-module P, the submodules eP and fP are an A-module and a B-module through $a \cdot x = (a,0)x$ and $b \cdot y = (0,b)y$. Then the functors

$$A\text{-}\mathbf{Mod} \times B\text{-}\mathbf{Mod} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} (A \times B)\text{-}\mathbf{Mod}$$

defined on objects by

$$\begin{aligned} \varphi(M,N) &= M \oplus N \\ \psi(P) &= (eP, fP) \end{aligned}$$

where $(a,b) \cdot (m,n) = (am, bn)$ on $M \oplus N$, and on morphisms by $\varphi(\alpha, \beta) = \alpha \oplus \beta$ and $\psi(\gamma) = (\gamma|_{eP}, \gamma|_{fP})$, form an equivalence of categories: there are natural isomorphisms $\psi \circ \varphi \cong \mathrm{id}$ and $\varphi \circ \psi \cong \mathrm{id}$.

***Proof*:**

Step 1: φ is a functor. The module axioms for $M \oplus N$ hold componentwise, and $(1,1)$ acts as the identity. For morphisms $\alpha \colon M \to M'$ and $\beta \colon N \to N'$, the map $\alpha \oplus \beta \colon (m,n) \mapsto (\alpha(m), \beta(n))$

is additive and satisfies $(\alpha \oplus \beta)((a,b)(m,n)) = (a\alpha(m), b\beta(n)) = (a,b)(\alpha \oplus \beta)(m,n)$. Identities and composites are preserved componentwise.

Step 2: ψ is a functor. Since e is central, $(a,b)(ep) = e((a,b)p)$, so eP is an $(A \times B)$-submodule of P, and $ex = x$ for $x \in eP$ because $e^2 = e$. The rule $a \cdot x = (a,0)x$ is additive in a and in x, satisfies $(aa') \cdot x = (a,0)(a',0)x = a \cdot (a' \cdot x)$, and $1 \cdot x = ex = x$, so eP is an A-module. The same computation with f makes fP a B-module. If $\gamma\colon P \to P'$ is $(A \times B)$-linear, then $\gamma(ep) = e\gamma(p)$ shows $\gamma(eP) \subseteq eP'$, and $\gamma((a,0)x) = (a,0)\gamma(x)$ shows that $\gamma|_{eP}$ is A-linear. Likewise $\gamma|_{fP}\colon fP \to fP'$ is B-linear, and restriction preserves identities and composites.

Step 3: $\psi \circ \varphi \cong \mathrm{id}$. For (M,N), $e(m,n) = (m,0)$ and $f(m,n) = (0,n)$, so $\psi(\varphi(M,N)) = (M \oplus 0, 0 \oplus N)$, where $a \cdot (m,0) = (am,0)$ and $b \cdot (0,n) = (0,bn)$. Hence $\eta_{(M,N)} = (m \mapsto (m,0),\ n \mapsto (0,n))$ is a pair of module isomorphisms, that is, an isomorphism $(M,N) \to \psi(\varphi(M,N))$. For a morphism (α,β), the map $\psi(\varphi(\alpha,\beta))$ sends $(m,0)$ to $(\alpha(m),0)$ and $(0,n)$ to $(0,\beta(n))$, so $\psi(\varphi(\alpha,\beta)) \circ \eta_{(M,N)} = \eta_{(M',N')} \circ (\alpha,\beta)$, and η is natural.

Step 4: $\varphi \circ \psi \cong \mathrm{id}$. For an $(A \times B)$-module P, define

$$\epsilon_P\colon eP \oplus fP \to P \qquad (x,y) \mapsto x + y$$

For $x \in eP$ and $y \in fP$ we have $(a,b)x = (a,b)ex = (a,0)x$ and $(a,b)y = (0,b)y$, so $\epsilon_P((a,b)(x,y)) = (a,0)x + (0,b)y = (a,b)(x+y)$ and ϵ_P is $(A \times B)$-linear. It is surjective because $p = ep + fp$, as $e + f = 1$. It is injective: if $x + y = 0$, then $0 = e(x+y) = x + efy = x$, using $ex = x$, $y = fy$ and $ef = 0$, and then $y = 0$. For an $(A \times B)$-linear $\gamma\colon P \to P'$,

$$\epsilon_{P'}\big(\varphi(\psi(\gamma))(x,y)\big) = \gamma(x) + \gamma(y) = \gamma(\epsilon_P(x,y))$$

so ϵ is a natural isomorphism $\varphi \circ \psi \to \mathrm{id}$. Steps 3 and 4 show that φ and ψ are mutually inverse equivalences, as we sought to show.

Proposition 1.2.12: Classifying Simple $(A \times B)$-Modules

Let P be a simple $(A \times B)$-module. Then

$$P \cong_{A\times B\text{-Mod}} M \oplus (0) \qquad \text{or} \qquad P \cong_{A\times B\text{-Mod}} (0) \oplus N$$

where M is a simple A-module and N is a simple B-module.

***Proof*:**
By proposition 1.2.11, $P \cong_{A\times B\text{-Mod}} M \oplus N$ where M is an A-module and N is a B-module. BUt then:

$$P \cong_{A\times B\text{-Mod}} (M \oplus (0)) \oplus ((0) \oplus N)$$

since P is simple, one of these has to be zero, and the other one has to be simple.

Corollary 1.2.13: Product Of Semi-Simple Rings

Let R and S be semi-simple rings. Then $R \times S$ is a semi-simple ring. Furtheromre, if $R_1, R_2, ..., R_n$ are the simple R-module up to isomorphism and $S_1, S_2, ..., S_m$ are the simple S-modules, then the simple $R \times S$-modules are of the form:

$$R_i \oplus (0) \qquad (0) \oplus (S_j)$$

***Proof*:**
By proposition 1.2.11, since R is a semi-simple R-module and S is a semi-simple S-module, $R \times S$ is a semi-simple $(R \times S)$-module, and so $R \times S$ is a semi-simple ring. Then by proposition 1.2.12, if $P \subseteq R \times S$ is a simple $(R \times S)$-module, then they it must be of the form:

$$R_i \oplus (0) \qquad (0) \oplus (S_j)$$

Corollary 1.2.14: Structure Of $\prod_i M_{N_i}(D_i)$: Artin Wedderburn part II

Let $R = M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$. Then

1. R is semi-simple
2. There exist exactly k simple R-modules up to isomorphism, each of the form:

$$0 \oplus \cdots \oplus D_i^{\oplus n_i} \oplus \cdots \oplus 0$$

***Proof*:**
The ring R is semi-simpleby corollary 1.2.13 using induction. Furthermore, each $M_{n_i}(D_i)$ has $D_i^{\oplus n_i}$ is it's unique simple $M_{n_i}(D_i)$-module, since there are k matrix rings in the productt, there must be exactly k simple R-modules, as we sought to show.

1.2.15 Remark We can as before get that

$$R^{\text{op}} \cong M_{n_1}(D_1^{\text{op}}) \times \cdots \times M_{n_k}(D_k^{\text{op}})$$

and hence R^{op} is left Noetherian/Artinian, which we know now is true if and only if R is right Noetherian/Artinian (Since a left R^{op})-module is a right R-module). By leftNoethArtRemark, we have that any semisimple ring is left-Noetherian/Artinian, and hence any semisimple ring is both left and right Noetherian/Artinian, even though R is not necesarily commutative!

Thus, we have the following statemnt, were the last thing we need to prove is uniquness

Theorem 1.2.16: Artin-Wedderburn

Let R be a ring. Then R is a semi-simple ring if and only if

$$R \cong_{\textbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_k is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i, and each D_i n_i are unique up to isomorphism

***Proof*:**
All that is left to prove is the uniqueness. Let:

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_s}(D_s) \cong_{\textbf{Ring}} R \cong_{\textbf{Ring}} M_{n'_1}(D'_1) \oplus \cdots \oplus M_{n_s}(D'_s)$$

By corollary 1.2.14, R has exactly s and t simple module, and so $s = t$. Hence, for any $D_i^{\oplus n_i}$, there must exist a $(D'_i)^{\oplus n'_i}$ such that

$$D_i^{\oplus n_i} \cong_R (D'_i)^{\oplus n'_i}$$

Since $D_i^{\oplus n_i} \cong_R (D'_i)^{\oplus n'_i}$, $\text{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\textbf{Ring}} \text{End}_{M_{n'_i}(D'_i)}((D'_i)^{\oplus n'_i})$. By corollary 1.2.8,

$$D_i^{\text{op}} \cong_{\textbf{Ring}} \text{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\textbf{Ring}} \text{End}_{M_{n'_i}(D'_i)}((D'_i)^{\oplus n'_i}) \cong_{\textbf{Ring}} (D'_i)^{\text{op}}$$

which, taking op on both sides, we see that:

$$D_i \cong_{\textbf{Ring}} D'_i$$

Finally, we need to show that $D^{\oplus n} \cong_R D^{\oplus m}$ implies $n = m$. Treating $D^{\oplus n}$ and $D^{\oplus m}$ as D-modules, then we get that each have a basis of dimension n and m. Importantly, the results of the Replacement Theorem (theorem 13.1.12) applies to divions rings (notice that commutativity was not necessary), and so corollary 13.1.13 and theorem 13.1.8 give us that $n = m$, completing the proof.

This theorem also shows us that if R is a [finitely-generated] semi-simple ring, then R is indeed the direct product of simple rings (since $M_{n_i}(D_i)$ is a simple ring). Notice too that this theorem just gave us the answer to the classification of all *finite simple rings*: namely they would have ot be of the form $M_n(\mathbb{F}_q)$[8] for some finite field of order q (which is a power of some prime p). Contrast this to how hard it is to find all finite simple groups! Furthermore, the finite dimensional assumption was key: the *Weyl Algebra*:

$$W = k\langle x, y\rangle/(xy - yx - 1)$$

is a simple algebra that is not the matrix ring of a division algebra.

Example 1.2.17: Artin Wedderburn

$\text{End}_{M_2(k)}(M_2(k)) = \text{End}_{M_2(k)}((k^2)^{\oplus 2}) = M_2(\text{End}_{M_2(k)}(k^2)) = M_2(k)$

[8] Why is the division ring a field? The answer lies in proposition 9.1.20

Corollary 1.2.18: Artin-Wedderburn For k-Algebras

Let R be a semi-simple k-algebra that is finite-dimensional over a field k. Then

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

where each D_i is a divsion k-algebra, finite dimensional as a k-vector space:

$$\dim_k R = \sum n_i^2 \dim_k(D_i)$$

and has exactly l simple R-modules as we have classified before.

***Proof*:**
By the Artin-Wedderburn theorem, we have that

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Let's now further assume R is a k-algebra for some field k, so that there is a map $k \hookrightarrow Z(R)$. Then by the above isomorphism:

$$\begin{aligned} k \mapsto Z(R) &\cong Z(M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)) \\ &\cong Z(M_{n_1}(D_1)) \times \cdots \times Z(M_{n_l}(D_l)) \\ &\cong Z(D_1) \times \cdots \times Z(D_l) \end{aligned}$$

where the last one comes form the fact that $Z(M_{n_i}(D_i))$ are the diagonal elements, and $z \mapsto zI$. This shows that D_i each have a natural k-algebra strucure induced by R, and so the above isomrophsim was already a k-algebra isomorphism:

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Finally, treating $M_{n_i}(D_i)$ as k-vector spaces we have:

$$M_{n_i}(D_i) \cong_{k\text{-}\mathbf{Vect}} D^{\oplus n_i^2}$$

which gives the desired result, completing the proof.

Corollary 1.2.19: Special Case: k Algebraically Closed

Let $k = \overline{k}$ be an algebraically closed field and D a division k-algebra that is finite-dimensional as a k-vector-space. Then $D = k$

***Proof*:**
Suppose $k \subsetneq D$, and pick any $\delta \in D \setminus k$. Then

$$k \subseteq k[\delta] \subseteq D$$

the key thing to observe is that $k[\delta]$ is still commutative!! This is essentially the same reasoning

for why any cyclic group is commutative! Since D is a finite-dimensional vector-sapce over k, some power of δ^n must be a k-linear combination of every smaller term. Since D is a division ring, every δ has an inverse, and so every element of $k[\delta]$ is in fact a unit. Hence $k[\delta]/k$ is an algebraic field extension. However, k is an *algebraically cloased field*, and hence $k[\delta] = k$. Since δ was arbitary, we must have that $D = k$, as we sought to show.

Corollary 1.2.20: Artin-Wedderburn For $\overline{k}$-Algebras

Let R be a semi-simple k-algebra that is finite-dimensional over an algebraically closed field $k = \overline{k}$. Then

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(k) \times \cdots \times M_{n_l}(k)$$

so that

$$\dim_k(R) = \sum_i^l n_i^2$$

and R has exactly l simple R-modules as we have classified before.

***Proof*:**
By corollary 1.2.18,

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

where each D_i is a division k-algebra that is finite-dimensional over k, and R has exactly l simple modules. By corollary 1.2.19, each D_i is k, so $R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(k) \times \cdots \times M_{n_l}(k)$, and $\dim_k(R) = \sum_i n_i^2 \dim_k(k) = \sum_i n_i^2$, as claimed.

As a consequence of this, if R is a semi-simple $\mathbb{C}$-algebra, the simple R-modules are all R-isomorphic to $\mathbb{C}^{n_i}$ for appropriate power n_i. As a final word of warning, it is natural to wonder if all simple rings are of the form $M_n(D_i)$

Exercise 1.2.1

1. Let k be a field, k^n be a direct product of fields a k^n-module over itslef. Let $0 \oplus \cdots \oplus k \oplus \cdots \oplus 0 = k_i \subseteq k^n$. Then $\mathrm{End}_{k^n}(k_i) \cong_{\mathbf{Ring}} k$, $\mathrm{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} k^n$, and $\mathrm{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} M_n(\mathrm{End}_{k^n}(k_i)) \cong_{\mathbf{Ring}} M_n(k)$. But that would imply $k^n \cong_{\mathbf{Ring}} M_n(k)$ – what's the contradiction?
2. Let k be an algebraically closed field, and let R be a finite-dimensional k-algebra. Let M be a simple R-module. Prove that if r is an element of the centre $Z(R)$ of R, there exists $c \in k$ such that $rm = cm$ for all $m \in M$

1.3 Central Simple Algebras

Definition 1.3.1: Algebra Representation

Let A be a finite dimensional (associative) algebra over k. Then a *representation* of A is a k-homomorphism:

$$\rho : A \to \mathrm{End}_k(V)$$

for some vector space V.

Recall that one definition of an R-module is given by a ring homomorphism $\rho : R \to \mathrm{End}_{\mathbf{Ab}}(M)$ for an abelian group M. If we restrict to k-homomorphisms and k-endomorphisms, we can define the A-module $\rho : A \to \mathrm{End}_k(V)$, hence we can think of representation of and algebra A as a left A-modules. This makes sense, as representation theory can be thought of the study of associative algebras (the study of group representation is the study of group algebras, the study of the representation of lie algebras is universal enveloping algebras, the study of the representation of quiver algebras is the study of path algebras, and so forth). Because of this, we shall see that many of the results for the representation of associative algebras outlined bellow to be familiar to anyone who has taken a basic course in representation theory.

Next, we define the object that shall represent the generalization of number rings:

Definition 1.3.2: Central Simple Algebra

Let A be an algebra over a field k. Then A is called *simple* if it is a simple ring. It is called a *central simple algebra* (CSA) if it is simple, is finite dimensional over k, and its center is k.

The idea behind this definition is that it generalizes field extensions F/k, for we have a distinguished field k, and we are interested in the "algebraic case", hence the extension ought to be finite. Contrasting to fields, a CSA need not be commutative and need be a division ring

Example 1.3.3: Simple and Central Simple Algebras

1. Naturally, $M_n(k)$ is a central simple algebra of dimension n^2
2. It is possible to be simple and have center k but not be finite: the Weyl algebra $k[x, \delta_x]$ is infinite dimensional over k.
3. $\mathbb{C}$ is a central simple algebra over $\mathbb{C}$, but not over $\mathbb{R}$ (as it's center contains more than $\mathbb{R}$
4. Let $A = \mathbb{H}$ be the quaternions (or Hamiltonians). Then this is a CSA over $\mathbb{R}$ with dimension 4. We shall soon define a quaternion algebra, and see that all 4-dimensional central simples algebra's over a field k is a quaternion algebra, and it shall be limited to 2-by-2 matrices or division algebras.

Let us now generalize some of basic representation theory results. Let E be an irreducible representation of A (take for example a minimal left ideal of A).

Lemma 1.3.4: Schur's Lemma - updated

Let E be an irreducible representation of a k-algebra A, and let $K \subseteq \mathrm{End}_k(E)$ be the endomorphisms that commute with the operations of A: it is the centralizer of A in $\mathrm{End}_k(E)$, $K = C_{\mathrm{End}_k(E)}(A)$. Then K is a division ring.

Note that this makes K an A-algebra

***Proof*:**
Take $u \in K$. We want to show that $u^{-1} \in K$. Consider $u^{-1}(0) = V \subseteq E$. For $x \in V$, we have:

$$0 = a \cdot u(x) = u(a \cdot x)$$

hence, V is stable under A, and as $u \neq 0$ we must have $V = 0$ as u is invertible in $\mathrm{End}_k(E)$. It follows that $u^{-1} \in K$.

Notice from this that we may say that the representation E must be faithful (the map must be injective). Indeed, the kernel of this representation would be an ideal of A but

Lemma 1.3.5: Minimal Centralizer of Representations

Let E be a semi-simple an faithful representation of A, $K = C_{\mathrm{End}_k(E)}(A)$, $B = C_{\mathrm{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\mathrm{End}_k(E)$). Then

$$A = B$$

***Proof*:**
Let $x \in E$ and $b \in B$. We shall show that there exists an $a \in A$ such that $a(x) = b(x)$.

Let $Ax = V \subseteq E$ be a stable subspace. Then V admits a (stable) compliment: $E = V \oplus V^{\perp}$. Let $\pi : E \to V$ be the natural projection map. Then certainly π is A-linear and commutes with K, namely $\pi \in K$. Then:

$$\pi(b(x)) = b(\pi(x)) = b(x)$$

namely, $b(x) \in V = aX$, implying $a(x) = b(x)$, as we sought to show.

Lemma 1.3.6: Generator lemma

Let E be a semi-simple an faithful representation of A, $K = C_{\mathrm{End}_k(E)}(A)$, $B = C_{\mathrm{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\mathrm{End}_k(E)$). Let $x_1, ..., x_n \in E$ and $b \in B$. Then there exits values $a_i \in A$ such that:

$$ax_1 = bx_1,\ ax_2 = bx_2 \qquad \cdots \qquad ax_n = bx_n$$

***Proof*:**
Let F be the direct sum space of n copies of representation E. Then F is formed of elements $(y_1, ..., y_n)$, $y_i \in E$, on which A acts by

$$a \cdot (y_1, ..., y_n) = (ay_1, ..., ay_n)$$

The representation F is still semi-simple. The centralizer of A in $\mathrm{End}_k(F)$ is formed of matrices (u_{ij}), $u_{ij} \in K$, and it follows that, if we let B operate on F by the formula:

$$b \cdot (y_1, ..., y_n) = (by_1, ..., by_n)$$

the elements of $\mathrm{End}_k(F)$ thus defined will still be in the centralizer of the centralizer of A (in the theory of semi-groups, this is sometimes called the bicommutant). By lemma 1.3.5 applying to the point $x = (x_1, ..., x_n) \in F$, and to $b \in B$ giving us the desired result.

Serre write K_n for $M_n(K)$ for a division ring K, but we shall stick to $M_n(D)$ where our division ring is D instead of K. If D contains k in its center, we have:

$$M_n(D) = M_n(k) \otimes D$$

through $(a_{ij}) \otimes d \mapsto (a_{ij}d)$.

Let A be a k-algebra. Then A acting on itself by left multiplication gives something called the *regular left representation*. If $A = M_n(D)$, it has basis e_{ij}, $1 \leq i, j \leq n$. The elements e_{i,j_0} for a fixed j_0 gives a subspace of $M_n(D)$, say $N_{j_0} \subseteq M_n(D)$. The subspace N_{j_0} is a minimal left ideal (namely, an irreducible representation). Then we see that for any other choice of j_0, say j_1, we get an isomorphic irreducible representation N_{j_1}. The same holds for every finite-dimensional simple k-algebra.

Lemma 1.3.7: Decomposition Of A Simple Algebra Into Left Ideals

Let A be a nonzero finite-dimensional simple k-algebra. Then there are a simple left ideal I of A and elements $a_1, \ldots, a_m \in A$ such that $A = Ia_1 \oplus \cdots \oplus Ia_m$ and $Ia_j \cong I$ as A-modules for every j.

***Proof*:**
Left ideals of A are k-subspaces, since $\lambda x = (\lambda 1)x$ for $\lambda \in k$.

Step 1: A simple left ideal. As $A \neq 0$ is finite-dimensional, it has a nonzero left ideal I of smallest k-dimension among the nonzero left ideals. Every nonzero A-submodule of I is a left ideal of dimension at most $\dim_k I$, hence equal to I, so I is a simple A-module.

Step 2: A is a sum of copies of I. For $a \in A$, the set Ia is a left ideal, and $x \mapsto xa$ is a surjective A-module homomorphism $I \to Ia$. Its kernel is a submodule of the simple module I, so either $Ia = 0$ or $Ia \cong I$. The sum $J = \sum_{a \in A} Ia$ is a left ideal with $Jb \subseteq \sum_{a \in A} I(ab) \subseteq J$ for every $b \in A$, so J is a two-sided ideal, and $J \supseteq I \cdot 1 = I \neq 0$. As A is simple, $J = A$.

Step 3: A direct decomposition. Among the finite families $a_1, \ldots, a_m$ of elements of A such that each Ia_j is nonzero and the sum $Ia_1 + \cdots + Ia_m$ is direct (the empty family is one), choose one with m largest, which is possible because $m \dim_k I \leq \dim_k A$. Let $D = Ia_1 \oplus \cdots \oplus Ia_m$. For $a \in A$ with $Ia \neq 0$, the intersection $Ia \cap D$ is a submodule of the simple module Ia. If it were 0, the sum $D + Ia$ would be direct, and the family $a_1, \ldots, a_m, a$ would contradict the maximality of m. Hence $Ia \subseteq D$ for every $a \in A$, and $A = J = D$. Each Ia_j is nonzero, so $Ia_j \cong I$ by Step 2, as claimed.

Theorem 1.3.8: Representation of Simple Algebra

Let A be a finite-dimensional simple k-algebra. Then every representation of A is isomorphic to a direct sum of irreducible representations.

***Proof*:**
If $A = 0$, every A-module is 0, the empty direct sum, so assume $A \neq 0$.

Step 1: A is a semi-simple ring. By lemma 1.3.7, A is the internal direct sum of the simple submodules Ia_j, so A is a semi-simple A-module by lemma 1.1.7 and a semi-simple ring by corollary 1.1.9.

Step 2: Representations. A representation $\rho\colon A \to \mathrm{End}_k(V)$ is the left A-module V with $a \cdot v = \rho(a)v$, and its subrepresentations are exactly its A-submodules, so its irreducible subrepresentations are its simple submodules. Since A is a semi-simple ring, V is a semi-simple A-module by definition 1.1.2, that is a direct sum of simple modules by lemma 1.1.7. So every representation of A is a direct sum of irreducible representations, as claimed.

Corollary 1.3.9: Representations of a Simple Algebra Are Determined by Dimension

Let A be a finite-dimensional simple k-algebra. Then:

1. All representations of A are semi-simple (completely reducible in Serre's words)
2. Two representations of A which have the same dimension are isomorphic

***Proof*:**
Part (1) is theorem 1.3.8. For (2) we may assume $A \neq 0$, since otherwise every representation is 0. Let I and $a_1, \ldots, a_m$ be as in lemma 1.3.7, so that $A = Ia_1 \oplus \cdots \oplus Ia_m$ with each $Ia_j \cong I$.

Step 1: Every simple A-module is isomorphic to I. Let S be a simple A-module and $0 \neq s \in S$. The homomorphism $A \to S$, $a \mapsto as$, sends 1 to $s \neq 0$, so its restriction to some Ia_j is nonzero, and this restriction $Ia_j \to S$ is an isomorphism by lemma 1.1.5. Hence $S \cong Ia_j \cong I$.

Step 2: The dimension determines the representation. By part (1) and Step 1, a representation V is isomorphic to a direct sum $\bigoplus_{t \in T} I$ of copies of I, so $\dim_k V = |T| \dim_k I$ with $0 < \dim_k I < \infty$. If $\dim_k V$ is finite, then $|T| = \dim_k V / \dim_k I$, and if it is infinite, then $|T| = \dim_k V$. Two representations of the same dimension are therefore direct sums of the same number of copies of I, hence isomorphic, as claimed.

Let us now turn towards the tensor product of simple algebras.

Lemma 1.3.10: Centralizer of Tensor Product of Algebras

Let A, A' be two k-algebras, B, B' be sub k-algebras of A, A' (respectively) with C, C' their centralizers on A, A' (respectively). Then the centralizer of $B \otimes B'$ in $A \otimes A'$ is $C \otimes C'$.

***Proof*:**
Let us first find the centralizer of $B \otimes 1$. For this, let a_i' be a basis of A'. Any $x \in A \otimes A'$ can be written uniquely in the form:

$$x = \sum_i a_i \otimes a_i' \qquad \text{for appropriate } a_i \in A$$

If x commutes with $b \otimes 1$, then $a_i b = b a_i$ that is $a_i \in C$ for all i. We thus see that the centralizer of $B \otimes 1$ is $C \otimes A'$. Similarly, that of $1 \otimes B'$ is $A \otimes C'$. It follows that the centralizer of $B \otimes B'$ is equal to the intersection of the centralizers of $B \otimes 1$ and $1 \otimes B'$, namely

$$(C \otimes A') \cap (A \otimes C') = C \otimes C'$$

as we sought to show.

Thus, the centre of the tensor product of two algebras is the tensor product of the centre of the two algebras. Applying this to the special case of $A = M_n(k) \otimes D$ for some division ring D, then:

$$Z(A) = k \otimes Z(D)$$

Corollary 1.3.11: Centre of simple algebra is a Field

Let A be a simple k-algebra, finitely generated over k. Then $Z(A)$ is a field.

***Proof*:**
Consider $Z(A)$. As A is a finitely generated k-algebra, $Z(A)$ is a finitely generated k-algebra. For any $0 \neq z \in Z(A)$, consider zA. As z is central,we have that zA is two-sided ideal. As A is simple, it must be that $zA = A$, that is z is there exists a $w \in A$ such that $za = 1$. Therefore, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field. Since z was an arbitrary nonzero element, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field.

We shall next require to see the interaction between the tensor product of central simple algebras. We require the following lemma:

Theorem 1.3.12: Ideal Generation Tensor Product

Let A be a k-algebra and K' a left division ring with center k. Then every ideal $N \subseteq A \otimes K'$ is generated (as a left module over K') by its intersection with $A \otimes 1$.

Note that neither A nor K' are assumed finite over k

***Proof*:**
We will first note that we can make K' operate on the left on $A \otimes K'$ by:

$$k'(a \otimes k'') = a \otimes k'k''$$

If N is an ideal of the algebra $A \otimes K'$, it is in particular a K'-module subspace of $A \otimes K'$.

Let a_i be a basis of A over k; the elements $a_i \otimes 1$ equally form a basis of the K'-module $A \otimes K'$. Let x be a primitive element of N with respect to this basis (Bourbaki, Alg.II, paragraph 5):

$$x = \sum_i a_i \otimes k'_i$$

Consider the element $x \cdot m$ ($m \in K'$); we have $x \cdot m \in N$ (N is an ideal), and on the other hand:

$$x \cdot m = \sum_i a_i \otimes k'_i m$$

It then follows from the properties of primitive elements that we have:

$$k'_i m = n k'_i \text{ for all } i \quad (n \in K')$$

Since one of the k'_i is equal to 1, we have $m = n$, and the preceding equality simply means that k'_i commutes with all $m \in K'$, i.e., that $k'_i \in k$. We can then write: $x = a \otimes 1$, by setting: $a = \sum_i k'_i a_i$. We have thus shown that every primitive element x is in $A \otimes 1$; the theorem follows immediately since every vector space is generated by its primitive elements.

Theorem 1.3.13: Tensor Product CSA

Let A, A' be simple k-algebras finitely generated over k, where one of them is a CSA. Then $A \otimes A'$ is a simple k-algebra

***Proof*:**
Suppose that $A' = M_n(k) \otimes K'$ is central, in other words that the center of the left field K' is reduced to k. It will suffice to show that $A \otimes K'$ is a simple algebra. Indeed, if this is demonstrated, $A \otimes K' = M_m(k) \otimes K''$, and $A \otimes A' = M_m(k) \otimes M_n(k) \otimes K'' = M_{mn}(k) \otimes K'' = M_{nm}(K'')$. The algebra $A \otimes A'$ will thus be isomorphic to a matrix algebra for which simplicity is immediately verified. Then the general case of $A \otimes K'$ being simple follow from theorem 1.3.12, completing the proof.

Corollary 1.3.14: Tensor Product of Two CSA's

The tensor product of two CSA is a CSA

***Proof*:**
In the previous proof, it was enough for one of them to be a CSA.

Corollary 1.3.15: Opposite Algebra Tensoring

Let A be a CSA over k and A^{op} the opposite algebar. Then:

$$A \otimes A^{\text{op}} \cong M_{n^2}(k)$$

***Proof*:**
Let us denote by E the algebra A considered as a vector space over k. We can make A act naturally on E by the left or right by multiplication to create a left or right A-module. This amounts to identifying A and A^{op} as sub-algebras of the algebra $\text{End}_k(E)$ of k-endomorphisms of E. Since the algebras A and A^{op} commute in $\text{End}_k(E)$, we can define a canonical homomorphism of $A \otimes A^{\text{op}}$ into $\text{End}_k(E)$ which extends the isomorphisms: $A \to \text{End}_k(E)$ and $A^{\text{op}} \to \text{End}_k(E)$. The algebra $A \otimes A^{\text{op}}$ being simple, this homomorphism is an isomorphism; moreover, as;

$$[A \otimes A^{\text{op}} : k] = [A : k]^2 = n^2 = [\text{End}_k(E) : k]$$

this isomorphism maps $A \otimes A^{\text{op}}$ onto $\text{End}_k(E) = M_{n^2}(k)$, as we sought to show.

1.3.1 Brauer Group

Let A be a CSA over k. As A is simple, we get by Artin-Wedderburn that it is isomorphic to $M_n(D) \cong M_n(k) \otimes D$. For example, if $n = 1$ and $k = \mathbb{R}$ we get (by a classical theorem from algebraic topology) that $A \cong \mathbb{R}$ or $A \cong \mathbb{H}$ (the quaternions). If $n > 1$, we get $M_n(\mathbb{R})$ and $M_n(\mathbb{H})$ (which are no longer division algebras). We see that the division ring is one of the "characterizing" feature of the CSA (the other being its matrix ring), especially that up to Morita-equivalence we get the same category. We thus have the following natural equivalence we may define:

Definition 1.3.16: Brauer Equivalence

Let k-**CSA** be a collection of central simple algebras over k. Then define the equivalence relation $A \sim A'$ where the two are equivalent if and only if their associated division rings are isomorphic.

Proposition 1.3.17: Brauer Group

The Brauer equivalence on a collection of CSA's forms a commutative group under the tensor product. This group will be denoted $\text{Br}(k)$.

In Serre, this group is denoted G_k.

***Proof*:**
By corollary 1.3.14, it is closed under the tensor product, the equivalence class containing k is certainly the identity, and corollary 1.3.15 shows that every equivalence class has an inverse. Associativity and commutativity is a classical result for tensor products.

Example 1.3.18: Brauer Groups

1. Let $k = \overline{k}$. Then $\mathrm{Br}(\overline{k}) = 0$ by corollary 1.2.20.
2. If $k = \mathbb{R}$, then we shall show in the next section that $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2\mathbb{Z}$, namely we shall have $\mathbb{R}$ and $\mathbb{H}$, the quaternions.
3. Let k be a finite field. Then $\mathrm{Br}(k) = 0$, as all finite division rings are fields ([4, Wedderburn's Little Theorem]).
4. If $k = \mathbb{Q}_p$, then $\mathrm{Br}(k) \cong \mathbb{Q}/\mathbb{Z}$ (Serre references Algebren - Chap. VII - 2nd paragarph).
5. If $k = \mathbb{Q}$, $\mathrm{Br}(\mathbb{Q})$ will be a subgroup of

$$\mathbb{Z}/2\mathbb{Z} \oplus (\mathbb{Q}/\mathbb{Z})^{\oplus \mathbb{N}}$$

which can be thought of as a result similar to that in number theory where we study the "global field" $\mathbb{Q}$ by looking at all its localizations (note that $\mathbb{Z}/2\mathbb{Z}$ comes from $\mathrm{Br}(\mathbb{R})$!) Serre references Deuring for the proof (see Sere p.8)

Theorem 1.3.19: Field Extensions and CSA

Let A be a CSA over k, L/k a field extension (either finite or infinite). Then $A \otimes L$ is simple

***Proof*:**
If A is a division ring, then all ideals of $A \otimes L$ will be given with in part with the intersection of the ideals of L, which is simply (0), making $A \otimes L$ simple.

If A is any CSA, write $A \cong M_n(k) \otimes D$. Then:

$$A \otimes L \cong M_n(k) \otimes (D \otimes L) \overset{!}{=} M_n(k) \otimes M_m(D')$$

where for $\overset{!}{=}$, D' is the division ring naturally given by $D \otimes_k L$! But this gives the result.

Corollary 1.3.20: Index of CSA

Let A be a simple, central and finite algebra over k. Then $[A : k]$ is a perfect square.

***Proof*:**
Let L be an algebraically closed extension of k. The algebra $A \otimes L$ can be endowed with an algebra structure over L, and we have:

$$[A \otimes L : L] = [A : k]$$

But $A \otimes L$ is a simple algebra, central over L, and finite over L. It then follows from example 1.3.18[1] that it is a matrix algebra over L, and $[A \otimes L : L]$ is indeed a perfect square.

Looking at the field extension of a k-algebra, it is natural to consider that $- \otimes_k L$ gives a homomorphism

of Brauer groups $\varphi_{k,L} : \mathrm{Br}(k) \to \mathrm{Br}(L)$, namely due to the following result from linear algebra:

$$(A \otimes_k L) \otimes_L (A' \otimes_k L) \cong (A \otimes_k A') \otimes_k L$$

The kernel $\ker \varphi_{k,L}$ consists of all k-algebra's that become trivial (matrix algebras) when extending to L. Serre calls these algebras the algebras that *admit L as a field of decomposition.* (or admit a decomposition by L). More recently, the term *splitting field* is adopted due to the parallels to number theory. Naturally, if $L = \overline{k}$, then every algebra admits a decomposition as $\mathrm{Br}(\overline{k}) = 0$. We only require a finite extension:

Theorem 1.3.21: Splitting Field is Finite

The Brauer group $\mathrm{Br}(k)$ is the union of $\ker \varphi_{k,L}$ going over each L/k where L is a finite field extension. In particular, there exists a finite extension L/k such that for a CSA A over k, $[A \otimes_k L] = [1]$ in $\mathrm{Br}(L)$.

***Proof*:**
By example 1.3.18[1] we have that for a CSA A over $\overline{k}$, $A \otimes_k \overline{k} \cong M_n(D)$. Then the right hand side is generated by finitely many elements of the left hand side. We need only take the elements that map to the generators of the right hand side and to choose the appropriate field elements to get find L/k, as we sought to show.

1.4 Build up to Quaternion Algebras

Theorem 1.4.1: Skolem-Noether Theorem

Let A, B be CSA's over k, and let $f, g : A \to B$ be k-algebra isomorphism. Then there exists a unit $b \in B$ such that:

$$g(a) = bf(a)b^{-1}$$

In particular, every automorphism is an inner automorphism

***Proof*:**
Let's first prove the special case of $B = M_n(k)$. Then the isomorphisms $f, g : A \to M_n(k)$ define actions on k^n, or said differently these define two A-modules (with two different actions): V_f, V_g. Then using f, g we have an A-module isomorphism $g \circ f^{-1} : V_f \to V_g$. But any isomorphism between vector-spaces is representable as (unit) matrix in $M_n(k)$, and so we we get our value by linear algebra.

For the general case, we know that $B \otimes_k B^{\mathrm{op}} \cong M_n(k)$ and $A \otimes_k B^{\mathrm{op}}$ is simple as B^{op} is still simple (use theorem 1.3.13). As $B \otimes_k^B$ op is a (finite dimensional) matrix algebra, we may apply our special case to:

$$f \otimes 1,\ g \otimes 1 : A \otimes_k B^{\mathrm{op}} \to B \otimes_k B^{\mathrm{op}} \cong M_n(k)$$

which shows that:

$$(f \otimes 1)(a \otimes c) = b(g \otimes 1(a \otimes c)b^{-1}$$

which is true for all $a \in A$ and $c \in B^{\text{op}}$. Thus, it is true for $a = 1$, giving us:

$$1 \otimes c = b(1 \otimes c)b^{-1}$$

for all $c \in B^{\text{op}}$. Thus,

$$b \in Z_{B \otimes_k B^{\text{op}}}(k \otimes B^{\text{op}}) = B \otimes k$$

Hence, as constants move through the tensor we get that $b = b \otimes_k 1$. And so, by setting $z = 1$, we find:

$$f(a) = b' g(a) {b'}^{-1}$$

as we sought to show.

Theorem 1.4.2: Centralizer of Subalgebras

Let A be a CSA over k, $B \subseteq A$ a simple subalgebra, $C = C_A(B)$ the centralizer of B in A. Then C is simple, $B = C_A(C)$, and:

$$[B : k][C : k] = [A : k]$$

***Proof*:**
Let E be B with only the structure of a k-vector space. Then B acts on E via left multiplication; this action isomorphically embeds B into $\text{End}_k(E)$. Then the centralizer of B in $\text{End}_k(E)$ is nothing more than the algebra of right translations by E, that is B^{op}.

Given this, we can consider the simple algebra $A \otimes_k \text{End}_k(E)$. We can now embed B into this tensored algebra in two ways: via $B \otimes 1$ or $1 \otimes B$. Their centralizers are $C \otimes \text{End}_k(E)$ and $A \otimes B^{\text{op}}$ respectively.

By theorem 1.4.1, there exists an inner automorphism of $A \otimes_k \text{End}_k(E)$ mapping $B \otimes 1$ to $1 \otimes B$; it is then a simple algebra exercise to see that this inner automorphism induces a map on the centralizers. As a consequence, as $A \otimes B^{\text{op}}$ is simple, so is $C \otimes \text{End}_k(E)$, which must make C simple.

Furthermore:

$$[C \otimes \text{End}_k(E) : k] = [C : k][B : k]^2 \qquad \text{and} \qquad [A \otimes B^{\text{op}} : k][A : k][B : k]$$

From which we get $[A : k] = [B : k][C : k]$. What's left to show is the centralizer of C in A is B. Say that B' is this centralizer. Then $B \subseteq B'$. Then as $[A : k] = [B : k][C : k]$, we get:

$$[B' : k] = [B : k]$$

But then as they contain we must get $B = B'$, completing the proof.

2

The Jacobson Radical and Non-Semisimple Algebras

The Artin-Wedderburn theorem (theorem 1.2.16) is a complete answer to a narrow question: it classifies the semisimple rings and, in the process, shows that a semisimple k-algebra is nothing more than a finite product of matrix rings over division algebras. When R is semisimple, every module splits into simples, every short exact sequence splits, and the entire representation theory of R collapses onto the finite list of its simple modules. This is the best possible situation, and it is also the exception. The algebras whose representation theory occupies most of this book are precisely the ones that fail to be semisimple: the group algebra $k[G]$ when $\mathrm{char}(k)$ divides $|G|$ (where Maschke's theorem breaks down), the truncated polynomial algebra $k[x]/(x^n)$, universal enveloping algebras of Lie algebras, and the path algebra of a quiver with an oriented cycle. For all of these, some module fails to be a direct sum of simples, some sequence refuses to split, and Wedderburn has nothing to say.

This chapter builds the tools that take over where Wedderburn stops. The first of these, and the subject of the present section, is the *Jacobson radical* $\mathrm{Jac}(A)$: the two-sided ideal that measures, in a single invariant, exactly how far A is from being semisimple. It is the obstruction that vanishes precisely when A is semisimple, and dividing it out always produces a semisimple ring $A/\mathrm{Jac}(A)$ whose simple modules are the same as those of A. In the sections that follow, the radical becomes the engine behind the theory of indecomposable modules and the Krull-Schmidt theorem (which replaces "decomposition into simples" with "decomposition into indecomposables"), and behind the construction of projective covers, which repair the failure of projectivity that semisimplicity had guaranteed for free. The radical is where the non-semisimple story begins.

2.1 The Jacobson Radical

Throughout chapter 1 the simple modules did all the work: a semisimple ring was assembled out of its minimal left ideals, and every module was a direct sum of simples. When a ring is *not* semisimple, the simple modules are still present, but they no longer see everything. Some elements of the ring act as zero on every simple module without being zero themselves, and these elements form an ideal that records the discrepancy. To isolate them, we begin one level down, with modules, and ask which part of a module is invisible to all of its simple quotients.

The maximal submodules of a module M are exactly the kernels of the surjections $M \twoheadrightarrow S$ onto simple modules: a submodule $N \subseteq M$ is maximal precisely when M/N is simple, and every simple quotient of M is of the form M/N for such an N (the classification of simples as quotients by maximal left ideals is lemma 1.1.3). An element lying in every maximal submodule is one that every simple quotient sends to zero. Collecting these gives the radical of a module.

Definition 2.1.1: Radical of a Module

Let M be a left A-module. The *radical* of M, written $\operatorname{rad}(M)$, is the intersection of all maximal submodules of M:

$$\operatorname{rad}(M) = \bigcap_{\substack{N \subseteq M \\ N \text{ maximal}}} N$$

If M has no maximal submodules, the intersection is empty and $\operatorname{rad}(M) := M$ by convention.

Since each maximal submodule is the kernel of a map onto a simple module, an equivalent description is

$$\operatorname{rad}(M) = \bigcap_{\substack{\varphi\colon M \to S \\ S \text{ simple}}} \ker(\varphi)$$

so that $\operatorname{rad}(M)$ is the set of elements killed by *every* homomorphism from M into a simple module. The quotient $M/\operatorname{rad}(M)$ is the largest quotient of M that embeds into a product of simple modules; it is the *semisimple quotient* of M, the part of M that its simple quotients can detect. Everything inside $\operatorname{rad}(M)$ is invisible to them.

Example 2.1.2: The Radical of a Cyclic $\mathbb{Z}$-Module

Take $A = \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. The maximal submodules of M correspond to the maximal ideals of $\mathbb{Z}$ containing $12\mathbb{Z}$, that is to the primes dividing 12, namely 2 and 3. The corresponding maximal submodules are $2\mathbb{Z}/12\mathbb{Z}$ and $3\mathbb{Z}/12\mathbb{Z}$, with simple quotients $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$. Their intersection is

$$\operatorname{rad}(\mathbb{Z}/12\mathbb{Z}) = \frac{2\mathbb{Z}}{12\mathbb{Z}} \cap \frac{3\mathbb{Z}}{12\mathbb{Z}} = \frac{6\mathbb{Z}}{12\mathbb{Z}}$$

so $\operatorname{rad}(M) = 6\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, and the semisimple quotient is $M/\operatorname{rad}(M) \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. The radical has stripped $\mathbb{Z}/12\mathbb{Z}$ down to its squarefree quotient: the extra factors of 2 and 3 that no simple quotient can see are exactly what lies in $6\mathbb{Z}/12\mathbb{Z}$.

Applying this construction to the ring itself, viewed as a module over itself, produces the central object of the chapter. The radical of A as a left A-module is the intersection of its maximal *left* ideals.

Definition 2.1.3: Jacobson Radical

Let A be a ring. The *Jacobson radical* of A, written $\mathrm{Jac}(A)$, is the radical of A regarded as a left module over itself, that is the intersection of all maximal left ideals of A:

$$\mathrm{Jac}(A) = \mathrm{rad}({}_AA) = \bigcap_{\substack{\mathfrak{m}\subseteq A \\ \mathfrak{m} \text{ maximal left ideal}}} \mathfrak{m}$$

The definition is one-sided: it uses left ideals, and a priori $\mathrm{Jac}(A)$ need only be a left ideal. One of the first surprises is that the resulting set is symmetric, a two-sided ideal that could equally have been defined with maximal *right* ideals. This symmetry, and several other faces of the same object, are the content of the next theorem. Before stating it, recall that an element $u \in A$ is *left-invertible* if there exists $v \in A$ with $vu = 1$.

Theorem 2.1.4: Characterizations of the Jacobson Radical

Let A be a ring and let $a \in A$. The following subsets of A coincide, and their common value is $\mathrm{Jac}(A)$:

1. the intersection of all maximal left ideals of A;
2. the intersection of the annihilators $\mathrm{Ann}_A(S)$ of all simple left A-modules S;
3. the set $\{a \in A \,|\, 1 - xa \text{ is left-invertible for all } x \in A\}$.

Consequently, $\mathrm{Jac}(A)$ is a two-sided ideal of A, and it is the largest two-sided ideal of A that annihilates every simple left A-module.

The three descriptions look at the same ideal from three angles. The first is the definition. The second recasts it representation-theoretically: $\mathrm{Jac}(A)$ is exactly the set of elements that act as zero on every simple module, which is why it is invisible to the simple representation theory. The third is an intrinsic ring-theoretic criterion that never mentions modules at all, and it is the one that makes the two-sidedness fall out for free, since it is symmetric in a way the definition is not.

***Proof*:**
We prove (1) = (2), then (2) = (3), and finally read off the two-sidedness and maximality statements.

Step 1: (1) = (2). Let J_1 denote the intersection of all maximal left ideals and J_2 the intersection of the annihilators of all simple left modules. If S is a simple left A-module, then for any $0 \neq s \in S$ we have $S = As$ (as As is a nonzero submodule of the simple module S), and by corollary 1.1.4 the map $a \mapsto as$ gives $S \cong A/\mathrm{Ann}_A(s)$ with $\mathrm{Ann}_A(s)$ a maximal left ideal. The annihilator of the whole module is

$$\mathrm{Ann}_A(S) = \bigcap_{0\neq s\in S} \mathrm{Ann}_A(s)$$

an intersection of maximal left ideals, so $J_1 \subseteq \mathrm{Ann}_A(S)$ for every simple S, giving $J_1 \subseteq J_2$. Conversely, if $\mathfrak{m}$ is any maximal left ideal, then $S = A/\mathfrak{m}$ is a simple left module and $\mathrm{Ann}_A(S) \subseteq \mathfrak{m}$ (since $\mathrm{Ann}_A(S)$ kills the coset $1 + \mathfrak{m}$, so $\mathrm{Ann}_A(S) \cdot 1 \subseteq \mathfrak{m}$). Hence $J_2 \subseteq \mathfrak{m}$ for every maximal left ideal $\mathfrak{m}$, giving $J_2 \subseteq J_1$. Thus $J_1 = J_2$.

Step 2: (2) $\subseteq$ (3). Let $a \in J_2$, so a annihilates every simple left module, and fix $x \in A$. We must show $1 - xa$ is left-invertible. If not, then $A(1-xa)$ is a proper left ideal, hence contained in some maximal left ideal $\mathfrak{m}$ (using that $1 \notin A(1-xa)$ and Zorn's lemma). The quotient $S = A/\mathfrak{m}$ is simple, so a annihilates it: $a \cdot (1 + \mathfrak{m}) = 0$, meaning $a \in \mathfrak{m}$. But $xa \in \mathfrak{m}$ as well, since $\mathfrak{m}$ is a left ideal, and $1 - xa \in \mathfrak{m}$ since $1 - xa \in A(1-xa) \subseteq \mathfrak{m}$. Adding, $1 = (1-xa) + xa \in \mathfrak{m}$, contradicting that $\mathfrak{m}$ is proper. Hence $1 - xa$ is left-invertible for every x, so $a \in J_3$, the set in (3).

Step 3: (3) $\subseteq$ (2). Let $a \in J_3$, and suppose toward a contradiction that a does *not* annihilate some simple left module S. Then $as \neq 0$ for some $s \in S$, and since As is a nonzero submodule of the simple module S we have $As = S$. In particular $s \in As$, so $s = xas$ for some $x \in A$, giving $(1-xa)s = 0$. By hypothesis $1 - xa$ is left-invertible, say $v(1-xa) = 1$; applying v gives $s = v(1-xa)s = 0$, contradicting the choice of s. Hence a annihilates every simple module, so $a \in J_2$.

Combining the three steps, $J_1 = J_2 = J_3 = \mathrm{Jac}(A)$.

Step 4: two-sidedness and maximality. As the intersection of the left ideals $\mathrm{Ann}_A(S)$, description (2) shows $\mathrm{Jac}(A)$ is a left ideal. But each $\mathrm{Ann}_A(S)$ is in fact a two-sided ideal: if $a \in \mathrm{Ann}_A(S)$ and $b \in A$, then for all $s \in S$ we have $(ab)s = a(bs) = 0$, so $ab \in \mathrm{Ann}_A(S)$. An intersection of two-sided ideals is two-sided, so $\mathrm{Jac}(A)$ is a two-sided ideal. Finally, if I is any two-sided ideal annihilating every simple left module, then $I \subseteq \mathrm{Ann}_A(S)$ for every simple S, hence $I \subseteq \bigcap_S \mathrm{Ann}_A(S) = \mathrm{Jac}(A)$; thus $\mathrm{Jac}(A)$ is the largest such ideal, completing the proof.

The left-invertibility criterion (3) deserves a second look, because it is what makes the radical robust. It says a lies in the radical exactly when $1 - xa$ can never be turned into a non-unit by any left multiplier x. A standard manipulation upgrades "left-invertible" to "two-sided invertible" for these particular elements: if $a \in \mathrm{Jac}(A)$ and $v(1-xa) = 1$, then $v = 1 + vxa = 1 - (-vx)a$, and since $a \in \mathrm{Jac}(A)$ the element v is itself left-invertible by the same criterion; a left-invertible element with a left-invertible left inverse is a unit. So for $a \in \mathrm{Jac}(A)$ the element $1 - xa$ is a two-sided unit for every x, and by the right-handed version of the whole theorem $1 - ax$ is a unit as well. This is the working form of the radical: $\mathrm{Jac}(A)$ is the set of elements a such that $1 - xay$ is a unit for all $x, y \in A$, and it is exactly this that will drive Nakayama's lemma below.

Description (2) also settles the two-sided nature of the radical without any of the asymmetry of the original definition, and it makes precise the sense in which the radical is invisible to representation theory. If A is semisimple, every module is a sum of simples, so an element killing all simples kills everything: $\mathrm{Jac}(A) = 0$. The content of the non-semisimple theory is that $\mathrm{Jac}(A)$ can be nonzero, and it is this ideal that the simple modules fail to detect.

Before turning to that theory, the radical of a ring is anchored by two computations. The first is the archetype of a local non-semisimple algebra.

Example 2.1.5: The Radical of a Truncated Polynomial Algebra

Let k be a field and let $A = k[x]/(x^n)$ with $n \geq 1$, writing $\bar{x}$ for the image of x. This is a commutative local ring: an element $f(\bar{x}) = c_0 + c_1\bar{x} + \cdots + c_{n-1}\bar{x}^{n-1}$ is a unit if and only if $c_0 \neq 0$, since when $c_0 = 0$ the element is nilpotent ($\bar{x}^n = 0$ forces $f(\bar{x})^n = 0$), and when $c_0 \neq 0$ one inverts using the geometric series, which terminates because $\bar{x}^n = 0$:

$$(c_0 + \bar{x}g)^{-1} = c_0^{-1}\big(1 + c_0^{-1}\bar{x}g\big)^{-1} = c_0^{-1}\sum_{j=0}^{n-1}(-c_0^{-1}\bar{x}g)^j$$

The non-units are therefore exactly the elements with $c_0 = 0$, which is the ideal $(\bar{x})$. In a commutative ring the maximal ideals are the maximal left ideals, and here there is a single maximal ideal, namely $(\bar{x})$, because $A/(\bar{x}) \cong k$ is a field while any element outside $(\bar{x})$ is a unit. Hence

$$\mathrm{Jac}(A) = (\bar{x}) = \{c_1\bar{x} + c_2\bar{x}^2 + \cdots + c_{n-1}\bar{x}^{n-1} \big| c_i \in k\}$$

This ideal is nilpotent: $(\bar{x})^n = (\bar{x}^n) = 0$, so already $\mathrm{Jac}(A)^n = 0$. The quotient $A/\mathrm{Jac}(A) = A/(\bar{x}) \cong k$ is a field, hence semisimple, which is the general phenomenon established in theorem 2.1.9. The single simple A-module is $k = A/(\bar{x})$, and $\bar{x}$ acts on it as zero, confirming that $\mathrm{Jac}(A) = (\bar{x})$ annihilates every simple module.

The second computation is the model of a non-commutative, non-local radical, where the one-sided origin of the definition is easiest to see in matrices.

Example 2.1.6: The Radical of Upper-Triangular Matrices

Let k be a field and let $A = T_n(k)$ be the ring of $n \times n$ upper-triangular matrices over k. Write $N \subseteq A$ for the strictly upper-triangular matrices (zero on and below the diagonal). The claim is $\mathrm{Jac}(A) = N$.

N annihilates every simple module. For each $1 \leq i \leq n$ there is a one-dimensional A-module $S_i = k$ on which an upper-triangular matrix $a = (a_{pq})$ acts by its ith diagonal entry, $a \cdot v = a_{ii}v$; this is a ring homomorphism $A \to k$ because the (i, i) entry of a product of upper-triangular matrices is the product of the (i, i) entries. The S_i are pairwise non-isomorphic simple A-modules. Any strictly upper-triangular matrix has all diagonal entries zero, so it acts as 0 on each S_i; thus $N \subseteq \mathrm{Ann}_A(S_i)$ for all i, and $N \subseteq \mathrm{Jac}(A)$ by theorem 2.1.4.

Nothing outside N lies in the radical. If $a \in A \setminus N$, some diagonal entry a_{ii} is nonzero, so a acts invertibly on S_i, and a does not annihilate the simple module S_i. By part (2) of theorem 2.1.4, $a \notin \mathrm{Jac}(A)$. Combining, $\mathrm{Jac}(A) = N$.

As a check on the intrinsic criterion, N is nilpotent with $N^n = 0$ (each multiplication by a strictly upper-triangular matrix pushes nonzero entries one further diagonal above the main diagonal), so for $a \in N$ the element $1 - xa$ has the form $1 -$ (nilpotent) and is a unit via the terminating geometric series $\sum_{j \geq 0}(xa)^j$. The quotient is $A/N \cong k^n$, the diagonal matrices, which is semisimple, and there are exactly n simple modules $S_1, \ldots, S_n$, matching the n factors of k^n.

The two examples share a feature: in each, $\mathrm{Jac}(A)$ turned out to be nilpotent, and the geometric series that inverts $1 - xa$ terminated precisely because a power of the radical vanished. This is not special to small examples. For the finite-dimensional and, more generally, left Artinian algebras that concern us, the radical is always nilpotent. The route to that fact runs through Nakayama's lemma, which converts the invertibility criterion into a statement about modules being forced to zero.

The lemma answers a question that recurs throughout module theory: if multiplying a module by the radical does not shrink it, must the module already be zero? The commutative version, over a local ring or with a general ideal contained in the radical, is proved in the companion volume [3] and will not be re-derived here; the module form over an arbitrary ring is the form needed here, and its proof uses only the invertibility criterion from theorem 2.1.4.

Lemma 2.1.7: Nakayama's Lemma

Let A be a ring and M a finitely generated left A-module.

1. If $\mathrm{Jac}(A)\,M = M$, then $M = 0$.
2. If $N \subseteq M$ is a submodule with $M = N + \mathrm{Jac}(A)\,M$, then $N = M$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

1. Suppose $M \neq 0$ and $JM = M$. Since M is finitely generated, choose a generating set $m_1, \ldots, m_r$ with r minimal, so no fewer than r elements generate M; as $M \neq 0$ we have $r \geq 1$. From $M = JM$ we may write the last generator as

$$m_r = \sum_{i=1}^{r} a_i m_i \qquad \text{with } a_i \in J$$

   Isolating m_r gives $(1-a_r)m_r = \sum_{i=1}^{r-1} a_i m_i$. Because $a_r \in J = \mathrm{Jac}(A)$, the element $1-a_r$ is a unit (the working form of theorem 2.1.4: $1-a_r$ is left-invertible, and for radical elements this upgrades to a two-sided inverse). Multiplying on the left by $(1-a_r)^{-1}$ expresses m_r as an A-combination of $m_1, \ldots, m_{r-1}$, so these $r-1$ elements already generate M, contradicting the minimality of r. Hence $M = 0$.

2. Apply part (1) to the finitely generated module M/N. Its finite generation is inherited from M (the images of generators of M generate the quotient). The hypothesis $M = N + JM$ gives, in the quotient, $M/N = J\,(M/N)$, since the image of JM is $J(M/N)$ and the image of N is 0. By part (1), $M/N = 0$, that is $N = M$, completing the proof.

The finite-generation hypothesis is not optional. Over the local ring $A = \mathbb{Z}_{(p)}$ (integers localized at p, whose radical is $p\mathbb{Z}_{(p)}$) the module $M = \mathbb{Q}$ satisfies $pM = M$ yet $M \neq 0$; it fails to be finitely generated, and the minimal-generator argument has nothing to grip. Part (2) is the form used in practice: it says that generators of $M/\mathrm{Jac}(A)M$ lift to generators of M. Once $\mathrm{Jac}(A)M$ is known (below) to be a "small" submodule, this becomes the statement that a minimal generating set of M is detected by the semisimple quotient $M/\mathrm{Jac}(A)M$, which is the seed of the theory of projective covers in a later section.

We now specialize to the setting where the radical is best behaved. A ring A is *left Artinian* if it satisfies the descending chain condition on left ideals; every finite-dimensional algebra over a field is left Artinian, since a descending chain of left ideals is a descending chain of subspaces and dimension cannot drop forever. In this setting the radical is not only an intersection of maximal left ideals but a nilpotent ideal, and it is the largest one.

Theorem 2.1.8: The Radical Is Nilpotent for Artinian Rings

Let A be a left Artinian ring. Then $\mathrm{Jac}(A)$ is nilpotent: there exists $m \geq 1$ with $\mathrm{Jac}(A)^m = 0$. Moreover, $\mathrm{Jac}(A)$ contains every nilpotent left ideal and every nilpotent two-sided ideal of A, so it is the largest nilpotent one-sided or two-sided ideal.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

Step 1: J is nilpotent. The descending chain of two-sided ideals

$$J \supseteq J^2 \supseteq J^3 \supseteq \cdots$$

is a descending chain of left ideals, so by the descending chain condition it stabilizes: there is m with $I := J^m = J^{m+1} = J\,I$. Suppose for contradiction that $I \neq 0$. Consider the set Σ of left ideals $L \subseteq A$ with $IL \neq 0$; it is nonempty because $I \cdot I = J^{2m} = I \neq 0$ shows $I \in \Sigma$ (using $J^{2m} = (J^m)^2 = I \cdot I$ and $J^{2m} = J^m = I$ since the chain stabilized). By the descending chain condition, Σ has a minimal element L_0. Choose $x \in L_0$ with $Ix \neq 0$; then $Ix \subseteq L_0$ is a left ideal with $I(Ix) = I^2x = Ix \neq 0$ (as $I^2 = I$), so $Ix \in \Sigma$, and minimality forces $Ix = L_0$. In particular $x \in L_0 = Ix$, so $x = ax$ for some $a \in I \subseteq J$. Then $(1-a)x = 0$, and since $a \in J$ the element $1 - a$ is a unit, giving $x = 0$ and hence $Ix = 0$, a contradiction. Therefore $I = J^m = 0$.

Step 2: J contains every nilpotent one-sided or two-sided ideal. Let I be a nilpotent left ideal, say $I^t = 0$. For any $a \in I$ and any $x \in A$, the element xa lies in I (a left ideal absorbs left multiplication), so $(xa)^t \in I^t = 0$; thus xa is nilpotent. A nilpotent element y with $y^t = 0$ makes $1 - y$ a unit, with inverse the terminating sum $1 + y + y^2 + \cdots + y^{t-1}$. Hence $1 - xa$ is a unit, in particular left-invertible, for every $x \in A$, so $a \in \mathrm{Jac}(A)$ by part (3) of theorem 2.1.4. This holds for every $a \in I$, so $I \subseteq J$. A nilpotent two-sided ideal is in particular a nilpotent left ideal, so the same conclusion applies, completing the proof.

For a left Artinian ring, then, "the Jacobson radical" and "the largest nilpotent ideal" name the same object; the two definitions of the radical current in the literature (intersection of maximal left ideals versus largest nilpotent ideal) agree exactly when A is Artinian. The nilpotence is what powers the terminating geometric series seen in the truncated-polynomial and upper-triangular examples, and it is the property that fails for general rings. It also gives a concrete way to detect non-semisimplicity: a single nonzero nilpotent element of a two-sided ideal already forces the radical to be nonzero. This is exactly the obstruction that appears for modular group algebras.

Having pinned down $\mathrm{Jac}(A)$ as a nilpotent ideal, the pieces are in place for the structural payoff. Dividing A by its radical removes precisely the elements that the simple modules could not see, and what remains is semisimple.

Theorem 2.1.9: Semisimplicity of the Quotient by the Radical

Let A be a left Artinian ring. Then $A/\mathrm{Jac}(A)$ is a semisimple ring, and hence Artin-Wedderburn (theorem 1.2.16) applies to it. Moreover, the simple left A-modules are exactly the simple left $A/\mathrm{Jac}(A)$-modules: every simple A-module is annihilated by $\mathrm{Jac}(A)$ and so factors through $A/\mathrm{Jac}(A)$, and every simple $A/\mathrm{Jac}(A)$-module is a simple A-module via the quotient map $A \twoheadrightarrow A/\mathrm{Jac}(A)$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $\bar{A} = A/J$.

Step 1: $\bar{A}$ is semisimple. By corollary 1.1.9 it suffices to show that $\bar{A}$ is semisimple as a left module over itself, that is that $\mathrm{rad}(\bar{A}) = 0$. The maximal left ideals of $\bar{A}$ are exactly the images $\mathfrak{m}/J$ of the maximal left ideals $\mathfrak{m}$ of A (which all contain $J = \mathrm{Jac}(A)$), by the correspondence

theorem. Their intersection is

$$\mathrm{Jac}(\bar{A}) = \bigcap_{\mathfrak{m} \supseteq J} \mathfrak{m}/J = \Big(\bigcap_{\mathfrak{m}} \mathfrak{m}\Big) \Big/ J = J/J = 0$$

So $\mathrm{Jac}(\bar{A}) = 0$. It remains to convert this into semisimplicity, which is where the Artinian hypothesis is spent: $\bar{A}$ is Artinian (a quotient of the Artinian ring A), and for an Artinian ring $\mathrm{Jac}(\bar{A}) = 0$ forces $\bar{A}$ to be semisimple. Concretely, since $\bar{A}$ is Artinian we may pick finitely many maximal left ideals $\mathfrak{m}_1/J, \ldots, \mathfrak{m}_r/J$ whose intersection is already 0 (take a minimal finite subintersection, which exists by the descending chain condition and equals the full intersection $\mathrm{Jac}(\bar{A}) = 0$). The map

$$\bar{A} \longrightarrow \bigoplus_{i=1}^{r} \bar{A}/(\mathfrak{m}_i/J)$$

is injective, since its kernel is $\bigcap_i (\mathfrak{m}_i/J) = 0$. Each summand $\bar{A}/(\mathfrak{m}_i/J)$ is a simple $\bar{A}$-module, so $\bar{A}$ embeds as a submodule of a semisimple module and is therefore semisimple by corollary 1.1.8. Thus $\bar{A}$ is a semisimple $\bar{A}$-module, and by corollary 1.1.9 a semisimple ring. Artin-Wedderburn now applies to $\bar{A}$.

Step 2: the simple modules coincide. Let S be a simple left A-module. By part (2) of theorem 2.1.4, $J = \mathrm{Jac}(A)$ annihilates every simple A-module, so $JS = 0$. Hence the A-action on S factors through the quotient $\bar{A} = A/J$: setting $(\bar{a})s := as$ is well-defined because J acts as zero. The A-submodules and $\bar{A}$-submodules of S are the same subsets (a subgroup of S is A-stable if and only if it is $\bar{A}$-stable, since the actions agree), so S is simple as an $\bar{A}$-module precisely because it is simple as an A-module.

Conversely, let T be a simple left $\bar{A}$-module. Pulling back the action along the surjection $\pi\colon A \twoheadrightarrow \bar{A}$, that is defining $a \cdot t := \pi(a)\, t$, makes T an A-module. Its A-submodules are exactly its $\bar{A}$-submodules (again the two actions have the same invariant subgroups, because π is surjective), so T is simple as an A-module. These two passages are mutually inverse, so the simple A-modules and the simple $\bar{A}$-modules are identified, as we sought to show.

This is the structural role of the radical: it is the exact obstruction one divides out to recover the Wedderburn world. The simple modules of A are unchanged by the passage to $A/\mathrm{Jac}(A)$, so the finite list of simples that a semisimple ring enjoys is available for *any* Artinian ring, read off from its semisimple quotient. What the radical takes away is not the simples but the way they fit together: over A the simples can be glued into indecomposable modules that do not split, and the gluing data lives in $\mathrm{Jac}(A)$. The theorem says that for an Artinian A there are finitely many simple A-modules, one for each simple factor of the Wedderburn decomposition of $A/\mathrm{Jac}(A)$, and this is the starting point for the character theory and the block theory developed later.

The special case $\mathrm{Jac}(A) = 0$ returns to chapter 1.

Corollary 2.1.10: Semisimplicity via the Radical

Let A be a left Artinian ring. Then A is semisimple if and only if $\mathrm{Jac}(A) = 0$.

***Proof*:**
If A is semisimple, then by corollary 1.1.9 it is a semisimple left module over itself, so its radical (the intersection of its maximal left ideals) is 0; that is, $\mathrm{Jac}(A) = \mathrm{rad}({}_AA) = 0$. Indeed, a semisimple module is a direct sum of simples, and its radical, the intersection of maximal submodules, is 0: writing $A = \bigoplus_i I_i$ as a sum of minimal left ideals (lemma 1.1.10), each $\bigoplus_{j\neq i} I_j$ is a maximal submodule, and their intersection is 0.

Conversely, if $\mathrm{Jac}(A) = 0$, then $A = A/\mathrm{Jac}(A)$, and by theorem 2.1.9 the quotient $A/\mathrm{Jac}(A)$ is semisimple; hence A itself is semisimple, completing the proof.

The corollary makes the radical the precise measure of the failure of semisimplicity: A is semisimple exactly when the measurement returns zero, and Artin-Wedderburn (theorem 1.2.16) describes what A looks like in that case, namely a product $M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$. For a general Artinian A, the same product describes $A/\mathrm{Jac}(A)$ rather than A, and the difference between A and this product is stored in the nilpotent ideal $\mathrm{Jac}(A)$. The truncated-polynomial algebra $k[x]/(x^n)$ with $n \geq 2$ is the simplest case where the measurement is nonzero: its radical $(\bar{x})$ is a nonzero nilpotent ideal, so it is not semisimple, even though its semisimple quotient is the field k.

The most consequential source of non-semisimple algebras in representation theory is the group algebra $k[G]$ when the characteristic of k divides the order of the group. Maschke's theorem guarantees semisimplicity precisely when $\mathrm{char}(k) \nmid |G|$; the radical detects, and measures, the failure in the opposite case.

Example 2.1.11: The Radical of a Modular Group Algebra
Let $G = \mathbb{Z}/p\mathbb{Z}$ be cyclic of order p and let k be a field. Writing g for a generator, the group algebra is

$$k[G] \cong k[t]/(t^p - 1)$$

via $g \mapsto t$, since g has order p. The structure of $\mathrm{Jac}(k[G])$ depends entirely on whether $\mathrm{char}(k)$ divides $|G| = p$.

The modular case $\mathrm{char}(k) = p$. In characteristic p the Frobenius identity gives $t^p - 1 = (t-1)^p$, so

$$k[G] \cong k[t]/\big((t-1)^p\big) \cong k[s]/(s^p) \qquad (s := t-1)$$

the last isomorphism being the change of variable $s = t - 1$. This is precisely the truncated polynomial algebra of example 2.1.5 with $n = p$, so its radical is

$$\mathrm{Jac}(k[G]) = (s) = (g-1)$$

the ideal generated by $g - 1$, which is the augmentation ideal (the kernel of the augmentation map $k[G] \to k$ sending $g \mapsto 1$). It is nilpotent, $(g-1)^p = g^p - 1 = 0$, and nonzero, so $k[G]$ is not semisimple: Maschke's theorem fails exactly because $p = \mathrm{char}(k)$ divides $|G|$. The unique simple $k[G]$-module is the trivial module k, on which g acts as the identity, so $g - 1$ acts as zero, confirming $\mathrm{Jac}(k[G]) = (g-1)$ annihilates the only simple module.

The coprime case $\mathrm{char}(k) \nmid p$. If the characteristic does not divide p, then the polynomial $t^p - 1$ is separable (its derivative pt^{p-1} is nonzero and shares no root with it), so it has no repeated factors. Over a field containing a primitive pth root of unity ζ it splits as $t^p - 1 = \prod_{i=0}^{p-1}(t - \zeta^i)$, and the Chinese remainder theorem gives

$$k[G] \cong \prod_{i=0}^{p-1} k[t]/(t - \zeta^i) \cong k^p$$

a product of p copies of k, which is semisimple with radical $\mathrm{Jac}(k[G]) = 0$. (Over a field without the roots of unity, $t^p - 1$ still factors into distinct irreducible separable factors and $k[G]$ is a product of finite separable field extensions of k, again semisimple with zero radical.) This is Maschke's theorem in the coprime case, seen through the radical: separability of $t^p - 1$ is exactly the condition $\mathrm{char}(k) \nmid |G|$, and it forces $\mathrm{Jac}(k[G]) = 0$.

The contrast in the two cases is the whole point of the chapter in miniature. When $\mathrm{char}(k) \nmid |G|$ the group algebra is semisimple, its radical is zero, and the representation theory of G over k is governed by Wedderburn and character theory. When $\mathrm{char}(k) \mid |G|$ the radical becomes a nonzero nilpotent ideal, the group algebra acquires indecomposable modules that are not simple, and one enters modular representation theory, where the radical, the indecomposables, and the projective covers built in the following sections are the working tools.

2.2 Indecomposable Modules and Krull-Schmidt

The previous chapter rested on a single organizing principle: over a semisimple ring, every module splits into simple pieces, and Artin-Wedderburn turns that splitting into a matrix description of the ring itself. The trouble is that semisimplicity is a strong hypothesis. The ring $k[x]/(x^2)$ is a two-dimensional k-algebra, and the module $k[x]/(x^2)$ over itself has a proper nonzero submodule $(x)/(x^2)$, yet that submodule has no complement: there is no way to write $k[x]/(x^2) = (x)/(x^2) \oplus N$. So this module does not break into simples at all. Once the semisimple world is left behind, the simple modules stop being the building blocks, and the decomposition theory of lemma 1.1.7 collapses.

What survives is a weaker notion of "building block". A module that cannot be split into two nonzero summands is *indecomposable* (definition 1.1.1), and every module of finite length is at least a finite direct sum of indecomposables, for the trivial reason that one keeps splitting until nothing splits further. The content of this section is that this crude decomposition is in fact canonical: the list of indecomposable summands, taken up to isomorphism, is uniquely determined by the module. This is the Krull-Schmidt theorem, and it is the correct replacement for the semisimple decomposition when simples are no longer available. The engine driving uniqueness is a structural fact about the endomorphism ring of an indecomposable module of finite length, that it is a *local ring*, which is itself a consequence of Fitting's lemma. So the section runs Fitting's lemma to the local-ring theorem to Krull-Schmidt, each result feeding the next.

Throughout, A is an algebra over a field k (the arguments need only that modules have finite length, so the results apply verbatim to modules of finite length over any ring). Modules are left modules. The starting point is worth isolating, since it is the gap between this chapter and the last. A simple module is indecomposable, because a simple module has no proper nonzero submodule at all, let alone a complemented one. The converse fails, and it fails already in dimension two: the $k[x]$-module $k[x]/(x^2)$ has the unique proper nonzero submodule $(x)/(x^2)$, so it is not simple, but that submodule is not a summand (any complement would have to be one-dimensional and x-stable, and $(x)/(x^2)$ is the only such subspace), so the module is indecomposable. An indecomposable module of finite length can therefore sit strictly between "simple" and "decomposable", and it is exactly this middle ground that Krull-Schmidt organizes.

The first tool is a splitting result for a single endomorphism. Given $\varphi \in \mathrm{End}_A(M)$, the iterated kernels $\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \cdots$ increase and the iterated images $\mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^2) \supseteq \cdots$ decrease. When M has finite length, both chains must stabilize, and Fitting's lemma says that at a common stabilization point the module splits into the eventual kernel and the eventual image.

Lemma 2.2.1: Fitting's Lemma

Let M be an A-module of finite length and let $\varphi \in \mathrm{End}_A(M)$. Then there exists a positive integer n such that

$$M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$$

Consequently, if M is indecomposable, then every $\varphi \in \mathrm{End}_A(M)$ is either nilpotent or an automorphism.

***Proof*:**
Write $\ell = \ell(M)$ for the length of M, which is finite by hypothesis.

Step 1: The kernel and image chains stabilize. The submodules $\ker(\varphi^i)$ form an increasing chain and the submodules $\mathrm{im}(\varphi^i)$ form a decreasing chain:

$$\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \cdots \qquad \mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^2) \supseteq \cdots$$

A module of finite length is both Noetherian and Artinian, so an increasing chain of submodules cannot strictly ascend more than ℓ times and a decreasing chain cannot strictly descend more than ℓ times. Setting $n = \ell$ makes both chains stationary from index n onward: for all $m \geq n$,

$$\ker(\varphi^m) = \ker(\varphi^n) \qquad \mathrm{im}(\varphi^m) = \mathrm{im}(\varphi^n)$$

Indeed, once the kernel chain repeats, it is stationary forever after: if $\ker(\varphi^i) = \ker(\varphi^{i+1})$ and $x \in \ker(\varphi^{i+2})$, then $\varphi(x) \in \ker(\varphi^{i+1}) = \ker(\varphi^i)$, so $x \in \ker(\varphi^{i+1}) = \ker(\varphi^i)$, giving $\ker(\varphi^{i+2}) = \ker(\varphi^{i+1})$; iterating shows $\ker(\varphi^j) = \ker(\varphi^i)$ for all $j \geq i$. A chain that strictly increased past step ℓ would produce more than ℓ strict inclusions, violating finite length, so the first repeat occurs by step ℓ. The same bound applies to the image chain, and $n = \ell$ dominates both stabilization thresholds.

Step 2: The two submodules intersect trivially. Fix n as above and take $y \in \ker(\varphi^n) \cap \mathrm{im}(\varphi^n)$. Write $y = \varphi^n(x)$ for some $x \in M$. Then $\varphi^n(y) = 0$ gives $\varphi^{2n}(x) = 0$, so $x \in \ker(\varphi^{2n}) = \ker(\varphi^n)$ by the stabilization of the kernel chain. Hence $y = \varphi^n(x) = 0$, and $\ker(\varphi^n) \cap \mathrm{im}(\varphi^n) = 0$.

Step 3: The two submodules span M. Take any $z \in M$. Since $\mathrm{im}(\varphi^n) = \mathrm{im}(\varphi^{2n})$, the element $\varphi^n(z) \in \mathrm{im}(\varphi^n)$ also lies in $\mathrm{im}(\varphi^{2n})$, so $\varphi^n(z) = \varphi^{2n}(w)$ for some $w \in M$. Then

$$z = \varphi^n(w) + \big(z - \varphi^n(w)\big)$$

where $\varphi^n(w) \in \mathrm{im}(\varphi^n)$, and the second term lies in $\ker(\varphi^n)$ because $\varphi^n(z - \varphi^n(w)) = \varphi^n(z) - \varphi^{2n}(w) = 0$. Thus $M = \ker(\varphi^n) + \mathrm{im}(\varphi^n)$, and together with Step 2 this gives the internal direct sum $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$.

Step 4: The dichotomy for indecomposable M. Suppose now that M is indecomposable. The decomposition $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$ forces one of the two summands to be M and the other to be 0. If $\ker(\varphi^n) = M$, then $\varphi^n = 0$, so φ is nilpotent. If instead $\ker(\varphi^n) = 0$ and $\mathrm{im}(\varphi^n) = M$, then φ is injective, since $\ker(\varphi) \subseteq \ker(\varphi^n) = 0$, and φ is surjective, since $\mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^n) = M$. An injective and surjective endomorphism is an automorphism. In either case φ is nilpotent or an automorphism, completing the proof.

The decomposition in Fitting's lemma is the "eventual kernel" and "eventual image" of φ, and

it works for any single endomorphism of any finite-length module, indecomposable or not. The sharp consequence appears only in the indecomposable case: an indecomposable module of finite length admits no interesting endomorphisms in the sense that the only two behaviors available to a $\varphi \in \mathrm{End}_A(M)$ are "eventually kills everything" (nilpotent) and "moves everything bijectively" (automorphism). There is no room in between, precisely because any intermediate behavior would produce a nontrivial Fitting decomposition and split the module. This is the mechanism that will make the endomorphism ring so rigid.

To see the dichotomy in a case where it can be checked by hand, take the endomorphism given by multiplication by x on $M = k[x]/(x^3)$. Multiplication by x has kernel $(x^2)/(x^3)$ and image $(x)/(x^3)$, which are not complementary. But the lemma is about a *power*: φ^3 is multiplication by $x^3 = 0$, so $\ker(\varphi^3) = M$ and $\mathrm{im}(\varphi^3) = 0$, giving the trivial decomposition $M = M \oplus 0$. This is the nilpotent branch of the dichotomy, and indeed multiplication by x is nilpotent. Multiplication by $1 + x$, on the other hand, is invertible (with inverse $1 - x + x^2$), landing in the automorphism branch. Every element of $\mathrm{End}_{k[x]}(M)$ is one or the other.

The endomorphism ring $\mathrm{End}_A(M)$ of an indecomposable finite-length module is close to being a division ring but not quite: Schur's lemma (lemma 1.1.5) gave a division ring only for *simple* M, and indecomposable is weaker. The right substitute is that $\mathrm{End}_A(M)$ is a *local ring*. A ring is local when its non-units form a two-sided ideal, equivalently when it has a unique maximal left ideal; the intuition is that "being a non-unit" is closed under addition, so failures of invertibility do not conspire to produce an invertible element. Fitting's lemma is exactly what closes the non-units under addition here, because a non-unit of $\mathrm{End}_A(M)$ is precisely a nilpotent endomorphism, and controlling sums of nilpotents is where the argument lives.

Theorem 2.2.2: Endomorphism Rings of Indecomposables are Local

Let M be a nonzero indecomposable A-module of finite length. Then $\mathrm{End}_A(M)$ is a local ring: the set of non-units of $\mathrm{End}_A(M)$ is a two-sided ideal, and it consists exactly of the nilpotent endomorphisms of M.

***Proof*:**

Write $E = \mathrm{End}_A(M)$ and let $\mathfrak{m} \subseteq E$ be the set of endomorphisms that are not units, that is, not automorphisms of M.

Step 1: The non-units are the nilpotents. By lemma 2.2.1, every $\varphi \in E$ is either an automorphism or nilpotent, and these are mutually exclusive (a nilpotent $\varphi \neq 0$ kills some nonzero element, and $\varphi = 0$ is not invertible since $M \neq 0$). Hence $\mathfrak{m}$ is exactly the set of nilpotent endomorphisms.

Step 2: $\mathfrak{m}$ is closed under left and right multiplication by E. Let $\varphi \in \mathfrak{m}$, so φ is not an automorphism, and let $\psi \in E$ be arbitrary. If $\psi\varphi$ were an automorphism, then φ would be injective (as $\psi\varphi$ is) and ψ surjective; an injective endomorphism of a finite-length module is bijective (its image has the same length as M, hence equals M), so φ would be an automorphism, a contradiction. Thus $\psi\varphi \in \mathfrak{m}$. Symmetrically, if $\varphi\psi$ were an automorphism, then φ would be surjective, hence bijective by the length argument, again contradicting $\varphi \in \mathfrak{m}$. So $\varphi\psi \in \mathfrak{m}$, and $\mathfrak{m}$ absorbs multiplication on both sides.

Step 3: $\mathfrak{m}$ is closed under addition. This is the step that uses indecomposability. Let $\varphi, \psi \in \mathfrak{m}$ and suppose, for contradiction, that $\varphi + \psi$ is an automorphism. Set

$$\alpha = (\varphi + \psi)^{-1}\varphi \qquad \beta = (\varphi + \psi)^{-1}\psi$$

so that $\alpha+\beta=(\varphi+\psi)^{-1}(\varphi+\psi)=\mathrm{id}_M$. By Step 2, $\mathfrak{m}$ absorbs left multiplication, so $\alpha,\beta\in\mathfrak{m}$; in particular α is nilpotent, say $\alpha^r=0$. From $\beta=\mathrm{id}_M-\alpha$ the standard geometric-series identity gives an inverse for β:

$$(\mathrm{id}_M-\alpha)\big(\mathrm{id}_M+\alpha+\alpha^2+\cdots+\alpha^{r-1}\big)=\mathrm{id}_M-\alpha^r=\mathrm{id}_M$$

so $\beta=\mathrm{id}_M-\alpha$ is an automorphism of M. But $\beta\in\mathfrak{m}$ means β is not an automorphism, a contradiction. Hence $\varphi+\psi\in\mathfrak{m}$.

Step 4: Conclusion. Steps 2 and 3 show $\mathfrak{m}$ is a two-sided ideal of E, and Step 1 identifies it as the set of non-units. A ring whose non-units form a two-sided ideal is local: any element outside $\mathfrak{m}$ is a unit, so $\mathfrak{m}$ is the unique maximal left ideal (a proper left ideal contains no units, hence lies in $\mathfrak{m}$). Therefore $E=\mathrm{End}_A(M)$ is local, as we sought to show.

The theorem sharpens Schur's lemma in exactly the direction needed. For a simple module, lemma 1.1.5 gave a division ring, where every nonzero element is a unit. For an indecomposable module of finite length, one gets the next best thing: a local ring, where the non-units are corralled into a single ideal and every element outside it is a unit. The residue $\mathrm{End}_A(M)/\mathfrak{m}$ is a division ring, and one can think of the local ring as a division ring with a nilpotent "fuzz" $\mathfrak{m}$ attached. The content used below is the geometric series in Step 3, which turns "id_M minus a nilpotent" into a unit; this is what will let one element of a decomposition be swapped for another in the exchange argument below.

The running example makes the local structure explicit.

Example 2.2.3: The Endomorphism Ring of a Truncated Polynomial Module
Fix $n\geq 1$ and let $M=k[x]/(x^n)$, viewed as a module over $A=k[x]$ (equivalently over $k[x]/(x^n)$, which acts through the same operators). This module is indecomposable of length n: its submodules are exactly the ideals $(x^j)/(x^n)$ for $0\leq j\leq n$, forming a single chain

$$0=(x^n)/(x^n)\subseteq(x^{n-1})/(x^n)\subseteq\cdots\subseteq(x)/(x^n)\subseteq M$$

and a chain of submodules has no way to split off a complementary summand (a decomposition $M=M_1\oplus M_2$ with both $M_i\neq 0$ would give two submodules neither containing the other).

Every A-endomorphism of M is multiplication by some polynomial: since M is generated by the image of 1, a homomorphism φ is determined by $\varphi(1)=\overline{f(x)}$, and then $\varphi(\overline{g})=\overline{gf}$. So

$$\mathrm{End}_A(M)\cong k[x]/(x^n)$$

as rings, the isomorphism sending φ to $\varphi(1)$. This is a local ring with maximal ideal $\mathfrak{m}=(x)/(x^n)$: an element $\overline{f}=\overline{a_0+a_1x+\cdots+a_{n-1}x^{n-1}}$ is a unit if and only if $a_0\neq 0$ (with inverse computed by the geometric series when $a_0=1$), and it is nilpotent if and only if $a_0=0$. The non-units are precisely the nilpotents, matching theorem 2.2.2, and $\mathrm{End}_A(M)/\mathfrak{m}\cong k$ is the residue division ring (here a field, since A is commutative).

The local endomorphism ring is now used to prove the main theorem. Existence of a decomposition into indecomposables is easy, by induction on length; the content is uniqueness. The classical route to uniqueness is an *exchange argument*: given two decompositions of the same module, one shows that each indecomposable summand of the first can be swapped into the second, one at a time, without changing the ambient module, and the local endomorphism rings are exactly what make each swap

possible. The theorem is due to Krull and Schmidt for finite groups and finite-length modules, with Azumaya's extension to any module that is a direct sum of submodules with local endomorphism rings.

Theorem 2.2.4: Krull-Schmidt(-Azumaya)

Let M be a nonzero A-module of finite length. Then:

1. M is a finite direct sum of indecomposable submodules,
$$M = M_1 \oplus M_2 \oplus \cdots \oplus M_r$$
with each M_i indecomposable.
2. This decomposition is unique up to isomorphism and reordering: if
$$M = M_1 \oplus \cdots \oplus M_r = N_1 \oplus \cdots \oplus N_s$$
are two decompositions into indecomposable submodules, then $r = s$ and there is a permutation σ of $\{1, \dots, r\}$ with $M_i \cong N_{\sigma(i)}$ for every i.

***Proof*:**

1. *Existence.* Argue by induction on the length $\ell(M)$. If M is indecomposable, then $M = M$ is already such a decomposition, with one summand. Otherwise $M = M_1 \oplus M_2$ with both M_i nonzero, and then $\ell(M_1), \ell(M_2) < \ell(M)$ because length is additive over direct sums and both summands are nonzero (hence of positive length). By the inductive hypothesis each M_i is a finite direct sum of indecomposables, and concatenating the two lists expresses M as a finite direct sum of indecomposables. The base case $\ell(M) = 1$ is a simple, hence indecomposable, module. This proves existence.
2. *Uniqueness.* Suppose
$$M = M_1 \oplus \cdots \oplus M_r = N_1 \oplus \cdots \oplus N_s$$
with all M_i and N_j indecomposable of finite length, so each $\mathrm{End}_A(M_i)$ and $\mathrm{End}_A(N_j)$ is local by theorem 2.2.2. Induct on r. The strategy is to exchange M_1 for one of the N_j, peel it off both sides, and apply the inductive hypothesis to what remains.

   Let $\iota_i \colon M_i \hookrightarrow M$ and $\pi_i \colon M \twoheadrightarrow M_i$ be the inclusion and projection for the first decomposition, and $\iota'_j \colon N_j \hookrightarrow M$, $\pi'_j \colon M \twoheadrightarrow N_j$ those for the second. These satisfy
$$\sum_{i=1}^{r} \iota_i \pi_i = \mathrm{id}_M \qquad \sum_{j=1}^{s} \iota'_j \pi'_j = \mathrm{id}_M$$
   Restricting the second identity to M_1 and projecting back to M_1 gives, in $\mathrm{End}_A(M_1)$,
$$\mathrm{id}_{M_1} = \pi_1 \Big(\sum_{j=1}^{s} \iota'_j \pi'_j \Big) \iota_1 = \sum_{j=1}^{s} (\pi_1 \iota'_j)(\pi'_j \iota_1)$$
   Now $\mathrm{End}_A(M_1)$ is local, so its non-units form an ideal and cannot sum to the unit id_{M_1}; at least one summand $(\pi_1 \iota'_j)(\pi'_j \iota_1)$ must be a unit of $\mathrm{End}_A(M_1)$. After reindexing the N_j

we may assume this happens for $j = 1$: the composite

$$\theta = (\pi_1 \iota_1')(\pi_1' \iota_1) \colon M_1 \longrightarrow M_1$$

is an automorphism of M_1, where $\pi_1' \iota_1 \colon M_1 \to N_1$ and $\pi_1 \iota_1' \colon N_1 \to M_1$.

Exchange step. Consider the maps

$$f = \pi_1' \iota_1 \colon M_1 \to N_1 \qquad g = \pi_1 \iota_1' \colon N_1 \to M_1$$

so $gf = \theta$ is an automorphism of M_1. Then f is a split monomorphism, with left inverse $\theta^{-1} g$, since $(\theta^{-1} g) f = \theta^{-1}(gf) = \mathrm{id}_{M_1}$. The endomorphism $e = f\theta^{-1} g \colon N_1 \to N_1$ is then idempotent:

$$e^2 = f\theta^{-1}(gf)\theta^{-1} g = f\theta^{-1}\theta\theta^{-1} g = f\theta^{-1} g = e$$

and its image is $\mathrm{im}(f)$ (from $ef = f\theta^{-1} gf = f$), so $N_1 = \mathrm{im}(e) \oplus \ker(e) = \mathrm{im}(f) \oplus \ker(e)$ is the splitting associated to the idempotent e. But N_1 is indecomposable and $\mathrm{im}(f) \neq 0$ (since f is injective and $M_1 \neq 0$), which forces $\mathrm{im}(f) = N_1$, making f surjective as well. Hence $f \colon M_1 \to N_1$ is an isomorphism, and in particular

$$M_1 \cong N_1$$

Peeling off the summand. It remains to remove M_1 and N_1 from the two decompositions and match the rest. Set $M' = M_2 \oplus \cdots \oplus M_r$, so $M = M_1 \oplus M'$ and M' is the kernel of the projection $\pi_1 \colon M \to M_1$. The claim is

$$M = N_1 \oplus M'$$

viewing $N_1 = \iota_1'(N_1) \subseteq M$ as the submodule it names in the second decomposition. The key observation is that the restriction of π_1 to N_1 is an isomorphism onto M_1: on N_1 this restriction is $\pi_1 \iota_1' = g$, and $g = \theta f^{-1}$ is a composite of the automorphism θ with the isomorphism f^{-1}, hence itself an isomorphism $N_1 \to M_1$. Granting that $\pi_1|_{N_1} \colon N_1 \to M_1$ is an isomorphism, the claim follows:

- *Trivial intersection.* If $y \in N_1 \cap M'$, then $\pi_1(y) = 0$ because $M' = \ker(\pi_1)$, and $\pi_1|_{N_1}$ is injective, so $y = 0$. Thus $N_1 \cap M' = 0$.
- *Spanning.* Given $z \in M$, set $a = \pi_1(z) \in M_1$. Since $\pi_1|_{N_1}$ is onto M_1, choose $y \in N_1$ with $\pi_1(y) = a$. Then $\pi_1(z - y) = a - a = 0$, so $z - y \in \ker(\pi_1) = M'$, and $z = y + (z - y) \in N_1 + M'$.

Hence $M = N_1 \oplus M'$, establishing the claim.

Induction. Quotienting $M = N_1 \oplus M'$ by the summand N_1 gives $M/N_1 \cong M'$, and quotienting the original $M = N_1 \oplus N'$ by the same submodule N_1 gives $M/N_1 \cong N'$. Therefore

$$M' \cong N'$$

Now M' carries the decomposition $M_2 \oplus \cdots \oplus M_r$ into $r - 1$ indecomposables, and the isomorphism $M' \cong N'$ transports the decomposition $N_2 \oplus \cdots \oplus N_s$ onto M' as a second decomposition into indecomposables. Applying the inductive hypothesis to M', whose first decomposition has $r - 1$ summands, yields $r - 1 = s - 1$, so $r = s$, together with a bijection matching $M_2, \ldots, M_r$ against $N_2, \ldots, N_s$ up to isomorphism. Combined with $M_1 \cong N_1$ from the exchange step, this produces the permutation σ with $M_i \cong N_{\sigma(i)}$ for all i. The base case $r = 1$ has $M = M_1$ indecomposable, forcing $s = 1$ and $N_1 = M$, completing the induction and the proof.

The theorem says that the finite-length representation theory of *any* algebra, semisimple or not, has canonical building blocks: the indecomposable modules. Over a semisimple algebra these are the simple modules and Krull-Schmidt recovers the decomposition of lemma 1.1.7, but the statement here needs no semisimplicity. What it does need is finite length, and the two places that hypothesis enters are worth marking. Finite length gave the Fitting decomposition (via the ascending and descending chain conditions), which produced the local endomorphism rings; and finite length is what let "injective endomorphism" upgrade to "automorphism" at several points. Drop finiteness and uniqueness can fail. A concrete failure: over the ring $R = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$ of real polynomial functions on the 2-sphere, the module R^3 splits both as $R \oplus P$, where P is the (indecomposable, non-free) module of tangent vector fields, and as $R \oplus R \oplus R$; the two decompositions have non-isomorphic summands, so uniqueness is lost once the summands are no longer of finite length. Krull-Schmidt is thus not a formality but a theorem with content tied to its hypotheses.

The exchange argument also explains *why* the answer is unique, not just *that* it is. The reason each M_i has a unique isomorphism partner is that when the identity of M_1 is written as a sum of composites through the N_j, locality of $\mathrm{End}_A(M_1)$ forbids the sum from being "spread out" among non-units; one composite must already be invertible, and that single composite pins down the partner. Locality is doing the work that a division ring did in the simple case, and it is the exact amount of structure required.

Reading off the multiplicities gives the invariant form of the theorem that is used in practice.

Corollary 2.2.5: Uniqueness of Multiplicities

Let M be a nonzero A-module of finite length. For each isomorphism class of indecomposable module I, the number of summands isomorphic to I appearing in a decomposition of M into indecomposables is independent of the decomposition. That is, writing

$$M \cong I_1^{\oplus m_1} \oplus I_2^{\oplus m_2} \oplus \cdots \oplus I_t^{\oplus m_t}$$

with the I_j pairwise non-isomorphic indecomposables, the classes $I_1, \ldots, I_t$ and the multiplicities $m_1, \ldots, m_t$ are uniquely determined by M.

***Proof*:**
By theorem 2.2.4, any two decompositions of M into indecomposables have the same number of summands and are matched by a permutation σ respecting isomorphism type. Grouping the summands of a decomposition by isomorphism class, the multiplicity of a class I is the number of indices i with $M_i \cong I$. Since σ carries each M_i to an $N_{\sigma(i)} \cong M_i$ and is a bijection, it restricts to a bijection between the summands isomorphic to I in the first decomposition and those isomorphic to I in the second. Hence the two decompositions assign I the same multiplicity, and this holds for every isomorphism class I. The set of classes that occur and their multiplicities are therefore invariants of M, completing the proof.

The multiplicities m_j are the numbers one computes when decomposing a representation, and the corollary is what guarantees the computation has a well-defined answer regardless of the method used to obtain it. In the language of the Grothendieck-style count, M determines a formal $\mathbb{N}$-linear combination $\sum_j m_j[I_j]$ of indecomposable classes, and two modules are isomorphic if and only if these combinations agree. This is the precise sense in which the indecomposables are a basis for the finite-length modules of A.

The full classification is transparent for $A = k[x]/(x^n)$, where the indecomposables can be listed and the uniqueness of multiplicities checked against an explicit computation.

Example 2.2.6: Modules over the Truncated Polynomial Algebra $k[x]/(x^n)$

Fix $n \geq 1$ and let $A = k[x]/(x^n)$. A finite-dimensional A-module is the same datum as a finite-dimensional k-vector space V together with a k-linear operator T, the action of the class of x, satisfying $T^n = 0$, so that T is nilpotent of nilpotency index at most n. Being finite-dimensional over k, every such module has finite length, so Krull-Schmidt applies.

The indecomposables. Since A is a quotient of the principal ideal domain $k[x]$, the structure theorem for finitely generated modules over $k[x]$ describes every finite-length A-module as a direct sum of cyclic modules $k[x]/(x^m)$. Only the exponents $1 \leq m \leq n$ occur, because x^n annihilates the module. Each

$$J_m := k[x]/(x^m) \qquad 1 \leq m \leq n$$

is indecomposable: as in example 2.2.3, its submodules form the single chain $0 \subseteq (x^{m-1}) \subseteq \cdots \subseteq (x) \subseteq J_m$, which admits no splitting into two nonzero summands. So the indecomposable A-modules are exactly

$$J_1, J_2, \ldots, J_n$$

up to isomorphism, with $\dim_k J_m = m$; these are the Jordan blocks for the nilpotent operator, J_m being a single Jordan block of size m with eigenvalue 0. There are precisely n of them.

A decomposition and its uniqueness. Take $n = 3$ and the module $M = A \oplus J_1$, that is, $V = k^4$ with the operator

$$T = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

a single Jordan block of size 3 in the top-left together with a 1×1 zero block. As an A-module,

$$M \cong J_3 \oplus J_1$$

with multiplicities $m_3 = 1$, $m_1 = 1$, and $m_2 = 0$. corollary 2.2.5 asserts these multiplicities are forced by M, and they can be read off invariantly from the ranks of powers of T, which do not depend on any chosen decomposition. The number of Jordan blocks of size at least j equals $\operatorname{rank}(T^{j-1}) - \operatorname{rank}(T^j)$, so the multiplicity of J_m is

$$m_m = \operatorname{rank}(T^{m-1}) - 2\operatorname{rank}(T^m) + \operatorname{rank}(T^{m+1})$$

Here $\operatorname{rank}(T^0) = \operatorname{rank}(\mathrm{id}) = 4$, $\operatorname{rank}(T) = 2$, $\operatorname{rank}(T^2) = 1$, and $\operatorname{rank}(T^3) = 0$. Substituting,

$$m_1 = 4 - 2 \cdot 2 + 1 = 1, \qquad m_2 = 2 - 2 \cdot 1 + 0 = 0, \qquad m_3 = 1 - 2 \cdot 0 + 0 = 1$$

recovering $M \cong J_3 \oplus J_1$. Since the ranks $\operatorname{rank}(T^j)$ are intrinsic to the operator T (hence to the module M), the multiplicities are intrinsic too, exactly as Krull-Schmidt predicts. Any presentation of M as a direct sum of J_m's, arrived at by any change of basis, must yield one copy of J_3, one copy of J_1, and no copy of J_2.

2.3 Idempotents, Projective Covers, and the Regular Representation

The Artin–Wedderburn theorem told us how to read off the simple modules of a semisimple algebra: they are exactly the summands of the regular module A, because a semisimple A splits as a direct sum of minimal left ideals and each minimal left ideal is simple. Once A is not semisimple this picture breaks. The regular module A still decomposes into indecomposable summands (the algebra is finite-dimensional, so a dimension count forces any module to be a finite direct sum of indecomposables), but those summands are no longer simple. They are the *indecomposable projective modules*, and the simple modules sit one level down, as their tops. This section builds the dictionary that replaces the semisimple one: the summands of A are cut out by a decomposition $1 = e_1 + \cdots + e_n$ of the identity into primitive orthogonal idempotents, each summand Ae_i is an indecomposable projective covering a single simple module, and the whole configuration is governed, through the quotient $A/\mathrm{Jac}(A)$, by the semisimple theory of chapter 1.

Throughout this section A is a finite-dimensional algebra over a field k (so in particular A is left Artinian and $\mathrm{Jac}(A)$ is nilpotent). Modules are left modules. The tool that transports information from the semisimple quotient $A/\mathrm{Jac}(A)$ back up to A is the lifting of idempotents, so that is where we begin.

The projection $\pi\colon A \to A/\mathrm{Jac}(A)$ collapses the radical. A decomposition of $A/\mathrm{Jac}(A)$ into a sum of simple summands is a statement about idempotents in $A/\mathrm{Jac}(A)$, and to pull such a decomposition back to A we need to produce idempotents in A that reduce to the given ones. The obstruction is that a preimage of an idempotent need not be idempotent. The nilpotence of $\mathrm{Jac}(A)$ removes the obstruction: a Newton-style correction converts any element that is idempotent modulo $\mathrm{Jac}(A)$ into a genuine idempotent lying over it.

Theorem 2.3.1: Lifting Idempotents Modulo the Radical

Let A be a finite-dimensional k-algebra and write $J = \mathrm{Jac}(A)$. Let $\pi\colon A \to A/J$ be the quotient map.

1. If $\bar{e} \in A/J$ satisfies $\bar{e}^2 = \bar{e}$, then there exists an idempotent $e \in A$ with $\pi(e) = \bar{e}$.
2. If $\bar{e}_1, \ldots, \bar{e}_m \in A/J$ are pairwise orthogonal idempotents with $\bar{e}_1 + \cdots + \bar{e}_m = \bar{1}$, then there exist pairwise orthogonal idempotents $e_1, \ldots, e_m \in A$ with $\pi(e_i) = \bar{e}_i$ and $e_1 + \cdots + e_m = 1$.

***Proof*:**

Since A is left Artinian, $J = \mathrm{Jac}(A)$ is nilpotent by theorem 2.1.8; fix N with $J^N = 0$.

Step 1: Lifting a single idempotent. Choose any $a \in A$ with $\pi(a) = \bar{e}$. Then $a^2 - a \in J$, since $\pi(a^2 - a) = \bar{e}^2 - \bar{e} = 0$. The idea is to correct a inside the coset $a + J$ until the error $a^2 - a$ is pushed into higher and higher powers of J, where it eventually vanishes. Set

$$e = 3a^2 - 2a^3$$

which lies in $a + J$, since $e - a = (3a^2 - 2a^3) - a = -(2a - 1)(a^2 - a) \in J$, so $\pi(e) = \bar{e}$. A direct

computation gives

$$e^2 - e = (3a^2 - 2a^3)^2 - (3a^2 - 2a^3) = (a^2 - a)^2\,(4a^2 - 4a - 3)$$

Writing $u = a^2 - a \in J$, this says $e^2 - e \in (u^2) \subseteq J^2$. Thus the operation $a \mapsto 3a^2 - 2a^3$ sends an element whose defect $a^2 - a$ lies in J^r to an element whose defect lies in J^{2r}, while staying in the same coset modulo J. Starting from $a_0 = a$ with defect in J^1 and iterating,

$$a_{t+1} = 3a_t^2 - 2a_t^3$$

produces $a_t \in a + J$ whose defect $a_t^2 - a_t$ lies in J^{2^t}. Once $2^t \geq N$ we have $a_t^2 - a_t \in J^N = 0$, so $e := a_t$ is a genuine idempotent with $\pi(e) = \bar{e}$. This proves (1).

Step 2: Lifting an orthogonal family. We argue by induction on m. For $m = 1$ the statement is $e_1 = 1$, which is immediate. Suppose the result holds for families of size $m-1$, and let $\bar{e}_1, \dots, \bar{e}_m$ be pairwise orthogonal idempotents summing to $\bar{1}$. Apply the inductive hypothesis to the shorter orthogonal family $\bar{e}_1, \dots, \bar{e}_{m-2}, \bar{e}_{m-1} + \bar{e}_m$ (its members are pairwise orthogonal idempotents summing to $\bar{1}$, since $\bar{e}_{m-1}$ and $\bar{e}_m$ are orthogonal). This yields pairwise orthogonal idempotents $e_1, \dots, e_{m-2}, f \in A$ lifting them, with $e_1 + \cdots + e_{m-2} + f = 1$ and $\pi(f) = \bar{e}_{m-1} + \bar{e}_m$.

It remains to split f. Consider the algebra fAf, which has identity element f. Its radical is $\mathrm{Jac}(fAf) = fJf$, and the reduction $fAf \to fAf/fJf$ identifies with the corner $\bar{f}(A/J)\bar{f}$ of the semisimple algebra A/J, where $\bar{f} = \bar{e}_{m-1} + \bar{e}_m$. In that corner, $\bar{e}_{m-1}$ and $\bar{e}_m$ are orthogonal idempotents summing to the identity $\bar{f}$. Applying part (1) inside the finite-dimensional algebra fAf to the idempotent $\bar{e}_{m-1} \in fAf/fJf$ produces an idempotent $e_{m-1} \in fAf$ with $\pi(e_{m-1}) = \bar{e}_{m-1}$. Set $e_m = f - e_{m-1}$. Then e_m is idempotent, since

$$e_m^2 = (f - e_{m-1})^2 = f - 2e_{m-1} + e_{m-1} = f - e_{m-1} = e_m$$

using $fe_{m-1} = e_{m-1}f = e_{m-1}$ (because $e_{m-1} \in fAf$) and $e_{m-1}^2 = e_{m-1}$; moreover $\pi(e_m) = \bar{f} - \bar{e}_{m-1} = \bar{e}_m$, and $e_{m-1}e_m = e_{m-1}f - e_{m-1}^2 = e_{m-1} - e_{m-1} = 0$, with $e_m e_{m-1} = 0$ symmetrically. Finally $e_{m-1}, e_m \in fAf$ are orthogonal to each of $e_1, \dots, e_{m-2}$, since f is, and $e_1 + \cdots + e_{m-2} + e_{m-1} + e_m = e_1 + \cdots + e_{m-2} + f = 1$. This completes the induction and the proof.

The polynomial $3a^2 - 2a^3$ is the second Newton iterate for the equation $x^2 - x = 0$; the quadratic doubling of the exponent is exactly Newton's quadratic convergence, made to terminate by nilpotence rather than to converge by a metric. The content of the theorem is that the arithmetic of idempotents in A is completely controlled by the arithmetic of idempotents in the semisimple quotient $A/\mathrm{Jac}(A)$: any way of splitting the identity in $A/\mathrm{Jac}(A)$ lifts to a way of splitting it in A, so the summand structure of the regular module is governed from below by Artin–Wedderburn. The next result turns this into a decomposition of A itself.

Before stating it, recall the mechanism from example 1.2.10: an idempotent $e \in A$ produces a direct sum decomposition $A = Ae \oplus A(1-e)$ of the regular module, because every $a \in A$ writes uniquely as $a = ae + a(1-e)$ with $ae \in Ae$ and $a(1-e) \in A(1-e)$. More generally an orthogonal decomposition $1 = e_1 + \cdots + e_n$ gives $A = Ae_1 \oplus \cdots \oplus Ae_n$. The summand Ae_i is indecomposable precisely when e_i cannot be split further, which is the meaning of *primitive*. Pushing the splitting as far as it will go decomposes the regular module into indecomposables.

Theorem 2.3.2: Primitive Idempotent Decomposition of the Regular Module

Let A be a finite-dimensional k-algebra. Then the identity admits a decomposition

$$1 = e_1 + e_2 + \cdots + e_n$$

into pairwise orthogonal primitive idempotents, and correspondingly the regular module decomposes as

$$A = Ae_1 \oplus Ae_2 \oplus \cdots \oplus Ae_n$$

where each Ae_i is an indecomposable projective left A-module. For an idempotent $e \in A$ with $e \neq 0$, the summand Ae is indecomposable if and only if e is primitive, if and only if the corner algebra eAe has no idempotents other than 0 and e.

***Proof*:**

Step 1: Idempotent decompositions correspond to direct-sum decompositions. Let $e \in A$ be an idempotent. Direct summands of the module Ae correspond to idempotents of its endomorphism ring: if $Ae = P \oplus Q$, the projection onto P along Q is an idempotent $\varepsilon \in \mathrm{End}_A(Ae)$, and conversely an idempotent $\varepsilon \in \mathrm{End}_A(Ae)$ splits $Ae = \varepsilon(Ae) \oplus (1-\varepsilon)(Ae)$. By the same computation as in lemma 1.2.3, the map $eAe \to \mathrm{End}_A(Ae)$ sending eae to right multiplication $x \mapsto x(eae)$ is a ring isomorphism

$$\mathrm{End}_A(Ae) \cong (eAe)^{\mathrm{op}}$$

The idempotents of $(eAe)^{\mathrm{op}}$ are the idempotents of eAe. So Ae decomposes nontrivially if and only if eAe has an idempotent other than 0 and $e = 1_{eAe}$. Writing such an idempotent as g and its complement $e - g$ (which is idempotent and orthogonal to g, by the computation in Step 2 of theorem 2.3.1), the decomposition $Ae = Ag \oplus A(e-g)$ corresponds to the refinement $e = g + (e-g)$ of e into orthogonal idempotents. Thus e is primitive (cannot be written as a sum of two nonzero orthogonal idempotents) if and only if eAe has no idempotent besides $0, e$, if and only if Ae is indecomposable. This proves the final three-way equivalence.

Step 2: The decomposition terminates. Start from 1. If 1 is primitive, take $n = 1$. Otherwise $1 = f + f'$ with f, f' nonzero orthogonal idempotents, giving $A = Af \oplus Af'$ with both summands nonzero. Since A is finite-dimensional and the summands Af, Af' are proper (each has strictly smaller k-dimension than A, as the other is nonzero), repeating the splitting on any summand that is still decomposable strictly decreases dimension at each step and so must stop. When it stops, every summand Ae_i is indecomposable, so by Step 1 each e_i is primitive. The resulting $e_1, \ldots, e_n$ are pairwise orthogonal (an idempotent split off from a corner Ae is orthogonal to every idempotent orthogonal to e) and sum to 1.

Step 3: Projectivity. Each Ae_i is a direct summand of the free module A, hence projective. A direct summand of a free module is projective because the splitting $A = Ae_i \oplus A(1-e_i)$ exhibits Ae_i as a retract of A, and retracts of free modules satisfy the lifting property defining projectivity. The general statement is proved in the companion volume, [4, Equivalencies for Projective Modules]. This gives the decomposition of A into indecomposable projectives, completing the proof.

The decomposition $A = \bigoplus_i Ae_i$ is the non-semisimple replacement for the Wedderburn splitting of a semisimple algebra into minimal left ideals. In the semisimple case each Ae_i was already simple, and the e_i were the matrix units picking out columns of the matrix blocks (proposition 1.2.5). In

general each Ae_i is only indecomposable, and it will turn out to have a whole composition series rather than length one. The indecomposable summands that appear are the objects the rest of the section studies, so they deserve a name.

Definition 2.3.3: Indecomposable Projective Module

Let A be a finite-dimensional k-algebra and $1 = e_1 + \cdots + e_n$ a decomposition of the identity into primitive orthogonal idempotents. The indecomposable projective left A-modules are the summands

$$P_i = Ae_i$$

Its *radical* is $\operatorname{rad}(P_i) = \operatorname{Jac}(A)\, P_i$, and its *top* (or *head*) is the quotient

$$\operatorname{top}(P_i) = P_i/\operatorname{rad}(P_i) = Ae_i/\operatorname{Jac}(A)Ae_i$$

The module P_i is called the *projective cover* of its top; the top is a simple module.

The claim that the top is simple is the hinge of the whole correspondence, so we verify it at once. Reducing modulo $\operatorname{Jac}(A)$ turns the primitive idempotent e_i into a primitive idempotent $\bar{e}_i \in A/\operatorname{Jac}(A)$, and over the semisimple algebra $A/\operatorname{Jac}(A)$ a primitive idempotent cuts out a simple module.

Example 2.3.4: The Top of Ae_i

Let $A = k[x]/(x^3)$, a local algebra with $\operatorname{Jac}(A) = (x)/(x^3)$. Here 1 is the only nonzero idempotent (a commutative local ring has no idempotents besides 0 and 1), so the identity decomposition is trivial, $n = 1$ and $e_1 = 1$, giving a single indecomposable projective $P_1 = A$. Its radical is $\operatorname{rad}(P_1) = \operatorname{Jac}(A)\, A = (x)/(x^3)$, and its top is

$$P_1/\operatorname{rad}(P_1) = \big(k[x]/(x^3)\big)\big/\big((x)/(x^3)\big) \cong k[x]/(x) = k$$

a one-dimensional, hence simple, module. So the single projective $P_1 = A$ (of dimension 3) covers the single simple module k (of dimension 1); the two extra dimensions live in the radical $\operatorname{rad}(P_1)$, whose composition factors are again copies of k. This is the smallest example in which a projective indecomposable is strictly larger than the simple it covers.

The passage $P_i \mapsto \operatorname{top}(P_i)$ from indecomposable projectives to simple modules, and its reverse, is the organizing principle of non-semisimple representation theory. Simple modules are visible in the semisimple quotient $A/\operatorname{Jac}(A)$, where Artin–Wedderburn counts them; primitive idempotents in A are the summands of the regular module; and lifting idempotents connects the two. The following theorem assembles these into a single correspondence.

Theorem 2.3.5: Simples, Indecomposable Projectives, and Primitive Idempotents

Let A be a finite-dimensional k-algebra, and let $1 = e_1 + \cdots + e_n$ be a decomposition of the identity into primitive orthogonal idempotents. There are mutually inverse bijections among the following three finite sets:

1. isomorphism classes of simple left A-modules;
2. isomorphism classes of indecomposable projective left A-modules;
3. conjugacy classes of primitive idempotents of A, where e and e' are conjugate if $e' = ueu^{-1}$ for a unit $u \in A^\times$.

The bijection from (2) to (1) sends $P_i \mapsto S_i := P_i/\mathrm{rad}(P_i)$; the bijection from (3) to (2) sends the class of e_i to $P_i = Ae_i$. If A is split, that is every A-linear endomorphism of a simple A-module is a scalar (for instance if $k = \bar{k}$), the number of classes equals the number of blocks in the Artin–Wedderburn decomposition of $A/\mathrm{Jac}(A)$, and

$$A/\mathrm{Jac}(A) \cong M_{d_1}(k) \times \cdots \times M_{d_r}(k), \qquad d_i = \dim_k S_i$$

where $S_1, \ldots, S_r$ are the simple modules and P_i appears $d_i = \dim_k S_i$ times in $A = \bigoplus_i P_i^{\oplus m_i}$.

***Proof*:**

Write $J = \mathrm{Jac}(A)$ and $\bar{A} = A/J$, which is semisimple by theorem 2.1.9.

Step 1: Simple modules are the same for A and $\bar{A}$. A simple A-module S is annihilated by J: the submodule $JS \subseteq S$ is either 0 or S, and $JS = S$ is impossible by Nakayama's lemma (lemma 2.1.7, applied to the cyclic module S: $JS = S$ would force $S = 0$). Hence $JS = 0$ and S is an $\bar{A}$-module; conversely every $\bar{A}$-module is an A-module through $\pi\colon A \to \bar{A}$. Since a submodule for A is the same as a submodule for $\bar{A}$ on a module killed by J, simplicity is preserved in both directions. Thus the simple A-modules are exactly the simple $\bar{A}$-modules, and by corollary 1.1.12 there are finitely many, one for each Wedderburn block of $\bar{A}$.

Step 2: $\mathrm{top}(P_i)$ is simple, and $P_i \mapsto \mathrm{top}(P_i)$ lands in (1). Reducing the decomposition $1 = e_1 + \cdots + e_n$ modulo J gives $\bar{1} = \bar{e}_1 + \cdots + \bar{e}_n$, a decomposition into orthogonal idempotents of $\bar{A}$. Applying the right-exact functor $\bar{A} \otimes_A -$ (equivalently reducing mod J) to $P_i = Ae_i$ gives

$$P_i/JP_i = Ae_i/JAe_i \cong \bar{A}\bar{e}_i$$

as $\bar{A}$-modules. Now $\bar{e}_i$ is primitive in $\bar{A}$: an orthogonal splitting $\bar{e}_i = \bar{g} + \bar{h}$ in $\bar{A}$ would lift, by theorem 2.3.1 applied inside the corner e_iAe_i, to an orthogonal splitting of e_i in A, contradicting primitivity of e_i. Over the semisimple algebra $\bar{A}$ a primitive idempotent generates a simple module, since $\bar{A}\bar{e}_i$ is an indecomposable summand of the semisimple module $\bar{A}$ and indecomposable semisimple means simple. Hence $\mathrm{top}(P_i) = P_i/\mathrm{rad}(P_i) = \bar{A}\bar{e}_i =: S_i$ is simple.

Step 3: The map (2) $\to$ (1) is a bijection. Distinct indecomposable projectives have distinct tops: if $Ae_i \cong Ae_j$ as A-modules then reducing mod J gives $\bar{A}\bar{e}_i \cong \bar{A}\bar{e}_j$, so $S_i \cong S_j$; conversely if $S_i \cong S_j$ then the two primitive idempotents $\bar{e}_i, \bar{e}_j$ generate isomorphic simple summands of $\bar{A}$, hence are conjugate in $\bar{A}$ (two primitive idempotents of a semisimple algebra generating isomorphic simples lie in the same Wedderburn block and differ by an inner automorphism), and a conjugating unit lifts to a unit of A because $\pi\colon A^\times \to \bar{A}^\times$ is surjective (an element of A

mapping to a unit of $\bar{A}$ is itself a unit, since J is nilpotent: if $\bar{a}$ is invertible then a has a left and right inverse modulo the nilpotent ideal J, and $1 + J \subseteq A^\times$). Conjugate idempotents give isomorphic projectives $Ae_i \cong Ae_j$ via right multiplication by the conjugating unit. So $P_i \cong P_j$ if and only if $S_i \cong S_j$, giving a bijection between (2) and (1). Every simple module arises as some S_i: a simple $\bar{A}$-module is a summand of $\bar{A} = \bigoplus_i \bar{A}\bar{e}_i$, hence isomorphic to some $\bar{A}\bar{e}_i = S_i$.

Step 4: Conjugacy classes of primitive idempotents. By the last paragraph, $Ae_i \cong Ae_j$ as modules if and only if e_i and e_j are conjugate in A: one direction lifts a conjugation, the other reduces an isomorphism to a conjugation via $\mathrm{End}_A(A) \cong A^{\mathrm{op}}$ (lemma 1.2.3), under which a module isomorphism $Ae_i \to Ae_j$ is right multiplication by an element u with $ue_iu^{-1} = e_j$. Thus the class of e_i under conjugacy corresponds to the isomorphism class of P_i, giving the bijection between (3) and (2).

Step 5: The split case. When $k = \bar{k}$, the algebra A is split: a simple module S is a quotient of A, hence finite-dimensional, so $\mathrm{End}_A(S)$ is a finite-dimensional division k-algebra by lemma 1.1.5, equal to k by corollary 1.2.19.

Now let A be split. By corollary 1.2.18 applied to the semisimple algebra $\bar{A}$, we have $\bar{A} \cong \prod_{l=1}^{r} M_{d_l}(D_l)$ with each D_l a finite-dimensional division k-algebra, and $\bar{A}$ has exactly r simple modules, the column modules $D_l^{\oplus d_l}$ of the blocks, on which the other blocks act by zero. By Step 1 these are the simple A-modules, and their A-endomorphisms are their $M_{d_l}(D_l)$-endomorphisms, which form the ring D_l^{op} by corollary 1.2.8. The split hypothesis gives $\dim_k D_l = 1$, so $D_l = k$ and the simple modules are $S_l = k^{d_l}$, with $\dim_k S_l = d_l$.

For the multiplicities, let $P_1, \dots, P_r$ be indecomposable projectives with $\mathrm{top}(P_i) = S_i$ (Step 3), and let m_i be the number of j with $Ae_j \cong P_i$, so that $A = \bigoplus_i P_i^{\oplus m_i}$. By Step 3, m_i is also the number of j with $\bar{A}\bar{e}_j \cong S_i$. For simple modules S and T, the space $\mathrm{Hom}_{\bar{A}}(S, T)$ is one-dimensional if $S \cong T$ and zero otherwise, by lemma 1.1.5 and the split hypothesis. So for any decomposition of $\bar{A}$ into finitely many simple submodules, $\dim_k \mathrm{Hom}_{\bar{A}}(S_i, \bar{A})$ is the number of summands isomorphic to S_i. The decomposition $\bar{A} = \bigoplus_j \bar{A}\bar{e}_j$ of Step 2 gives the number m_i. The decomposition of $\bar{A} \cong \prod_l M_{d_l}(k)$ in which each block $M_{d_l}(k)$ is the direct sum of its d_l columns, each isomorphic to k^{d_l}, gives the number d_i. So $m_i = d_i = \dim_k S_i$, as we sought to show.

The correspondence is the reason the count of simple modules is finite and computable: it equals the number of Wedderburn blocks of $A/\mathrm{Jac}(A)$, which Artin–Wedderburn hands us. In the semisimple case the three sets coincide with the columns of the matrix blocks and the map $P_i \mapsto S_i$ is the identity; the passage to a general finite-dimensional algebra keeps the same indexing set but separates the projective P_i from the simple S_i it covers. The word "cover" is precise, and the next definition makes it so. A cover should be the most economical projective surjecting onto a module, with no wasted projective sitting over the radical.

Definition 2.3.6: Superfluous Submodule and Projective Cover

Let A be a ring and M an A-module. A submodule $L \subseteq M$ is *superfluous* (or *small*) if for every submodule $N \subseteq M$ the equality $L + N = M$ forces $N = M$. A surjection $f\colon P \to M$ is *essential* if $\ker f$ is superfluous in P.
A *projective cover* of M is a projective module P together with an essential surjection $f\colon P \to M$; that is, f is surjective, P is projective, and $\ker f$ is superfluous in P.

The condition that $\ker f$ be superfluous is what pins the cover down. Any module has a surjection from a projective (a free module surjects onto it), but without the superfluous condition one could pad the source with a free summand mapping to zero and never have a canonical choice. Superfluousness forbids exactly this padding: no proper submodule of P maps onto M, so P is as small a projective as can surject onto M. Over a finite-dimensional algebra the radical gives a ready source of superfluous submodules, and this makes covers exist and be unique.

Theorem 2.3.7: Existence of Projective Covers

Let A be a finite-dimensional k-algebra and M a finitely generated (equivalently, finite-length) left A-module. Then M has a projective cover $f\colon P \to M$, unique up to isomorphism over M. For a simple module $S = S_i$, the projective cover is the indecomposable projective $P_i = Ae_i$ with its quotient map $P_i \twoheadrightarrow P_i/\mathrm{rad}(P_i) = S_i$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

Step 1: JP is superfluous in a finitely generated P. Let P be finitely generated and suppose $JP + N = P$ for a submodule $N \subseteq P$. Then $J \cdot (P/N) = (JP + N)/N = P/N$, so the finitely generated module P/N satisfies $J \cdot (P/N) = P/N$. Nakayama's lemma (the module form, lemma 2.1.7; concretely, using the nilpotence of J: if $J^N = 0$ and $JQ = Q$ then $Q = JQ = J^2Q = \cdots = J^NQ = 0$) forces $P/N = 0$, i.e. $N = P$. Hence JP is superfluous in P.

Step 2: Construction of the cover. The top $\bar{M} = M/JM$ is a module over the semisimple algebra $\bar{A} = A/J$, hence semisimple: $\bar{M} \cong \bigoplus_i S_i^{\oplus c_i}$ for multiplicities $c_i \geq 0$. Set

$$P = \bigoplus_i P_i^{\oplus c_i}$$

with each $P_i = Ae_i$ the indecomposable projective covering S_i. Then $P/JP = \bigoplus_i (P_i/JP_i)^{\oplus c_i} = \bigoplus_i S_i^{\oplus c_i} = \bar{M}$, so the projection $P \to P/JP = \bar{M}$ agrees with $M \to \bar{M}$ on tops. Because P is projective, the surjection $M \twoheadrightarrow \bar{M}$ lifts through $P \to \bar{M}$ to a homomorphism $f\colon P \to M$ with $M \to \bar{M}$ equal to $P \xrightarrow{f} M \to \bar{M}$. The image $f(P)$ satisfies $f(P) + JM = M$ (their images in $\bar{M}$ agree and equal all of $\bar{M}$), and JM is superfluous in M by Step 1 applied to M; hence $f(P) = M$ and f is surjective.

Step 3: f is essential. Let $K = \ker f$. Suppose $K + N = P$ for a submodule N. Then $f(N) = f(P) = M$, so $N + JP \supseteq N + \ker(P \to \bar{M})$ surjects onto $\bar{M} = P/JP$, giving $N + JP = P$; since JP is superfluous (Step 1), $N = P$. Thus K is superfluous and $f\colon P \to M$ is a projective cover.

Step 4: Uniqueness. Let $f\colon P \to M$ and $f'\colon P' \to M$ be two projective covers. Projectivity of P'

lifts f' through f to $g\colon P' \to P$ with $fg = f'$. Then $f(g(P')) = f'(P') = M$, so $g(P') + \ker f = P$; since $\ker f$ is superfluous, $g(P') = P$ and g is surjective. As P' is projective, the surjection g splits, $P' = \ker g \oplus P''$ with $P'' \cong P$; but $\ker g \subseteq \ker f' = \ker(fg)$ is superfluous in P' and a superfluous direct summand is zero (a summand L with complement C satisfies $L + C = P'$, forcing $C = P'$ hence $L = 0$). So g is an isomorphism over M.

Step 5: The simple case. For $M = S_i$ the top $\bar{M} = S_i$ has $c_i = 1$ and all other multiplicities 0, so the construction returns $P = P_i$ with $f\colon P_i \twoheadrightarrow P_i/\mathrm{rad}(P_i) = S_i$, the stated cover, completing the proof.

The theorem certifies the terminology in definition 2.3.3: $P_i \to S_i$ really is *the* projective cover, and every finite-length module is covered by an explicit direct sum of the P_i read off from its top. Uniqueness makes the assignment $M \mapsto P(M)$ well defined and turns the correspondence of theorem 2.3.5 into a computational tool: to find the projective cover of any module, compute its top $M/\mathrm{Jac}(A)M$, decompose that semisimple module into simples, and sum the corresponding P_i. What the projective P_i contributes beyond the simple top S_i is recorded in its composition series, and tabulating those multiplicities across all i produces the central numerical invariant of the algebra.

Definition 2.3.8: Cartan Matrix of an Algebra

Let A be a finite-dimensional k-algebra with simple modules $S_1, \dots, S_r$ and indecomposable projectives $P_1, \dots, P_r$, indexed so that $P_i/\mathrm{rad}(P_i) = S_i$. The *Cartan matrix* of A is the $r \times r$ matrix $C = (C_{ij})$ of nonnegative integers

$$C_{ij} = [P_i : S_j]$$

the multiplicity of the simple module S_j among the composition factors of a composition series of P_i.

The multiplicities $[P_i : S_j]$ are well defined by the Jordan–Hölder theorem, which guarantees the composition factors of a finite-length module are independent of the chosen composition series. The Cartan matrix measures how far A is from semisimple in a single array: when A is semisimple every $P_i = S_i$ has length one, so $C_{ij} = \delta_{ij}$ is the identity matrix, and any off-diagonal or large diagonal entry records simples fused together by the radical. The diagonal entry $C_{ii} = [P_i : S_i]$ is at least 1 (the top contributes one copy of S_i), and $C_{ii} > 1$ signals a self-extension living in the radical. The next example computes the whole apparatus of this section for the smallest non-semisimple algebra with two non-isomorphic simple modules, so that the Cartan matrix is not diagonal.

Before the example, one further point for completeness: the entire theory has a dual half. Everything above was built from the regular module A as a source of projectives; dualizing over k turns projectives into injectives and covers into envelopes.

2.3.9 Remark: The Dual Theory of Injective Envelopes Dual to a projective cover is an *injective envelope*: an injective module E with an essential injection $M \hookrightarrow E$, meaning every nonzero submodule of E meets the image of M nontrivially. Over a finite-dimensional algebra A these exist and are unique, and the duality $D = \mathrm{Hom}_k(-, k)$ carries left A-modules to right A-modules (equivalently left A^{op}-modules), exchanging projective covers of A-modules with injective envelopes of A^{op}-modules. The indecomposable injective A-modules are thus $D(e_i A)$, and each has a simple *socle* (its largest semisimple submodule) rather than a simple top; the socle plays for injectives the role the top plays for projectives. The general theory of injective and projective modules over a ring, including existence of injective envelopes via injective hulls, is developed in the companion volume [4]; the present section uses only the projective side.

Example 2.3.10: Projective Covers over the Upper-Triangular Algebra $T_2(k)$
Let

$$A = T_2(k) = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in k \right\}$$

the algebra of 2×2 upper-triangular matrices over a field k, a three-dimensional k-algebra. We work out its idempotents, indecomposable projectives, simple modules, projective covers, and Cartan matrix.

Idempotents. The matrix units

$$e_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

are orthogonal idempotents with $e_1 + e_2 = 1$. Each is primitive: the corner algebra $e_1 A e_1 = k\, e_1 \cong k$ has no idempotents but $0, e_1$, and likewise $e_2 A e_2 \cong k$, so by theorem 2.3.2 both Ae_1 and Ae_2 are indecomposable. Thus $1 = e_1 + e_2$ is a primitive orthogonal decomposition and there are two indecomposable projectives.

The radical. The Jacobson radical is the strictly upper-triangular part,

$$\mathrm{Jac}(A) = \begin{pmatrix} 0 & k \\ 0 & 0 \end{pmatrix}$$

a one-dimensional nilpotent ideal with $\mathrm{Jac}(A)^2 = 0$, and the quotient is

$$A/\mathrm{Jac}(A) \cong \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \cong k \times k$$

a product of two copies of k, confirming via theorem 2.3.5 that there are exactly two simple modules, both one-dimensional.

The indecomposable projectives. Left multiplication by A on e_i gives, with $E_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$,

$$P_1 = Ae_1 = \begin{pmatrix} k & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_2 = Ae_2 = \begin{pmatrix} 0 & k \\ 0 & k \end{pmatrix}$$

so $\dim_k P_1 = 1$ and $\dim_k P_2 = 2$, and indeed $\dim_k A = 1 + 2 = 3 = \dim_k P_1 + \dim_k P_2$, consistent with $A = P_1 \oplus P_2$.

The simple modules and covers. The tops are

$$S_1 = P_1/\mathrm{rad}(P_1) = P_1/\mathrm{Jac}(A)P_1 = P_1 \cong k, \qquad S_2 = P_2/\mathrm{rad}(P_2) = P_2/\mathrm{Jac}(A)P_2 \cong k$$

Here $\mathrm{Jac}(A)P_1 = 0$, so $P_1 = S_1$ is already simple and is its own projective cover, an essential surjection $P_1 \xrightarrow{\cong} S_1$. For P_2, the radical $\mathrm{Jac}(A)P_2 = \left(\begin{smallmatrix} 0 & k \\ 0 & 0 \end{smallmatrix}\right)$ is one-dimensional, and

$$\mathrm{rad}(P_2) = \begin{pmatrix} 0 & k \\ 0 & 0 \end{pmatrix} \cong S_1, \qquad P_2/\mathrm{rad}(P_2) \cong S_2$$

So P_2 is a length-two module with a nonsplit exact sequence

$$0 \to S_1 \to P_2 \to S_2 \to 0$$

and the quotient map $P_2 \twoheadrightarrow S_2$ is the projective cover of S_2, with kernel $\mathrm{rad}(P_2) = S_1$ superfluous in P_2 (it lies inside $\mathrm{Jac}(A)P_2$). The two simple modules S_1 and S_2 are not isomorphic, since e_1 and e_2 act as 1 on one and 0 on the other; the algebra is not semisimple, as $\mathrm{Jac}(A) \neq 0$.

The Cartan matrix. Reading off composition factors: P_1 has the single factor S_1, while P_2 has factors S_2 (its top) and S_1 (its radical). Hence

$$[P_1 : S_1] = 1, \quad [P_1 : S_2] = 0, \quad [P_2 : S_1] = 1, \quad [P_2 : S_2] = 1$$

and the Cartan matrix is the upper-triangular integer matrix

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Its determinant is 1 and its off-diagonal entry $C_{21} = 1$ records the single copy of S_1 that the radical of P_2 contributes: the exact numerical trace of the failure of $T_2(k)$ to be semisimple, where the Cartan matrix would have been the identity.

3

Representation of Finite Groups

All modules will be left R-modules for a not necessarily commutative ring R. Furthermore, all actions will be non-trivial actions unless specified otherwise.

3.1 Representations of Finite Groups

Similarly to how we studied groups and by how they acted on themselves through group-actions (ex. how a group acted on itself by conjugation gave us Sylow's Theorem), we will now try to study groups using *linear algebra*. Many really difficult results in classical group theory will become much easier when studied through this new "representation" of groups (and hence why it is called *representation theory*). Furthermore, vector-spaces afford many more structures than a module structures (ex. a complex vector-space is the starting point for the Nullstellensatz, Complex Geometry, Analysis, and much more!) Hence, one large advantage of Representation theory will be the intersection of many mathematical disciplines which contribute towards our study (and they all feed back on each other).

In this part of the book, G will always be a finite group[1], and k will always be a field. The module-theoretic set-up of this section works over any field, but the results that follow do not: Maschke's theorem and everything downstream of it need $\mathrm{char}(k) \nmid |G|$, and the character theory of section 3.2 needs $k = \mathbb{C}$. Each result states the hypothesis it needs, and section 3.5$\star$ takes up what happens when $\mathrm{char}(k)$ divides $|G|$. We write $\mathrm{Aut}_{k\text{-}\mathbf{Vect}}(V) = \mathrm{GL}(V)$ for the automorphisms of the k-vector space V. Once a basis is chosen and $\dim(V) = n < \infty$, this is isomorphic to $\mathrm{GL}_n(k)$, the group of invertible $n \times n$ matrices over k.

[1] Not sure about where to address infinite groups

Definition 3.1.1: Representation

Let G be a finite group and V a vector space over k. Then a *linear representation* is any homomorphism $\varphi : G \to \mathrm{GL}(V)$. The *degree* of the representation is the dimension of V. A *matrix representation* of G is any homomorphism $\varphi : G \to \mathrm{GL}_n(k)$

Usually, when we say representation we will mean *linear representation*. If a vector-space V has a representation of G, we will often call it a G-representation. To be as rigorous as possible we would say we have a representation of G on V over k, if the context requires it. Just like for group actions, a representation is *faithful* if it is injective. Representations need not be faithful, and most are not: the trivial representation, the sign representation of S_n, and anything inflated from a proper quotient all have kernel.

As usual, we can curry or uncurry an action[2]: we can take $\varphi : G \to \mathrm{GL}(V)$ and make it $G \times V \to V$ where $g : V \to V$ with $v \mapsto g \cdot v$ is linear over k for all g. It is left for the curious to see that this is indeed the proper uncurrying of this action.

Using the definition of $\varphi : G \to \mathrm{GL}(V)$, it is easy to define a representation homomorphism mirroring the definition of group-actions and modules (which are ring actions on abelian groups):

Definition 3.1.2: Representation Homomorphism

Let V, W be two vectors spaces over k with a linear representation by G. Then $\varphi : V \to W$ a *representation homomorphism* if φ k-linear and for any $g \in G$:

$$\varphi(g \cdot v) = g \cdot \varphi(v)$$

such a homomorphism will be called G-linear.

From this perspective, it might be hard to see how our previous discussion on semi-simple rings might give us any insight. We thus introduce a perspective better suited to the results of chapter 1: every representation of G on V over k is in bijective correspondence with all $k[G]$-modules over V, where $k[G]$ represents the *group ring* as seen in section 9.2. This second perspective will also make it easier to define a couple of examples, and so we will delay giving examples till the correspondence has been established. As a reminder, a group ring $k[G]$ over a ring k (which in our case will always be a field) is the set of all formal sums:

$$\sum_{g_i \in G} a_i g_i$$

where addition and multiplication are defined in the natural way[3]:

$$\sum_i^n a_i g_i + \sum b_i g_i = \sum_i (a_i + b_i) g_i$$

$$\left(\sum_i^n a_i g_i\right)\left(\sum b_i g_i\right) = \sum_\ell^n \left(\sum_{\substack{i,j \\ g_i g_j = g_\ell}} a_i b_j \right) g_\ell$$

[2] See remark 4.1.3

[3] A word on this really being a natural way?

Another common notation for group rings is kG, where the brackets are dropped. Several facts about group rings are used below:

1. The group G naturally injects into $k[G]$ by identifying g_i with $1g_i$ which is the formal sum $\sum_j \delta_{ij} g_j$ (where δ_{ij} is the Kronecker delta).
2. The field k naturally injects in $k[G]$ by corresponding an element $a \in k$ with $ag_1 = ae$ where $g_1 = e$ is the identity. If it is clear from context, we will sometimes simply write $a \in k[G]$ if it is understood $a = ae = ag_1$.
3. There is a natural k action on $k[G]$ defined by
$$b \cdot \left(\sum_i a_i g_i \right) = \sum_i (ba_i) g_i$$
due to this, $k[G]$ is a vector space over k with the elements of G as a basis, and hence is finite dimensional if $|G| < \infty$.
4. By definition, The elements of k identified as kg_1 commute with all the elements of $k[G]$, hence it is in the center of $k[G]$: $k \subseteq Z(k[G])$. Thus, $k[G]$ is in fact a k-algebra.
5. The group ring is left adjoint to the unit-group functor $A \mapsto A^\times$ from k-algebras to groups, that is $\mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k[G], A) \cong \mathrm{Hom}_{\mathbf{Grp}}(G, A^\times)$, hence is free with respect to groups, justifying the name "formal sum" for it's elements.

Using these, we show how representations of G on V over k correspond to $k[G]$-modules: let $\varphi : G \to \mathrm{GL}(V)$ be a representation, with $G = \{g_1, g_2, ..., g_n\}$. Thus, each $\varphi(g_i)$ is linear transformation (in particular an automorphism). Define the $k[G]$ action on V by defining:

$$\left(\sum_i^n a_i g_i \right) \cdot v = \left(\sum_i^n a_i \varphi(g_i) \right)(v) = \sum_i^n a_i \varphi(g_i)(v)$$

Then this defines a $k[G]$-module (as should be verified if it does not seem like it). A particular reminder for those who are checking that $\varphi(g_i g_j) = \varphi(g_i) \circ \varphi(g_j)$, since φ is a group homomorphism onto $\mathrm{GL}(V)$ under composition. Notice how this action in fact extends the action of k on $k[G]$ since $a \cdot v = av$ (similarly to how the $F[x]$ action extended the action of F on V in section 14.3).

Conversely, let's say V is an $k[G]$-module. Seeking to make V a vector space over k and finding a group homomorphism $\varphi : G \to \mathrm{GL}(V)$, the first thing is that V is already a K vector-space by limiting to the action of $k \subseteq k[G]$. Furthermore, for each $g \in G$, we can obtain a map $\varphi : V \to V$ by defining:

$$\varphi(g)(v) = g \cdot v$$

Then $\varphi(g)$ is a linear transformation: for $\alpha, \beta \in k$, $v, w \in V$

$$\begin{aligned} \varphi(g)(\alpha v + \beta w) &= g \cdot (\alpha v + \beta w) \\ &= \alpha(g \cdot v) + \beta(g \cdot w) \\ &= \alpha\varphi(g)(v) + \beta\varphi(g)(w) \end{aligned}$$

and by the associativity axiom of modules, we get

$$\varphi(g_i g_j)(v) = (\varphi(g_i) \circ \varphi(g_j))(v)$$

Thus, $\varphi : G \to \mathrm{GL}(V)$ is indeed a group homomorphism, and we have:

$$\left\{ V \text{ an } k[G]\text{-module} \right\} \rightleftarrows \left\{ \begin{array}{c} V \text{ a vector space over } k \\ \text{and} \\ \varphi : G \to \mathrm{GL}(V) \text{ a representation} \end{array} \right\}$$

Note that this also means that this also shows that linear representations also have kernel objects, since modules have kernels. Hence, the kernel of a G-representation homomorphism between V and W is G-invariant.

Definition 3.1.3: Affording Representation

Let V be a $k[G]$-module. Then given the previous discussion, we say that V *affords* the representation φ of G.

Given this new perspective, we can also characterize sub-actions. If we have a $k[G]$-module V, then a $k[G]$-submodule $W \subseteq V$ would have to be a k-submodule, and hence a subspace, but it also has to be closed under the action of g on W, i.e. $g \cdot W \subseteq W$. Hence, any $k[G]$-submodule is a G-stable subspace of V. Conversely, if $W \subseteq V$ is G-stable, then it is closed under the action of G and is a subspace, and hence a $k[G]$-submodule. Hence

$$\left\{ W \subseteq V \text{ a } k[G]\text{-submodule} \right\} \rightleftarrows \left\{ W \subseteq V \text{ a } G\text{-stable subspace} \right\}$$

Next, we should know when two representations are equivalent. Recall that when studying a linear transformation T on V to itself, a basis for V can be found that gives a "canonical form" to the matrix representation of T. Changing basis for V kept V the same, but changed the matrix representation of T up to similarity (i.e. changed the isomorphism from $\mathrm{GL}(V)$ to $\mathrm{GL}_n(k)$). The following is the analogous term:

Definition 3.1.4: Equivalent Representation

Let φ and ψ be two representations of G over k. Then we say these two represetnations are *equivalent* (or *similar*) if the $k[G]$-module affording them are isomorphic as modules. If two representations are not equivalent, they are called *inequivalent* (or *not similar*).

When the field is known[a], if V and W are equivalent (i.e. isomorphic as $k[G]$-modules), then we write $V \cong_G W$.

[a] In most cases, the field will be $\mathbb{C}$

Note that two different representations can mean two different vector-spaces over k. If $\varphi : G \to \mathrm{GL}(V)$ and $\psi : G \to \mathrm{GL}(W)$ are equivalent representations, it would make sense if there is a relation between V and W. If $T : V \to W$ is the $k[G]$-module isomorphism between them, It is a k-module isomorphism, and hence $V \cong W$ as vector spaces, implying $\dim(V) = \dim(W)$. Furthermore, by definition of T, $T(g \cdot v) = g \cdot T(v)$. By definition of the action, we have that $T(\varphi(g)(v)) = \psi(g)(T(v))$, i.e.

$$T \circ \varphi(g) = \psi(g) \circ T \quad \Leftrightarrow \quad T \circ \varphi(g) \circ T^{-1} = \psi(g) \qquad \forall g \in G$$

That is, T is the *simultaneous change of basis* for all $\varphi(g)$, $g \in G$. In terms of matrices, there must exist a fixed invertible matrix P such that for all $g \in G$,

$$P\varphi(g)P^{-1} = \psi(g)$$

The function T (or matrix P) are said to *intertwine* φ and ψ (it gives the rules on how to relate φ and ψ).

Example 3.1.5: Representations

1. Let G be any finite group, V any 1 dimensional vector-space over k. Define the $k[G]$-module structure on V by
$$g \cdot v = v$$
for all $g \in G$, $v \in V$. This called the *trivial representation* of G. This module affords the representation $\varphi : G \to \mathrm{GL}(V)$ by $\varphi(g) = I$ (the corresponding matrix representation from G to $\mathrm{GL}_1(k)$ sends $g \mapsto I$).

2. Let G be any finite group and k any field. Let $V = k[G]$ and consider it as a $k[G]$-module over itself. Then $\dim_k(V) = |G| = n$, with the basis being the elements of G. Then we see that each $g \in G$ permute the basis elements:
$$g \cdot g_i = gg_i = g_j \in G$$
Thus, the matrix representation for the action by the element g is an $n \times n$ matrix with 1 in the ij component if $g \cdot g_j = g_i$ (hence n 1's) and zero everywhere else. This representation of g is called the *regular representation* of G. Note how the underlying field did not matter: the coefficients in the matrices can only ever by 1 or 0, and we would simply say "Let φ be the regular representation of G" without specifying which field the group-ring is.

3. Let $G = S_n$ ($n \in \mathbb{N}_{>0}$) and k be any field. Then let V be a vector-space over k of dimension n, $(e_1, e_2, ..., e_n)$ a basis for V, and define the action:
$$\sigma \cdot e_i = e_{\sigma(i)}$$
that is, σ permute the basis of V. Through this, we see that we afford a faithful representation $\varphi : S_n \hookrightarrow \mathrm{GL}(V)$. Furthermore, since any finite group is a subgroup of S_n for an appropriate n, we see that a group of order n we can define $\varphi : G \to S_n \to \mathrm{GL}(V)$, giving another perspective on how to embed any finite group into a matrix group. Notice too how we made a $k[G]$-module given the action of a $k[S_n]$-module. This is called the *permutation representation* of G.

   Note that this means that *any* finite groups acts on a vector space, and by using this construction we see that any finite group G of size n can be embedded into $\mathrm{GL}_n(k)$ showing how much structure the matrix groups embody.

4. Generlizing the last result, if $\psi : H \to \mathrm{GL}(V)$ is any representation and $\varphi : G \to H$ is a group homomorphism, then $\psi \circ \varphi$ is a representation of G on V.

5. Recall that in section 20.1.3 we defined a *group character* to be a homomorphism from G to $L^\times$. This is in fact a 1-dimensional representation of G, since $\mathrm{GL}_1(L) = L^\times$. For a concrete

example, let $G = C_n$ be the cyclic group of order n and $\zeta_n \in \mathbb{C}^\times$ a *primitive* nth root of unity, and let a be a generator of G. Then the map:

$$\chi : G \to \mathbb{C}^\times \qquad a^i \mapsto \zeta_n^i$$

is a faithful representation of G (primitivity is what makes it faithful).

6. Let $G = D_4$, the dihedral group of order 8 (we write $|D_n| = 2n$). We already have the regular representation of dimension 8 and permutation representation of dimension 8, but can we do better? In fact we can. In order to see this, let's first write out D_4 in terms of generators and relations:

$$D_4 = \langle r, s \mid r^4 = s^2 = 1,\ rs = sr^{-1} \rangle$$

Let $k = \mathbb{R}$ and consider $\mathrm{GL}_2(\mathbb{R})$. Matrices satisfying these relations determine a representation of degree 2. It is faithful precisely when their images generate a group of order 8, which one checks directly for the matrices below. This can be laborious, but with some work we can find:

$$r \mapsto \begin{pmatrix} \cos(2\pi/4) & -\sin(2\pi/4) \\ \sin(2\pi/4) & \cos(2\pi/4) \end{pmatrix} \qquad s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Let $G = Q_8$. We will show how different fiels can produce different represetnations. First, recall the following two generator and relation set for Q_8:

$$Q_8 = \langle i, j \mid i^4 = j^4 = 1,\ i^2 = j^2,\ i^{-1}ji = j^{-1} \rangle$$

$$Q_8 = \langle i, j, k \mid i^2 = j^2 = k^2 = ijk \rangle$$

Working over $\mathbb{C}$ in $\mathrm{GL}_2(\mathbb{C})$, we can use the first representation an find:

$$i \mapsto \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \qquad j \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

can you find one if $k = \mathbb{R}$ and $\mathrm{GL}_4(\mathbb{R})$? (Hint: think of the quaternions $\mathbb{H}$)

7. Let $H \trianglelefteq G$ which is further an *elementary* abelian p-group for some prime p (that is, abelian of exponent p). Without exponent p the scalar action below is not well defined. Then $V = H$ is a vector-space over $\mathbb{F}_p$ with the action of $a \cdot v = v^a$. Then define the action of G on H by conjugation: $g \cdot h = ghg^{-1}$. This action is well-defined since $gh^a g^{-1} = (ghg^{-1})^a$. The kernel of this action is $C_G(H)$. Such an action is called a *module representation*.

8. Let V be the $k[S_n]$-module which affords the permutation representation. We will find 2 S_n-invariant subspaces: Let $N \subseteq V$ be the subspace defined by:

$$N = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n\}$$

Then this a subspace that is also S_n-invariant. As practice, show that if $n \geq 3$, then N is the unique 1-dimension subspace which is S_n-stable. This submodule is called the *trace* submodule of V.

Another natural S_n-stable subspace is the following: take $I \subseteq V$ such that

$$I = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0\}$$

notice that $I = \ker(\varphi)$ where $\varphi : V \to \mathbb{C}$, $\varphi(\alpha_1 e_1 \alpha_2 e_2 \cdots \alpha_n e_n) = \sum_i \alpha_i$, and so I is $n-1$ dimensional. Since φ is called the *augmentation map* (was in section 7.3 of D&F), I is called the *augmentation module*.

9. If V is a regular representation of G so that $V = k[G]$, then V has the same two natural G-invariant subspaces:

$$N = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n\}$$
$$I = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0\}$$

these are both in fact *two-sided ideals*, even if G is not abelian. Since they appear naturally, they are usually given a name: N is called the *trace ideal* and I is called the *augment ideal*.

Corollary 3.1.6: Schur's Lemma For Representations

Let V, W be irreducible G-representations (over $\mathbb{C}$) with represetnations ρ_V and ρ_W. Then:

1. If $V \ncong_G W$, then there does not exist a non-trivial G-linear map between V and W
2. Let's say $V = W$, are finite dimensional vecor-spaces with $\rho_V = \rho_W$, then the only non-trivial G-linear maps from V to W are scalars: $\lambda I : V \to W$, $\lambda I(v) = \lambda v$.

***Proof*:**

1. Since V and W are irreducible G-representations, they are simple $k[G]$-modules, and hence we apply the original Schur's lemma.
2. Let $V = W$ of the same dimension with matching actions $\rho_V = \rho_W$, and let $f : V \to W$. Since f is a linear transformation over $\mathbb{C}$, there always exists an eigenvalue, say λ so that $f(x) = \lambda x$. Consider:
$$\varphi = f - \lambda I$$
so that $\varphi(x) = 0$. Since the difference of two G-linear maps is G-linear, then $\ker(\varphi)$ is a G-invariant subspace. But V is irreducible, and $\ker(\varphi) \neq 0$ since it contains the nonzero eigenvector x, and hence must be all of V, and so $\varphi(V) = 0$, telling us that $0 = f(x) - \lambda I(x)$, i.e.
$$f(x) = \lambda x$$
completing the proof.

Recall that if V is a simple R-module over a semi-simple ring R, then $\mathrm{End}_R(V) \cong_{\mathbf{Ring}} D^{\mathrm{op}}$ is a division ring, and that any finite-dimensional division $\mathbb{C}$-algebra is in fact equal to $\mathbb{C}$. This lemma now gives an explicit isomorphism between these two!

Since we are working over a vector-space, it is often convenient when possible to bring in an inner product that give further relations between our objects. The key the following inner product is that it is invariant on G-actions:

Definition 3.1.7: G-Invariant Inner Product

Let $k = \mathbb{R}$ or $\mathbb{C}$ and let V be a finite-dimensional $k[G]$-module. (An inner product presupposes an ordering or a conjugation on the scalars, so this definition is not available over a general field.) Then a *G-invariant inner product* is an inner product $\langle \cdot, \cdot \rangle$ along with the additional property of:

$$\langle g \cdot v, g \cdot w \rangle = \langle v, w \rangle$$

Instead of giving examples of this inner product, we will soon show that when G is finite and $|G|$ is invertible in k, any inner product on V can be averaged into a G-invariant one (hint: can you see how to use the "averaging" idea in Machke's Theorem to do so? The answer is given in corollary 3.1.12, page 82).

Now, the study of linear representations in themseves can difficult; we are essentially studying all possible $\mathbb{C}[G]$-module. We would rather be able to decompose them into smaller pieces, and show that any represetnation of G is some how build form these simpler pieces (similarly to our exploration of semi-simple modules). The rest of this chapter is dedicated to seeing if we can bring Artin-Wedderburn's theorem and all of it's consequences to understanding linear representations of finite groups:

Definition 3.1.8: Decomposing Representations

A representation is called *simple, reducible, indecomposable, decomposable*, or *semisimple* if the $k[G]$-module affording it has the corresponding property

Recall that indecomposable in general does not imply irreducible. The example under definition 1.1.1 could is in fact a representation of $\mathbb{Z}$ defined by:

$$n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

which is reducible (since it is in block upper-triangular form) but indecomposable: $\text{span}(e_1)$ is the only invariant line, so no basis puts it in block-diagonal form. Notice that any 1-dimensional representation is automatically an irreducible, since any $k[G]$-submodule of k would be a k-subspace of k, but k is already irreducible as vector-space over k.

To get an idea of how $k[G]$-submodules interact with the definitions above, let V be an n-dimensional representation of G affording the represetnation ρ, and $W \subseteq V$ an m-dimensional $k[G]$-submodule (i.e. m-dimensional G-invariant subspace) with $0 \neq m \neq n$. Then choosing a basis $\{w_1, w_2, ..., w_m\}$ for W, we can extend it to a basis $\{w_1, w_2, ..., w_m, v_{m+1}, ..., v_n\}$. Since W is G-invariant, for any $w \in W$, $\rho(g)(w) \in W$. In matrix representation, $\varphi(g)(w^i) = \rho^i_j w^j$, and the coefficients of $v^{m+1}, ..., v_n$ are zero. Hence we get the matrix representation of $\rho(g)$ as:

$$\begin{pmatrix} \alpha(g) & \beta(g) \\ 0 & \gamma(g) \end{pmatrix}$$

Notice too that:

$$\begin{aligned}\begin{pmatrix}\alpha(gh) & \beta(gh)\\ 0 & \gamma(gh)\end{pmatrix} &= \rho(gh)\\ &= \rho(g)\rho(h)\\ &= \begin{pmatrix}\alpha(g) & \beta(g)\\ 0 & \gamma(g)\end{pmatrix}\begin{pmatrix}\alpha(h) & \beta(h)\\ 0 & \gamma(h)\end{pmatrix}\\ &= \begin{pmatrix}\alpha(g)\alpha(h) & \alpha(g)\beta(h)+\beta(g)\gamma(h)\\ 0 & \gamma(g)\gamma(h)\end{pmatrix}\end{aligned}$$

which shows that α and γ are themselves representations. The converse is immediate and needs no computation: if every matrix is block upper triangular in a given basis, the span of the first m basis vectors is G-invariant. As an example of this, let V be the permutation representation of S_3. Then the set of all vectors that are invariant to the action of $G = S_3$ is $V^G = \text{span}(e_1 + e_2 + e_3)$. Let $W = \text{span}(e_1 + e_2 + e_3)$. Since $G \curvearrowright W$ is of the form $g \cdot w = w$, we see that W is in fact the *trivial representation*. Since V is semi-simple, we have that $V = V^G \oplus U$ for some G-stable complement U. Making some new basis for V, say $(e_1 + e_2 + e_3, e_2, e_3)$, we see that $S_3 \curvearrowright V$ will have matrix representations that are of the above form:

$$\begin{pmatrix}1 & * & *\\ 0 & * & *\\ 0 & * & *\end{pmatrix}$$

If V were irreducible, then no such basis would exist. If $V = U \oplus W$, then we would get a block diagonal form. It turns out that all representations of *finite* groups over a field of characteristic zero, or of characteristic not dividing $|G|$, are block-diagonal, i.e. are semisimple. (Finiteness is essential: the representation of $\mathbb{Z}$ displayed above is over $\mathbb{C}$ and is not semisimple.)

Theorem 3.1.9: Maschke's Theorem

Let G be a finite group and let k be a field whose characteristic does not divide $|G|$ or $\text{char}(k) = 0$. Then if V is a $k[G]$-module, it is a semi-simple $k[G]$-module.

The condition that $\text{char}(k) \nmid |G|$ is equivalent to $|G|$ being invertible in k, which is what the proof below divides by. Note how this theorem is "surprising": a finite group G certainly cannot be reduced to a direct sum of simple groups, only built from them as iterated extensions, and (as of writing this book) we do not know how to construct all finite groups via such extensions. Hence, having a very computable decomposition into irreducible representations provides a very potent tool to analyze the structure of finite groups.

***Proof*:**
We will use lemma 1.1.7[3] to show that for any $U \subseteq V$, there exists a complement W such that $V = U \oplus W$.

We will find a projection onto U: $\pi : V \to U$. Thus, we need to show it has the following properties:

1. $\pi(u) = u$ for all $u \in U$
2. $\pi(\pi(v)) = \pi(v)$ for all $v \in V$[a]

To see why this is sufficient, let $\pi\colon V \to U$ be a $k[G]$-module homomorphism with these properties. Let $W = \ker(\pi)$. Then $W \subseteq V$ is a submodule, and W is a complement of U. First, let $v \in U \cap W$. Then since $v \in U$, $\pi(v) = v$ and since $v \in W$, $\pi(v) = 0$, hence $v = 0$, meaning $U \cap W = 0$. Next, let $v \in V$ be an arbitrary element, and write $v = \pi(v) + (v - \pi(v))$. By definition, $\pi(v) \in U$, and by the 2nd property:

$$\pi(v - \pi(v)) = \pi(v) - \pi(\pi(v)) = \pi(v) - \pi(v) = 0$$

and hence $v - \pi(v) \in W$. Hence $v \in U + W$, and so $V = U + W$, hence $V \cong U \oplus W$. Hence, it suffices to find an $k[G]$-module projection π.

Let $U \subseteq V$ be a $k[G]$-submodule. Then it is already a k-subspace, and so it has a vector-space direct sum complement $W_0 \subseteq V$ (take a basis β for U, extend it to a basis $\mathcal{B}$ for V, then take $\mathcal{B} - \beta$) Thus, $V = U \oplus W_0$ as a vector-space. However, W_0 need not be G-stable (i.e. a $k[G]$-submodule), for example $G = C_2$ on $\mathbb{R}^2$ with the reflection across $y = x$ which has the G-subspace $y = x$ and complement spanned by $(1, 0)$, which is not G-invariant. Let $\pi_0\colon V \to U$ be the vector-space projection map:

$$\pi_0(u + w_0) = u$$

To find a G-stable complement, we average π_0 over G:

$$g\pi_0 g^{-1} : V \to U \qquad g\pi_0 g^{-1}(v) = g \cdot \pi_0(g^{-1} \cdot v)$$

Since π_0 maps V to U, and U is a $k[G]$-submodule, U is stable under this G-action, hence $g\pi_0 g^{-1}$ is well-defined. Note that both g and g^{-1} act as k-linear transformations, so $g\pi_0 g^{-1}$ is a k-linear transformation. Furthermore, if $u \in U$, then so must $g^{-1}u$, so by definition of π_0, $\pi_0(g^{-1}u) = g^{-1}u$. Thus, for all $g \in G$:

$$g\pi_0 g^{-1}(u) = u$$

hence, $g\pi_0 g^{-1}$ is also a vector-space projection of V onto U!

Let $n = |G|$, and viewing n as an element of k ($n = 1 + \cdots + 1$), since $\mathrm{char}(k) \nmid |G|$ we have $n \neq 0$, i.e. $n \in k^\times$. Thus, n has an inverse in k, meaning we can define:

$$\pi = \frac{1}{n} \sum_{g \in G} g\pi_0 g^{-1}$$

We can immediately deduce that π has the following properties:

1. By definition, π is still k-linear from V to U.
2. Restricting π to U would restrict each summand in the definition of π, and so as we have pointed out earlier each summand would be the identity map. Hence, we get $1/n$ times n copies of u, meaning $\pi|_U = \mathrm{id}$
3. π is well-defined, hence $\pi(v) \in U$, and since π restricts to the identity on U we get $\pi(\pi(v)) = \pi(v)$, i.e. $\pi^2 = \pi$

It remains to show that π is in fact a $k[G]$-module homomorphism (i.e. is $k[G]$ linear). It suffices to show that for any $h \in G$, $\pi(hv) = h\pi(v)$. This is simply a computation:

$$\begin{aligned}
\pi(hv) &= \frac{1}{n}\sum_{g\in G} g\pi_0(g^{-1}hv) \\
&= \frac{1}{n}\sum_{g\in G} h(h^{-1}g)\pi_0(g^{-1}hv) \\
&\overset{!}{=} \frac{1}{n}\sum_{\substack{x=h^{-1}g \\ g\in G}} h(x\pi_0(x^{-1}v)) \\
&= h\frac{1}{n}\sum_{\substack{x=h^{-1}g \\ g\in G}} (x\pi_0(x^{-1}v)) \\
&= h\pi(v)
\end{aligned}$$

where $\overset{!}{=}$ is the reindexing $x = h^{-1}g$, valid because left translation by h^{-1} is a bijection of G, so as g runs over G so does x. Hence, π is indeed a $k[G]$-homomorphism, and so it is a projection map, completing the proof.

[a] In fact, this is implied by 1

Thus, if we take any representation V over k of G, and consider the $k[G]$-module, then we know through $k[G]$ and G being finite and k a field (so V is a vector-space) that there are only *finitely many* simple $k[G]$-modules (i.e. finitely many irreducible representations of G), meaning we only have a finite number of "building blocks" from which we can build) any representation for G! Even better, the way we put together irreducible representation is through direct sums the simplest of ways of putting together algebraic objects! Then through the Artin-Wedderburn decomposition of $k[G]$ we obtain the list of irreducibles, and Maschke's theorem decomposes any representation V of G into them, giving us a simple finite list of irreducible representations per finite group! Note that if $p = \mathrm{ch}(k) > 0$ and $p \big| |G|$, then $k[G]$ is *not* semisimple, or if G is infinite, it need not be semi-simple

Example 3.1.10: Non-Semisimple Group Rings

For the infinite case, the action of $\mathbb{Z}$ on k^2 presented above is a non-semisimple $k[\mathbb{Z}]$-module. For finite groups, cyclic groups already exhibit both behaviors. Let $G = C_n$ be cyclic of order n, generated by g.

1. The k-algebra map $k[a] \to k[C_n]$, $a \mapsto g$, is surjective because g generates C_n, and its kernel contains $a^n - 1$. Both $k[a]/(a^n - 1)$ and $k[C_n]$ have dimension n over k, with bases $1, a, \ldots, a^{n-1}$ and $1, g, \ldots, g^{n-1}$, so the induced map $k[a]/(a^n - 1) \to k[C_n]$ is an isomorphism.
2. Suppose $\mathrm{char}(k) \nmid n$. The derivative na^{n-1} of $a^n - 1$ is nonzero and shares no root with $a^n - 1$, so $a^n - 1 = f_1 \cdots f_r$ with $f_1, \ldots, f_r$ distinct monic irreducible polynomials, which are pairwise coprime in the principal ideal domain $k[a]$. The Chinese Remainder Theorem [4, Chinese Remainder Theorem] gives

   $$k[C_n] \cong k[a]/(f_1) \times \cdots \times k[a]/(f_r)$$

   a product of fields. A field is M_1 of a division ring, so this is an Artin-Wedderburn

decomposition of $k[C_n]$, and $k[C_n]$ is semisimple by theorem 1.2.16.

3. Suppose $\text{char}(k) = p$. In $k[C_p]$ the element $g-1$ is nonzero, while $(g-1)^p = g^p - 1 = 0$ because the binomial coefficients $\binom{p}{i}$ vanish in k for $0 < i < p$. A product of fields has no nonzero nilpotent elements, so $k[C_p]$ is not a product of fields. Since $k[C_p]$ is commutative, each factor $M_{n_i}(D_i)$ of an Artin-Wedderburn decomposition of it would be commutative, which forces $n_i = 1$ (as $E_{12}E_{21} \neq E_{21}E_{12}$ when $n_i \geq 2$) and D_i a field. Hence $k[C_p]$ is not semisimple, by theorem 1.2.16.

If $k = \mathbb{C}$, then Maschke's Theorem always applies, and we can further use Artin Wedderburn over closed k-algebras (corollary 1.2.20) to deduce a relatively simple structure! Hence, from now on, we will always have $k = \mathbb{C}$, meaning all of our representations of finite groups will be over the complex vector-field of finite dimension.

Corollary 3.1.11: Matrix Representation Consequences

Let G be finite, let V be finite dimensional, and let $\rho : G \to \text{GL}(V)$ be a representation over a field k where $\text{char}(k) \nmid |G|$. Then there exists a basis for V such that for all $g \in G$, the matrix $\rho(g)$ is of block-diagonal form:

$$\begin{pmatrix} \rho_1(g) & & & \\ & \rho_2(g) & & \\ & & \ddots & \\ & & & \rho_m \end{pmatrix}$$

where ρ_i is an irreducible representation of G, $1 \leq i \leq m$.

***Proof*:**
Let V be the $k[G]$-module representing ρ. By Maschke's Theorem, we may write $V = U_1 \oplus \cdots \oplus U_m$. where U_i is simple $k[G]$-module. For each U_i, choose a basis β_i. Let $\beta = \bigcup_i^m \beta_i$. then each matrix representation of $\rho(g)$ is of the form in the corollary, with $\rho(g)|_{U_i} = \rho_i(g)$, completing the proof.

Corollary 3.1.12: Defining G-Invariant Inner Product

Let $k = \mathbb{R}$ or $\mathbb{C}$, let G be finite, let V be a finite-dimensional $k[G]$-module, and let $\langle \cdot, \cdot \rangle$ be any inner product on V. Then this inner product induces a G-invariant inner product

***Proof*:**
If $\langle \cdot, \cdot \rangle$ is an inner product, then:

$$\langle v, w \rangle_G = \frac{1}{|G|} \sum_{g \in G} \langle gv, gw \rangle$$

is a well-defined inner product (as can be checked), and for any $h \in G$

$$\begin{aligned}\langle hv, hw\rangle_G &= \frac{1}{|G|}\sum_{g\in G}\langle ghv, ghw\rangle \\ &= \frac{1}{|G|}\sum_{x=gh\in G}\langle xv, xw\rangle \\ &= \langle v, w\rangle_G\end{aligned}$$

In other words, there is always an inner product on our space in which the group actions keeps all elements of the same length and the same angle between them, showing how we can interpret a group as somehow rotating and reflecting an object in a vector-space. Now that we know any represetnation of G is composed in some way by finitely many irreducible representations, it would be useful to be able to quickly deduce the Artin-Wedderburn decomposition of $\mathbb{C}[G]$:

Proposition 3.1.13: Properties Of Representations

Let $k = \mathbb{C}$ and G be finite. Let $S_1, S_2, ..., S_r$ be the irreducible G-representations (up to isomorphism). Then:

1. the number of irreducible G-representations is equal to the number of conjugacy classes in G
2. For every finite-dimensional representation V of G,

   $$V \cong S_1^{\oplus \ell_1} \oplus \cdots \oplus S_r^{\oplus \ell_r}$$

   where the $\ell_i \geq 0$ are uniquely determined, namely $\ell_i = \dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}[G]}(S_i, V)$. (An irreducible need not occur at all: the trivial representation of S_3 has $\ell_2 = \ell_3 = 0$.)
3. $\mathbb{C}[G] \cong \bigoplus_{i=1}^{r} \left(S_i^{\oplus \dim_k S_i}\right)$ as regular representation
4. $|G| = \sum(\dim_k S_i)^2$

***Proof*:**
part (3) and (4) come from corollary 1.2.20.

For part (1), we first use Artin-Wedderburn to conclude that $Z(\mathbb{C}[G]) = \prod_i^r \mathbb{C}$, so $\dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = r$, so the dimension of the center is equal to the number of irreducible G-representations.

Now, let $\mathcal{K}_1, ..., \mathcal{K}_s$ represent the conjugacy classes (recall they partition G). Define:

$$\sigma_i = \sum_{g\in\mathcal{K}_i} g$$

Note that any pair of elements σ_i and σ_j have no common terms since the conjugacy classes partition G. Furthermore, by definition, for any $g \in G$, $g\sigma_i g^{-1} = \sigma_i$, and so $\sigma_i \in Z(\mathbb{C}[G])$. If the σ_i's form a basis for $Z(\mathbb{C}[G])$, then we would have $s = r$. As a $\mathbb{C}$-vector-space, they are already linearly independent since the group elements from a basis, so it suffices to show they

span. If $\sum_{g\in G}\lambda_g g \in Z(\mathbb{C}[G])$. Then by definition for all $h \in G$,

$$\begin{aligned}
&\Leftrightarrow\ h\left(\sum_{g\in G}\lambda_g g\right) = \left(\sum_{g\in G}\lambda_g g\right)h \\
&\Leftrightarrow\ \left(\sum_{g\in G}\lambda_g g\right) = h^{-1}\left(\sum_{g\in G}\lambda_g g\right)h \\
&\Leftrightarrow\ \sum_{g\in G}\lambda_g g = \left(\sum_{g\in G}\lambda_g (h^{-1}gh)\right) \\
&\overset{!}{\Leftrightarrow}\ \sum_{g\in G}\lambda_g g = \sum_{g\in G}\lambda_{hgh^{-1}} g \\
&\Leftrightarrow \lambda_{hgh^{-1}} = \lambda_g \qquad \forall g \in G
\end{aligned}$$

since h arbitrary, we see that the coefficients of elements in the same conjugacy class remain with an element that's in that conjugacy class. Equivalently, the function $G \to \mathbb{C}$, $g \mapsto \lambda_g$ is constant on conjugacy classes. Thus:

$$Z(\mathbb{C}[G]) = \text{span}\left\{\sum_{g\in G} f([g])g\right\}$$

where f is a function from the conjugacy classes of G to $\mathbb{C}$. Hence $s = r$.

for (2), Since $\mathbb{C}[G]$ is a semi-simple ring, V is a semi-simple $\mathbb{C}[G]$-module, giving us existence of the decomposition by AW. For uniqueness, first note that:

$$\text{End}_{\mathbb{C}[G]}(S_i) = \mathbb{C}$$

since $\mathbb{C}$ is the only [finite-dimensional?] division $\mathbb{C}$-algebra. Then, we get by Schur's lemma:

$$\text{Hom}_{\mathbb{C}[G]}(S_i, V) \overset{\text{Schur}}{\cong} \text{Hom}_{\mathbb{C}[G]}(S_i, S_i)^{\oplus l_i} \cong \mathbb{C}^{l_i}$$

Corollary 3.1.14: Abelian Groups And Degree 1 Decomposition

1. If A is a *finite* abelian group, then every irreducible complex representation of A is 1-dimensional (i.e. a homomorphism from A to $\mathbb{C}^\times$), and A has $|A|$ inequivalent irreducible complex representations.
2. The number of inequivalent (irreducible) 1-dimensional complex representations of any finite group G is equal to $|G/G'|$

***Proof*:**

1. Since A is abelian, so is $\mathbb{C}[A]$. By the Artin-Wedderburn decomposition, it must be a product of matrices, which since $\mathbb{C}[A]$ abelian is so must each matrix ring, which is only

true if and only if $n_i = 1$. Then, by corollary 1.2.20, $\mathbb{C}[A]$ is a product of fields. It follows that any irreducible representation (i.e. simple $\mathbb{C}[A]$-module) is 1-dimensional. Conversely, if A were not abelian, $\mathbb{C}[A]$ cannot be a product of fields, hence one of the dimensions of the matrix rings *must* be greater than 1

2. Let G be an arbitrary group. We want to show that there is a bijective correspondence between the 1-dimension representations of G and the 1-dimensional representations of G/G'. $\rho : G/G' \to \mathbb{C}^\times$ is a 1-dimensional representations of G/G', then we have the natural map $G \to G/G'$, which shows that each 1-dimensional representations of G/G' is also a 1-dimensional representations of G. Convexly, if f $\rho : G \to \mathbb{C}^\times$ is a 1-dimensional representations of G, then by the universal property, this factors through:

$$\begin{array}{ccc} G/G' & \longrightarrow & \mathbb{C}^\times \\ \uparrow & \nearrow & \\ G & & \end{array}$$

showing that any 1-dimensional representation of G is associated to a 1-dimensional representations of G/G', showing there cannot be more 1-dimensional representations of G than there are of G/G', completing the proof.

A more general technique can be adopted from this proof: If $H \trianglelefteq G$ and V is an irreducible representation of G/H, then V is an irreducible representation of G.

Example 3.1.15: Decomposition Of Modules

1. DnF p. 847, and a bit about $\dim(V) = 1$ example right under definition

Finally, for those who read chapter 50, the following results are a glimpse at how representation theory is a "generalization" of number theory. First, if $H \leqslant G$ is a subgroup and (ρ, V) is an irreducible representation, $\rho|_H$ breaks up into the direct sum of representations, not necessarily of the same size or multiplicity. This is similar to how for a prime $\mathfrak{p} \subseteq F$ it may split to multiple primes with varying inertia degree and ramification numbers in a finite separable extension L/F. If L/F is furthermore normal (and hence Galois) then all the inertia degree's and ramification numbers match. In a similar vein:

Theorem 3.1.16: Clifford's Theorem

G be a group, $N \trianglelefteq G$ a normal subgroup, K a field, $\pi : G \to \mathrm{GL}_n(K)$ be an irreducible representation. Then the restriction $\pi|_N$ breaks up into a direct sum of irreducible representations of N of equal dimensions (i.e. it is semisimple and each representation is of the same dimension). These irreducible representations of N lie in one orbit for the action of G by conjugation on the equivalence classes of irreducible representations of N (i.e. they are isotypic components with the same multiplicity).

***Proof*:**
Write V for the space of π and let $W \subseteq V$ be an irreducible N-subrepresentation.

Step 1: $V|_N$ is semisimple. For $g \in G$ the subspace gW is N-stable, because N is normal: for $n \in N$ one has $n(gW) = g(g^{-1}ng)W = gW$, using $g^{-1}ng \in N$. It is irreducible over N, since $w \mapsto gw$ carries N-subrepresentations of W bijectively onto N-subrepresentations of gW, and the representation it affords is $n \mapsto \pi_W(g^{-1}ng)$, the conjugate π_W^g. In particular all the gW have the same dimension as W. Now $\sum_{g\in G} gW$ is stable under N by the above and under G because $h\sum_g gW = \sum_g (hg)W$, so it is a non-zero subrepresentation of the irreducible V and therefore equals V. A sum of irreducible subrepresentations is semisimple, so $V|_N$ is semisimple and all its irreducible constituents have $\dim W$.

Step 2: one orbit, with equal multiplicities. For an irreducible representation U of N let $V_U \subseteq V$ be its isotypic component in $V|_N$. If $g \in G$ then $v \mapsto gv$ carries V_U onto V_{U^g}, where $U^g(n) = U(g^{-1}ng)$: it is a bijection by Step 1 applied to each constituent, and it is an isomorphism of vector spaces, so V_U and V_{U^g} have equal dimension and hence U and U^g occur with equal multiplicity. Consequently G permutes the isotypic components, and the sum of those lying in a single G-orbit of isomorphism classes is a G-subrepresentation. It is non-zero, so it is all of V: the classes occurring form exactly one orbit, and the multiplicities agree across it, completing the proof.

Exercise 3.1.1

1. If χ_1 and χ_2 are both 1 dimensional representations, then $\chi_1 \cong \chi_2 \Leftrightarrow \chi_1 = \chi_2$. Use the property of equivalent representations to see this.
2. Let V be a non-trivial irreducible G-representation (i.e. not $\chi(g) = 1$ for all $g \in G$). Show that there are $\dim_{\mathbb{C}}(V^G) = 0$
3. By Caley's Theorem, all finite groups are linear. Do there exists infinite non-linear groups? (Hard: Do there exists finitely generated non-linear groups?)
4. Show that if ρ is irreducible and $N \trianglelefteq G$ is normal, and N fixes some *nonzero* vector $v \in V$, then N acts trivially on all of V.

3.2 Character Theory

We know now we can decompose a G-representation V into the direct sum of r different types of irreducible G-modules. How do we now proceed to find these? These can be quite unwieldy: if we have a 10 or 100-dimensional representation of a group G, then we have a matrix with 100 and 10′000 entries! We would then have to multiply these and figure out what relations they have between each other, quite a hassle! Furthermore, how do we do this in such way that is independent of equivalent representations? A convenient basis might be nice, but to find a nice basis for every $\rho(g) \in \rho(G)$, we'd have to find a basis that can *simultaneously* diagonalize every element in $\rho(G)$! If there was some invariant we can find on a representation that we can use to keep track of the decomposition information between representations that is invariant of representation, that would be useful to this endevaour. The following does exactly this (Move the discussion of trace here to build-up to it? Give some of D&F examples to demonstrate it does get quite ugly sometimes?)

Definition 3.2.1: Class Function and Character

Let G be finite and let V be a finite-dimensional representation of G over an arbitrary field F. Then

1. A function $f : G \to F$ that is constant on conjugacy classes is called a *class function* of G
2. If φ is a representation of G afforded by the $F[G]$-module V, then a function:

$$\chi_V : G \to F \qquad g \mapsto \text{tr}(g : V \to V) = \text{tr}(\varphi(g))$$

is called the *character* of φ, where $\text{tr}(\varphi(g))$ is the trace of φ with respect to some basis for V. A character is called *irreducible* if φ is irreducible (and reducible oherwise). The *degree* of a character is the degree of any representation affording it.

For ease of reference, recall these properties of the trace:

1. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ an $\text{tr}(cA) = c\text{tr}(A)$
2. $\text{tr}(A) = \text{tr}(A^t)$
3. $\text{tr}(AB) = \text{tr}(BA)$

Note that in general, a character is *not* a homomorphism from G into the additive or multiplicative structure of k. Since $\text{tr}(AB) = \text{tr}(BA)$, $\text{tr}(ABA^{-1}) = \text{tr}(B)$, we see that the trace (and hence the character) is independent of choice of basis, and so equivalent representations have the same character, meaning the character is indeed an invariant. Furthermore, for any representation φ of G,

$$\varphi(g^{-1}xg) = \varphi(g)^{-1}\varphi(x)\varphi(g)$$

hence, we can take the trace and use the 3rd property above to show that characters an an example of *class functions*. Furthermore, since the trace is linear, we can extend the domain of a character to be the trace over any element in $\mathbb{C}[G]$. If $F = \mathbb{C}$, then the collection of class function's form a complex vector-space, since the sum of two class functions is a class function, and multiplying by a scalar $z \in \mathbb{C}$ does not change that a class function is invariant.

Example 3.2.2: Character

1. Given the trivial representation of G (i.e. any 1-dim vector space where $g \mapsto I$), then if φ is the representating affording the trivial representation, it has $\chi_\varphi(g) = 1$ for all g. This is called the *trivial character*.
2. If we have a degree 1-representation, then the character and the representation are the same. Thus for abelian group, complex representations and their characters are the same. For example, take the representation $\varphi : C_p \to \mathbb{C}^\times$, $\varphi(a^i) = \zeta_p^i$. Then:

$$\chi(a^i) = \zeta_p^i$$

3. Let $\Pi : G \to S_n$ be a permutation representation of G and φ the linear of G of degree n on V where:

$$\varphi(g)(e_i) = e_{\Pi(g)(i)}$$

with respect to the given basis, $\varphi(g)$ has 1 in the ii-entry if $\Phi(g)$ fixes i, and zero otherwise. Thus, we can say that the character χ is:

$$\chi(g) = \#\{\text{number of fixed points of } g \text{ on } \{1, 2, ..., n\}\}$$

The character of the representation attached to any permutation action is called the *permutation character*. The case where G acts on itself by left multiplication gives the regular character below.

4. Consider any regular representation of G, φ. Then the character is:

$$\chi(g) = \begin{cases} 0 & g \neq 1 \\ |G| & g = 1 \end{cases}$$

This character is called the *regular character* of G (note how this character is not a group homomorphism into k or $k^\times$)

5. Take the representation $\varphi : D_n \to \mathrm{GL}_2(\mathbb{R})$ from example 3.1.5. Then:

$$\chi(r) = 2\cos(2\pi/n) \qquad \chi(s) = 0 \qquad \chi(1) = 2$$

where the last one is since $1 \in G$ must map to the identity matrix $I \in \mathrm{GL}_2(\mathbb{R})$.

From now on, all our representation will be complex.

Lemma 3.2.3: Properties Of Character

Let $\rho_V : G \to \mathrm{GL}(V)$ be a complex representation. Then:

1. $\chi_V(1) = \dim_{\mathbb{C}}(V)$
2. $\rho_V(g)$ is diagonalizable for all $g \in G$ and $\chi(g)$ is integral over the integers: $\chi(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$
3. If $g \in G$ has order n, then $\chi_V(g)$ is a sum of $\dim(V)$ number of nth roots of 1
4. $|\chi_V(g)| \leq \chi_V(1)$ for all $g \in G$
5. $\chi_V(g^{-1}) = \overline{\chi_V(g)}$

***Proof*:**

1. We showed this in the above examples.
2. The criterion we need from linear algebra is that $A \in M_n(\mathbb{C})$ is diagonlizable if and only if the minimal polynomial has no repeated roots, which can be implied if $M^k = I$ for some k, which it certainly is in our case since every element is finite.

3. If $g^n = 1$, then $\rho_V(g)^n = 1$, so by (2)

$$\rho_V(g) \sim \begin{pmatrix} z_1 & & \\ & \ddots & \\ & & z_k \end{pmatrix}$$

where $z_i^n = 1$ for all $1 \leq i \leq k$. Furthermore, this shows that these trace values in the matrix representation are always integral over the integers, being the sum of integral elements.

4. $$|\chi_V(g)| = \left|\sum z_i\right| \leq \sum |z_i| = \dim_{\mathbb{C}}(V) = \chi_V(1)$$

5. Since $\rho_V(g^{-1}) = \rho_V(g)^{-1}$ and

$$\rho_V(g)^{-1} \sim \begin{pmatrix} z_1^{-1} & & \\ & \ddots & \\ & & z_k^{-1} \end{pmatrix}$$

$\chi_V(g^{-1}) = \sum_i z_i^{-1}$, but $\|z_i\| = 1$, so $z_i^{-1} = \overline{z_i}$, so

$$\chi_V(g^{-1}) = \sum_i z_i^{-1} = \sum_i \overline{z_i} = \overline{\chi_V(g)}$$

Now, if we have some G-representation V, it has some character χ. In the following, we will show how we can always break-down a character to be the sums of characters on the irreducible G-representations:

Proposition 3.2.4: Linearly Independent Characters

Let G be a finite group and V a semi-simple $k[G]$-module, and φ the representation that V affords. Part (1) holds over any field. Part (2) requires $k = \mathbb{C}$ (or an algebraically closed field of characteristic 0), and is *false* over a splitting field of characteristic p dividing $|G|$, where the irreducible characters are fewer than the conjugacy classes. Then:

1. The character of V is the sum of the characters of the components appearing in the direct sum decomposition.
2. The irreducible characters of G are linearly independent set in the set of class functions. Furthermore, the irreducible characters of G form a basis of the vector-space of class functions.

***Proof*:**
For the 1st point, By the Artin-Wedderburn decoposition, we have:

$$V \cong (S_1)^{n_1} \oplus \cdots \oplus (S_k)^{n_k}$$

where S_i are simple $k[G]$-modules (irreducible G-representations). If we focus on the case where

$M = S_1 \oplus S_2$, then we can choose a basis for S_1 and S_2 which will give us:

$$\varphi(g) = \begin{pmatrix} \varphi_1(g) & 0 \\ 0 & \varphi_2(g) \end{pmatrix}$$

where φ_1, φ_2 are the representations afforded by M_1 and M_2 respectively. Then if ψ_1 and ψ_2 are the characters of φ_1 and φ_2 an χ is the character of φ then we get:

$$\chi = \psi_1 + \psi_2$$

using induction get's us the result.

For the 2nd point, first notice that the class functions defined by being 1 on a conjugacy class and 0 otherise forms a basis for class functions of dimension r. It thus suffices to sho that the r irreducible characters are independent. To that end, define:

$$\widetilde{\chi_V} : \mathbb{C}[G] \to \mathbb{C} \qquad x \mapsto \mathrm{tr}(x : V \to V)$$

Note that this map is $\mathbb{C}$-linear since $\mathbb{C}$ commutes with $\mathbb{C}[G]$, and $\widetilde{\chi_V}|_G = \chi_V$. Now, say $S_1, ..., S_r$ are simple $\mathbb{C}[G]$-modules and:

$$\sum_i^r \lambda_i \chi_{S_i} = 0 \qquad \lambda_i \in \mathbb{C}$$

Then this is equivalent to:

$$\sum_i^r \lambda_i \widetilde{\chi_{S_i}}(g) = 0 \qquad \forall g \in G$$

and since the maps are $\mathbb{C}$-linear, by linearity:

$$\sum_i^r \lambda_i \widetilde{\chi_{S_i}}(x) = 0 \qquad \forall x \in \mathbb{C}[G]$$

which by plugging into the definition we get:

$$\sum_i^r \lambda_i \mathrm{tr}(x : S_i \to S_i) = 0 \qquad \forall x \in \mathbb{C}[G]$$

Next, by Artin Wederburn, we have the decomposition:

$$C[G] \cong M_{n_1}(\mathbb{C}) \times \cdots \times M_{n_r}(\mathbb{C})$$

where each $M_{n_i}(\mathbb{C}) \cong S_i^{\oplus n_i}$, with $S_i = \mathbb{C}^{\oplus n_i}$. Using this, take the element:

$$x_i = (0, ..., I, ..., 0)$$

where I is the identity matrix[a]. Then $x_i : S_j \to S_j$ is 0 if $i \neq j$, and the identity marix if $i = j$. Thus, the trace is:

$$\mathrm{tr}(x_i : S_j \to S_j) = \begin{cases} 0 & i \neq j \\ n_j & i = j \end{cases}$$

and since n_j is at least 1, we must have that

$$\lambda_1 = \cdots = \lambda_r = 0$$

where we took advantage of the fact we are in characeristic 0

[a] If the field had finite characteristic, it is better to take the elementary matrix E_{11}

3.2.5 Remark We can prove this directly without using Artin-Wedderburn, which we will

Corollary 3.2.6: Detecting Isomorphic Representations Using Characters

Let $k = \mathbb{C}$ and let V_1, V_2 be representations that are semi-simple modules. (Characteristic 0 is needed: over $\overline{\mathbb{F}}_2$ the trivial $\mathbb{Z}/3$-module T and $T^{\oplus 3}$ have the same character, since $3 = 1$ in k, but are not isomorphic.)

1. Every character of G is uniquely of the form

$$\sum_i^r \lambda_i \chi_i \qquad \lambda_i \in \mathbb{Z}_{\geq 0}$$

   where χ_i are irreducible characters.

2. the representations V_1 and V_2 are isomorphic if and only if $\chi_{V_1} = \chi_{V_2}$

***Proof*:**

1. This comes straight from the fact that irreducible characters form a basis, and that any G-representation V will be some direct sum of a finite list of known simple G-modules
2. If $V_1 \cong_G V_2$, there is a single change of basis P intertwining the two actions, so that if A and B are basis matrices, $A = PBP^{-1}$. But the trace is independent of conjugation, and so the characters are the same. For the othe way around, treating a representation as a $k[G]$-module, then by Machke's theorem, we *know* the list simple $k[G]$-modules and the actions of $k[G]$ on them. Since any $k[G]$-module can be broken down into a direct summand of simlpe $k[G]$-modules, and the two chracters are the same, the list of $k[G]$-modules and their multiplicity are the same.

Thus, we have translated the problem of finding the irreducible representations of a group G to finding the irreducible characters of G! Since a character of a representation is determined by the irreducible characters, we give a way to systematize the tracking of irreducible characters:

Definition 3.2.7: Character Table

Let $\varphi : G \to \mathrm{GL}(V)$ be a representaiton of G an χ_V it's character. Then we call the *character table* a table with the rows being irreducible characters and columns being an element in the conjugacy class. Common practice is to make the first column the identity conjugacy class and the first row the trivial representation

Example 3.2.8: Character Table

Consider $G = S_3$ Then we can start creating it's character table:

	e	$(1\ 2)$	$(1\ 2\ 3)$
1	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since G has 3 conjugacy classes, exactly 3 irreducible characters of G. How would we go about finding it? The comning propositions work towards this.

To be able to find the missing character, and more generally find new characters for a character table, we may look to construct new character given old one, and hope that we break down those characters into a linear combination of irreducible characters. In a sense, we are finding a way to create represetnations without doing it "by hand". We will do this in two way: the first is by taking advatange of group-actions, since are relatively easy to define and their representations are matrices with 1's and 0's. Since all groups embed into a symmetric group, this can be done systematically. We then see how to work directly with known representations to create new representations.

Definition 3.2.9: Precaution Representation

Let G act on a finite set X. Define $\mathbb{C}X$ to be the set of formal sums:

$$\mathbb{C}X = \left\{ \sum_{x \in X} \lambda_x x \;\middle|\; \lambda_x \in \mathbb{C} \right\}$$

So that we get a $\mathbb{C}$-vector space with $\dim_{\mathbb{C}}(\mathbb{C}X) = |X|$. Then we can make this a G-represetnation by defining:

$$g \cdot \left(\sum \lambda_x x \right) = \sum \lambda_x (g \cdot x)$$

The 3 usual permutation representations are $G \curvearrowright G$ by left multiplication (which gives the regular representation) or conjugation, and $G \curvearrowright \{*\}$ trivially (which gives the trivial representation).

Lemma 3.2.10: Character Of Permutation Representation

Let G act on X, $G \curvearrowright X$. Then

$$\chi_{\mathbb{C}X}(g) = \#\{x \in X \mid gx = x\}$$

***Proof*:**
FIx $g \in G$. Then the set of x's is a basis for the representation. To make the computation simpler, we will find a convinient order for them. In particular, by choosing any $x_1 \in X$, we can find the orbit of the element:

$$x_1 \xrightarrow{g} x_2 \longrightarrow \cdots \longrightarrow x_k \qquad x_{k+1} \xrightarrow{g} \cdots$$

so the basis representation will look like:

$$\begin{pmatrix} 0 & & & \\ 1 & 0 & & \\ & & \ddots & \\ 0 & 1 & & \\ & & 1 & 0 \end{pmatrix}$$

so we get that the trace is 0 if the order of the orbit is greater than 1, hence:

$$\chi_{\mathbb{C}X}(g) = \#\{g\text{-orbits of size } 1\}$$

An immediate consequence of this is that we have another way of computing the character of the regular representation

$$\chi_{\mathbb{C}G}(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$$

which makes sense, since the $\dim_{\mathbb{C}}(\mathbb{C}G) = |G|$.

Next, we turn towards a more systematic way of constructing characters by creating them out of pervious ones, that is, given some representations V and W of a group G, we can make new representations by creating new representations from V and G:

Definition 3.2.11: Constructing Representations from smaller ones

Let V, W be G-representations. Then

1. we can make a $V \oplus W$ representation by:

$$g \cdot (v, w) = (g \cdot v, g \cdot w) \ [(\rho_V(g)v, \rho_W(g)w)]$$

2. we can make $V \otimes_{\mathbb{C}} W$ a representations by:

$$g \cdot (v \otimes w) \mapsto (gv) \otimes (gw)$$

3. we can make $\mathrm{Hom}_{\mathbb{C}}(V, W)$ a representations by:

$$g \cdot f : V \to W \quad v \mapsto g \cdot f(g^{-1} \cdot v)$$

4. we can make $V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$ a representaiton by:

$$g \cdot f : V \to \mathbb{C}, \ v \mapsto f(g^{-1} \cdot v)$$

Note that the last construction is the same as the 3rd if we take $\mathbb{C}$ as the trivial representation. Before we show what the character's of these new representations are, we first find a relatively simpler way of representing the hom representation of V and W:

Lemma 3.2.12: Hom-Tensor Representation Dual

Let V, W be G representations. Then:

$$\mathrm{Hom}_{\mathbb{C}}(V,W) \cong_G V^* \otimes_{\mathbb{C}} W$$

as G-representations.

Note that this is not true in general, even for vector-spaces. It is important that the spaces are finite-dimensional. This is a different statement from the hom-tensor adjunction of theorem 15.3.9, not a strengthening of it: the adjunction holds for arbitrary modules, whereas the lemma here needs $\dim_{\mathbb{C}} V < \infty$. It is the more usable description for the case of finite-dimensional vector-spaces.

***Proof*:**
By the universal property:

$$\theta : V^* \otimes_{\mathbb{C}} W \to \mathrm{Hom}_{\mathbb{C}}(V,W) \qquad \lambda \otimes w \mapsto (v \mapsto \lambda(v)w)$$

you should check that this is well-define (note that it is bilinear).

Now, pick basis $e_1, e_2, ..., e_n$ for V, $f_1, f_2, ..., f_m$ for W, and the natural dual $e_1^*, ..., e_n^*$ for V^*. Then we get a basis for $V^* \otimes W$:

$$(e_i^* \otimes f_j) \qquad 1 \le i \le n,\ 1 \le j \le m$$

Now, notice that:

$$\theta(e_i^* \otimes f_j) \qquad v \mapsto e_i^*(v) f_j \sim \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & 0 \\ 0 & & 1 & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} = E_{ij}$$

which is the matrix whose only nonzero entry is a 1 in the position sending the basis vector e_i to f_j, and zero everywhere else. This is clearly a basis, hence θ sends a basis to a basis, and hence θ is a vector-space isomorphism. We next check that θ is G-linear:

$$\begin{aligned} \theta(g \cdot (\lambda \otimes w)) &= \theta(g\lambda \otimes gw) \\ &\equiv v \mapsto (g\lambda)(v)gw \\ &= v \mapsto (g\lambda)(g^{-1}v)w \\ &= g(\lambda(g^{-1}v)w) \\ &\equiv g(\theta(\lambda \otimes w)g^{-1}v) \\ &= g \cdot \theta(\lambda \otimes w) \end{aligned}$$

Lemma 3.2.13: Creating New Characters

1. $\chi_{V\oplus W} = \chi_V + \chi_W$
2. $\chi_{V\otimes W} = \chi_V \cdot \chi_W$
3. $\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V}\chi_W$
4. $\chi_{V^*} = \overline{\chi_V}$

***Proof*:**

1. obvious
2. It is enough to show that if $f : V \to V$ and $g : W \to W$ are linear functions, then:
$$\mathrm{tr}(f \otimes h) = \mathrm{tr}(f)\mathrm{tr}(h)$$
Now if we pick basis (e_i) for V and (f_j) for W, then $f(e_i) = \sum a_{ij}e_j$ and $h(f_k) = \sum b_{lk}f_l$. Then:
$$(f \otimes h)(e_i \otimes f_k) = \left(\sum_j a_{ij}e_j\right) \otimes \left(\sum_l b_{lk}f_l\right) = \sum_{j,l} a_{ij}b_{lk}(e_j \otimes f_l)$$
The diagonal coefficient, that of $e_i \otimes f_k$ itself, is $a_{ii}b_{kk}$, so summing over the basis gives $\mathrm{tr}(f \otimes h) = \sum_{i,k} a_{ii}b_{kk} = (\sum_i a_{ii})\,(\sum_k b_{kk}) = \mathrm{tr}(f)\mathrm{tr}(h)$
3. Recall that for $f \in V^*$,
$$(gf)(v) = f(g^{-1} \cdot v)$$
that is
$$\rho_{V^*}(g)(f) = f \circ \rho_V(g^{-1})$$
that is, it is the transpose:
$$\rho_{V^*}(g) = (\rho_V(g^{-1}))^t$$
So, since the transpose matrix has the same trace, we have
$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}$$
we then apply lemma 3.2.12 and part (3)to get:
$$\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V} \cdot \chi_W$$
4. $\chi_{V^*} = \chi_{\mathrm{Hom}(V,\mathbb{C})} = \overline{\chi_V} \cdot 1 = \overline{\chi_V}$.

With this, we can create new characters! However, we have yet to establish how we may find the irreducible components of these chracters. For example, if we take $V \otimes V$, we may find that $\chi_{V\otimes V}$ is a linear combination of some characters already present in our table, but then we have a left-over part. How would we determine if the left-over is irreducible? We will show this by defining an *inner product* on the vector-space of class functions, and show that the irreducible form an *orthogonal basis*!

Definition 3.2.14: Inner Product On Class functions

Let $\chi_1, \chi_2 : G \to \mathbb{C}$ be class functions. Then define:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_1(g)\overline{\chi_2(g)} \in \mathbb{C}$$

Then this is a [Hermitian] *inner product*.

It should be verified that this is indeeed an inner product. Since class functions are by definition constant on conjugacy classes, we only need to run through representatives of conjgacy classe times the order of each conjugacy class:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{\substack{\mathcal{K} \in \text{conj}(G) \\ g \in \mathcal{K}}} |\mathcal{K}| \chi_1(g)\overline{\chi_2(g)} \in \mathbb{C}$$

where we pick a single represetnative $g \in \mathcal{K}$ and sum over the number of conjugacy classes, i.e. $|\text{conj}(G)|$. Looking a the inner product like this is desirable to make computation easier (ex. if taking the inner product of a character with itself, we take the *modulus* squared $|\chi(g)|^2$ of each entry in a row (the modulus matters: character values need not be real), multiply by the size of it's conjugcay class). Notice how this inner product mirrors the G-invariant inner product we induced in corollary 3.1.12 if we let $\langle \cdot, \cdot \rangle$ be the usual conjugate dot product on $\mathbb{C}^n$. (TBD)

Lemma 3.2.15: Inner Product With Trivial Character

Let V be a G-representation. Then if $\mathbb{1}$ is the trivial representation,

$$\langle \chi_V, \chi_{\mathbb{1}} \rangle = \dim_{\mathbb{C}}(V^G)$$

where $V^G = \{v \in V \mid g \cdot v = v \ \forall g \in G\}$ is the subspace fixed by G.

***Proof*:**
Define the linear map $\pi : V \to V$ where

$$v \mapsto \frac{1}{|G|} \sum_{g \in G} g \cdot v$$

then:

$$\pi|_{V^G} = \text{id}$$

and

$$\text{im}(\pi) \subseteq V^G$$

which implies $\pi^2 = \pi$, i.e. a projection. By linear linear algebra:

$$V = \ker(\pi) \oplus \text{im}(\pi)$$

where on the kernel, $\pi = 0$, and on the image $\pi = \text{id}$. Thus,

$$\frac{1}{|G|} \sum_g \chi_V(g) = \frac{1}{|G|} \sum_g \text{tr}(g : V \to V) = \text{tr}(\pi) = \dim_{\mathbb{C}}(V^G)$$

but the left hand side is $\langle \chi_V, \chi_{\mathbb{1}} \rangle$, completing the proof.

Corollary 3.2.16: Row Orthogonality

If V and W are any representations of G, then

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}}(\hom_{\mathbb{C}[G]}(W, V))$$

In particualr, if V an W are irreducible, then

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & V \cong W \\ 0 & \text{otherwise} \end{cases}$$

that is, irreducible character's are orthogonal to each other.

***Proof*:**
We reduce to the previous lemma:

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)} \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}_{\mathbb{C}}(W,V)}(g) \\ &= \langle \chi_{\mathrm{Hom}_{\mathbb{C}}(W,V)}, 1 \rangle \\ &= \dim_{\mathbb{C}}(\mathrm{Hom}_{\mathbb{C}}(W, V)^G) \end{aligned}$$

Note that $f \in \mathrm{Hom}_{\mathbb{C}}(W, V)$ is G-invariant, so $g \cdot f = gfg^{-1} = f$ for all $g \in G$ (recall the definition of $g \cdot f$ on $\mathrm{Hom}_{\mathbb{C}}(W, V)$). Thus this is true iff

$$gf = fg$$

Thus

$$gf(w) = f(gw) \qquad \forall g \forall w \in W$$

so f is $\mathbb{C}[G]$-linear so f is in the $\mathrm{Hom}_{\mathbb{C}[G]}(W, V)$.

There is also a more concrete proof worth remarking: since characters are linear combination's of irreducible characters, the direct product of irreducible modules translates to the sum of an irreducible character with the multiplicity of the irreducible module afording the representation being the multiplicity of the ireducibile characters. Hence:

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \langle \sum_i m_i \chi^{(i)}, \sum_j m_j \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i n_j \langle \chi^{(i)}, \chi^{(j)} \rangle \end{aligned}$$

where we have now reduced it to $\langle \chi^{(i)}, \chi^{(j)} \rangle$. Expanding the inner product and using the hom-set definition, we can apply Schur's Lemma and get:

$$\langle \chi^{(i)}, \chi^{(j)} \rangle = \delta_{ij}$$

Thus, our formula becomes:

$$\begin{aligned}\langle \chi_V, \chi_W \rangle &= \langle \sum_i m_i \chi^{(i)}, \sum_j n_j \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i n_j \langle \chi^{(i)}, \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i n_j \delta_{ij} = \sum_i m_i n_i\end{aligned}$$

To find $\langle \chi^{(i)}, \chi^{(i)} \rangle$ for an irreducible character, seeing it through multiplicites, then we know that an irreducible character corresponds to an irreducible representation of multiplicity one. (This is a mnemonic, not an independent proof: the identity $\langle \chi_V, \chi_V \rangle = \sum_i m_i^2$ is itself a consequence of the orthonormality established above.) So if in particular:

$$\langle \chi^{(i)}, \chi^{(i)} \rangle = m_i^2 = 1 \cdot 1 = 1$$

Thus, the irreducible character produce and orthogonal basis of class functions!

Corollary 3.2.17: Consequence of Orthogonality

Let $V = \bigoplus_i^r S_i^{l_i}$ where each S_i irreducible and $S_i \ncong S_j$. Then

1. the multipliity of l_i of S_i is: $l_i = \langle \chi_V, \chi_{S_i} \rangle$
2. $\langle \chi_V, \chi_V \rangle = \sum_i l_i^2$
3. $\hom_G(W, V) \cong \hom_G(V, W)^*$

***Proof*:**

1. Use the Artin-Wedderburn decomposition and the fact that direct sums of modules becomes sums of characters
2. this is just taking advantage of the summing formula
3. This is taking advantage of the fact that $\langle \chi, \psi \rangle = \overline{\langle \psi, \chi \rangle}$.

The second line is useful enough to be given it's own name:

Definition 3.2.18: Character Norm

Let V be a G-representation and let $\langle \cdot, \cdot \rangle$ be the inner product on the class functions of G. Then for any class function $\psi = \sum_i \lambda_i \chi_i$,

$$\|\psi\| = \left(\sum_i |\lambda_i|^2 \right)^{1/2} = \langle \psi, \psi \rangle^{1/2}$$

We will use all this informaiton to deduce new characters:

Example 3.2.19: Orthogonality Relation
Let's go back to our table for S_3:

	e	$(1\ 2)$	$(1\ 2\ 3)$
1	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since S_3 has 3 conjugacy classes, it can at most have one more irreducible character. What can this character be? We may start by looking at the degree-three permutation representation of S_3, the one on $\mathbb{C}X$ for $X = \{1, 2, 3\}$ with its natural action, and take its character χ_V. Then:

$$\langle \chi_V, 1 \rangle = 1$$
$$\langle \chi_V, \chi_{\text{sng}} \rangle = 0$$

showing the permutation character χ_V has one copy of the trivial representation (which makes sense, since S_3 is transitive on X, so the fixed space is spanned by the single vector summing the basis). Since

$$\chi_V(e) = 3 \qquad \chi_V((1\ 2)) = 1 \qquad \chi_V((1\ 2\ 3)) = 0$$

then we can take a "compliment" character by subtracting the trivial character, which defines a character which gives:

$$\chi^{\perp}(e) = 2 \qquad \chi^{\perp}(1\ 2) = 0 \qquad \chi^{\perp}(1\ 2\ 3) = -1$$

So, if there were to be another irreducible character, it would be of this form. But does there exist a degree 2 irreducible representation with this character? There is: the class function $(2, 0, -1)$ has norm $\langle \chi^{\perp}, \chi^{\perp} \rangle = (1 \cdot 4 + 3 \cdot 0 + 2 \cdot 1)/6 = 1$, so it is irreducible. Counting also confirms it: we saw that the number of irreducible representations is equal to the number of conjugacy classes, meaning the number of characters must be equal to the number of irreducible representations.

More generally, there can be a few approaches to finding it. One is to tensor two characters we know and see if it is irreducible, or one of it's linear components is irreducible. Another is to find the compliment of a represetnation of an existing character (since $k[G]$-modules are semi-simple). This compliment method is in general less desirable, and usually harder, but is managable if working with permutation representations.To demonstrate, let V be the degree three permutation representation of S_3[a] with basis $\{e_1, e_2, e_3\}$. Then if $\mathbb{1}$ is the trivial action, $V^{\mathbb{1}} = \text{span}(e_1 + e_2 + e_3)$, Thus, by Machke's theorem, $V^{\mathbb{1}}$ has a compliment. To see it, we simply have to extend $e_1 + e_2 + e_3$ to a basis whose last two vectors span its orthogonal complement, for example $(e_1 + e_2 + e_3, e_2 - e_1, e_3 - e_1)$ (these last two are not orthogonal to each other, but they do span the complement). Then we can calculate what the action ρ does on this basis. For example:

$$\begin{aligned} \rho(1\ 2)(e_1 + e_2 + e_3) &= e_1 + e_2 + e_3 \\ \rho(1\ 2)(e_2 - e_1) &= e_1 - e_2 = -(e_2 - e_1) \\ \rho(1\ 2)(e_3 - e_1) &= e_3 - e_2 = -(e_2 - e_1) + (e_3 - e_1) \end{aligned}$$

If we write out ρ for an element in each conjugacy class, we will find that its trace will be

$$\chi^{\perp}(e) = 2 \qquad \chi^{\perp}((1\ 2)) = 0 \quad \chi^{\perp}((1\ 2\ 3)) = -1$$

showing we have found the desired irreducible representation! Finally, each irreducible occurs in the regular representation with multiplicity equal to its degree (proposition 3.1.13), so $\mathbb{C}S_3 \cong \mathbb{1} \oplus \text{sgn} \oplus (V^\perp)^{\oplus 2}$ as representations, of dimension $1+1+4=6$. As algebras Artin-Wedderburn gives $\mathbb{C}S_3 \cong \mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$

Now that we know the character table for S_3, we can construct any other character of representations of S_3, and hence any other representation of S_3. For example, we know that $V^\perp \otimes V^\perp$ has as character $\chi_{V^\perp} \cdot \chi_{V^\perp}$. Using this, we can compute that:

$$\langle V^\perp \otimes V^\perp, \mathbb{1} \rangle = 1$$
$$\langle V^\perp \otimes V^\perp, \text{sgn} \rangle = 1$$
$$\langle V^\perp \otimes V^\perp, V^\perp \rangle = 1$$

showing that

$$\chi_{V^\perp \otimes V^\perp} = \chi_{\mathbb{1}} + \chi_{\text{sgn}} + \chi_{V^\perp}$$

[a]This is the representation on $\mathbb{C}X$ for $X = \{1,2,3\}$ with its natural S_3-action, not the regular representation, which has degree $|S_3| = 6$.

Recall from linear algebra that if the rows are orthogonal in a square matrix, then so would be the column (and vice-versa). We bring to light this relation for the character table. As in linear algebra, it comes down to proper normalizing of a different basis: recall the class functions $\delta_{\mathcal{K}} : G \to \mathbb{C}$ where:

$$\delta_{\mathcal{K}}(g) = \begin{cases} 1 & g \in \mathcal{K} \\ 0 & \text{otherwise} \end{cases}$$

this naturally forms a basis for the class equations too. The following theorem will also show how to go between this basis and the irreducible character basis:

Theorem 3.2.20: Column Orthogonality

If $\chi_1, \chi_2, ..., \chi_r$ are the irreducible characters of G and $g_1, g_2, ..., g_r$ represent the conjugacy classes of G, then

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \delta_{st} \frac{|G|}{|\text{conj}(g_s)|}$$

which by Orbt-stabalizer,

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \delta_{st} |C_G(g_s)|$$

If $g_s, g_t \in \text{conj}(g)$, then we get that

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \sum_i |\chi_i(g_s)|^2 = |C_G(g_s)|$$

in particular, we must have:

$$\sum_i |\chi_i(e)|^2 = |G|$$

***Proof*:**
Define $f_s : G \to \mathbb{C}$,

$$g \mapsto \begin{cases} 1 & g \in \text{conj}(g_s) \\ 0 & \text{otherwise} \end{cases}$$

which is natural basis for the class functions. since it's a class function, we know we can write it as the sum of orhtogonal ireducibile characters:

$$f_s = \sum_i^r \lambda_{is}\chi_i \qquad \lambda_{is} \in \mathbb{C}$$

since these are orthornormal, we know that:

$$\lambda_{is} = \langle f_s, \chi_i \rangle$$

so computing:

$$\begin{aligned} \lambda_{is} &= \langle f_s, \chi_i \rangle \\ &= \frac{1}{|G|} \sum_{g \in G} f_s(g)\overline{\chi_i(g)} \\ &= \frac{1}{|G|} \sum_{g \in \text{conj}(g_s)} \overline{\chi_i(g)} \\ &= \frac{|\text{conj}(g_s)|}{|G|}\overline{\chi_i(g_s)} \end{aligned}$$

so

$$f_s = \frac{\text{conj}(g_s)|}{|G|} \sum_i^r \overline{\chi_i(g_s)}\chi_i$$

we know evaluate at g_t.

Since both the rows and columns of a character table are all orthogonal to each other, the character table is in fact invertible!

Example 3.2.21: Using Column Orthogonality
Let $G = S_4$. We want to find it's character table. We first pick what conjugacy classes to index the columns by, say:

conj. class	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
size	1	6	8	3	6

Next, we can immediately deduce that $\chi_{\mathbb{1}}$ and χ_{sgn} are irreducible characters, giving us the current begining of a character table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	−1	1	1	−1

To find more characters, we can first take the pemutation character of $\mathbb{C}X$. The character of each conjugacy class is the number of fixed points, giving us:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{C}X}$	4	2	1	0	0

We can see that the trivial character is part of linear decomposition of $\chi_{\mathbb{C}X}$, since:

$$\langle \chi_{\mathbb{C}X}, \chi_{\mathbb{1}} \rangle = \big(1(4 \cdot 1) + 6(2 \cdot 1) + 8(1 \cdot 1) + 3(0 \cdot 1) + 6(0 \cdot 1)\big)/24 = 24/24 = 1$$

and so $\chi_{\mathbb{C}X} - \chi_{\mathbb{1}}$ is a 3 dimensional character, let's label this χ_3. Then $\|\chi_3\|^2 = 1$, showing that it's irreducible, hence we expand our table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
$\chi_{\mathbb{C}X} - \chi_{\mathbb{1}} = \chi_3$	3	1	0	-1	-1

To find the next character, we can multiply χ_{sgn} and χ_3 (note we can do χ_3 with itself, but it will be 9 dimensional, which we will show in the next proposition cannot be an irreducible representation). Let$\chi_4 = \chi_{\text{sgn}} \cdot \chi_3$. Then $\|\chi_4\|^2 = 1$, meaning it's irreducible, and so we add another row to our table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	1	0	-1	-1
$\chi_{\text{sng}} \cdot \chi_3 = \chi_4$	3	-1	0	-1	1

Finally, since $|G| = \sum_i \chi_i(e)^2$, we can deduce that $\chi_5(e) = 2$. We cannot do this same trick exactly for the other columns since we will get the value up to a complex number z with $\|z\| = 1$. To find the rest of the character values, we can either use column orthogonality For example, to find $\chi_5(1\ 2)$, noticet that the inner product of the first and second column is:

$$1 \cdot 1 + (-1) \cdot 1 + 3 \cdot 1 + 3 \cdot (-1) = 0$$

and so $\chi_5((1\ 2))$ must be 0. For the next column, we'd get:

$$1 \cdot 1 + 1 \cdot 1 + 3 \cdot 0 + 3 \cdot 0 = 2$$

and so, since $\chi_5(1)\chi_5((1\ 2\ 3)) = 2\chi_5((1\ 2\ 3))$ must cancel this 2, we get that $\chi_5((1\ 2\ 3)) = -1$. Continuing this way, we get that the rest of the table will be:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	1	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

We can verify this is indeed irreducible by taking $\|\chi_5\|^2 = 1$.

Another method would be to use the regular representation, where we get:

$$\chi_{\mathbb{C}[G]} = \sum_i \chi_i(1)\chi_i$$

These can be verified by taking advantage of the fact that the "column norm" $\sum_i |\chi_i(g)|^2$ should give you $|C_G(g)|$, which is $|G|$ divided by the class size printed on top of the table. For $g = (1\ 2)$, for instance, the column gives $1+1+1+1+0 = 4 = 24/6$.

As a fun aside, the representation with character χ_4 is the rotation action on a cube: if $C \subseteq \mathbb{R}^3$ is the set of vertices of a cube, then χ_4 is the character of the action of S_4 on $\mathbb{R}^3$ by all rotations preserving C, where S_4 permutes the four long diagonals. A transposition acts as a 180° edge rotation and a 4-cycle as a 90° face rotation, with traces -1 and 1, matching the row. It is χ_4 and not χ_3 because a rotation has determinant 1, and $\det \chi_3 = \text{sgn}$.

Notice how the dimension of each irreducible representation divided the order of the group. This is in fact true in general. It is not a trivial fact, the only "order" information we have shown so far is that the sums of the squares of the dimensions of the ireducibile representation sum to $|G|$. It is also relatively eays to show that if $W \subseteq V$ is an irreducible $\mathbb{C}[G]$-module then $(\dim W)^2 \leq |G|$, which is immediate from part (4) above. The following is thus a much stronger result:

Proposition 3.2.22: Dimension Theorem of Irreducible Representations

Let V be an irreducible G-representation. Then:

$$\dim_{\mathbb{C}}(V) \mid |G|$$

that is, the dimension of V divides the order of G.

***Proof*:**
Let $g_1, g_2, ..., g_r$ be representatives of the conjugacy class. Recall the following trick from proposition 3.1.13:

$$e_i = \left(\sum_{g \in \text{conj}(g_i)} g \right) \in Z(\mathbb{C}[G])$$

Note that e_i in fact lies in $e_i \in Z(\mathbb{Z}[G])$, since $ge_ig^{-1} = e_i$ (by definition), and so extending linearly all $xe_ix^{-1} = e_i$. Now we have a commutative ring that is a finitely generated $\mathbb{Z}$-module (one way to see this is as we have done in proposition 3.1.13, another way is to notice that $\mathbb{Z}[G]$ is a finitely generated $\mathbb{Z}$-module and $\mathbb{Z}$ is Noetherian, so the center is finitely generated)

Thus, by proposition 26.1.4, $\mathbb{Z} \subseteq Z(\mathbb{Z}[G])$ is an *integral extension*. Next, we have a ring homomorphism:

$$\varphi : Z(\mathbb{C}[G]) \to \text{End}_{\mathbb{C}[G]}(V) \overset{!}{\cong} \mathbb{C} \qquad z \mapsto (z : V \to V)$$

where $\overset{!}{=}$ comes from corollary 1.2.19 and Schur's lemma, and $z(v) = zv$. Hence each $z \in Z(\mathbb{C}[G])$ acts on V as a scalar $\varphi(z) \in \mathbb{C}$, that is as $\varphi(z) \cdot I$. (Note this scalar may well be 0, for instance whenever $\chi_V(g_i) = 0$, so φ does *not* land in $\text{GL}(V)$.) Acting by a scalar is all we need, and it is exactly the restriction of the original G-action ρ to $Z(\mathbb{C}[G])$. Hence $\rho(e_i) = \lambda_i I$ for appropriate

$\lambda_i = \varphi(e_i)$, meaning $\chi(e_i) = \text{tr}(e_i : V \to V) = n\varphi(e_i)$ where $n = \dim(V)$ since by what we have just shown, $\varphi(e_i)$ acts as scalar multiplication on the representation V.

Now, since $e_i \in Z(\mathbb{C}[G])$ is integral over $\mathbb{Z}$, $\varphi(e_i) \in \mathbb{C}$ is integral over $\mathbb{Z}$, that is $\varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}} \subseteq \mathbb{C}$ (i.e. $\varphi(e_i)$ is algebraic). Next, we can express $\chi(e_i)$ in a different way:

$$\sum_{g \in \text{conj}(g_i)} \chi_V(g) = |\text{conj}(g_i)| \cdot \chi_V(g_i) = \chi(e_i)$$

so:

$$\frac{\text{conj}(g_i)}{n} \chi_V(g_i) = \varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

which is telling us that if we divide it by n, we still get an algebraic integer! Taking a nice linear combination and multiplying by $\overline{\chi_V(g_i)}$ we get:

$$\sum_i^r \frac{\text{conj}(g_i)}{n} \chi_V(g_i) \overline{\chi_V(g_i)}$$

Then we have just shown that $\frac{\text{conj}(g_i)}{n}\chi_V(g_i)$ is an algebraic integer, and $\overline{\chi_V(g_i)}$ is an algebraic integer (being the sum of roots of unity). Now, notice this is the form of the inner product, giving us:

$$\frac{|G|}{n}\langle \chi_V, \chi_V \rangle = \frac{|G|}{n} \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

as a V irreducible. Thus, we have that $\frac{|G|}{n}$ is some algebraic integer. We further have that it is certainly a rational number ($|G|, n \in \mathbb{N}$), but then we have:

$$\frac{|G|}{n} \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} \overset{!}{=} \mathbb{Z}$$

where $\overset{!}{=}$ comes from the fact that if R is a Unique Factorization Domain, $\alpha \in \text{Frac}(R)$, α is integral over R if and only if $\alpha \in R$, completing the proof.

This will make the computation of characters of abelian groups paticularly simple! (see rep theory of finite groups: An introductory Appraoch p. 47!)

Exercise 3.2.1

1. Let G act on the n-dimensional vector-space V by basis permutation. Show that $\dim(V^G)$ equals the number of orbits of G.

3.2.1 More Character Tables

In this section, we go over a few more exmaples of character tables in order to see some techniques used to find them. For any abelian, group, the character table is straight forward. As a really simple example, consider $\mathbb{Z}/3\mathbb{Z}$:

	1	1	1
	0	1	2
χ_0	1	1	1
χ_1	1	ζ_3	ζ_3^2
χ_2	1	ζ_3^2	ζ_3

If we take $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, we will get a 9 by 9 table. The table bellow fills in a few of the characters (note that these were found by taking $\chi_i(g_i)\chi_j(g_j)$)

	1	1	1	1	1	1	1	1	1
$\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$	$(0,0)$	$(1,0)$	$(2,0)$	$(0,1)$	$(1,1)$	$(2,1)$	$(0,2)$	$(1,2)$	$(2,2)$
$\chi_{(0,0)}$	1	1	1	1	1	1	1	1	1
$\chi_{(1,0)}$	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2
$\chi_{(1,1)}$	1	ζ_3	ζ_3^2	ζ_3	ζ_3^2	1	ζ_3^2	1	ζ_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Note that the root ζ_3 might change if we take the direct product of cyclic groups of differnet orders, for example the character table of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ has roots of unity of order 3 and 6 (as you can verify).

More generally, if G is abelian, we break G down into a product of cyclic groups and construct the table like above. Thus, we focus on non-abelian groups. We have already found the character table for S_3:

S_3	e	(1 2)	(1 2 3)
$\chi_{\mathbb{1}}$	1	1	1
χ_{sgn}	1	-1	1
$\chi^{\perp}$	2	0	-1

and the character table for S_4:

	1	6	8	3	6
S_4	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	1	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

Next, let $G = D_4$. Then: $D' = \{e, r^2\}$, so G has 4 degree 1-irreducible representations. We also found in example 3.1.5 that D_4 has a degree 2 irreducible representation. These are all of the representations, sinc $1 + 1 + 1 + 1 + 2^2 = 8$. To find the 1-dimensional irreducible representations, notice that $D_4/D_4' \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, meaning we can find the irreducible representations of $\mathbb{Z}/2\mathbb{Z}$, which are:

$\mathbb{Z}/2\mathbb{Z}$	0	1
χ_0	1	1
χ_1	1	-1

and take the tensor product to create:

$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$
$\chi_{(0,0)}$	1	1	1	1
$\chi_{(1,0)}$	1	-1	1	-1
$\chi_{(0,1)}$	1	1	-1	-1
$\chi_{(1,1)}$	1	-1	-1	1

and using this, we can complete the table for D_4:

D_4	e	r	r^2	σ	σr
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	-1	1	-1	1
χ_5	2	0	-2	0	0

More generally, for D_n, the character table will conssite of characters of degree 1 and 2.

Let's now let $G = Q_8$. Taking the presentation:

$$Q_8 = \langle i, j | i^4 = 1,\ i^2 = j^2,\ i^{-1}ji = j^{-1} \rangle$$

with $k = ij$ and $i^2 = -1$, we get the conjugacy classes can be represented by $1, -1, i, j, k$, with sizes 1, 1, 2, 2, 2. Since $Q_8/Q_8' \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, there are 4 degree 1 characters. The final irreducible character thus *must* have degree 2; let's label if χ_5. We have in fact explicitely found this character before in example 3.1.5, but we can find it without actually manually finding a representation. Let φ be an irreducible representation of degree 2. Since $-1 \in Z(Q_8)$, by Schur's lemma $\varphi(-1)$ is a 2×2 scalar matrix of order 2, hence it is $\pm I$, and so $\chi_5(-1) = \pm 2$. Next, if $\chi_5(i) = a$, $\chi_5(j) = b$, and $\chi_5(k) = c$, then since χ_5 is orthogonal, we have that:

$$1 = \langle \chi_5, \chi_5 \rangle = \frac{1}{8}(2^2 + (\pm 2)^2 + 2a\overline{a} + 2b\overline{b} + 2c\overline{c})$$

and since $a\overline{a}, b\overline{b}, c\overline{c} \in \mathbb{R}_{\geq 0}$, they must all be zero. Furthermore, we must have that:

$$0 = \langle \chi_1, \chi_5 \rangle = \frac{1}{8}\big(2 + (\pm 2) + 0 + 0 + 0\big)$$

implying that $\chi_5(-1) = -2$. Thus, we have that the character table is:

Q_8	1	i	-1	j	k
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	-1	1	-1	1
χ_5	2	0	-2	0	0

Notice that this character table is identical to the character table of D_4. Hence, identical character tables does not imply isomorphic groups! Finally, we could have found the last row of the above table by applying column orthogonality to find the missing values, once we knew $\chi_5(1) = 2$.

Next, let $G = A_4$. Since we found the character table for S_4, we can use it it to find the character table of A_4. We first note that A_4 has 4 conjugacy classes (represented by 1, (1 2)(3 4), (1 2 3) and (1 3 2)), and $[A_4 : A_4'] = 3$, meaning A_4 has 3 chacacters of degree 1, and the only other character must be of degree 3. Next, using the orthogonality relation, notice that χ_3 in the table for S_4 when restricted to A_4 is still irreducible (this is certainly not generally the case, for ex. the degree 2 representaiton of S_3 will become reducible since any proper subgroup of S_3 is abelian). Overall, we get:

A_4	1	(1 2)(3 4)	(1 2 3)	(1 3 2)
χ_1	1	1	1	1
χ_2	1	1	ζ_3	ζ_3^2
χ_3	1	1	ζ_3^2	ζ_3
χ_4	3	-1	0	0

Finally, we will do the character table for S_5:

S_5	1	(1 2)	(1 2 3)	(1 2 3 4)	(1 2 3 4 5)	(1 2)(3 4)	(1 2)(3 4 5)
χ_1	1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	1	-1
χ_3	4	2	1	0	-1	0	-1
χ_4	4	-2	1	0	-1	0	1
χ_5	5	-1	-1	1	0	1	-1
χ_6	5	1	-1	-1	0	1	1
χ_7	6	0	0	0	1	-2	0

TBD

3.2.2 Burnside's Theorem

We now move towards Burnside's Theorem, which shows that any group of order $p^a q^b$ for primes p, q and numbers $a, b \in \mathbb{N}$ is a solvable group! This proof is relatively simple in the context of representation theory, but it took over 70 years before a proof was found using just group theory! This is a good example of the power perspective: simply switching to the context of representation theory and seeing groups as embedded into $GL(V)$ let us prove a claim that from another perspective is much much harder!

Lemma 3.2.23: Burnside build-up Lemma

Suppose that V is an irreducible representation is of dimension n (i.e. $\chi_V(1) = n$). If $\gcd(n, |\text{conj}(g)|) = 1$, then:

$$|\chi_V(g)| = 0 \text{ or } n$$

***Proof*:**
By the above proof, we have:

$$\frac{|\text{conj}(g)|}{n}\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

where $\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$ as well, so by Bézout ($1 = an + b|\text{conj}(g)|$, using $\gcd(n, |\text{conj}(g)|) = 1$)

$$\frac{1}{n}\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

Let $\alpha = \frac{1}{n}\chi_V(g)$. Furthermore, notice that:

$$|\alpha| = \left|\frac{\chi_V(g)}{\chi_V(1)}\right| \leq 1$$

Now, let N be the Galois closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$ and write $\Gamma = \text{Gal}(N/\mathbb{Q})$ for its Galois group (not to be confused with the group G of the lemma). Let

$$\beta = \prod_{\sigma \in \Gamma} \sigma(\alpha)$$

Then certainly β is in the fixed field of the Galois group, $\beta \in N^{\Gamma} = \mathbb{Q}$. Notice that each $\sigma(\alpha)$ satisfies the same monic polynomials as that of α, and so $\sigma(\alpha) \in \overline{\mathbb{Z}}^{\mathbb{C}}$, and so $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}}$, but then $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} = \mathbb{Z}$.

Now, recall that:

$$\alpha = \frac{\chi_V(g)}{n} = \frac{\zeta_1 + \zeta_2 + ... + \zeta_n}{n}$$

which are roots of unity, which means that:

$$\sigma(\alpha) = \frac{\sigma(\zeta_1) + \cdots + \sigma(\zeta_n)}{n}$$

so $|\sigma(\zeta_1)| \leq 1$, so $|\beta| \leq 1$. Since β is an integer, we either get that $|\beta| \in \{0, 1\}$, If $|\beta| = 0$, then $\alpha = 0$, and if $|\beta| = 1$, then $|\alpha| = 1$, and so plugging back into our original equation we get that:

$$\chi_V \in \{0, n\}$$

as we sought to show.

We can also treat this lemma as another check that we have given the correct character table (verify it for the character table of S_4 we found in example 3.2.21).

Theorem 3.2.24: Burnside's Theorem

Let $p \neq q$ be prime numbers, $a, b \in \mathbb{N}$. If $|G| = p^a q^b$, then G is solvable.

Notice how this immediately implies that a non-abelian simple group must have at least three distinct prime divisors, so its order is at least $2 \cdot 3 \cdot 5 = 30$. The true minimum is 60, attained by A_5. There is in fact no simple group of order 30. Recall also that if $|G| = p^a$, then we would need that $a = 1$ (namely, any p-group has a non-trivial center)

***Proof*:**
Take the composition series:

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$$

where G_i/G_{i-1}. Then we will show that if G is simple, then G is *abelian*. For the sake of contradiction, suppose that G is non-abelian and simple with order $p^a q^b$ (where we may as well assume $a, b \in \mathbb{N}_{>0}$). Let $\chi_1, \chi_2, ..., \chi_r$ be the irreducible representation for the group G, and $\chi_1 = \chi_{\mathbb{1}}$. Let $Z_i = \{g \in G \mid |\chi_i(g)| = \chi_i(1)\}$. By the equality case of the triangle inequality this says exactly that $\rho_i(g)$ acts as a scalar, so Z_i is the kernel of $G \to \mathrm{PGL}(V_i)$ and in particular

$$Z_i \trianglelefteq G$$

Fix i and suppose $Z_i = G$. Then G acts by scalars on V_i, so every subspace of V_i is a subrepresentation and hence $\dim V_i = 1$. So χ_i is a homomorphism

$$\chi_i : G \to \mathbb{C}^\times$$

with $\ker \chi_i \trianglelefteq G$. Since G is simple, either $\ker \chi_i = 1$, in which case G embeds in the abelian group $\mathbb{C}^\times$ and is abelian, a contradiction; or $\ker \chi_i = G$, in which case χ_i is trivial and $i = 1$. Thus, it must be that:

$$Z_i = 1 \qquad \forall i > 1$$

Now, take the sylow q-subgroup Q: (notice we are using that $a, b > 0$)

$$1 \subsetneq Q \subsetneq G$$

So Q is a q-group, and so has a non-trivial center: $Z(Q) \neq 1$. Take $h \in Z(Q) \setminus \{e\}$. Hence $Q \leqslant C_G(h)$. , so $[G : C_G(h)] \big| [G : Q] = p^a$, where $[G : C_G(h)] = |\mathrm{conj}(h)$. Now recall we have shown that:

$$\chi_i(1) \big| |G| = p^a q^b$$

Now, if $p \nmid \chi_i(1)$, then by the previous line we have that $\chi_i(1) \big| q^b$. So, $\gcd(\chi_i(1), |\mathrm{conj}(h)|) = 1$, and so by lemma 3.2.23

$$\chi_i(h) = 0 \qquad \text{or} \qquad \chi_i(1)$$

But if $|\chi_i(h)| = \chi_i(1)$, $h \in Z_i$, and so $i = 1$

So, if $p \nmid \chi_i(1)$ and $i > 1$, then $\chi_i(h) = 0$. Using this, we can get our contradiction: as h is not the trivial element, we get:

$$\begin{aligned}
0 &= \sum_i \overline{\chi_i(h)} \chi_i(1) \\
&= \sum_i \chi_i(h) \chi_i(1) && \text{(conjugating, as each } \chi_i(1) \text{ is real)} \\
&\overset{!}{=} 1 + \sum_{\substack{i>1 \\ p \big| \chi_i(1)}} \underbrace{\chi_i(1)}_{\in p\mathbb{Z}} \underbrace{\chi_i(h)}_{\in \overline{\mathbb{Z}}^{\mathbb{C}}} \\
&= 1 + p\alpha && \text{some } \alpha \in \overline{\mathbb{Z}}^{\mathbb{C}}
\end{aligned}$$

where the one in the $\overset{!}{=}$ equality came from when $i = 1$. Thus,

$$\alpha = -1/p \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} = \mathbb{Z}$$

a contradiction, completing the proof.

(huge list of character tables: commented)

3.3 Induced Representation

This is another way of constructing representations. We saw that if $H \leqslant G$ and V is a G representation, then we can make V an H-representation through the inclusion map $H \hookrightarrow G \to \mathrm{GL}(V)$. We may ask if we may go the other way: given an H representation W, can we find a G-representation? If φ is the representations afforded by H, can we find a Φ such that $\Phi|_H = \varphi$? It turns out we cannot: take $A_3 \subseteq S_3$ and consider a faithful representation of degree 1 (recall it's a cyclic group). Then notice that any 1 dimensional representations of S_3 will have to have A_3 in it's kernel, meaning we cannot extend the action of A_3 to an action of S_3. A little more generally, if we let G be a simple group, $H \leqslant G$, and φ a representation of H where $\ker(\varphi)$ is a proper nontrivial subgroup of H, then if we somehow extend the representation φ of H to Φ of G, then $\ker(\Phi) \cap H = \ker(\varphi)$, so $\ker(\Phi)$ is nontrivial (it contains $\ker\varphi$) and proper (else $\ker\varphi = H$), contradicting that G is simple. So instead, we will not be extending representation from H to G, but *inducing* a representation in a natural way. This method of inducing a representation will work even in the two above cases, meaning an induced representation will not always be an extension.

Let $H \leqslant G$, and consider some H-representation V, which can be represented as a $k[H]$-module V. Recall that we can "extend scalars" and define a new $k[G]$-module:

$$k[G] \otimes_{k[H]} V$$

This is extension of scalars along the inclusion $k[H] \hookrightarrow k[G]$.

Definition 3.3.1: Induced Representation

Let $H \leqslant G$ of a finite group G and let V be a $k[H]$-module affording the representation φ. Then the $k[G]$-module $k[G] \otimes_{k[H]} V$ is called the *induced module* of V, the representation is called the *induced representation*, and if χ is the character of φ, then the character of the induced representation is called the *induced character*, and is written as $\mathrm{Ind}_H^G(\chi)$.

Another intuition behind defining the induced module is the following: If W is an H-representation, then let our new module be:

$$\mathrm{Ind}_H^G(W) = \{f \in \mathrm{Hom}_{\mathbf{Set}}(G, W) \mid f(hg) = h \cdot f(g),\ \forall g \in G, h \in H\}$$

Then $\mathrm{Ind}_H^G(W)$ is clearly a complex vector space under pointwise addition and scalar multiplication, and the G-action on this space would be:

$$\gamma \cdot f : G \to W \qquad g \mapsto f(g\gamma)$$

It should be verified that this is indeed G-linear. In this case, we get that:

$$\mathrm{Ind}_H^G(W) = \mathrm{Hom}_{\mathbb{C}[H]}(\mathbb{C}[G], W)$$

This second construction is the *coinduced* module, and it is a genuinely different functor from the induced module $k[G] \otimes_{k[H]} V$ in general. The two agree precisely because $[G : H]$ is finite: $k[G]$ is free of finite rank as a right $k[H]$-module, and for a finitely generated free module the natural map

$k[G] \otimes_{k[H]} V \to \mathrm{Hom}_{k[H]}(k[G], V)$ is an isomorphism. Since G is finite throughout this section, we may and do use the two interchangeably, and we take the tensor construction of definition 3.3.1 as the definition of record. The dimension of $\mathrm{Ind}_H^G(W)$ is

$$\dim_{\mathbb{C}}(\mathrm{Ind}_H^G(W)) = [G : H] \dim_{\mathbb{C}}(W)$$

with the $\mathbb{C}$-vector space isomorphism being:

$$\mathrm{Ind}_H^G(W) \to W^{\oplus [G:H]} \qquad f \mapsto f(g_1), ..., f(g_{[G:H]})$$

where we treat $G = \coprod_i Hg_i$ for the *right* cosets Hg_i (the equivariance $f(hg) = hf(g)$ makes f constant-up-to-H on right cosets, so those are the ones its values are indexed by). Note that every block and character formula later in this section is indexed by *left* cosets g_iH instead, matching the tensor model $k[G] = \bigoplus_i g_i k[H]$. The two indexings are exchanged by $g \mapsto g^{-1}$, and we will say which is meant wherever it matters. By letting $g_1 = e$ being the identity, there is a natural canonical map:

$$\mathrm{Ind}_H^G(W) \to W \qquad f \mapsto f(e)$$

This motivates the following adjoint of the restriction functor:

Theorem 3.3.2: Frobenius Reciprocity

Let $H \leqslant G$. Then if V is a G-representation and W and H-representation, then:

$$\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W) \cong_{\mathbb{C}} \mathrm{Hom}_H(V|_H, W)$$

where $V|_H$ is the restriction of the action to H

***Proof*:**
We will explicitly construct inverse bijections Φ from left to right and Θ from right to left.
Let $\alpha : \mathrm{Ind}_H^G(W) \to W$, $f \mapsto f(1)$. Then α is H-linear. Then let

$$\Phi(f) = \alpha \circ f$$

Then $\Phi(f)$ is H-linear (check $\mathbb{C}$-linearity of Φ). For the other way, let $\psi : V|_H \to W$ be H-linear,

$$\Theta(\psi) : V \to \mathrm{Ind}_H^G(W) \qquad v \mapsto (g \mapsto \psi(gv))$$

Then this is well-defined. What's left is to check they are inverses of each other

3.3.3 Left Vs Right Adjoint This proof shows that induction is right-adjoint to restriction. It is also *left* adjoint: for a finite group, $\mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) \cong \mathrm{Hom}_H(W, V|_H)$ as well. A pair of functors (F, U) where F is both left and right adjoint to U is sometimes called a *Frobenius pairing*, and this is exactly what the coincidence of induction and coinduction at finite index buys.
The two-sided adjunction is special to finite index. For an infinite group the tensor construction $k[G] \otimes_{k[H]} -$ is left adjoint to restriction and the coinduced module $\mathrm{Hom}_{k[H]}(k[G], -)$ is right adjoint, and these are different functors. In the smooth representation theory of locally profinite groups the convention is that Ind denotes the coinduced (right adjoint) functor and c-Ind, *compact induction*, denotes the left adjoint (see [6]).

Corollary 3.3.4: Frobenius Reciprocity On Characters

Let G be finite, $k = \mathbb{C}$, and $H \leqslant G$. Let χ be the character of a G-representation V and ψ the character of an H-representation W. Then:

$$\langle \chi|_H, \psi \rangle_H = \langle \chi, \mathrm{Ind}_H^G(\psi) \rangle_G$$

***Proof*:**
Both sides count the same dimension. First note that for representations A, B of a finite group over $\mathbb{C}$ we have $\dim_{\mathbb{C}} \mathrm{Hom}(A, B) = \dim_{\mathbb{C}} \mathrm{Hom}(B, A)$: by corollary 3.2.16 these are $\langle \chi_B, \chi_A \rangle$ and $\langle \chi_A, \chi_B \rangle$, and writing A and B in terms of irreducibles (theorem 3.1.9) both equal $\sum_i a_i b_i$ for the multiplicities a_i, b_i.

Now apply corollary 3.2.16 to each side:

$$\langle \chi, \mathrm{Ind}_H^G(\psi) \rangle_G = \dim_{\mathbb{C}} \mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) = \dim_{\mathbb{C}} \mathrm{Hom}_G(V, \mathrm{Ind}_H^G W)$$

using the symmetry just noted, and this last space is $\mathrm{Hom}_H(V|_H, W)$ by theorem 3.3.2. Applying the symmetry and corollary 3.2.16 once more,

$$\dim_{\mathbb{C}} \mathrm{Hom}_H(V|_H, W) = \dim_{\mathbb{C}} \mathrm{Hom}_H(W, V|_H) = \langle \chi|_H, \psi \rangle_H$$

as claimed.

Example 3.3.5: Induced Representations

1. $\mathrm{Ind}_G^G(W) = W$
2. $\mathrm{Ind}_{\{e\}}^G(\mathbb{C}) = \mathbb{C}[G]$
3. $\mathrm{Ind}_H^G(\mathbb{1}) = \mathbb{C}[G/H]$

Another way of thinking of this result is that if V is an irreducible representation of G and W is an irreducible representation of H, then the number of times V occurs in $\mathrm{Ind}_H^G(W)$ is equal to the number of times W occurs in $V|_H$.

Proposition 3.3.6: Properties Of Induced Representation

Let $K \leqslant H \leqslant G$, W be an K-representation, $\mathrm{Ind}_K^H(W)$ the induced H representation, $\mathrm{Ind}_H^G(\mathrm{Ind}_K^H(W))$ the induced G-representation. Then:

1. If χ, χ' are characters of K-representations,
$$\mathrm{Ind}_K^H(\chi + \chi') = \mathrm{Ind}_K^H(\chi) + \mathrm{Ind}_K^H(\chi')$$
2. Induction is transitive:
$$\mathrm{Ind}_H^G(\mathrm{Ind}_K^H(W)) = \mathrm{Ind}_K^G(W)$$

***Proof*:**
For (1), let W_1, W_2 be K-representations affording χ, χ'. Tensoring is additive in its second argument, so

$$k[H] \otimes_{k[K]} (W_1 \oplus W_2) \cong \big(k[H] \otimes_{k[K]} W_1\big) \oplus \big(k[H] \otimes_{k[K]} W_2\big)$$

as $k[H]$-modules, and the character of a direct sum is the sum of the characters.

For (2), the point is associativity of the tensor product together with $k[H] \otimes_{k[H]} k[G] \cong k[G]$ as $(k[G], k[H])$-bimodules. Explicitly,

$$\mathrm{Ind}_H^G\big(\mathrm{Ind}_K^H(W)\big) = k[G]\otimes_{k[H]}\big(k[H]\otimes_{k[K]}W\big) \cong \big(k[G]\otimes_{k[H]}k[H]\big)\otimes_{k[K]}W \cong k[G]\otimes_{k[K]}W = \mathrm{Ind}_K^G(W)$$

where the first isomorphism is associativity and the second is the unit isomorphism. Each map is $k[G]$-linear because G acts by left multiplication on the leftmost factor throughout, which every map above respects.

Theorem 3.3.7: Computing Induced Representation

Let $H \leqslant G$ of a finite group G, and $g_1, g_2, ..., g_m$ be representatives for distinct left cosets of H in G. Let V be an H-representation of degree n. Then the induced G-representation $W = k[G] \otimes_{k[H]} V$ is an nm degree representation, and if Φ repersented the induced representation, then there exists a basis for W such that W affords the matrix representation defined for each $g \in G$ by:

$$\Phi(g) = \begin{pmatrix} \varphi(g_1^{-1}gg_1) & \cdots & \varphi(g_1^{-1}gg_m) \\ \vdots & \vdots & \cdots \\ \varphi(g_m^{-1}gg_1) & \cdots & \varphi(g_m^{-1}gg_m) \end{pmatrix}$$

where each $\varphi(g_i^{-1}gg_j)$ is an $n \times n$ block, with $\varphi(g_i^{-1}gg_j) = 0$ if $g_i^{-1}gg_j \notin H$.

***Proof*:**
First, $k[G]$ is a free right $k[H]$-module with decomposition:

$$k[G] = g_1k[H] \oplus \cdots \oplus g_mk[H]$$

Since tensor products commute with direct sums, we get:

$$W = k[G] \otimes_{k[H]} V \cong (g_1 \otimes V) \oplus \cdots \oplus (g_m \otimes V)$$

Since k is in the center of $k[G]$, the above isomorphism is also a k-vector space isomorphism. Thus, if $v_1, v_2, ..., v_n$ is a basis for V, then $\{g_i \otimes v_j \mid 1 \leq i \leq m,\ 1 \leq j \leq n\}$ is a basis for W, and hence $\dim_k(W) = nm$.

To show $\Phi(g)$ is of the given form, order the basis into m of size n as:

$$g_1 \otimes v_1, g_1 \otimes v_2, ..., g_1 \otimes v_n, g_2 \otimes v_1, ..., g_m \otimes v_n$$

Using this order, we can compute the matrix $\Phi(g)$ for each g acting on W given this basis. Fixing j and g, let i be the unique index with $gg_j \in g_i H$ and write $gg_j = g_i h$ with $h \in H$. Then for every k:

$$g(g_j \otimes v_k) = gg_j \otimes v_k = g_i h \otimes v_k = g_i \otimes hv_k = \sum_{t=1}^{n} a_{tk}(h)(g_i \otimes v_t)$$

where a_{tk} is the t, k coefficient of the matrix h acting on V with respect to the basis $v_1, v_2, ..., v_n$. Notice that the action of g on W maps the jth block of n basis vectors of W to the ith block of basis vectors, and then has the matrix $\varphi(h)$ as that block. Since $h = g_i^{-1} g g_j$, this fully describes $\Phi(g)$, as we sought to show.

Corollary 3.3.8: Computing The Induced Character

Let $H \leqslant G$ where G is a finite group, V be an H-representation with character χ, and $\mathrm{Ind}_H^G(\chi)$ the induced character. Then:

$$\mathrm{Ind}_H^G(\chi)(g) = \sum_{i=1}^{m} \chi(g_i^{-1} g g_i) \qquad \text{for all } g \in G$$

where $g_1, \ldots, g_m$ is a complete set of representatives for the left cosets of H in G, $m = [G : H]$, and $\chi(g_i^{-1} g g_i) := 0$ if $g_i^{-1} g g_i \notin H$. (The right-hand side does not depend on the chosen representatives: replacing g_i by $g_i h$ conjugates $g_i^{-1} g g_i$ by $h \in H$, which preserves membership in H and leaves χ unchanged, χ being a class function on H.) Hence, $\mathrm{Ind}_H^G(\chi)(g) = 0$ if g is not conjugate of some element in H in G, in particular if $H \trianglelefteq G$ then $\mathrm{Ind}_H^G(\chi)$ is zero on all elements $G - H$.

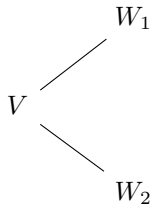
***Proof*:**
Let φ be the representation afforded by V. By theorem 3.3.7, $k[G] \otimes_{k[H]} V$ has a basis in which g acts by the block matrix $\Phi(g)$ whose (i, j) block is $\varphi(g_i^{-1} g g_j)$, the block being zero when $g_i^{-1} g g_j \notin H$. By definition 3.3.1, $\mathrm{Ind}_H^G(\chi)(g)$ is the trace of $\Phi(g)$, which is the sum of the traces of the diagonal blocks $\varphi(g_i^{-1} g g_i)$. Each of these traces is $\chi(g_i^{-1} g g_i)$ when $g_i^{-1} g g_i \in H$ and 0 otherwise, which gives the formula. If g is conjugate to no element of H, every $g_i^{-1} g g_i$ lies outside H and the sum is 0. If $H \trianglelefteq G$ and $g \notin H$, then $g_i^{-1} g g_i \notin H$ for every i, since $g_i^{-1} g g_i \in H$ would give $g \in g_i H g_i^{-1} = H$, so $\mathrm{Ind}_H^G(\chi)(g) = 0$, as claimed.

Proposition 3.3.9: Induced Index Two Induced Representation

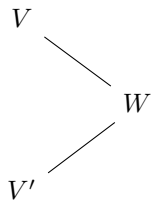
Let $H \leqslant G$ where $[G : H]$ has index two[a] (ex. $A_n \leqslant S_n$ or $C_n \leqslant D_{2n}$). Then if V is a finite-dimensional irreducible representation of G, either $V|_H$ is irreducible or $V|_H = W_1 \oplus W_2$ where $W_1 \ncong W_2$, are both ireducibile, and have the same dimension. Moreover, the second option implies that $V \cong V \otimes \chi$ where $\chi : G \twoheadrightarrow G/H \cong \{\pm 1\}$ (i.e. a 1-dimensional representation)

[a] Recall that this makes H a normal subgroup

In the case where $V|_H$ is not irreducible, we can imagine it splitting into a diagram like so:



while in the case where $V|_H$ is irreducible, what we actually have (and did not have time to prove), is that we have *two* irreducible representation's of G (V and V') that both get restricted to W:



and then the induced representation of W is the direct sum of V and V'.

***Proof*:**
Let $W \subseteq V|_H$ be an irreducible H-representation. Then Since $H \trianglelefteq G$, gW is is an H-stable space: gW is some subspace of V (it satisfies the subspace axioms since the action is linear), and for any $h \in H$, $h \cdot gW = hgW = g(g^{-1}hg)W = gW$, since $g^{-1}hg \in H$ by normality of H and W is H-stable. Now, gW is irreducible: the map $w \mapsto gw$ is a k-linear bijection $W \to gW$ satisfying $h(gw) = g((g^{-1}hg)w)$, so it intertwines the H-action up to conjugation. Since $g^{-1}Hg = H$ by normality, it carries H-submodules to H-submodules in both directions, so gW is irreducible. Finally, the structure of gW only depends on that of the coset gH

Thus, fix $\gamma \in G \setminus H$ so that

$$\sum_{g \in G} gW = W + \gamma W \subseteq V$$

where the $\subseteq$ comes from the fact that this is a sum of two subspaces. Now, notice that the left hand side is a G-stable space which is nonzero, and V is irreducible so has no proper nonzero G-stable subspace. Therefore we actually have the equality:

$$W + \gamma W = V$$

meaning the $H + gH = G$ is parallelled in the action! We now have two possibilities: either $W = \gamma W$, and so $V|_H = W$, and we have $\dim V = \dim W$ (which is case (a)).

On the other hand, if $W \neq \gamma W$, then $W \cap \gamma W = (0)$ (since they are both irreducible), and so $V = W \oplus \gamma W$ (i.e. direct sum as H-representations), and so V is the direct sum of two irreducible H-representations.

Now, by Frobenius reciprocity, we get:

$$\langle W, V|_H \rangle_H = \langle \mathrm{Ind}_H^G(W), V \rangle_G$$

Since W occurs in $V|_H$ by definition, we get that $1 \leq \langle W, V|_H \rangle_H$, while since V is irreducible and, continuing in case (b), $\dim \mathrm{Ind}_H^G(W) = [G:H] \dim W = 2 \dim W = \dim V$, the representation V can occur at most once, so $\langle \mathrm{Ind}_H^G(W), V \rangle_G \leq 1$, and hence:

$$1 = \langle W, V|_H \rangle_H = \langle \mathrm{Ind}_H^G(W), V \rangle_G = 1$$

Thus, from the first one, we get that $W \ncong \gamma W$ (or else the left hand side would be equal to 2), and we must have $V = \mathrm{Ind}_H^G(W)$ (or else the right hand side would be equal to something greater than 1)

If part (b) holds, then

$$(V \otimes \chi)|_H \cong V|_H$$

and

$$V \cong \mathrm{Ind}_H^G(W) \cong V \otimes \chi$$

Conversely, if $V \cong V \otimes \chi$, then $\chi_V(g) = \chi_V(g)\chi(g)$ for all $g \in G$. Since $\chi(g) = -1$ for $g \notin H$, this gives $\chi_V(g) = -\chi_V(g)$, so $\chi_V(g) = 0$ for all $g \notin H$. And so:

$$\|\chi_{V|_H}\|_H^2 = 2\|\chi_V\|_G^2$$

and so $\langle \chi_{V|_H}, \chi_{V|_H} \rangle_H = 2$, which rules out a single irreducible (norm 1) and $W \oplus W$ (norm 4), forcing $V|_H = W_1 \oplus W_2$ with $W_1 \ncong W_2$ irreducible, as we sought to show.

Theorem 3.1.16 says that the constituents of $\pi|_N$ form a single G-orbit. Once induction is available, that orbit description turns into a construction: the representation is induced from the subgroup stabilizing one of the classes. This is the form in which Clifford's theorem is normally applied.

Corollary 3.3.10: Clifford Induction From the Stabilizer

Let G be a group, $N \trianglelefteq G$, K a field, and let (π, V) be an irreducible representation of G on a finite-dimensional K-space. Let $W \subseteq V$ be an irreducible N-subrepresentation, let

$$H = \{g \in G \mid \pi_W^g \cong \pi_W\}, \qquad \pi_W^g(n) = \pi_W(g^{-1}ng)$$

be the stabilizer of its class, and let V_W be the π_W-isotypic component of $V|_N$. Then $N \subseteq H$, the subspace V_W is an irreducible H-subrepresentation, and

$$V \cong \mathrm{Ind}_H^G V_W$$

If $\pi|_N$ has more than one isomorphism class among its constituents, then $H \neq G$ and π is induced from a proper subgroup.

***Proof*:**
That $N \subseteq H$ is immediate, since $\pi_W^n \cong \pi_W$ for $n \in N$ by conjugating the isomorphism. Since V is finite dimensional, $V|_N$ has finitely many constituents, so the orbit below is finite and $[G : H]$ is finite. No finiteness of G itself is needed anywhere. By theorem 3.1.16 the classes occurring in $V|_N$ form one G-orbit, and $g \mapsto (U \mapsto U^g)$ is a left action, so the orbit is in bijection with G/H and $gV_W = V_{\pi_W^g}$ depends only on the coset gH. Summing the isotypic decomposition of the semisimple $V|_N$ over that orbit,

$$V = \bigoplus_{gH \in G/H} gV_W$$

This decomposition exhibits V as an induced representation. By definition 3.3.1 the induced representation is $\operatorname{Ind}_H^G V_W = K[G] \otimes_{K[H]} V_W$. The map

$$\Theta \colon K[G] \otimes_{K[H]} V_W \longrightarrow V, \qquad g \otimes v \longmapsto gv$$

is well defined, because $gh \otimes v$ and $g \otimes hv$ both go to ghv for $h \in H$, and it is G-equivariant by construction. It is surjective, since its image contains every gV_W and those span V by the display. Finally $\dim_K K[G] \otimes_{K[H]} V_W = [G : H] \dim_K V_W$, which is $\dim_K V$ by the display again, so Θ is an isomorphism.

Irreducibility of V_W over H. If $V_0 \subseteq V_W$ were a non-zero proper H-subrepresentation then $\bigoplus_{gH} gV_0$ would be a non-zero proper G-subrepresentation of V, the sum being direct and G-stable because V_0 is H-stable and gV_0 depends only on gH. That contradicts irreducibility of V.

For the last sentence, suppose more than one isomorphism class occurs in $\pi|_N$. The classes are in bijection with G/H, so $[G : H] > 1$ and H is proper, completing the proof.

3.4 *Application to Harmonic Analysis

In this section, we take a brief moment to explore the representation of locally compact abelian groups, such as $\mathbb{R}$. This section will assume some understanding of topological groups, and is mainly to motivate for the reader the connection between the representation theory of this chapter with *Fourier analysis* and more broadly *Harmonic analysis*.

Every group G is always a topological group with the discrete topology (see [13, the appendix on topological groups]). The group G is finite iff it is a compact discrete group. Many groups of interest are either compact (like the torus or compact lie groups), or locally compact (like the real line). The representation theory of these groups rely on the correct framing of the representation in the finite case. For the case of infinite dimension *non-abelian* groups, the build-up is a good deal too complicated for this expository section[4], however many of the major ideas for the abelian are simple enough to exposit, and even better naturally motivate the Fourier transform and abstract harmonic analysis through the lens of representation theory.

Let G be a finite abelian group so that all complex irreducible representations are 1-dimensional, and hence characters and necessarily a homomorphism. As G is finite, say $n = |G|$, then $\chi(g^n) = \chi(e) = 1$ for all $g \in G$, and so $\chi^n = \mathrm{id}$, showing that $\chi : G \to \mathbb{C}^*$ restricts to a map $\chi : G \to S^1$. This shows

[4] see [8] for the EYNTKA series on geometric representation theory

that all characters are *unitary representations*. All characters for a finite group are necessarily unitary representations, while for non-finite groups there can be non-unitary representations[5]. Such groups which have a (general) representation theory that does not generalize as well as infinite unitary representations (many representations are indecomposable but not irreducible, for the details see the companion Lie groups notes). Thus, for this sections we stick to maps to S^1 with the natural multiplication structure as the subgroup of $\mathbb{C}^*$: label this collection $\hat{G} = \mathrm{Hom}_{\mathbf{Ab}}(G, S^1)$. As the codomain is a group, $\mathrm{Hom}_{\mathbf{Ab}}(G, S^1)$ has a group structure given by point-wise multiplication.

Definition 3.4.1: Pontryagin Group

Let G be locally compact abelian group. Then the *Pontryagin dual* of G is the group:

$$\mathrm{Hom}_{\mathbf{Top}}(G, S^1)$$

of *continuous* homomorphisms, with the operation being point-wise multiplication and the compact-open topology. The topology is what makes $\widehat{G}$ itself locally compact abelian, and so what makes the double dual $\widehat{\widehat{G}}$ meaningful; for finite G it is discrete and can be ignored.

This is just another way of representing the characters of the group. Let us stick to the special case where G is a finite abelian group and re-frame the previous theory using the Pontryagin group:

Proposition 3.4.2: Pontryagin Dual

Let G be a finite abelian group, $\hat{G} = \mathrm{Hom}_{\mathbf{Ab}}(G, S^1)$ the group of characters under point-wise multiplication. Then there is an isomorphism:

$$G \cong \widehat{G}$$

Furthermore, there is a natural isomorphism:

$$G \cong \hat{\hat{G}}$$

If G is infinite, it is rarely the case that $G \cong \widehat{G}$ (think $\widehat{L^p} \cong L^q$). Indeed when G is compact and infinite it never is, since $\widehat{G}$ is then discrete. What is most important is that $\widehat{G}$ contains the necessary information to recover G, as the double-dual recovers G. The operation $\widehat{(-)}$ passes back and forth between a group and a unitary representation, and since the representations in play are unitary, that collection is itself a group[6]. The passage between the two is thus:

$$\text{Group} \overset{\widehat{(-)}}{\rightleftarrows} \text{Unitary Rep.}$$

***Proof*:**
As G is finite abelian,

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

[5] $\mathrm{SL}_2(\mathbb{R})$ acting on $\mathbb{R}^2$ need not be unitary

[6] the collection of representations also form a ring called the *representation ring* and has fascinating ties to K-theory and intersection theory, but it is again too much for this section and the connections are discussed in [2, Part II] of the EYTNKA series

where $n_1 \big| n_2 \big| \cdots \big| n_k$. Choose generators, $a_1, ..., a_k$. Define the indicator characters on the generators:

$$\chi_i(a_j) = \begin{cases} e^{\frac{2\pi i}{n_i}} & i = j \\ 1 & i \neq j \end{cases}$$

These certainly generate $\hat{G}$. Define the map on the generators:

$$\Phi(a_i) = \chi_i$$

This is an isomorphism.

For the double-dual, take the natural map (distinct from the Φ above):

$$\Psi(g_i) = \mathrm{ev}_{g_i}$$

where as usual $\mathrm{ev}_{g_i}(\chi) = \chi(g_i)$. This is easy to check that is an isomorphism, completing the proof.

Now, to study a space is to study all the (continuous) functions on it. Here the space is a finite abelian group. Let G have the discrete topology so that the set of continuous functions from G to $\mathbb{C}$ is all set functions. The goal is a $\mathbb{C}$-algebra isomorphism between $\mathbb{C}[G]$ and $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. These two sets are clearly bijective via the natural mapping:

$$\sum_i a_i g_i \mapsto (g_i \mapsto a_i)$$

And this map preserves addition if addition is defined point-wise in the hom-set:

$$\begin{aligned} &\Phi\left(\sum_i a_i g_i + \sum_i b_i g_i\right) \\ &= \Phi\left(\sum_i (a_i + b_i) g_i\right) \\ &\mapsto (g_i \mapsto a_i + b_i) \\ &= (g_i \mapsto a_i) + (g_i \mapsto b_i) \\ &= \Phi\left(\sum_i a_i g_i\right) + \Phi\left(\sum_i b_i g_i\right) \end{aligned}$$

But it does *not* preserve multiplication if the operation is pointwise multiplication:

$$\begin{aligned}
&\Phi\left(\sum_{g\in G} a_i g \sum_{g\in G} b_i g\right) \\
&= \Phi\left(\sum_{g\in G}\sum_{g_k g_\ell = g} (a_k b_\ell) g\right) \\
&\mapsto (g_i \mapsto \sum_{g_k g_\ell = g_i} (a_k b_\ell)) \\
&\neq (g_i \mapsto a_i b_i) \\
&= (g_i \mapsto a_i)\cdot(g_i \mapsto b_i) \\
&= \Phi\left(\sum_i a_i g_i\right)\cdot\Phi\left(\sum_i b_i g_i\right)
\end{aligned}$$

This is because the points g_i act on each other. There is a natural multiplication function defined on $\mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C})$ called *convolution*:

Definition 3.4.3: Convolution

Let $p, q \in \mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C})$. Then their *convolution* is define pointwise as:

$$(p * q)(g) = \sum_{g_i g_j = g} p(g_i)q(g_j) = \sum_{h\in G} p(h)q(h^{-1}g)$$

Lemma 3.4.4: Isomorphism With Convolution

Let G be a finite abelian group. Then:

$$(\mathbb{C}[G],+,\cdot) \cong_{k\text{-}\mathbf{Alg}} (\mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C}),+,*)$$

where the left hand side is the usual group ring and the right hand side has convolution as multiplication

***Proof*:**
The first two lines of the multiplication calculation above evaluate $\Phi(x\cdot y)$ at g_i as $\sum_{g_k g_\ell = g_i} a_k b_\ell$, which is exactly $(\Phi(x) * \Phi(y))(g_i)$ for the convolution just defined; additivity and bijectivity were checked above.

Now, $(\mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C}),+,*)$ is a $|G|$-dimensional $\mathbb{C}$ vector-space with the natural basis given by the indicator function:

$$\delta_h(g) = \begin{cases} 1 & g = h \\ 0 & g \neq h \end{cases}$$

Namely for any $f : G \to \mathbb{C}$ there is a unique decomposition given the functionals defined by each $g \in G$:

$$f = \sum_{h\in G} f(h)\delta_h \qquad \text{so that} \qquad f(g) = \sum_{h\in G} f(h)\delta_h(g) = f(g)\cdot 1 = f(g)$$

The key feature of this basis is that it acts like evaluation when taking the inner product. Define:

$$\langle f, g\rangle = \sum_{h\in G} f(h)\overline{g(h)}$$

Then $\{\delta_g\}_{g\in G}$ is an orthogonal basis and:

$$\langle f, \delta_{g_i}\rangle = f(g_i)\cdot\overline{\delta_{g_i}(g_i)} = f(g_i)\cdot 1 = f(g_i) = a_i$$

This is a special case of the Riesz representation theorem: every linear functional here is inner-product representable, and ev_{g_i} is represented by δ_{g_i}. This inner product can be thought as expressly chosen so that the collection δ_{g_i} forms an orthonormal basis, or conversely given this standard inner product, the indicator function will always be an orthonormal basis. Now $\hat{G}$ also has $|G|$ elements, so the characters are at least the right number to be a second basis, but they are a different family of functions from the δ_g, not the image of them under any isomorphism, and nothing so far says they are independent, let alone orthogonal. The key insight of Fourier analysis is that they are!

Proposition 3.4.5: Character Form Orthogonal Basis: Fourier Basis

The characters $\widehat{G}$ form an orthogonal basis for $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Normalizing by $1/\sqrt{|G|}$ forms an orthonormal basis.

***Proof*:**
If $i = j$:

$$\begin{aligned}\langle \chi_i, \chi_i\rangle &= \sum_{g\in G}\chi_i(g)\overline{\chi_i(g)}\\ &= \sum_{g\in G}|\chi_i(g)|^2\\ &= \sum_{g\in G}1^2\\ &= |G|\end{aligned}$$

if $i \neq j$, Then as $\overline{\chi_j} = \chi_j^{-1}$ (as we are mapping to the unit circle), we get that $\chi' = \chi_i\chi_j^{-1}$ is a non-trivial character, so there is some h such that $\chi'(h) \neq 1$. Let

$$\langle \chi_i, \chi_j\rangle = S = \sum_{g\in G}\chi'(g)$$

Then:

$$\chi'(h)S = \chi'(h)\sum_{g\in G}\chi'(g) = \sum_{g\in G}\chi'(h)\chi'(g) = \sum_{g\in G}\chi'(hg)$$

As $hG = G$, this is just a re-arranging of terms, hence:

$$\chi'(h)S = \sum_{g\in G}\chi'(g) = S \qquad \text{so} \qquad S(\chi'(h) - 1) = 0$$

Now, by assumption $\chi'(h) \neq 1$, so $\chi'(h) - 1 \neq 0$, so as $\mathbb{C}$ is an integral domain it must be that $S = 0$, implying:

$$\langle \chi_i, \chi_j\rangle = 0$$

as we sought to show.

This inner product can be thought of as extracting the "phase" information.

Example 3.4.6: Phase-Space Basis

Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \{(0,0),(1,0),(0,1),(1,1)\}$. A character $\chi : G \to \mathbb{C}^*$ is determined by its values on the generators $g_1 = (1,0)$ and $g_2 = (0,1)$. Since both generators have order 2, necessarily $\chi(g_1)^2 = \chi(g_1^2) = \chi(0,0) = 1$ and likewise for g_2. Thus, the character's values on the generators must be either 1 or -1. This gives $2 \times 2 = 4$ possible characters, indexed by pairs $(k_1, k_2) \in \{0,1\}^2$.

The four characters χ_{k_1,k_2} are defined by their values on the generators:

$$\chi_{k_1,k_2}(1,0) = (-1)^{k_1} \qquad \text{and} \qquad \chi_{k_1,k_2}(0,1) = (-1)^{k_2}$$

From this, the value for any element $(a,b) \in G$ is $\chi_{k_1,k_2}(a,b) = (-1)^{k_1 a + k_2 b}$.

This gives the following character table, which explicitly lists the functions that form our new "basis":

$\hat{G}$	**(0,0)**	**(1,0)**	**(0,1)**	**(1,1)**
χ_{00}	1	1	1	1
χ_{10}	1	-1	1	-1
χ_{01}	1	1	-1	-1
χ_{11}	1	-1	-1	1

The orthogonality relation from proposition 3.4.5 can be checked directly. Computing $\langle \chi_{10}, \chi_{11} \rangle$:

$$\begin{aligned}
\langle \chi_{10}, \chi_{11} \rangle &= \sum_{g \in G} \chi_{10}(g)\overline{\chi_{11}(g)} \\
&= \chi_{10}(0,0)\chi_{11}(0,0) + \chi_{10}(1,0)\chi_{11}(1,0) + \chi_{10}(0,1)\chi_{11}(0,1) + \chi_{10}(1,1)\chi_{11}(1,1) \\
&= (1)(1) + (-1)(-1) + (1)(-1) + (-1)(1) \\
&= 1 + 1 - 1 - 1 = 0
\end{aligned}$$

Using the above example, let us change perspectives on how to interpret this change of basis. The function $f : G \to \mathbb{C}$ takes the "points" $g \in G$ and maps them (continuously) to the complex numbers, and those points are evaluated by the Riesz representation of f against the indicator functions that form an orthonormal basis. But the characters give another orthogonal basis for $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ against which the inner product can be taken, orthonormal once scaled by $1/\sqrt{|G|}$. The characters $\chi \in \hat{G}$ may thus be read as points in their own right: points carrying "phase" information rather than the "location" information that the indicator functions carry. This motivates the following definition:

Definition 3.4.7: Fourier Transform (Gelfand Transform)

Let G be a finite abelian group with the discrete topology. Then the *Fourier transform* is:

$$(\widehat{-}) : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}) \qquad f \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

where the image of f is often written as:

$$\hat{f}(\chi) = \langle f, \chi \rangle = \sum_{g \in G} f(g)\overline{\chi(g)}$$

The name *Gelfand transform* comes from the fact that the Fourier transform is traditionally about real periodic functions, or Lebesgue measurable functions. It was not until later that Pontryagin, van Kampen, Weil and Gelfand generalized it to the context of locally compact abelian groups, where the above is the special case where G is finite.

Now, certainly as vector spaces $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \cong_{\mathbb{C}} \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$ mapping:

$$(g \mapsto \langle f, \delta_g \rangle) \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

But does this map preserve the inner product, and more so can the convolution product of $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ be preserved given the appropriate choice of product on $\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$? The answer is yes, and can be thought of as the corner-stone result of harmonic analysis on discrete abelian groups:

Theorem 3.4.8: Fourier Reciprocity

The map given in definition 3.4.7 is a $\mathbb{C}$-algebra isomorphism,

$$(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *) \cong (\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}), +, \cdot)$$

with $\cdot$ in $(\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}), +, \cdot)$ being point-wise multiplication. It scales the inner product by $|G|$, so the normalisation $\mathcal{U}(f) = \frac{1}{\sqrt{|G|}}\hat{f}$ is unitary.

Recall that unitary means $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$. In a conventional real analysis course:

- the $\mathbb{C}$-algebra homomorphism is called the *Convolution Theorem*,
- the unitary property is called *Plancherel's Theorem*,
- invertibility (i.e. bijectivity) is called the *Fourier Inversion Theorem*.

***Proof*:**

1. *1. Linearity:* Let $f, h \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ and $a, b \in \mathbb{C}$.

$$\begin{aligned}
\widehat{af + bh}(\chi) &= \langle af + bh, \chi \rangle && \text{by definition} \\
&= \sum_{g \in G} (af + bh)(g)\overline{\chi(g)} \\
&= a \sum_{g \in G} f(g)\overline{\chi(g)} + b \sum_{g \in G} h(g)\overline{\chi(g)} && \text{by linearity of sums} \\
&= a\langle f, \chi \rangle + b\langle h, \chi \rangle \\
&= a\hat{f}(\chi) + b\hat{h}(\chi)
\end{aligned}$$

Since this holds for all $\chi \in \hat{G}$, the transform is a linear map.

2. $\mathbb{C}$-algebra Homomorphism (The Convolution Theorem): We must show that the Fourier transform converts convolution into pointwise multiplication.

$$\begin{aligned}
\widehat{f * h}(\chi) &= \langle f * h, \chi \rangle \\
&= \sum_{k \in G} (f * h)(k)\overline{\chi(k)} && \text{by definition of inner product} \\
&= \sum_{k \in G} \left(\sum_{g \in G} f(g)h(g^{-1}k) \right) \overline{\chi(k)} && \text{by definition of convolution} \\
&= \sum_{g \in G} f(g) \sum_{k \in G} h(g^{-1}k)\overline{\chi(k)} && \text{swapping order of summation}
\end{aligned}$$

Let $\ell = g^{-1}k$, so $k = g\ell$. As k runs over G, so does ℓ. The inner sum becomes:

$$\sum_{\ell \in G} h(\ell)\overline{\chi(g\ell)} = \sum_{\ell \in G} h(\ell)\overline{\chi(g)\chi(\ell)} = \overline{\chi(g)} \sum_{\ell \in G} h(\ell)\overline{\chi(\ell)} = \overline{\chi(g)}\hat{h}(\chi)$$

Substituting this back into the main expression:

$$\begin{aligned}
\widehat{f * h}(\chi) &= \sum_{g \in G} f(g) \left(\overline{\chi(g)}\hat{h}(\chi) \right) \\
&= \hat{h}(\chi) \sum_{g \in G} f(g)\overline{\chi(g)} \\
&= \hat{h}(\chi)\hat{f}(\chi)
\end{aligned}$$

Thus, $\widehat{f * h} = \hat{f} \cdot \hat{h}$, proving the map is an algebra homomorphism.

3. Isomorphism (The Fourier Inversion Theorem): We show the map is invertible by constructing its inverse. Let $\mathcal{F}(f) = \hat{f}$. We claim the inverse is given by $\mathcal{F}^{-1}(\hat{f})(g) =$

$\frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g)$.

$$\begin{aligned}
\frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g) &= \frac{1}{|G|}\sum_{\chi\in\hat{G}}\left(\sum_{h\in G} f(h)\overline{\chi(h)}\right)\chi(g) && \text{by definition of } \hat{f}\\
&= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(g)\overline{\chi(h)} && \text{swapping order of summation}\\
&= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(gh^{-1})
\end{aligned}$$

The inner sum is an example of the *Second Orthogonality Relation* and is a standard exercise the reader should do. In particular, the sum of all characters evaluated at a group element (giving all powers of a certain root of unity) is $|G|$ if the element is e and 0 otherwise. Thus $\sum_{\chi\in\hat{G}}\chi(gh^{-1}) = |G|\delta_{g,h}$.

$$\frac{1}{|G|}\sum_{h\in G} f(h)(|G|\delta_{g,h}) = \sum_{h\in G} f(h)\delta_{g,h} = f(g)$$

Since we can recover f uniquely from $\hat{f}$, the map is an isomorphism.

4. Unitary Property (Plancherel's Theorem): Note that here is where the transform picks up a factor of $|G|$, and so where the normalising constant is forced. Let us calculate the inner product for the given definition.

$$\begin{aligned}
\langle\hat{f},\hat{h}\rangle_{\hat{G}} &= \sum_{\chi\in\hat{G}}\hat{f}(\chi)\overline{\hat{h}(\chi)}\\
&= \sum_{\chi\in\hat{G}}\left(\sum_{g\in G} f(g)\overline{\chi(g)}\right)\overline{\left(\sum_{k\in G} h(k)\overline{\chi(k)}\right)}\\
&= \sum_{\chi\in\hat{G}}\left(\sum_{g\in G} f(g)\overline{\chi(g)}\right)\left(\sum_{k\in G}\overline{h(k)}\chi(k)\right)\\
&= \sum_{g\in G}\sum_{k\in G} f(g)\overline{h(k)}\sum_{\chi\in\hat{G}}\chi(k)\overline{\chi(g)} && \text{re-arranging terms}\\
&= \sum_{g\in G}\sum_{k\in G} f(g)\overline{h(k)}(|G|\delta_{g,k}) && \text{by Second Orthogonality}\\
&= |G|\sum_{g\in G} f(g)\overline{h(g)}\\
&= |G|\langle f,h\rangle_G
\end{aligned}$$

This proves *Plancherel's Theorem.* To make the map strictly unitary, one must normalize the transform. If we define the unitary transform $\mathcal{U}(f) = \frac{1}{\sqrt{|G|}}\hat{f}$, then it follows immediately that $\langle\mathcal{U}(f),\mathcal{U}(h)\rangle = \langle f,h\rangle$, completing the proof.

Example 3.4.9: Fourier Transform

Take $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ from example 3.4.6. Now for the Fourier transform of a concrete function. Let $f = \delta_{(1,0)}$, so that $f(1,0) = 1$ and is zero elsewhere. This is a basis vector in the standard "position" basis, and its coordinates in the character basis come from computing $\hat{f}(\chi) = \langle f, \chi \rangle$ for each character.

$$\begin{aligned}\hat{f}(\chi) = \langle \delta_{(1,0)}, \chi \rangle &= \sum_{g \in G} \delta_{(1,0)}(g)\overline{\chi(g)} \\ &= 1 \cdot \overline{\chi(1,0)} + 0 + 0 + 0 = \chi(1,0)\end{aligned}$$

The character table gives this value for each χ:

- $\hat{f}(\chi_{00}) = \chi_{00}(1,0) = 1$
- $\hat{f}(\chi_{10}) = \chi_{10}(1,0) = -1$
- $\hat{f}(\chi_{01}) = \chi_{01}(1,0) = 1$
- $\hat{f}(\chi_{11}) = \chi_{11}(1,0) = -1$

Thus, the original function $f = \delta_{(1,0)}$, which is a spike at one point in the "position" basis, has been transformed into a new function $\hat{f}$ on $\hat{G}$ with values $(1, -1, 1, -1)$.

Recall the inversion formula from the proof of theorem 3.4.8: $f(g) = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi)\chi(g)$ (here $|G| = 4$). Let's check the value at $g = (1,0)$:

$$\begin{aligned}f(1,0) &= \frac{1}{4}\left(\hat{f}(\chi_{00})\chi_{00}(1,0) + \hat{f}(\chi_{10})\chi_{10}(1,0) + \hat{f}(\chi_{01})\chi_{01}(1,0) + \hat{f}(\chi_{11})\chi_{11}(1,0)\right) \\ &= \frac{1}{4}\left((1)(1) + (-1)(-1) + (1)(1) + (-1)(-1)\right) \\ &= \frac{1}{4}(1+1+1+1) = 1\end{aligned}$$

Now the value at $g = (0,1)$, which should be 0:

$$\begin{aligned}f(0,1) &= \frac{1}{4}\left(\hat{f}(\chi_{00})\chi_{00}(0,1) + \hat{f}(\chi_{10})\chi_{10}(0,1) + \hat{f}(\chi_{01})\chi_{01}(0,1) + \hat{f}(\chi_{11})\chi_{11}(0,1)\right) \\ &= \frac{1}{4}\left((1)(1) + (-1)(1) + (1)(-1) + (-1)(-1)\right) \\ &= \frac{1}{4}(1-1-1+1) = 0\end{aligned}$$

This now connects to representation theory. So far, for a finite abelian group G, the Fourier transform has been established as a unitary $\mathbb{C}$-algebra isomorphism between the algebra of functions on G with convolution, $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *)$, and the algebra of functions on the dual group $\hat{G}$ with pointwise multiplication. The representation theory perspective shows that Fourier analysis is the simplest case of an Artin-Wedderburn decomposition. The key is to view the $\mathbb{C}$-group algebra $\mathbb{C}[G]$ with the natural G-action given by left multiplication (convolution), namely $\rho : G \to \mathrm{GL}(\mathbb{C}[G])$ given by:

$$\rho(g) : f \mapsto g * f$$

From this viewpoint, the study of the $\mathbb{C}$-algebra $(\mathbb{C}[G], +, *)$ and the $\mathbb{C}[G]$-module $\mathbb{C}[G]$ is the same thing that is the study of the regular representation of G. Now, this representation can be broken down into the direct sum of 1-dimensional irreducible representations. Namely, for a finite abelian group, the set of irreducible representations is exactly the dual group: $\mathrm{Irr}(G) = \widehat{G}$, giving:

$$\mathbb{C}[G] \cong \bigoplus_{\chi \in \widehat{G}} V_\chi$$

where each V_χ is a 1-dimensional subspace corresponding to the character χ.

This decomposition is made concrete by the Fourier transform. The Fourier Inversion Formula, $f = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi)\chi$, is the literal expression of this decomposition. It writes the function f as a sum of components, where each term $\hat{f}(\chi)\chi$ is the projection of f onto the 1-dimensional irreducible subspace V_χ. Then the fourier coefficient $\hat{f}(\chi)$ is the coordinate of f in the irreducible component corresponding to χ.

The projection functions $p_\chi = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} g$ can be constructed inside $\mathbb{C}[G]$, and project onto the irreducible subspace V_χ. Convolving any function f with p_χ annihilates all other "frequency components", leaving the part of f that lies in V_χ.

Hence, the Fourier basis is *the* basis that diagonalizes the natural action of the group on its own algebra of functions.

If G is a *non-abelian* finite group, the identification $\mathrm{Irr}(G) = \hat{G}$ breaks and with it the simple transform: $\mathrm{Hom}(G, S^1)$ remains a group, but it only sees the abelianisation $\widehat{G^{\mathrm{ab}}}$, while the object one wants, $\mathrm{Irr}(G)$, carries no natural group structure. However, the representation theory perspective remains: the generalized Fourier transform (given by the *Peter-Weyl Theorem*) decomposes a function f on the group into a sum of *matrix coefficients* corresponding to these higher-dimensional matrix representations.

3.4.1 Generalization to Locally Compact Abelian Groups

What happens for abelian groups like $(\mathbb{R}, +)$, $(\mathbb{Z}, +)$, or the circle group $(S^1, \cdot)$? These are examples of *Locally Compact Abelian (LCA)* groups. Moving from the finite to the infinite setting means replacing finite sums with integrals, and integrating on an arbitrary group first requires a notion of "volume" or "measure". The shape of the answer is already visible in the finite case: read the delta functions δ_h as *measures*, and every f becomes a sum of weighted point masses. The right object, then, is a measure on the group that the group operation cannot see. The essential new ingredient is the *Haar measure*. For any LCA group G, there exists a measure μ that gives a consistent way to integrate functions on the group.

Definition 3.4.10: Haar Measure

A *Haar measure* on an LCA group G is a measure μ on the Borel σ-algebra of G that is left-translation-invariant. That is, for any measurable set $E \subseteq G$ and any element $g \in G$:

$$\mu(gE) = \mu(E)$$

This measure is unique up to a positive scaling constant. For abelian groups, it is also right-translation-invariant.

See [15] for details on how to construct such a measure on any LCA that is unique up to scaling. This measure is the direct generalization of taking the order of the group.

Example 3.4.11: Haar Measure

1. For a finite group with the discrete topology, the Haar measure is the counting measure, and $\mu(G) = |G|$.
2. For $(\mathbb{R}, +)$, the Haar measure is the standard Lebesgue measure dx.
3. For $(S^1, \cdot)$, the Haar measure can be taken as $d\theta$.
4. For $(\mathbb{Z}, +)$, the Haar measure is again the counting measure.

In the infinite case points can no longer be represented by indicator functions; the *Dirac delta distributions* take on that role instead. See [14] for the theory of distributions.

With the Haar measure in hand, the main objects of study can be redefined by replacing every sum over G with an integral with respect to $d\mu$.

1. *The Group Algebra $L^1(G)$:* The space of all functions becomes too large. The natural replacement for the convolution algebra $\mathbb{C}[G]$ is the space of absolutely integrable functions, $L^1(G, d\mu)$. For two functions $f, h \in L^1(G)$, their convolution is defined by the integral:

$$(f * h)(g) = \int_G f(x)h(x^{-1}g)d\mu(x)$$

   This space $(L^1(G), +, *)$ forms a Banach algebra.

2. *The Dual Group $\widehat{G}$:* It was proven that the correct codomain should be S^1. The dual group is the set of all *continuous* group homomorphisms into S^1.

$$\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$$

   The dual group $\widehat{G}$ is itself an LCA group. It is crucial to note that, unlike the finite case, $G \cong \widehat{G}$ is the exception rather than the rule, holding for $\mathbb{R}$ but failing for $\mathbb{Z}$ and S^1:

$$\widehat{\mathbb{R}} \cong \mathbb{R}, \qquad \widehat{\mathbb{Z}} \cong S^1, \qquad \widehat{S^1} \cong \mathbb{Z}$$

   This highlights why it is essential to treat $\hat{G}$ as a distinct space. The theory that governs this relationship is *Pontryagin Duality*, which states that the double-dual is canonically isomorphic to the original group: $\widehat{\widehat{G}} \cong G$.

3. *The Fourier Transform:* The definition of the Fourier transform is the direct analogue of the finite case, replacing the sum with an integral. For a function $f \in L^1(G) \cap L^2(G)$, its Fourier transform is a function $\hat{f}$ on the dual group $\hat{G}$:

$$\hat{f}(\chi) = \int_G f(g)\overline{\chi(g)}d\mu(g)$$

   Similarly, the inner product that defines the Hilbert space structure on $L^2(G)$ is:

$$\langle f, h\rangle = \int_G f(g)\overline{h(g)}d\mu(g)$$

With these definitions, the entire suite of theorems proven in the finite case carries over:

Theorem 3.4.12: Fourier Reciprocity on LCA Groups

Let G be an LCA group with Haar measure μ. Then there exists a unique Haar measure ν on the dual group $\widehat{G}$ for which the following hold. (Unlike the finite case these are statements about two different maps: $L^2(G) \to L^2(\widehat{G})$ is a unitary isomorphism, while $L^1(G) \to C_0(\widehat{G})$ is an injective algebra map that is not onto.)

1. *The Fourier Inversion Theorem:* For a sufficiently well-behaved function f (e.g., $f \in L^1 \cap L^2$ such that $\hat{f} \in L^1$), the original function can be recovered by an integral over the dual group:
$$f(g) = \int_{\widehat{G}} \hat{f}(\chi)\chi(g)d\nu(\chi)$$
2. *Plancherel's Theorem:* The Fourier transform extends uniquely from $L^1 \cap L^2$ to a unitary isomorphism of Hilbert spaces, $\mathcal{F} : L^2(G, \mu) \to L^2(\widehat{G}, \nu)$. This means it preserves the inner product:
$$\langle \hat{f}, \hat{h} \rangle_{L^2(\widehat{G})} = \langle f, h \rangle_{L^2(G)}$$
3. *The Convolution Theorem:* For $f, h \in L^1(G)$, the Fourier transform maps convolution to pointwise multiplication:
$$\widehat{f * h}(\chi) = \hat{f}(\chi) \cdot \hat{h}(\chi)$$

***Proof*:**
This is the Pontryagin duality theorem; a full proof via the Gelfand transform belongs to the harmonic analysis of locally compact abelian groups and is not reproduced in this starred section.

The correspondence between the finite and general LCA settings can be summarized as follows:

Finite Abelian Case	**LCA Case**
Group G	LCA Group G
Order $\lvert G \rvert$	Haar Measure μ
Group Algebra $\mathbb{C}[G]$	Banach Algebra $L^1(G, \mu)$
Hilbert Space $\mathrm{Hom}(G, \mathbb{C})$	Hilbert Space $L^2(G, \mu)$
Summation $\sum_{g \in G}$	Integration $\int_G d\mu(g)$
Dual Group $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$	Dual Group $\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$
Inner Product $\sum f(g)\overline{h(g)}$	Inner Product $\int f(g)\overline{h(g)}d\mu(g)$
Fourier Transform $\sum f(g)\overline{\chi(g)}$	Fourier Transform $\int f(g)\overline{\chi(g)}d\mu(g)$

Further interesting facts can be found here: https://mathoverflow.net/questions/37021/is-fourier-analysis-a-special-case-of-representation-theory-or-an-analogue

3.4.2 The Algebraic-Geometric Meaning of the Dual Group

In algebraic geometry, a fundamental principle is that the geometry of a space can be recovered from its algebra of functions. For an affine space k^n over an algebraically closed k, the corresponding

algebra is the polynomial ring $k[x_1, \ldots, x_n]$, and the Nullstellensatz makes the points of the space correspond to the maximal ideals of this algebra. (Over a non-closed field the correspondence fails: $(x^2+1) \subseteq \mathbb{R}[x]$ is maximal and is no point of $\mathbb{R}^1$.) Specifically, for any point $p = (a_1, \ldots, a_n) \in k^n$, the evaluation map:

$$\mathrm{ev}_p : k[x_1, \ldots, x_n] \to k \qquad f \mapsto f(p)$$

is an algebra homomorphism whose kernel, $m_p = (x_1 - a_1, \ldots, x_n - a_n)$, is a maximal ideal. The quotient $k[x_1, \ldots, x_n]/m_p \cong k$ recovers the value of the function at that point.

A similar setup is now in place for a finite abelian group G, with the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *, +)$ taking the role of the function algebra. A natural question arises: can we recover the "points" of our space from the maximal ideals of this algebra? Let's first attempt a direct analogy, assuming the points are the elements of the group G itself. For a point $g_i \in G$ there is a point-evaluation map, which is linear:

$$\mathrm{ev}_{g_i} : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto f(g_i)$$

The kernel of this map is the vector subspace of functions that are zero at g_i. However, for this subspace to be an ideal, it must be closed under convolution. Let $f \in \ker(\mathrm{ev}_{g_i})$ (so $f(g_i) = 0$) and let h be any function in our algebra. Consider their convolution:

$$(h * f)(g_i) = \sum_{k \in G} h(k) f(k^{-1} g_i)$$

The value of f at other points, $f(k^{-1}g_i)$, can be non-zero, and in fact the subspace really does fail to be an ideal. Take $G = \mathbb{Z}/2\mathbb{Z} = \{e, a\}$ and $g_i = e$. Then $\delta_a \in \ker(\mathrm{ev}_e)$, but $\delta_a * \delta_a = \delta_{a^2} = \delta_e$, which is 1 at e. So the kernel of the point-evaluation map is *not* an ideal with respect to convolution. This implies the points of G do not correspond to the maximal ideals of its convolution algebra.

The key insight is that the correct "geometric points" for the convolution algebra are not the elements of G, but the characters of the dual group $\widehat{G}$.

Proposition 3.4.13: Characters and Maximal Ideals

Let G be a finite abelian group. The maximal ideals of the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ are in bijective correspondence with the characters $\chi \in \widehat{G}$.

***Proof*:**

For each character $\chi \in \widehat{G}$, define a Fourier evaluation map:

$$\widehat{\mathrm{ev}}_\chi : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto \hat{f}(\chi)$$

This map takes a function and returns its Fourier coefficient at the frequency χ. By 3.4.8 $\widehat{\mathrm{ev}}_\chi$ is a $\mathbb{C}$-algebra homomorphism. Since $\widehat{\mathrm{ev}}_\chi$ is a surjective algebra homomorphism onto a field ($\mathbb{C}$), its kernel, which we denote by I_χ, is a maximal ideal:

$$I_\chi = \ker(\widehat{\mathrm{ev}}_\chi) = \{f \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \big| \hat{f}(\chi) = 0\}$$

By the First Isomorphism Theorem for rings, we have:

$$\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})/I_\chi \cong \mathbb{C}$$

This shows that every character $\chi \in \widehat{G}$ defines a maximal ideal. Conversely, after applying the Fourier transform we get the usual multiplication and ring structure, and then an approach similar to the Gelfand-Kolmogorov theorem gives the desired result.

> For the infinite case the same picture holds with one caveat: $L^1(G,\mu)$ is non-unital unless G is discrete, carrying only an approximate identity, so it is the maximal *modular* ideals that correspond to the homomorphisms into $\mathbb{C}$ (via the *Gelfand-Mazur Theorem*, which identifies the quotient by such an ideal with $\mathbb{C}$).

Thus, with the convolution algebra $\mathbb{C}[G]$ as our primary object, the natural geometric space that emerges is not the group G itself, but its Pontryagin dual, $\widehat{G}$. The Fourier transform is the mechanism that allows us to evaluate the functions of our algebra at the points of this "true" underlying space. This perspective, generalized by Gelfand, is the foundation of non-commutative geometry.

Concepts	**Classical Algebraic Geometry**
Algebra	Polynomial Ring $(k[x_1,\dots,x_n],\cdot)$
Product	Polynomial Multiplication
Geometric Space	Affine Space k^n
A "Point" p	A tuple $(a_1,\dots,a_n)\in k^n$
Evaluation Map	$\mathrm{ev}_p : f \mapsto f(p)$
Maximal Ideal for p	$\ker(\mathrm{ev}_p) = (x_i - a_i)$

Concepts	**Harmonic Analysis on G**
Algebra	Group Algebra $(\mathbb{C}[G],*)$
Product	Convolution
Geometric Space	The Dual Group $\widehat{G}$
A "Point" p	A character $\chi\in\widehat{G}$
Evaluation Map	Fourier "Evaluation" $\widehat{\mathrm{ev}}_\chi : f \mapsto \hat{f}(\chi)$
Maximal Ideal for p	$\ker(\widehat{\mathrm{ev}}_\chi) = I_\chi$

3.5⋆ Modular Representations and the Failure of Semisimplicity

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Two chapters have approached $k[G]$ from opposite sides. Chapter 3 has assumed throughout that $\mathrm{char}(k)\nmid|G|$, where theorem 3.1.9 makes every module semisimple and the character theory of section 3.2 computes what the modules are. Chapter 2 built machinery for algebras where semisimplicity fails: the radical $\mathrm{Jac}(A)$ measuring the failure, Krull-Schmidt replacing simple summands by indecomposable ones, and projective covers replacing the projections that no longer exist. The group algebra $k[G]$ with $\mathrm{char}(k)=p$ dividing $|G|$ is the object that machinery was built for, and this section applies it.

There is a temptation to read theorem 3.1.9 as having settled the characteristic-p case. The condition $p\big||G|$ does say exactly when semisimplicity fails,[7] and example 2.1.11 computed the resulting radical for $G=\mathbb{Z}/p\mathbb{Z}$ down to the last element. What the theorem does not say is what replaces the semisimple picture when it fails, and that question, for a general finite group, is open. The subject

[7] Theorem 3.1.9 gives one direction. The other is quick: the element $\nu=\sum_{g\in G} g$ satisfies $h\nu=\nu=\nu h$ for every $h\in G$, so $k\nu$ is a two-sided ideal, and $\nu^2=|G|\nu=0$ when p divides $|G|$. Since $k[G]$ is finite dimensional it is left Artinian, so a nonzero nilpotent ideal forces $\mathrm{Jac}(k[G])\neq 0$ by theorem 2.1.8, and $k[G]$ is not semisimple by corollary 2.1.10.

devoted to it is *modular representation theory*, treated in *Modular Representation Theory* [7]. The word *modular* here refers to reduction modulo a prime and has nothing to do with modular forms or with the modular arithmetic of $\mathbb{Z}/n\mathbb{Z}$.

It is worth being precise about what is open, because the framework is not. Krull-Schmidt (theorem 2.2.4) applies to $k[G]$ for any field, so the regular module has a decomposition into indecomposable summands that is unique up to isomorphism and reordering, theorem 2.3.5 matches those summands with the simple modules, and theorem 2.3.7 identifies them as the projective covers. The structural questions are answered. What is not known, for a general G and p, are the *invariants*: the dimensions and characters of the simple $k[G]$-modules. Determining them for the symmetric groups is a well-known open problem. Worse, the indecomposable modules themselves are unclassifiable for most groups, in a sense theorem 3.5.7 below makes precise.

We begin with the coarsest decomposition of $k[G]$, which costs nothing beyond chapter 2 and already organizes everything that follows.

Definition 3.5.1: Block of an Algebra

Let A be a finite-dimensional k-algebra and let $Z(A)$ denote its centre. A *block idempotent* of A is a primitive idempotent of the algebra $Z(A)$, and for such an idempotent b the two-sided ideal $B = bA$ is called a *block* of A.

The word *block* is overworked, and three of its meanings have already appeared in this book. A Jordan block is a matrix in the normal form of example 2.2.6; the Artin-Wedderburn factors $M_{n_i}(D_i)$ of a semisimple algebra are called blocks in the statement of theorem 2.3.5; and definition 3.5.1 is the sense used from here on. The last two agree on semisimple algebras, where the block idempotents are exactly the identity elements of the Artin-Wedderburn factors, so no earlier statement changes meaning. They diverge as soon as $\mathrm{Jac}(A) \neq 0$, which is the case this section is about.[8]

Theorem 3.5.2: Block Decomposition

Let A be a finite-dimensional k-algebra. Then the identity has a decomposition

$$1 = b_1 + b_2 + \cdots + b_m$$

into pairwise orthogonal primitive idempotents of $Z(A)$, and this decomposition is unique up to reordering. The corresponding blocks give a direct sum decomposition

$$A = b_1A \oplus b_2A \oplus \cdots \oplus b_mA$$

of A into two-sided ideals, each of which is a k-algebra with identity element b_i. Every indecomposable left A-module M satisfies $M = b_iM$ for exactly one index i, and every block contains at least one simple module in this sense.

***Proof*:**
Existence. The centre $Z(A)$ is a commutative k-algebra, and it is finite dimensional because A is. Applying theorem 2.3.2 to the algebra $Z(A)$ gives a decomposition $1 = b_1 + \cdots + b_m$ of its

[8]The eponym is also overworked. Richard Brauer's name attaches both to the block theory of this section and to the Brauer group of section 1.3. The two objects are unrelated.

identity, which is the identity of A, into pairwise orthogonal primitive idempotents of $Z(A)$.

The decomposition of A. Each b_i is central, so $b_iA = Ab_i$ is a two-sided ideal, and it is closed under multiplication with b_i acting as a two-sided identity on it. Since the b_i are orthogonal with sum 1, every $a \in A$ satisfies $a = \sum_i b_i a$ with $b_i a \in b_i A$, and if $\sum_i b_i a_i = 0$ then multiplying by b_j gives $b_j a_j = 0$ for each j. The sum is therefore direct.

Uniqueness. Let $1 = c_1 + \cdots + c_l$ be a second such decomposition. For each i we have $b_i = \sum_j b_i c_j$, and each $b_i c_j$ is an idempotent of $Z(A)$ because b_i and c_j are commuting idempotents. These are pairwise orthogonal, since $(b_i c_j)(b_i c_{j'}) = b_i c_j c_{j'} = 0$ for $j \neq j'$. Primitivity of b_i forces exactly one summand to be nonzero, so $b_i = b_i c_j$ for a unique j, which is to say $b_i \leq c_j$ in the sense that $b_i c_j = b_i$. Running the same argument with the roles reversed gives, for that j, a unique i' with $c_j = c_j b_{i'}$. Then $b_i = b_i c_j$ and $c_j = c_j b_{i'}$ give $b_i b_{i'} = b_i c_j b_{i'} = b_i c_j = b_i \neq 0$, and orthogonality of distinct b's forces $i' = i$, whence $b_i = b_i c_j = c_j b_i = c_j$. So $i \mapsto j$ is a bijection matching the two decompositions.

Modules. Let M be an indecomposable left A-module. Each $b_i M$ is a submodule, because b_i is central: for $a \in A$ we have $a(b_i m) = b_i(am)$. The identity acts as the identity map on M, so $M = \sum_i b_i M$, and the sum is direct by the same orthogonality computation as above. Indecomposability leaves exactly one nonzero summand, so $M = b_i M$ for exactly one i.

Each block has a simple module. The map $A \to b_i A$, $a \mapsto b_i a$, is a surjective ring homomorphism, since b_i is a central idempotent, and it carries 1 to the identity b_i of $b_i A$. Now $b_i A$ is a nonzero finite-dimensional algebra, so it has a maximal left ideal and hence a simple module S. Pulling back along the surjection makes S a simple A-module with $b_i S = S$.

The theorem says that the study of $k[G]$-modules splits, with no loss and no overlap, into m independent problems, one per block. This is the first thing one does with a modular group algebra, and it is why the literature speaks of the representation theory "of a block".

The blocks of $k[G]$ can be located without factoring the algebra, which is what makes them computable. Each block idempotent b_i is central, so multiplication by b_i is an endomorphism of any A-module, and on a simple module it is either 0 or the identity because it is idempotent. A simple module is indecomposable, so theorem 3.5.2 gives $S = b_i S$ for exactly one index i: one block idempotent acts as the identity on S and the rest act as zero, which is what it means for S to belong to that block. Suppose now that k is a splitting field for G, so that $\mathrm{End}_{k[G]}(S) = k$ for every simple S. Then Schur's lemma makes every central element act on S by a scalar, and the resulting k-algebra map $\omega_S : Z(k[G]) \to k$ is called the *central character* of S. By the previous paragraph ω_S determines the block of S, since it returns 1 on that block's idempotent and 0 on the others. Now the class sums $\sigma_1, \ldots, \sigma_s$, one for each conjugacy class of G, form a k-basis of $Z(k[G])$ over any field: each σ_i is central because conjugation permutes each class, the σ_i are linearly independent because the group elements are, and $\sum_g \lambda_g g$ is central exactly when $g \mapsto \lambda_g$ is constant on conjugacy classes. That is the computation carried out inside the proof of proposition 3.1.13, which used nothing about the characteristic of the field. Every b_i is therefore a k-combination of class sums, so two simple modules whose central characters agree on every class sum lie in the same block. The criterion is a statement about the simple modules in characteristic p, which are themselves the unknowns. What turns block-finding into arithmetic with the ordinary character table is Brauer's refinement, which runs the test on the *ordinary* irreducible characters instead. For an ordinary irreducible χ and a class sum σ_C with representative g_C, the relevant scalar is

$$\omega_\chi(\sigma_C) = \frac{|C|\,\chi(g_C)}{\chi(1)}$$

an algebraic integer. Ordinary irreducible characters are sorted into blocks of $k[G]$ because the block idempotents lift from $k[G]$ to $R[G]$ once a p-modular system is fixed (definition 3.5.4), and two ordinary irreducibles lie in the same block when these scalars agree for every class modulo a maximal ideal above p. The division by $\chi(1)$ is not optional: for a simple module of dimension 1 a class sum acts by the plain sum of the scalars, and dropping the normalization for a character of higher degree gives the wrong answer. Example 3.5.3 uses the degree-1 shortcut precisely because both of its simple modules are one dimensional and are reductions of ordinary representations.

For a semisimple algebra the decomposition is the Artin-Wedderburn one and nothing is gained. The content is in the opposite direction: when $\mathrm{Jac}(A) \neq 0$ a single block can carry many simple modules, and untangling them is the work.

Example 3.5.3: The Blocks of $k[S_3]$ in Characteristic 3

Let k be a field with $\mathrm{char}(k) = 3$ and let $G = S_3$, so $p = 3$ divides $|G| = 6$ and theorem 3.1.9 does not apply. The centre $Z(k[S_3])$ has the three class sums as a basis:

$$\sigma_1 = e, \qquad \sigma_2 = (1\ 2) + (1\ 3) + (2\ 3), \qquad \sigma_3 = (1\ 2\ 3) + (1\ 3\ 2)$$

Write $N = \{e, (1\ 2\ 3), (1\ 3\ 2)\}$ for the normal subgroup of order 3, generated by $c = (1\ 2\ 3)$, and let I be the ideal of $k[S_3]$ generated by $c - 1$, which is two-sided because N is normal. By the computation of example 2.1.11 with $p = 3$ we have $(c-1)^3 = c^3 - 1 = 0$, so I is nilpotent. For any simple $k[S_3]$-module S the subset IS is a submodule, and nilpotency of I forces $IS \neq S$, so $IS = 0$ and N acts trivially on S. The simple $k[S_3]$-modules are therefore exactly the simple modules of

$$k[S_3/N] \cong k[\mathbb{Z}/2\mathbb{Z}]$$

inflated along the quotient map. Since $\mathrm{char}(k) = 3$ does not divide 2, theorem 3.1.9 makes $k[\mathbb{Z}/2\mathbb{Z}]$ semisimple, and $t^2 - 1 = (t-1)(t+1)$ has distinct roots, so $k[\mathbb{Z}/2\mathbb{Z}] \cong k \times k$. Hence $k[S_3]$ has exactly two simple modules, both of dimension 1: the trivial module $\mathbb{1}$ and the sign module sgn. In particular both are absolutely simple, so k is a splitting field and the central characters above are defined.

To locate them we compute central characters. Both simple modules are one dimensional, so a group element g acts on S by the scalar $\chi_S(g)$ and a class sum acts by the sum of those scalars over the class. The three-cycles are even permutations, so

$$\omega_{\mathbb{1}}(\sigma_2) = 3 \cdot 1 = 0, \qquad \omega_{\mathrm{sgn}}(\sigma_2) = 3 \cdot (-1) = 0, \qquad \omega_{\mathbb{1}}(\sigma_3) = \omega_{\mathrm{sgn}}(\sigma_3) = 2 \cdot 1 = 2$$

all computed in k, where $3 = 0$, and $\omega(\sigma_1) = 1$ for both. The two central characters agree on every class sum, so $\mathbb{1}$ and sgn lie in the same block. Every block contains a simple module by theorem 3.5.2, so there is no room for a second: $k[S_3]$ has exactly one block, itself.

This is the phenomenon the section is about. In characteristic 0 the algebra $\mathbb{C}[S_3]$ splits into three Artin-Wedderburn factors, one per irreducible character of example 3.2.8, and each factor is a matrix algebra that can be understood on its own. In characteristic 3 the algebra does not split at all, and the two simple modules are bound together in a single block, so the algebra is indecomposable as an algebra.

Having fixed the coarse decomposition, we turn to the numerical invariants. The first was already defined in general in definition 2.3.8, and the group algebra is its intended application: the Cartan matrix of $k[G]$ has entries $C_{ij} = [P_i : S_j]$, the multiplicity of the simple module S_j among the

composition factors of the projective cover of S_i. In characteristic 0 this matrix is the identity, because $P_i = S_i$ for a semisimple algebra, and it carries no information. In characteristic p it records how far the projectives are from the simples.

Computing it from the definition means knowing the projective modules, which is exactly what one does not have at the outset. The second invariant is what makes it accessible, and it is the one Brauer's program is built on. It compares characteristic p with characteristic 0 directly. Doing this properly requires a p-modular system: a complete discrete valuation ring R with residue field k of characteristic p and fraction field F of characteristic 0, chosen large enough that both $F[G]$ and $k[G]$ split. A finitely generated $F[G]$-module then contains an R-lattice stable under G, and reducing that lattice modulo the maximal ideal produces a $k[G]$-module. The composition factors of the reduction do not depend on which lattice was chosen, a theorem of Brauer and Nesbitt, and this independence is what makes the following definition well posed.

Definition 3.5.4: Decomposition Matrix

Let (F, R, k) be a splitting p-modular system for G, let $T_1, \dots, T_s$ be the simple $F[G]$-modules and $S_1, \dots, S_r$ the simple $k[G]$-modules. The *decomposition matrix* of G at p is the $s \times r$ matrix $D = (d_{ij})$ of nonnegative integers where d_{ij} is the multiplicity of S_j as a composition factor of the reduction, modulo the maximal ideal of R, of a full G-stable R-lattice in T_i. The lattice must be full, meaning that it spans T_i over F, or the reduction has the wrong dimension. The reduction is taken modulo the maximal ideal of R and not modulo p, because R is typically ramified over $\mathbb{Z}_p$, in which case the two differ.

The rows of D are indexed by the ordinary irreducible characters, which section 3.2 computes, and the columns by the modular simple modules, which are the unknowns. Each row records how one ordinary irreducible breaks up when reduced. This is Brauer's idea in one sentence: the characteristic-0 theory is completely understood, so the characteristic-p theory should be reachable by reduction, and D is the matrix that records the reduction. What makes the idea pay is that D determines the Cartan matrix of definition 2.3.8 outright.

Theorem 3.5.5: Cartan Matrix, Decomposition Matrix, and p-Regular Classes

Let (F, R, k) be a splitting p-modular system for G, with decomposition matrix D and Cartan matrix C. Then

$$C = D^{\mathsf{T}} D$$

In particular C is symmetric. Moreover the number of simple $k[G]$-modules equals the number of p-regular conjugacy classes of G, where a conjugacy class is p-regular when the order of its elements is prime to p.

We state these without proof. The first is a statement about the three Grothendieck groups attached to (F, R, k) and the maps c, d and e between them, the configuration known as the *cde triangle*: the relation $C = D^{\mathsf{T}} D$ is the matrix form of the factorization $c = de$ together with the fact that e is the transpose of d over a splitting system. The second is a separate theorem of Brauer, proved without the triangle. Developing that machinery needs the theory of lattices over a complete discrete valuation ring and the Brauer characters built from it, which is where [7] begins. Complete treatments are in Serre's *Linear Representations of Finite Groups* [16, Part III] and Webb's *A Course in Finite Group Representation Theory* [17, Chs. 9 and 10].

The counting statement is already enough to check the earlier computation. The 3-regular classes of S_3 are those whose elements have order prime to 3, namely $\{e\}$ and the class of transpositions, so there are two of them, matching the two simple modules found in example 3.5.3. It also puts the characteristic-0 count of proposition 3.1.13 in context: over $\mathbb{C}$ the number of simple modules is the number of *all* conjugacy classes, and passing to characteristic p discards exactly the classes of order divisible by p.

Example 3.5.6: The Decomposition and Cartan Matrices of S_3 at $p = 3$

The ordinary irreducible characters of S_3 are the trivial character $\chi_{\mathbb{1}}$, the sign character χ_{sgn}, and a character $\chi^{\perp}$ of degree 2. The first two are read off example 3.2.8, and $\chi^{\perp}$ is constructed in example 3.2.19 as the character of the complement $V^{\perp}$ of the trivial summand in the permutation representation, with $\chi^{\perp}(e) = 2$, $\chi^{\perp}((1\ 2)) = 0$ and $\chi^{\perp}((1\ 2\ 3)) = -1$. The two simple $k[S_3]$-modules are $S_1 = \mathbb{1}$ and $S_2 = \mathrm{sgn}$, both of dimension 1, as computed in example 3.5.3. Take the p-modular system $(F, R, k) = (\mathbb{Q}_3, \mathbb{Z}_3, \mathbb{F}_3)$, which is splitting for S_3: all three ordinary irreducibles are realizable over $\mathbb{Q}$, the degree-2 one by the explicit rational basis of example 3.2.19, and they stay irreducible after extending scalars to $\mathbb{C}$, so they are absolutely irreducible. Both modular simples are one dimensional, hence absolutely simple, so theorem 3.5.5 applies.

Reducing modulo 3, the trivial and sign representations are already defined over k and remain simple, so they contribute the rows $(1, 0)$ and $(0, 1)$. The representation $V^{\perp}$ reduces to a module M of dimension 2. Both simple modules are one dimensional, so M has exactly two composition factors, each S_1 or S_2. Taking determinants of the matrices of the group action identifies which. A permutation matrix has determinant the sign of the permutation, so the degree-three permutation representation has determinant sgn, and its trivial summand contributes 1, leaving $\det V^{\perp} = \mathrm{sgn}$. The determinant is a polynomial with integer coefficients in the matrix entries, so it commutes with reducing those entries modulo the maximal ideal, giving $\det M = \mathrm{sgn}$ as a character valued in k. In a basis adapted to a composition series the matrices are block triangular, so the determinant of a module is the product of the determinants of its composition factors, and the two factors of M have determinants multiplying to sgn. Two copies of S_1 would give $\mathbb{1}$ and two copies of S_2 would give $\mathrm{sgn}^2 = \mathbb{1}$, and $\mathrm{sgn} \neq \mathbb{1}$ over k because $\mathrm{char}(k) = 3 \neq 2$. So the factors are one S_1 and one S_2, giving the row $(1, 1)$. Hence

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \qquad \text{and} \qquad D^{\mathsf{T}} D = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

so by theorem 3.5.5 the Cartan matrix of $k[S_3]$ is

$$C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Reading the entries back: $C_{11} = [P_1 : S_1] = 2$ and $C_{12} = [P_1 : S_2] = 1$, so the projective cover of the trivial module has three composition factors and dimension 3. That is a check on the whole computation, since theorem 2.3.5 gives $k[S_3] = P_1 \oplus P_2$ with each P_i appearing $\dim_k S_i = 1$ time, and $3 + 3 = 6 = \dim_k k[S_3]$.

Two features of C are general. It is symmetric, which holds for any group algebra over a splitting field and fails for finite-dimensional algebras at large. And $\det C = 4 - 1 = 3$ is a power of p, which is also always true. Here it happens to equal the order of a Sylow 3-subgroup of S_3.

Note what the computation used: the ordinary character table, which is classical, and one piece of genuinely modular information, the way $\chi^{\perp}$ breaks up. That is the general shape of the problem.

Brauer's program provides information of exactly this kind, and it does not solve the problem. The relation $C = D^{\mathsf{T}}D$ determines C from D, and the counting theorem fixes the number of columns of D, but nothing here computes the entries of D itself. For the symmetric groups those entries are unknown in general.

The final obstruction is of a different kind, and it explains why the subject computes invariants and does not attempt a classification of modules.

Theorem 3.5.7: Higman's Theorem on Finite Representation Type

Let p be a prime, let k be a field of characteristic p, and let G be a finite group. Then $k[G]$ has only finitely many isomorphism classes of indecomposable modules if and only if the Sylow p-subgroups of G are cyclic.

The proof reduces to the case of a p-group and then to the two extremes, and it is given in [17, Theorem 11.4.3]. The negative half is the striking one. As soon as a Sylow p-subgroup fails to be cyclic it has $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ as a quotient, and $k[\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}]$ already has infinitely many indecomposable modules, of unbounded dimension. For such a group the phrase "classify the modules" has no reasonable answer, and the invariants of this section are what one computes instead. Contrast this with chapter 2, where Krull-Schmidt guaranteed a unique decomposition into indecomposables: the guarantee is real, and it does not imply that the list of indecomposables is finite or knowable.

What survives is a body of conjectures relating the invariants of a block to the structure of an associated p-subgroup, its *defect group*. Two of them are among the central open problems of the subject.

Conjecture 3.5.8: Alperin's Weight Conjecture

Let k be an algebraically closed field of characteristic p and G a finite group. A *weight* is a pair (Q, S) with Q a p-subgroup of G and S a simple projective $k[N_G(Q)/Q]$-module. Then the number of simple modules in a p-block of $k[G]$ equals the number of G-conjugacy classes of weights belonging to that block.

Conjecture 3.5.9: Broué's Abelian Defect Group Conjecture

Let k be an algebraically closed field of characteristic p, and let B be a block of $k[G]$ whose defect group P is abelian. Let B' be the Brauer correspondent of B, a block of $k[N_G(P)]$. Then the bounded derived categories of B and B' are equivalent as triangulated categories.

Neither statement is self-contained here: the *defect group* of a block and the *Brauer correspondent* are the central constructions of the local theory, and defining them is where [7] does its work. Both conjectures remain open as of writing, and both have been verified in many cases. Broué's conjecture is known for blocks with cyclic defect groups, and Alperin's has been reduced to a statement about finite simple groups. The subject is not static: Brauer's height zero conjecture, open since 1955, was proved in 2024, and the McKay conjecture has also been settled. Alperin's book *Local Representation Theory* [1] is the standard introduction to the local theory these conjectures live in, and Curtis and Reiner [9] remains the comprehensive reference for the subject as a whole.

A last connection. *Algebraic Groups and Representations* [8] also meets a failure of complete reducibility in characteristic p, for reductive algebraic groups, where the simple objects again carry

known labels and unknown characters. One warning about the vocabulary. That book has its own *decomposition matrix*, which records how a Weyl module $V(\lambda)$ breaks into simple modules in a single characteristic. It is a different matrix from the D of definition 3.5.4, which compares characteristic 0 with characteristic p for a finite group. The two subjects are relatives, and each owns its own side of the difficulty.

Exercise 3.5.1

1. Let P be a finite p-group and let $\mathrm{char}(k) = p$. Show that $k[P]$ has exactly one simple module, the trivial one, and exactly one block. Deduce that the Cartan matrix of $k[P]$ is the 1×1 matrix $(|P|)$.

2. Show that if $\mathrm{char}(k) \nmid |G|$ then the Cartan matrix of $k[G]$ is the identity matrix, and that the blocks of $k[G]$ are exactly the Artin-Wedderburn factors of theorem 1.2.16. Which step fails when $\mathrm{char}(k)$ divides $|G|$?

3. Repeat example 3.5.6 for S_3 at $p = 2$. Show that there are two simple $k[S_3]$-modules, of dimensions 1 and 2, compute the decomposition matrix, and deduce the Cartan matrix. Check that $\det C$ is again the order of a Sylow 2-subgroup.

4. Using theorem 3.5.7, determine for which primes p the group algebra $k[A_5]$ has finite representation type, where $\mathrm{char}(k) = p$ and $|A_5| = 60$.

4

Quiver Representations and Path Algebras

A quiver is a directed graph, and a representation of a quiver assigns a vector space to each vertex and a linear map to each arrow. A single vertex carrying a single loop gives a vector space with one endomorphism, and two vertices joined by two parallel arrows give a pair of linear maps between the same two spaces. Every finite-dimensional representation is a finite direct sum of indecomposable ones, so the indecomposable representations are the objects to classify. For the loop they include a nilpotent Jordan block of every size, and their list is infinite over every field. Chapter 3 ended with the same phenomenon for group algebras: in characteristic p, the algebra $k[G]$ has only finitely many indecomposable modules exactly when the Sylow p-subgroups of G are cyclic (theorem 3.5.7).

Gabriel proved in 1972 that a connected quiver has only finitely many indecomposable representations exactly when the graph obtained by forgetting the orientation of its arrows is one of the Dynkin diagrams A_n, D_n, E_6, E_7, E_8. The dimension vectors of the indecomposable representations are then exactly the positive roots of an integral quadratic form attached to the graph, its Tits form. At a dimension vector, the Tits form is the dimension of the group of base changes acting on the representations with that dimension vector, minus the dimension of the space of these representations, and over an infinite field a quiver with finitely many indecomposables has a Tits form that is positive at every nonzero dimension vector. The proof given here that a positive definite Tits form gives finitely many indecomposables is due to Bernstein, Gelfand and Ponomarev (1973). It uses reflection functors, which carry the indecomposable representations of one orientation of a graph, other than one simple representation, to those of another, and act on dimension vectors by the reflections that permute the roots.[1] Quivers also describe algebras: the path algebra kQ has the representations of Q as its modules, and by a second theorem of Gabriel every finite-dimensional algebra over an

[1] The diagrams A_n, D_n, E_6, E_7, E_8, together with B_n, C_n, F_4 and G_2, also classify the finite-dimensional simple Lie algebras over $\mathbb{C}$ [10, §§11.4, 14.2, 18.4]. The diagrams of type A, D, E are those whose roots all have one length, the only case that quivers produce (box 4.1.44).

algebraically closed field is Morita equivalent to a quotient of a path algebra by an admissible ideal (theorem 4.2.10).

4.1 Quivers and Their Decomposition

Definition 4.1.1: Quiver

A *quiver* is a quadruple $Q = (I, E, s, t)$ consisting of a set I of *vertices*, a set E of *arrows* (also called *edges*) and two maps $s, t \colon E \to I$. An arrow e with $s(e) = i$ and $t(e) = j$ is drawn as

$$\overset{i}{\bullet} \xrightarrow{e} \overset{j}{\bullet}$$

and written $e \colon i \to j$, and the vertices $s(e)$ and $t(e)$ are its *source* and its *target*. The quiver Q is *finite* if I and E are finite sets.

A quiver is usually specified by a drawing, which leaves the maps s and t implicit. An arrow may have the same source and target, and several arrows may join the same two vertices.

Example 4.1.2: Quivers

1. The quiver with $I = \{1\}$, $E = \{e\}$ and $s(e) = t(e) = 1$ is drawn as

$$\circlearrowleft 1$$

2. The quiver with $I = \{1, 2\}$, $E = \{e\}$, $s(e) = 1$ and $t(e) = 2$ is drawn as

$$1 \xrightarrow{e} 2$$

3. The quiver with $I = \{1, 2\}$, $E = \{e, f\}$, $s(e) = t(f) = 1$ and $t(e) = s(f) = 2$ is drawn as

$$1 \underset{f}{\overset{e}{\rightleftarrows}} 2$$

Definition 4.1.3: Path and Oriented Cycle

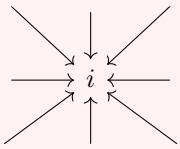
Let $Q = (I, E, s, t)$ be a quiver.

1. A *path* of *length* $k \geq 1$ in Q is a sequence of k arrows, written from right to left as
$$\gamma = a_k a_{k-1} \cdots a_1$$
with $t(a_l) = s(a_{l+1})$ for $1 \leq l \leq k-1$, so that a_1 is traversed first. Its *source* is $s(\gamma) = s(a_1)$, its *target* is $t(\gamma) = t(a_k)$, and γ is a path *from* $s(\gamma)$ *to* $t(\gamma)$.
2. The *trivial path* at a vertex i is the path e_i of length 0, which contains no arrows and has $s(e_i) = t(e_i) = i$.
3. An *oriented cycle* is a path γ of positive length with $s(\gamma) = t(\gamma)$. An oriented cycle of length 1, that is, an arrow a with $s(a) = t(a)$, is a *loop*.

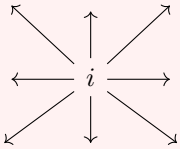
Definition 4.1.4: Sink And Source

Let $Q = (I, E, s, t)$ be a quiver, $e \in E$ and $i \in I$.

1. The arrow e is *incoming at* i if $t(e) = i$, *outgoing at* i if $s(e) = i$, and an *arrow at* i if it is incoming or outgoing at i.
2. The vertex i is a *sink* if no arrow is outgoing at i, as in



3. The vertex i is a *source* if no arrow is incoming at i, as in



4. If i is a sink or a source, then iQ denotes the quiver obtained from Q by reversing every arrow at i, that is, by exchanging the source and the target of each arrow at i.

4.1.1 Quiver Representation and Varieties

Definition 4.1.5: Quiver Representation

Let $Q = (I, E, s, t)$ be a quiver and $\mathbb{F}$ a field. A *representation* V *of* Q over $\mathbb{F}$ consists of

1. an $\mathbb{F}$-vector space $V(i)$ for each vertex $i \in I$, and
2. an $\mathbb{F}$-linear map $T_e \colon V(i) \to V(j)$ for each arrow $e \colon i \to j$.

The collection of all representations of Q over $\mathbb{F}$ is denoted $\mathrm{Rep}_{\mathbb{F}}(Q)$, or $\mathrm{Rep}(Q)$.

Example 4.1.6: Quiver Representations

1. A representation of the quiver
$$1 \xrightarrow{e} 2$$
is a pair of vector spaces $V(1)$, $V(2)$ with a linear map $T_e \colon V(1) \to V(2)$, so every linear map between two vector spaces is a representation of this quiver.

2. A representation of the quiver
$$\overset{f}{\circlearrowright}\ 1 \xrightarrow{e} 2$$
is a linear map $T_e \colon V(1) \to V(2)$ together with a linear map $T_f \colon V(2) \to V(2)$, which may be the identity or any other endomorphism of $V(2)$.

3. A representation of the quiver
$$1 \xrightarrow{e} 2 \xrightarrow{f} 3$$
is a pair of linear maps $V(1) \xrightarrow{T_e} V(2) \xrightarrow{T_f} V(3)$. Among them are the pairs exact at $V(2)$, those with $\mathrm{im}(T_e) = \ker(T_f)$. If $V(1) = V(2) = V(3)$, the two maps may be scalar multiplications. The representation $\mathbb{F} \to \mathbb{F}^2 \to \mathbb{F}$ with $T_e(x) = (x, 0)$ and $T_f(x, y) = x$ has $T_f \circ T_e = \mathrm{id}_{\mathbb{F}}$, although neither map is invertible.

Definition 4.1.7: Quiver Dimension

Let $Q = (I, E, s, t)$ be a quiver and V a representation of Q. The *dimension vector*, or *quiver dimension*, of V is the tuple
$$\dim_Q V = (\dim V(i))_{i \in I}$$
When every $V(i)$ is finite-dimensional and $I = \{1, \dots, n\}$, it is the vector $\dim_Q V = (\dim V(1), \dots, \dim V(n)) \in \mathbb{N}^n$.

Definition 4.1.8: Quiver Subrepresentation

Let $Q = (I, E, s, t)$ be a quiver and V a representation of Q. A *subrepresentation* W of V, written $W \subseteq V$, consists of a subspace $W(i) \subseteq V(i)$ for each vertex $i \in I$ such that $T_e(W(i)) \subseteq W(j)$ for every arrow $e\colon i \to j$. With the restricted maps $S_e = T_e|_{W(i)}\colon W(i) \to W(j)$ it is a representation of Q.

Definition 4.1.9: Quiver Morphisms

Let $Q = (I, E, s, t)$ be a quiver and let V and W be representations of Q, with maps T_e and S_e respectively. A *quiver morphism* $\varphi\colon V \to W$ is a family $\varphi = (\varphi(i))_{i\in I}$ of linear maps $\varphi(i)\colon V(i) \to W(i)$ such that for every arrow $e\colon i \to j$ the square

$$\begin{array}{ccc} V(i) & \xrightarrow{T_e} & V(j) \\ \downarrow{\scriptstyle \varphi(i)} & & \downarrow{\scriptstyle \varphi(j)} \\ W(i) & \xrightarrow{S_e} & W(j) \end{array}$$

commutes, that is, $S_e \circ \varphi(i) = \varphi(j) \circ T_e$. The set of quiver morphisms from V to W is denoted $\mathrm{Hom}_Q(V, W)$, or $\mathrm{Hom}(V, W)$ when Q is clear from the context. The composite of quiver morphisms $\varphi\colon V \to W$ and $\psi\colon W \to U$ is the family $\psi \circ \varphi = (\psi(i) \circ \varphi(i))_{i\in I}$, and the identity of V is the family $\mathrm{id}_V = (\mathrm{id}_{V(i)})_{i\in I}$.

Proposition 4.1.10: Structure Of $\mathrm{Hom}(V, W)$

Let $V, W \in \mathrm{Rep}(Q)$. Then $\mathrm{Hom}_Q(V, W)$ is a vector space over $\mathbb{F}$, with addition and scalar multiplication defined vertex by vertex.

***Proof*:**

A quiver morphism $\varphi\colon V \to W$ is the data of a linear map $\varphi(i)\colon V(i) \to W(i)$ for each vertex i, subject to the condition $S_e \circ \varphi(i) = \varphi(j) \circ T_e$ for every edge e from i to j. The collection of all vertex-indexed families of linear maps, $\prod_i \mathrm{Hom}_{\mathbb{F}}(V(i), W(i))$, is a vector space under vertexwise addition and scalar multiplication, so it suffices to show that $\mathrm{Hom}(V, W)$ is a subspace.

The zero family satisfies $S_e \circ 0 = 0 = 0 \circ T_e$, so it is a quiver morphism. If $\varphi, \psi \in \mathrm{Hom}(V, W)$ and $c \in \mathbb{F}$, then for every edge e from i to j, linearity of S_e gives

$$S_e \circ (\varphi(i) + c\psi(i)) = S_e \circ \varphi(i) + c\,(S_e \circ \psi(i)) = \varphi(j) \circ T_e + c\,(\psi(j) \circ T_e) = (\varphi(j) + c\psi(j)) \circ T_e$$

so the family $(\varphi(i) + c\psi(i))_i$ is again a quiver morphism. Hence $\mathrm{Hom}(V, W)$ contains 0 and is closed under addition and scalar multiplication, making it a subspace of $\prod_i \mathrm{Hom}_{\mathbb{F}}(V(i), W(i))$ and in particular a vector space over $\mathbb{F}$.

Example 4.1.11: Quiver Morphism

1. Let $V = \overset{\mathbb{F}}{\bullet} \xrightarrow{1} \overset{\mathbb{F}}{\bullet}$ and $W = \overset{\mathbb{F}}{\bullet} \xrightarrow{0} \overset{0}{\bullet}$ be representations of $1 \xrightarrow{e} 2$. The maps $\varphi(1) = 1$ and $\varphi(2) = 0$ form a quiver morphism $\varphi\colon V \to W$,

$$\begin{array}{ccc} V & & \mathbb{F} & \xrightarrow{1} & \mathbb{F} \\ \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle 0} \\ W & & \mathbb{F} & \xrightarrow{0} & 0 \end{array}$$

   since both composites $\mathbb{F} \to 0$ are zero. Every quiver morphism $\psi\colon W \to V$ is zero: its square at e reads $1 \circ \psi(1) = \psi(2) \circ 0 = 0$, so $\psi(1) = 0$, and $\psi(2)$ is defined on the zero space.

2. Let $V = \overset{\mathbb{R}}{\bullet} \xrightarrow{1} \overset{\mathbb{R}}{\bullet} \xrightarrow{1} \overset{\mathbb{R}}{\bullet}$ and $W = \overset{\mathbb{R}^2}{\bullet} \xrightarrow{A} \overset{\mathbb{R}^2}{\bullet} \xrightarrow{B} \overset{\mathbb{R}}{\bullet}$ be representations of $1 \to 2 \to 3$, with

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 1 \end{pmatrix}$$

   The matrices $\varphi(1) = \varphi(2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\varphi(3) = 2$ form a quiver morphism $\varphi\colon V \to W$, because $A\varphi(1) = \varphi(2) \cdot 1$ and $B\varphi(2) = 2 = \varphi(3) \cdot 1$. The matrices $\psi(1) = \psi(2) = \begin{pmatrix} 1 & 1 \end{pmatrix}$ and $\psi(3) = 1$ form a nonzero quiver morphism $\psi\colon W \to V$, because $1 \cdot \psi(1) = \psi(2)A$ and $1 \cdot \psi(2) = \psi(3)B$.

With the composites and identities of definition 4.1.9, the representations of Q form a category $\mathrm{Rep}(Q)$. For quiver morphisms $\varphi\colon V \to W$ and $\psi\colon W \to U$, where U has maps R_e, and an arrow $e\colon i \to j$, the squares of φ and ψ give

$$R_e \circ \psi(i) \circ \varphi(i) = \psi(j) \circ S_e \circ \varphi(i) = \psi(j) \circ \varphi(j) \circ T_e$$

so $\psi \circ \varphi$ is a quiver morphism. The family id_V is a quiver morphism because $T_e \circ \mathrm{id}_{V(i)} = \mathrm{id}_{V(j)} \circ T_e$, and composition is associative with the identities id_V as units because these laws hold for linear maps at each vertex. A quiver morphism is a morphism between representations of one quiver Q. A map from a quiver $Q = (I, E, s, t)$ to a quiver $Q' = (I', E', s', t')$ is a different notion, a pair of maps $m_I\colon I \to I'$ and $m_E\colon E \to E'$ with $m_I \circ s = s' \circ m_E$ and $m_I \circ t = t' \circ m_E$, and maps of quivers are not used in this chapter.

The *zero representation* 0 has the zero space at every vertex. Kernels and images of quiver morphisms are formed vertex by vertex.

Definition 4.1.12: Kernels And Images

Let $\varphi\colon V \to W$ be a quiver morphism. The *kernel* $\ker(\varphi)$ is the subrepresentation of V with $\ker(\varphi)(i) = \ker(\varphi(i))$, and the *image* $\mathrm{im}(\varphi)$ is the subrepresentation of W with $\mathrm{im}(\varphi)(i) = \mathrm{im}(\varphi(i))$, for every vertex i.

Both are subrepresentations in the sense of definition 4.1.8. Let $e\colon i \to j$ be an arrow. If $v \in \ker(\varphi(i))$, then $\varphi(j)(T_e v) = S_e(\varphi(i)v) = 0$, so T_e carries $\ker(\varphi(i))$ into $\ker(\varphi(j))$. For every $v \in V(i)$ the same square gives $S_e(\varphi(i)v) = \varphi(j)(T_e v)$, so S_e carries $\mathrm{im}(\varphi(i))$ into $\mathrm{im}(\varphi(j))$.

Definition 4.1.13: Quiver Isomorphisms

Let $V, W \in \mathrm{Rep}(Q)$. A quiver morphism $\varphi\colon V \to W$ is an *isomorphism* if there is a quiver morphism $\psi\colon W \to V$ with $\psi \circ \varphi = \mathrm{id}_V$ and $\varphi \circ \psi = \mathrm{id}_W$. The representations V and W are *isomorphic*, written $V \cong W$, if there is an isomorphism $V \to W$.

The components $\varphi(i)$ of an isomorphism are bijective, with inverses $\psi(i)$. Conversely, a quiver morphism φ whose components are all bijective is an isomorphism: composing $S_e \circ \varphi(i) = \varphi(j) \circ T_e$ with $\varphi(j)^{-1}$ on the left and with $\varphi(i)^{-1}$ on the right gives $\varphi(j)^{-1} \circ S_e = T_e \circ \varphi(i)^{-1}$, so the family $(\varphi(i)^{-1})_{i \in I}$ is a quiver morphism inverse to φ. In particular isomorphic representations have the same dimension vector.

Example 4.1.14: Quiver Isomorphism

Let Q be $1 \xrightarrow{e} 2$. Finite-dimensional representations V and W of Q, with maps T_e and S_e, are isomorphic if and only if $\dim V(1) = \dim W(1)$, $\dim V(2) = \dim W(2)$ and $\mathrm{rank}(T_e) = \mathrm{rank}(S_e)$.

If $\varphi\colon V \to W$ is an isomorphism, then $\varphi(1)$ and $\varphi(2)$ are linear isomorphisms, which gives the equalities of dimensions, and $S_e = \varphi(2) \circ T_e \circ \varphi(1)^{-1}$ has the same rank as T_e. Conversely, let $n_1 = \dim V(1)$, $n_2 = \dim V(2)$ and $r = \mathrm{rank}(T_e)$. By rank-nullity $\ker(T_e)$ has dimension $n_1 - r$, so $V(1)$ has a basis $u_1, \dots, u_{n_1}$ whose last $n_1 - r$ vectors form a basis of $\ker(T_e)$. The vectors $T_e u_1, \dots, T_e u_r$ span $\mathrm{im}(T_e)$, which has dimension r, so they are linearly independent, and they extend to a basis $T_e u_1, \dots, T_e u_r, w_{r+1}, \dots, w_{n_2}$ of $V(2)$. In these bases T_e has the matrix

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

of size $n_2 \times n_1$, where I_r is the $r \times r$ identity matrix. The same construction for W gives bases in which S_e has the same matrix. The linear isomorphisms $\varphi(1)\colon V(1) \to W(1)$ and $\varphi(2)\colon V(2) \to W(2)$ that send each chosen basis vector for V to the chosen basis vector for W in the same position satisfy $S_e \circ \varphi(1) = \varphi(2) \circ T_e$, since in the chosen bases both sides have the matrix above. So φ is a quiver morphism with bijective components, hence an isomorphism. The isomorphism classes of finite-dimensional representations of Q are therefore in bijection with the triples (n_1, n_2, r) of nonnegative integers with $r \le \min(n_1, n_2)$, the triple being realized by the matrix above.

A subrepresentation has a quotient, formed vertex by vertex.

Definition 4.1.15: Quotient Representation

Let $V \in \mathrm{Rep}(Q)$ and let $W \subseteq V$ be a subrepresentation. The *quotient representation* V/W has the space $V(i)/W(i)$ at each vertex i, and along each arrow $e\colon i \to j$ the map $V(i)/W(i) \to V(j)/W(j)$, $x + W(i) \mapsto T_e(x) + W(j)$. This map is well defined because T_e carries $W(i)$ into $W(j)$, W being a subrepresentation.

Proposition 4.1.16: 1st Isomorphism for Quivers

Let $V, W \in \mathrm{Rep}(Q)$ and $\varphi\colon V \to W$ be a quiver morphism. Then:

$$\mathrm{im}(\varphi) \cong V/\ker(\varphi)$$

***Proof*:**
Write $K = \ker(\varphi)$, a subrepresentation of V (definition 4.1.12). First, the quotient V/K of definition 4.1.15 carries the edge maps induced by those of V: for an edge e from i to j, set $\overline{T}_e(v + K(i)) = T_e v + K(j)$. This is well defined, since $v \in K(i) = \ker(\varphi(i))$ gives $\varphi(j)(T_e v) = S_e(\varphi(i)v) = 0$, so $T_e v \in K(j)$. Second, S_e maps $\mathrm{im}(\varphi(i))$ into $\mathrm{im}(\varphi(j))$, since $S_e(\varphi(i)v) = \varphi(j)(T_e v)$, so the edge maps of the subrepresentation $\mathrm{im}(\varphi) \subseteq W$ are the restrictions of the S_e.

For each vertex i, define

$$\overline{\varphi}(i)\colon V(i)/K(i) \to \mathrm{im}(\varphi(i)) \qquad v + K(i) \mapsto \varphi(i)v$$

This is well defined and injective because $\varphi(i)v = \varphi(i)v'$ holds exactly when $v - v' \in \ker(\varphi(i)) = K(i)$, it is linear, and it is surjective onto $\mathrm{im}(\varphi(i))$ by construction. So each $\overline{\varphi}(i)$ is a linear isomorphism.

The family $\overline{\varphi} = (\overline{\varphi}(i))_i$ is a quiver morphism $V/K \to \mathrm{im}(\varphi)$: for every edge e from i to j and every $v \in V(i)$,

$$\overline{\varphi}(j)\left(\overline{T}_e(v + K(i))\right) = \varphi(j)(T_e v) = S_e(\varphi(i)v) = S_e\left(\overline{\varphi}(i)(v + K(i))\right)$$

where the middle equality is the commuting square for φ. Composing this identity with $\overline{\varphi}(j)^{-1}$ on the left and with $\overline{\varphi}(i)^{-1}$ on the right yields $\overline{T}_e \circ \overline{\varphi}(i)^{-1} = \overline{\varphi}(j)^{-1} \circ S_e$, so the vertexwise inverses also form a quiver morphism $\mathrm{im}(\varphi) \to V/K$. The two morphisms are mutually inverse at every vertex, so $\mathrm{im}(\varphi) \cong V/\ker(\varphi)$ (definition 4.1.13).

Definition 4.1.17: Direct Sum Of Representations

Let $V, W \in \mathrm{Rep}(Q)$, with maps T_e and S_e. The *direct sum* $V \oplus W \in \mathrm{Rep}(Q)$ has

1. the space $(V \oplus W)(i) = V(i) \oplus W(i)$ at each vertex i, and
2. the map $T_e \oplus S_e\colon V(i) \oplus W(i) \to V(j) \oplus W(j)$, $(v, w) \mapsto (T_e v, S_e w)$, along each arrow $e\colon i \to j$.

If $U_1, \ldots, U_n$ are subrepresentations of V with $V(i) = U_1(i) \oplus \cdots \oplus U_n(i)$ at every vertex i, then V is the *internal direct sum* of $U_1, \ldots, U_n$, written $V = U_1 \oplus \cdots \oplus U_n$.

An internal direct sum is isomorphic to the direct sum of its summands. For $V = U_1 \oplus U_2$, the maps $U_1(i) \oplus U_2(i) \to V(i)$, $(u_1, u_2) \mapsto u_1 + u_2$, are bijective, and they form a quiver morphism $U_1 \oplus U_2 \to V$ because $T_e(u_1 + u_2) = T_e u_1 + T_e u_2$ with $T_e u_1 \in U_1(j)$ and $T_e u_2 \in U_2(j)$. A quiver morphism with bijective components is an isomorphism, as noted after definition 4.1.13. For $n \geq 3$ summands, the vertexwise sums $U'(i) = U_1(i) \oplus \cdots \oplus U_{n-1}(i)$ form a subrepresentation U' with $V = U' \oplus U_n$, so induction on n gives $V \cong U_1 \oplus \cdots \oplus U_n$.

Definition 4.1.18: Representation Space

Let $Q = (I, E, s, t)$ be a finite quiver with $I = \{1, \dots, n\}$, and let $\underline{d} = (d_1, \dots, d_n) \in \mathbb{N}^n$. The *representation space* of dimension vector $\underline{d}$ is

$$\mathrm{Rep}(Q, \underline{d}) := \prod_{e \in E} M_{d_{t(e)} \times d_{s(e)}}(\mathbb{F})$$

A point $A = (A_e)_{e \in E}$ of $\mathrm{Rep}(Q, \underline{d})$ is the representation of Q with $\mathbb{F}^{d_i}$ at the vertex i and the matrix A_e along the arrow e, so every point is a representation with dimension vector $\underline{d}$. As a vector space, $\mathrm{Rep}(Q, \underline{d})$ has dimension $\sum_{e \in E} d_{s(e)} d_{t(e)}$.

Choosing a basis of every vertex space turns a representation with dimension vector $\underline{d}$ into a point of $\mathrm{Rep}(Q, \underline{d})$, and different choices of bases change the point by the following action.

Proposition 4.1.19: Action On The Representation Space

Let $Q = (I, E, s, t)$ be a finite quiver with $I = \{1, \dots, n\}$, let $\underline{d} \in \mathbb{N}^n$, and let $G(\underline{d}) = \prod_{i \in I} \mathrm{GL}_{d_i}(\mathbb{F})$. For $g = (g_1, \dots, g_n) \in G(\underline{d})$ and $A \in \mathrm{Rep}(Q, \underline{d})$ set

$$(g \cdot A)_e = g_{t(e)} A_e g_{s(e)}^{-1} \qquad e \in E$$

This defines an action of $G(\underline{d})$ on $\mathrm{Rep}(Q, \underline{d})$.

***Proof*:**

Each $g_{t(e)} A_e g_{s(e)}^{-1}$ is a $d_{t(e)} \times d_{s(e)}$ matrix, so $g \cdot A \in \mathrm{Rep}(Q, \underline{d})$, and the identity element $(I_{d_1}, \dots, I_{d_n})$ fixes every A. For $g, h \in G(\underline{d})$, whose product is computed vertex by vertex, and every arrow e,

$$(g \cdot (h \cdot A))_e = g_{t(e)} h_{t(e)} A_e h_{s(e)}^{-1} g_{s(e)}^{-1} = (gh)_{t(e)} A_e (gh)_{s(e)}^{-1} = ((gh) \cdot A)_e$$

because $(g_{s(e)} h_{s(e)})^{-1} = h_{s(e)}^{-1} g_{s(e)}^{-1}$. Hence $g \cdot (h \cdot A) = (gh) \cdot A$, and the formula is an action, as claimed.

Theorem 4.1.20: Isomorphism Equivalence

Let Q be a finite quiver with vertex set $I = \{1, \dots, n\}$, let $\underline{d} \in \mathbb{N}^n$, and let $G(\underline{d})$ act on $\mathrm{Rep}(Q, \underline{d})$ as in proposition 4.1.19. Two points $A, B \in \mathrm{Rep}(Q, \underline{d})$ are isomorphic representations if and only if $B = g \cdot A$ for some $g \in G(\underline{d})$. Consequently, sending the orbit of A to the isomorphism class of A is a bijection from the set of $G(\underline{d})$-orbits in $\mathrm{Rep}(Q, \underline{d})$ onto the set of isomorphism classes of representations of Q with dimension vector $\underline{d}$.

***Proof*:**

A morphism φ from the representation A to the representation B is a family of matrices $\varphi(i) \in M_{d_i}(\mathbb{F})$ with $\varphi(t(e)) A_e = B_e \varphi(s(e))$ for every arrow e (definition 4.1.9). If φ is an isomorphism with inverse ψ (definition 4.1.13), then $\psi(i)\varphi(i) = \varphi(i)\psi(i) = I_{d_i}$, so $g = (\varphi(1), \dots, \varphi(n))$ lies

in $G(\underline{d})$, and the condition reads $B_e = g_{t(e)} A_e g_{s(e)}^{-1}$, that is, $B = g \cdot A$. Conversely, if $B = g \cdot A$, then $\varphi(i) = g_i$ satisfies $\varphi(t(e))A_e = B_e\varphi(s(e))$ and $\psi(i) = g_i^{-1}$ satisfies $\psi(t(e))B_e = A_e\psi(s(e))$, so φ and ψ are mutually inverse morphisms and the two representations are isomorphic.

The map from orbits to isomorphism classes is therefore well defined and injective. For surjectivity, let V be a representation with dimension vector $\underline{d}$, choose a basis of each $V(i)$, and let A_e be the matrix of T_e in the chosen bases. The coordinate isomorphisms $V(i) \to \mathbb{F}^{d_i}$ carry each T_e to A_e, so they form an isomorphism from V to the representation A, and the isomorphism class of V is the image of the orbit of A, completing the proof.

4.1.2 Decomposing Quivers

Definition 4.1.21: Simple Representation

Let Q be a quiver. A representation $V \in \mathrm{Rep}(Q)$ is *simple*, or *irreducible*, if $V \neq 0$ and the only subrepresentations of V are 0 and V.

A finite-dimensional vector space, which is a representation of the quiver with one vertex and no arrows, is a direct sum of one-dimensional subspaces, and these are simple. For quivers with arrows this fails: example 4.1.23 below exhibits a representation that is not a direct sum of simple subrepresentations. The summands that cannot be split further are the indecomposable representations.

Definition 4.1.22: Indecomposable Representation

Let Q be a quiver. A representation $V \in \mathrm{Rep}(Q)$ is *decomposable* if there exist nonzero subrepresentations $V_1, V_2 \subseteq V$ such that

$$V \cong V_1 \oplus V_2$$

and *indecomposable* otherwise. In particular the zero representation is indecomposable.

Example 4.1.23: Simple And Indecomposable Representations

Let Q be $1 \xrightarrow{e} 2$. The representations

$$0 \xrightarrow{0} \mathbb{F} \qquad \mathbb{F} \xrightarrow{0} 0$$

have total dimension 1, so each is simple, its only subrepresentations being 0 and itself, and indecomposable, since a direct sum of two nonzero representations has total dimension at least 2. The representation $V = \mathbb{F} \xrightarrow{1} \mathbb{F}$ is not simple. Its subrepresentations are exactly 0, $0 \xrightarrow{0} \mathbb{F}$ and V, because a subrepresentation W with $W(1) = \mathbb{F}$ has $W(2) \supseteq T_e(\mathbb{F}) = \mathbb{F}$.

However, V is indecomposable. If $V \cong V_1 \oplus V_2$ with V_1 and V_2 nonzero subrepresentations, then $\dim_Q V_1 + \dim_Q V_2 = \dim_Q V = (1, 1)$ with both summands nonzero, so one of V_1 and V_2 has dimension vector $(1, 0)$, and the list of subrepresentations of V contains none with this dimension vector.

So a representation need not be a direct sum of simple subrepresentations: the only simple subrep-

resentation of $\mathbb{F} \xrightarrow{1} \mathbb{F}$ is $0 \xrightarrow{0} \mathbb{F}$, and no subrepresentation complements it. Every representation of finite total dimension is nevertheless a direct sum of indecomposable ones.

Theorem 4.1.24: Quiver Representation Decomposition

Let Q be a quiver and V a representation of Q with $\sum_i \dim V(i) < \infty$. Then there are finitely many indecomposable subrepresentations $U_1, \dots, U_n$ of V such that

$$V = U_1 \oplus U_2 \oplus \cdots \oplus U_n$$

***Proof*:**
Write $\mathrm{Tdim}(V) = \sum_i \dim V(i)$ for the *total dimension*, and argue by strong induction on $\mathrm{Tdim}(V)$. If V is indecomposable, then $V = V$ is the required decomposition, with a single summand. This covers every V with $\mathrm{Tdim}(V) \leq 1$, because an isomorphism $V \cong V_1 \oplus V_2$ with V_1 and V_2 nonzero would give $\mathrm{Tdim}(V) = \mathrm{Tdim}(V_1) + \mathrm{Tdim}(V_2) \geq 2$.

Suppose V is decomposable, and let $\theta\colon V_1 \oplus V_2 \to V$ be an isomorphism with V_1 and V_2 nonzero (definition 4.1.22). An isomorphism carries subrepresentations to subrepresentations, so $V_1' = \theta(V_1 \oplus 0)$ and $V_2' = \theta(0 \oplus V_2)$ are nonzero subrepresentations of V with $V(i) = V_1'(i) \oplus V_2'(i)$ at every vertex, each $\theta(i)$ being a linear isomorphism. Both total dimensions are positive and they add up to $\mathrm{Tdim}(V)$, so each is smaller than $\mathrm{Tdim}(V)$, and by the inductive hypothesis V_1' and V_2' are finite direct sums of indecomposable subrepresentations. Together these summands decompose V, completing the proof.

The indecomposable representations of $1 \to 2 \to \cdots \to n$, and with them the decompositions of all its finite-dimensional representations, are determined in example 4.1.54 below.

4.1.3 The Tits Form and Its Roots

Let $Q = (I, E, s, t)$ be a finite quiver, meaning that I and E are finite sets, with $I = \{1, \dots, n\}$, and let $\underline{d} \in \mathbb{N}^n$. By theorem 4.1.20, the isomorphism classes of representations with dimension vector $\underline{d}$ are the orbits of $G(\underline{d})$ on $\mathrm{Rep}(Q, \underline{d})$. The space $\mathrm{Rep}(Q, \underline{d})$ has dimension $\sum_{e \in E} d_{s(e)} d_{t(e)}$, the group $G(\underline{d})$ consists of invertible matrices inside a space of dimension $\sum_{i \in I} d_i^2$, and the scalar tuples $(\lambda I_{d_1}, \dots, \lambda I_{d_n})$ act trivially. A dimension count therefore suggests that, for nonzero $\underline{d}$ and over an infinite field, finitely many orbits can cover $\mathrm{Rep}(Q, \underline{d})$ only if

$$\sum_{i \in I} d_i^2 - \sum_{e \in E} d_{s(e)} d_{t(e)} \geq 1$$

and the proof of theorem 4.1.46 makes this precise with transcendence degrees. The left side is the Tits form of Q at $\underline{d}$.

Definition 4.1.25: Graph

A *graph* Γ consists of a finite set I of *vertices* and, for every pair of vertices i, j, a number $n_{ij} = n_{ji} \in \mathbb{N}$ of *edges* joining i and j. An edge joining a vertex to itself is a *loop*, two distinct vertices i, j with $n_{ij} \geq 2$ are joined by a *multiple edge*, and two distinct vertices are *adjacent* if $n_{ij} \geq 1$.

1. The *underlying graph* of a finite quiver $Q = (I, E, s, t)$ has vertex set I, with n_{ij} the number of arrows e with $\{s(e), t(e)\} = \{i, j\}$ for $i \neq j$, and n_{ii} the number of arrows with $s(e) = t(e) = i$.
2. A graph Γ' with vertex set I' and edge numbers n'_{ij} is a *subgraph* of Γ if $I' \subseteq I$ and $n'_{ij} \leq n_{ij}$ for all $i, j \in I'$. It is the *full subgraph* on I' if $n'_{ij} = n_{ij}$ for all $i, j \in I'$.
3. A *path* in Γ is a sequence $(v_1, \ldots, v_m)$ of distinct vertices, $m \geq 1$, with v_k adjacent to v_{k+1} for $1 \leq k < m$, and it *joins* v_1 and v_m. A *cycle* is a path $(v_1, \ldots, v_m)$ with $m \geq 3$ and v_m adjacent to v_1.
4. Γ is *connected* if any two vertices are joined by a path, and the *degree* of a vertex is the number of vertices adjacent to it.

Two graphs are identified when they differ only by a renaming of their vertices. Paths and cycles in a graph ignore orientation, and they are distinct from the paths and oriented cycles of a quiver (definition 4.1.3).

Definition 4.1.26: Euler And Tits Form

Let $Q = (I, E, s, t)$ be a finite quiver, and for $i \in I$ let $\alpha_i \in \mathbb{Z}^I$ be the vector with i-th coordinate 1 and all other coordinates 0, so that $x = \sum_i x_i \alpha_i$. The *Euler form* of Q is the bilinear form on $\mathbb{R}^I$ given by

$$\langle x, y \rangle = \sum_{i \in I} x_i y_i - \sum_{e \in E} x_{s(e)} y_{t(e)}$$

its *symmetrization* is $(x, y) = \langle x, y \rangle + \langle y, x \rangle$, and the *Tits form* of Q is the quadratic form $q(x) = \langle x, x \rangle = \frac{1}{2}(x, x)$. For $x \in \mathbb{Z}^I$, write $x \geq 0$ if $x_i \geq 0$ for every i, and $x \not\geq 0$ otherwise. The dimension vector $\dim_Q V$ of a finite-dimensional representation (definition 4.1.7) is a vector $\dim_Q V \geq 0$ in $\mathbb{Z}^I$.

The Euler form depends on the orientation of the arrows, but its symmetrization does not.

Proposition 4.1.27: The Tits Form Depends Only On The Underlying Graph

Let Q be a finite quiver whose underlying graph Γ has edge numbers n_{ij}. For every $x \in \mathbb{R}^I$,

$$q(x) = \sum_{i \in I} (1 - n_{ii}) x_i^2 - \sum_{\{i,j\}} n_{ij} x_i x_j$$

where the second sum runs over the unordered pairs of distinct vertices, and

$$(\alpha_i, \alpha_i) = 2 - 2n_{ii} \qquad (\alpha_i, \alpha_j) = -n_{ij} \text{ for } i \neq j$$

In particular q and $(\ ,\ )$ depend only on Γ. The same formulas define the *Tits form* q_Γ and the symmetric form $(\ ,\ )_\Gamma$ of an arbitrary graph Γ.

***Proof*:**
In $\sum_e x_{s(e)} x_{t(e)}$, each of the n_{ii} loops at i contributes x_i^2, and each of the n_{ij} arrows between distinct vertices i and j contributes $x_i x_j$, whichever its direction. This gives the formula for $q(x) = \langle x, x \rangle$. For the symmetric form, $\langle \alpha_i, \alpha_j \rangle = \delta_{ij} - a_{ij}$, where a_{ij} is the number of arrows from i to j, so $(\alpha_i, \alpha_j) = 2\delta_{ij} - a_{ij} - a_{ji}$. For $i = j$ this is $2 - 2n_{ii}$, and for $i \neq j$ it is $-n_{ij}$. Neither formula involves the direction of an arrow, completing the proof.

Example 4.1.28: Euler And Tits Forms

1. Let Q be $1 \xrightarrow{e} 2 \xrightarrow{f} 3$ and Q' be $1 \xleftarrow{e} 2 \xrightarrow{f} 3$. Their Euler forms

$$\langle x, y \rangle_Q = \sum_{i=1}^{3} x_i y_i - x_1 y_2 - x_2 y_3 \qquad \langle x, y \rangle_{Q'} = \sum_{i=1}^{3} x_i y_i - x_2 y_1 - x_2 y_3$$

   differ, since $\langle \alpha_1, \alpha_2 \rangle_Q = -1$ while $\langle \alpha_1, \alpha_2 \rangle_{Q'} = 0$, but both Tits forms equal $x_1^2 + x_2^2 + x_3^2 - x_1 x_2 - x_2 x_3$, as proposition 4.1.27 predicts. Completing squares,

$$q(x) = \left(x_1 - \tfrac{1}{2} x_2\right)^2 + \left(x_3 - \tfrac{1}{2} x_2\right)^2 + \tfrac{1}{2} x_2^2$$

   so $q(x) > 0$ for every nonzero $x \in \mathbb{R}^3$: the right side vanishes only if $x_2 = 0$, and then $x_1 = x_3 = 0$.

2. Let K be the *Kronecker quiver*, with two arrows a, b from 1 to 2. Its underlying graph has $n_{12} = 2$, and $q(x) = x_1^2 + x_2^2 - 2x_1 x_2 = (x_1 - x_2)^2$ vanishes at $x = \alpha_1 + \alpha_2$. For this dimension vector, $\mathrm{Rep}(K, (1,1)) = \mathbb{F}^2$, and $(g_1, g_2) \in \mathbb{F}^\times \times \mathbb{F}^\times$ sends (A_a, A_b) to $(g_2 g_1^{-1} A_a, g_2 g_1^{-1} A_b)$. The orbit of a nonzero point is the set of nonzero points of the line through it, so over an infinite field there are infinitely many orbits.

Gabriel's theorem singles out the first family of graphs below. The second family carries vectors at which the Tits form vanishes.

Definition 4.1.29: Dynkin and Extended Dynkin Diagrams

For integers $a, b, c \geq 1$, the *star* $T(a, b, c)$ is the graph consisting of a central vertex and three paths of a, b and c vertices, its *arms*, with the first vertex of each arm adjacent to the central vertex, consecutive vertices of each arm adjacent, and no other edges. The graphs below have no loops and no multiple edges, except $\tilde{A}_0$ and $\tilde{A}_1$.

1. The *Dynkin diagrams of type A, D, E* are the path A_n on n vertices ($n \geq 1$), $D_n = T(1, 1, n-3)$ ($n \geq 4$), $E_6 = T(1, 2, 2)$, $E_7 = T(1, 2, 3)$ and $E_8 = T(1, 2, 4)$, drawn in fig. 4.1. The subscript is the number of vertices.

2. The *extended Dynkin diagrams* are the following graphs. The graph $\tilde{A}_0$ is one vertex with one loop, $\tilde{A}_1$ is two vertices joined by two edges, and $\tilde{A}_n$ ($n \geq 2$) is the cycle on $n+1$ vertices. The graph $\tilde{D}_4$ is a central vertex adjacent to four further vertices, and for $n \geq 5$ the graph $\tilde{D}_n$ is a path $(v_1, \ldots, v_{n-3})$ together with two further vertices adjacent to v_1 and two further vertices adjacent to v_{n-3}. Finally, $\tilde{E}_6 = T(2, 2, 2)$, $\tilde{E}_7 = T(1, 3, 3)$ and $\tilde{E}_8 = T(1, 2, 5)$. The subscript is one less than the number of vertices.

Each extended Dynkin diagram $\hat{\Gamma}$ carries the vector $\delta_{\hat{\Gamma}}$, with one positive integer for each vertex, recorded in fig. 4.2. Every entry of $\delta_{\tilde{A}_n}$ is 1. The four vertices of $\tilde{D}_n$ of degree one carry 1, and the others carry 2. In $\tilde{E}_6$, $\tilde{E}_7$ and $\tilde{E}_8$ the central vertex carries 3, 4 and 6 respectively. Read away from the central vertex, each arm of $\tilde{E}_6$ carries $2, 1$. The arm of $\tilde{E}_7$ with one vertex carries 2, and its two arms with three vertices carry $3, 2, 1$. The arms of $\tilde{E}_8$ with one, two and five vertices carry 3, then $4, 2$, then $5, 4, 3, 2, 1$.

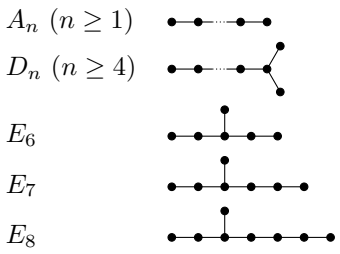


Figure 4.1: The Dynkin diagrams of type A, D, E

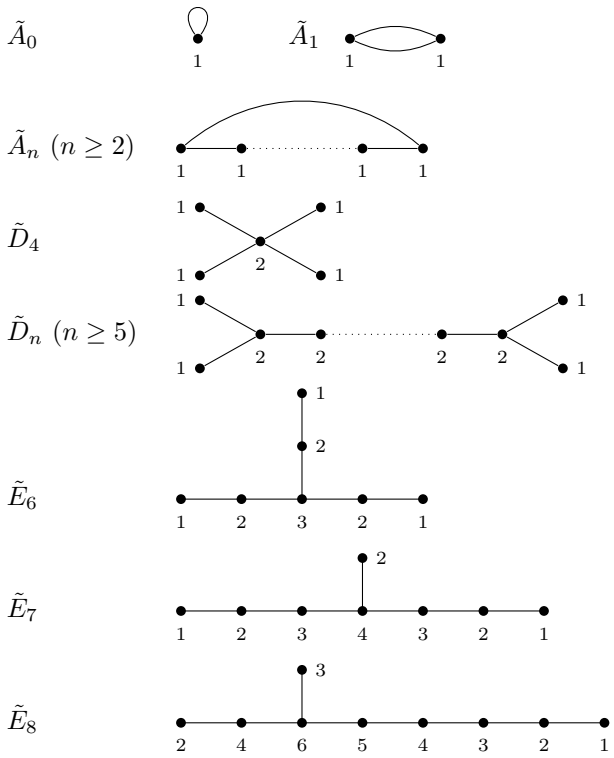


Figure 4.2: The extended Dynkin diagrams, each vertex labelled by its entry of δ

Each Dynkin diagram of type A, D, E is an extended Dynkin diagram with one vertex removed, and the vector δ of the larger graph writes the Tits form of the larger graph as a sum of squares.

Lemma 4.1.30: Null Vectors Of The Extended Dynkin Diagrams

Let $\hat{\Gamma}$ be an extended Dynkin diagram with vertex set $\hat{I}$, edge numbers n_{ij} and vector $\delta = \delta_{\hat{\Gamma}}$ (definition 4.1.29).

1. $(\delta, \alpha_i) = 0$ for every $i \in \hat{I}$, and $q(\delta) = 0$.
2. If $\hat{\Gamma} \neq \tilde{A}_0$, then for every $x \in \mathbb{R}^{\hat{I}}$

$$q(x) = \frac{1}{2} \sum_{\{i,j\}} n_{ij} \delta_i \delta_j \left(\frac{x_i}{\delta_i} - \frac{x_j}{\delta_j} \right)^2$$

the sum running over the unordered pairs of distinct vertices. Hence $q(x) \geq 0$, with equality if and only if x is a real multiple of δ.

***Proof*:**

1. For $\tilde{A}_0$, the single vertex has $(\alpha_1, \alpha_1) = 2 - 2 = 0$ by proposition 4.1.27, so $(\delta, \alpha_1) = 0$. Every other extended Dynkin diagram is loopless, and proposition 4.1.27 gives $(\delta, \alpha_i) = 2\delta_i - \sum_{j \neq i} n_{ij}\delta_j$, so the claim is that twice each entry of δ equals the sum of the entries at the adjacent vertices, each counted n_{ij} times. For $\tilde{A}_1$ this reads $2 \cdot 1 = 2 \cdot 1$, and for $\tilde{A}_n$ with $n \geq 2$ every vertex has two adjacent vertices and $2 \cdot 1 = 1 + 1$. In $\tilde{D}_n$, a vertex of degree one is adjacent only to a vertex carrying 2, and $2 \cdot 1 = 2$. For $n = 4$ the central vertex gives $2 \cdot 2 = 1 + 1 + 1 + 1$. For $n \geq 5$ the vertices v_1 and v_{n-3} are distinct, each is adjacent to two vertices carrying 1 and to one v_k, and $2 \cdot 2 = 1 + 1 + 2$, while each v_k with $1 < k < n - 3$ gives $2 \cdot 2 = 2 + 2$. In $\tilde{E}_6$, $\tilde{E}_7$ and $\tilde{E}_8$, the entries along each arm, preceded by the central entry and followed by 0, form an arithmetic progression, namely $3, 2, 1, 0$, then $4, 2, 0$ and $4, 3, 2, 1, 0$, then $6, 3, 0$ and $6, 4, 2, 0$ and $6, 5, 4, 3, 2, 1, 0$. Each arm vertex is adjacent exactly to its two neighbours in this progression (the 0 standing for no vertex), so twice its entry is the sum of theirs. At the central vertex, $2 \cdot 3 = 2 + 2 + 2$, $2 \cdot 4 = 2 + 3 + 3$ and $2 \cdot 6 = 3 + 4 + 5$. Finally, $q(\delta) = \frac{1}{2}(\delta, \delta) = \frac{1}{2}\sum_i \delta_i(\delta, \alpha_i) = 0$.

2. Now $\hat{\Gamma}$ has no loops, so $(x, x) = 2\sum_i x_i^2 - 2\sum_{\{i,j\}} n_{ij}x_i x_j$ by proposition 4.1.27. Part (1) gives $2\delta_i = \sum_{j \neq i} n_{ij}\delta_j$ for every i, and every δ_i is positive, so

$$2\sum_i x_i^2 = \sum_i \frac{x_i^2}{\delta_i} \sum_{j \neq i} n_{ij}\delta_j = \sum_{\{i,j\}} n_{ij}\left(\frac{\delta_j}{\delta_i}x_i^2 + \frac{\delta_i}{\delta_j}x_j^2\right)$$

Therefore

$$(x, x) = \sum_{\{i,j\}} n_{ij}\left(\frac{\delta_j}{\delta_i}x_i^2 - 2x_i x_j + \frac{\delta_i}{\delta_j}x_j^2\right) = \sum_{\{i,j\}} n_{ij}\delta_i\delta_j\left(\frac{x_i}{\delta_i} - \frac{x_j}{\delta_j}\right)^2$$

and halving gives the formula for $q(x)$. Every term is nonnegative, so $q(x) \geq 0$. If $q(x) = 0$, then $x_i/\delta_i = x_j/\delta_j$ for all adjacent i and j. The graph $\hat{\Gamma}$ is connected, so following a path from a fixed vertex shows that x_i/δ_i takes a single value t at every vertex, and $x = t\delta$. Conversely $q(t\delta) = t^2 q(\delta) = 0$ by part (1), completing the proof.

Lemma 4.1.31: Graphs Without Extended Dynkin Subgraphs

Let Γ be a connected graph, loops and multiple edges allowed, that contains none of the extended Dynkin diagrams as a subgraph. Then Γ is one of A_n $(n \geq 1)$, D_n $(n \geq 4)$, E_6, E_7, E_8.

***Proof*:**

Step 1: no loops, multiple edges or cycles. A loop at a vertex i, with the vertex i, is a subgraph $\tilde{A}_0$, and two of the edges of a multiple edge, with their two vertices, form a subgraph $\tilde{A}_1$. If $(v_1, \ldots, v_m)$ is a cycle, then these vertices, with the edges joining v_k to v_{k+1} and v_m to v_1, form a subgraph $\tilde{A}_{m-1}$. So Γ has no loops, no multiple edges and no cycles, and it is determined by its adjacency relation.

Step 2: every degree is at most 3. A vertex adjacent to four distinct vertices forms, with them

and the four edges, a subgraph $\tilde{D}_4$.

Step 3: at most one vertex has degree 3. Suppose $u \neq w$ both have degree 3, and choose a path $(u = v_1, \ldots, v_r = w)$, with $r \geq 2$, by connectedness. Choose two vertices $a \neq b$ adjacent to u and different from v_2, and two vertices $c \neq d$ adjacent to w and different from v_{r-1}. The vertex a is not on the path: $a \neq v_1$ because there are no loops, $a \neq v_2$ by choice, and $a = v_k$ with $k \geq 3$ would make $(v_1, \ldots, v_k)$ a cycle. The same holds for b, c and d. If $a = c$, then $(v_1, \ldots, v_r, a)$ is a cycle, since a is adjacent to v_r and to v_1, and similarly $a \neq d$, $b \neq c$, $b \neq d$. So the $r + 4$ vertices $v_1, \ldots, v_r, a, b, c, d$ are distinct, and with the edges of the path and the edges ua, ub, wc, wd they form a subgraph $\tilde{D}_{r+3}$, which is excluded.

Step 4: if no vertex has degree 3*, then* $\Gamma = A_n$. Choose a path $(v_1, \ldots, v_m)$ in Γ with m as large as possible. Suppose a vertex y is not among $v_1, \ldots, v_m$, choose a path $(y = w_1, \ldots, w_p = v_1)$, and let w_q be the first of its vertices among $v_1, \ldots, v_m$. Then $q \geq 2$, and $x = w_{q-1}$ is not among $v_1, \ldots, v_m$ and is adjacent to $w_q = v_k$ for some k. If $k = 1$ or $k = m$, adding x at that end gives a longer path, and if $1 < k < m$, then v_k is adjacent to v_{k-1}, v_{k+1} and x. Both are impossible, so the path contains every vertex. An edge joining v_i to v_j with $j \geq i+2$ would make $(v_i, \ldots, v_j)$ a cycle, so the edges are exactly those joining v_k to v_{k+1}, and $\Gamma = A_m$.

Step 5: if exactly one vertex z *has degree* 3*, then* Γ *is a star.* Let x_1, x_2, x_3 be the vertices adjacent to z, and let Γ' be the full subgraph on the vertices other than z. Every vertex $y \neq z$ is joined to some x_l by a path in Γ': in a path from y to z in Γ, the vertex before z is some x_l, and the part from y to x_l avoids z. No path in Γ' joins x_l to $x_{l'}$ with $l \neq l'$, since such a path followed by z would be a cycle in Γ. For $l = 1, 2, 3$, let C_l be the set of vertices joined to x_l by a path in Γ'. If $y \in C_l \cap C_{l'}$ with $l \neq l'$, take paths from x_l to y and from y to $x_{l'}$ in Γ', and let y' be the first vertex of the first path lying on the second. The first path up to y', followed by the second path from y', is a path in Γ' joining x_l to $x_{l'}$, which is excluded. So the sets C_1, C_2, C_3 are disjoint and cover the vertices other than z. No edge joins two of them: if $y \in C_l$ is adjacent to y', then a path from x_l to y, extended by y' or cut at y' if it already passes through y', joins x_l to y', so $y' \in C_l$. The full subgraph on each C_l is connected, has no cycle, and has every degree at most 2, so by the argument of Step 4 it is a path. In it, x_l has degree at most 1, because x_l has degree at most 2 in Γ and one of its edges goes to z, so x_l is an end of that path. Writing a_l for the number of vertices of C_l, $\Gamma = T(a_1, a_2, a_3)$.

Step 6: the arms. Order the arms so that $a_1 \leq a_2 \leq a_3$. If $a_1 \geq 2$, the central vertex and the first two vertices of each arm form a subgraph $T(2, 2, 2) = \tilde{E}_6$. If $a_1 = 1$ and $a_2 \geq 3$, the central vertex, the arm with one vertex and the first three vertices of the other arms form $T(1, 3, 3) = \tilde{E}_7$. If $a_1 = 1$, $a_2 = 2$ and $a_3 \geq 5$, the first five vertices of the longest arm give $T(1, 2, 5) = \tilde{E}_8$. The remaining possibilities are $T(1, 1, a_3) = D_{a_3+3}$, $T(1, 2, 2) = E_6$, $T(1, 2, 3) = E_7$ and $T(1, 2, 4) = E_8$. With Steps 2 to 5, every case has been treated, completing the proof.

Theorem 4.1.32: Positive Definite Forms And Dynkin Diagrams

Let Γ be a connected graph with vertex set I, loops and multiple edges allowed, and Tits form q. The following are equivalent.

1. Γ is one of the Dynkin diagrams A_n $(n \geq 1)$, D_n $(n \geq 4)$, E_6, E_7, E_8.
2. q is positive definite, that is, $q(x) > 0$ for every nonzero $x \in \mathbb{R}^I$.
3. $q(x) > 0$ for every nonzero $x \in \mathbb{Z}^I$ with $x \geq 0$.

***Proof*:**

1. *(1) implies (2).* Each graph in (1) is the full subgraph of an extended Dynkin diagram $\hat{\Gamma} \neq \tilde{A}_0$ on all its vertices but one, i_0. Removing a vertex of $\tilde{A}_n$ leaves the path A_n. Removing a vertex of degree one from $\tilde{D}_4$ leaves $T(1,1,1) = D_4$, and removing one of the two vertices of degree one adjacent to v_1 from $\tilde{D}_n$, $n \geq 5$, leaves the star with central vertex v_{n-3}, two arms with one vertex, and the arm $v_{n-4}, \ldots, v_1$ followed by the other vertex of degree one adjacent to v_1, with $n-3$ vertices, which is $T(1,1,n-3) = D_n$. Removing the last vertex of an arm with two vertices of $\tilde{E}_6 = T(2,2,2)$, of an arm with three vertices of $\tilde{E}_7 = T(1,3,3)$, or of the arm with five vertices of $\tilde{E}_8 = T(1,2,5)$ leaves E_6, E_7 or E_8. For $x \in \mathbb{R}^I$ let $\hat{x} \in \mathbb{R}^{\hat{I}}$ agree with x on I and vanish at i_0. As Γ is the full subgraph on $I = \hat{I} \setminus \{i_0\}$, the formula of proposition 4.1.27 gives $q_\Gamma(x) = q_{\hat{\Gamma}}(\hat{x})$, every term involving i_0 being 0. By lemma 4.1.30(2), $q_\Gamma(x) \geq 0$, with equality only if $\hat{x} = t\delta$ for some $t \in \mathbb{R}$. In that case $0 = \hat{x}_{i_0} = t\delta_{i_0}$ with $\delta_{i_0} > 0$, so $t = 0$ and $x = 0$. Hence q_Γ is positive definite.
2. *(2) implies (3).* Every nonzero $x \in \mathbb{Z}^I$ is a nonzero vector of $\mathbb{R}^I$.
3. *(3) implies (1).* Suppose Γ is not one of the graphs in (1). By lemma 4.1.31, Γ has a subgraph Γ' that is an extended Dynkin diagram, with vertex set $I' \subseteq I$ and edge numbers $n'_{ij} \leq n_{ij}$. Let $\delta = \delta_{\Gamma'}$, and let $x \in \mathbb{Z}^I$ agree with δ on I' and vanish elsewhere, so $x \geq 0$ and $x \neq 0$. Only the vertices of I' contribute to the formula of proposition 4.1.27, and every product $\delta_i \delta_j$ is positive, so

$$q_\Gamma(x) = \sum_{i \in I'} (1 - n_{ii})\delta_i^2 - \sum_{\{i,j\} \subseteq I'} n_{ij}\delta_i\delta_j \leq \sum_{i \in I'} (1 - n'_{ii})\delta_i^2 - \sum_{\{i,j\} \subseteq I'} n'_{ij}\delta_i\delta_j = q_{\Gamma'}(\delta)$$

and $q_{\Gamma'}(\delta) = 0$ by lemma 4.1.30(1). So (3) fails.

The three implications prove the equivalence.

In particular, a connected graph with positive definite Tits form has no loops, no multiple edges and no cycles. The integer vectors at which a Tits form takes the value 1 are its roots.

Definition 4.1.33: Roots, Simple Reflections And The Weyl Group Of A Graph

Let Γ be a graph with vertex set I, Tits form q and symmetric form $(\ ,\ )$.

1. A *root* of Γ is a vector $x \in \mathbb{Z}^I$ with $q(x) = 1$, and Φ_Γ is the set of roots. A root x is *positive* if $x \geq 0$ and *negative* if $-x \geq 0$, and Φ_Γ^+ is the set of positive roots.
2. For a vertex i at which Γ has no loop, the *simple reflection* at i is the linear map $s_i \colon \mathbb{R}^I \to \mathbb{R}^I$, $s_i(x) = x - (x, \alpha_i)\alpha_i$. Since $(\alpha_i, \alpha_i) = 2$ by proposition 4.1.27, $(s_i(x), \alpha_i) = (x, \alpha_i) - 2(x, \alpha_i) = -(x, \alpha_i)$, so $s_i(s_i(x)) = x$ and s_i is invertible.
3. If Γ has no loops, its *Weyl group* W_Γ is the group of linear automorphisms of $\mathbb{R}^I$ generated by the simple reflections, and for an ordering $i_1, \ldots, i_n$ of the n vertices, each listed once, the product $c = s_{i_n} s_{i_{n-1}} \cdots s_{i_1} \in W_\Gamma$ is a *Coxeter element*.

Lemma 4.1.34: Simple Reflections Preserve The Tits Form

Let Γ be a graph and i a vertex at which Γ has no loop.

1. For every $x \in \mathbb{R}^I$, the vector $s_i(x)$ differs from x only in its i-th coordinate. Moreover $s_i(\alpha_i) = -\alpha_i$ and $s_i(\mathbb{Z}^I) = \mathbb{Z}^I$.
2. For all $x, y \in \mathbb{R}^I$, $(s_i(x), s_i(y)) = (x, y)$ and $q(s_i(x)) = q(x)$.
3. $s_i(\Phi_\Gamma) = \Phi_\Gamma$.

***Proof*:**

1. The difference $s_i(x) - x = -(x, \alpha_i)\alpha_i$ is a multiple of α_i, and $s_i(\alpha_i) = \alpha_i - 2\alpha_i = -\alpha_i$ because $(\alpha_i, \alpha_i) = 2$. For $x \in \mathbb{Z}^I$, $(x, \alpha_i) = \sum_j x_j(\alpha_j, \alpha_i)$ is an integer by proposition 4.1.27, so $s_i(\mathbb{Z}^I) \subseteq \mathbb{Z}^I$, and $s_i(s_i(x)) = x$ (definition 4.1.33) gives equality.
2. By bilinearity and $(\alpha_i, \alpha_i) = 2$,
$$(s_i(x), s_i(y)) = (x, y) - 2(x, \alpha_i)(y, \alpha_i) + (x, \alpha_i)(y, \alpha_i)(\alpha_i, \alpha_i) = (x, y)$$
and taking $y = x$ gives $q(s_i(x)) = \frac{1}{2}(s_i(x), s_i(x)) = q(x)$.
3. By (1) and (2), s_i maps Φ_Γ into Φ_Γ, and since $s_i(s_i(x)) = x$, every root x is the image $s_i(s_i(x))$ of the root $s_i(x)$, completing the proof.

Proposition 4.1.35: Roots Of A Positive Definite Tits Form

Let Γ be a graph with vertex set $I = \{1, \dots, n\}$ whose Tits form q is positive definite.

1. Γ has no loops, and $\alpha_1, \dots, \alpha_n$ are roots.
2. Every root is positive or negative, so Φ_Γ is the disjoint union of Φ_Γ^+ and $-\Phi_\Gamma^+$.
3. If x is a root and $i \in I$ with $x \neq \pm\alpha_i$, then $(x, \alpha_i) \in \{-1, 0, 1\}$. The set Φ_Γ is finite.
4. Every element of W_Γ permutes Φ_Γ, and W_Γ is finite.

***Proof*:**

The formula of proposition 4.1.27 shows that q takes integer values on $\mathbb{Z}^I$, and q is positive on nonzero vectors, so $q(y) \geq 1$ for every nonzero $y \in \mathbb{Z}^I$. Bilinearity gives $q(x + y) = q(x) + q(y) + (x, y)$ for all $x, y \in \mathbb{R}^I$.

1. By proposition 4.1.27, $q(\alpha_i) = 1 - n_{ii}$, which is positive only if $n_{ii} = 0$. Hence there are no loops and $q(\alpha_i) = 1$.
2. Let x be a root, and write $x = x^+ - x^-$, where $x_i^+ = \max(x_i, 0)$ and $x_i^- = \max(-x_i, 0)$, so that $x^+, x^- \geq 0$ and no coordinate is nonzero in both. In $(x^+, x^-) = \sum_{i,j} x_i^+ x_j^- (\alpha_i, \alpha_j)$, every nonzero term has $i \neq j$, so $(\alpha_i, \alpha_j) = -n_{ij} \leq 0$ and $(x^+, x^-) \leq 0$. Then
$$1 = q(x) = q(x^+) + q(x^-) - (x^+, x^-) \geq q(x^+) + q(x^-)$$
If x^+ and x^- were both nonzero, the right side would be at least 2. So $x = x^+ \geq 0$ or $x = -x^-$ with $x^- \geq 0$. The two cases exclude each other, since $q(0) = 0$ makes 0 a non-root.
3. Let x be a root and $i \in I$. If $x = \pm\alpha_i$, then $(x, \alpha_i) = \pm 2$. Otherwise $x - \alpha_i$ and $x + \alpha_i$ are nonzero vectors of $\mathbb{Z}^I$, and $q(x \mp \alpha_i) = q(x) + q(\alpha_i) \mp (x, \alpha_i) = 2 \mp (x, \alpha_i)$ is positive, so $(x, \alpha_i) \in \{-1, 0, 1\}$. In every case $(x, \alpha_i) \in \{-2, \dots, 2\}$. The map $\rho\colon \mathbb{Z}^I \to \mathbb{Z}^n$, $\rho(x) = ((x, \alpha_1), \dots, (x, \alpha_n))$, is injective: if $\rho(x) = \rho(y)$, then $(x - y, \alpha_i) = 0$ for all i, so $(x - y, x - y) = \sum_i (x_i - y_i)(x - y, \alpha_i) = 0$ and $x = y$ by positive definiteness. Since ρ maps Φ_Γ into $\{-2, \dots, 2\}^n$, the set Φ_Γ has at most 5^n elements.
4. By lemma 4.1.34(3), each s_i permutes Φ_Γ, and $s_i^{-1} = s_i$, so every element of W_Γ permutes Φ_Γ. Restriction to Φ_Γ is therefore a group homomorphism from W_Γ to the group of permutations of Φ_Γ. An element of its kernel fixes every root, in particular every α_i by (1), and the α_i form a basis of $\mathbb{R}^I$, so it is the identity. Hence W_Γ embeds in the permutation group of the finite set Φ_Γ and is finite, completing the proof.

Example 4.1.36: The Roots Of A_n

Number the vertices of the path A_n as $1, \dots, n$ in order, and for $x \in \mathbb{R}^n$ set $x_0 = x_{n+1} = 0$. Then
$$q(x) = \sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} = \frac{1}{2} \sum_{i=0}^{n} (x_{i+1} - x_i)^2$$

because expanding the right side gives $\frac{1}{2}\big(\sum_{i=1}^{n} x_i^2 + \sum_{i=1}^{n} x_i^2 - 2\sum_{i=1}^{n-1} x_i x_{i+1}\big)$, the terms involving x_0 and x_{n+1} being 0. For $x \in \mathbb{Z}^n$, the condition $q(x) = 1$ says that the integer sequence $0 = x_0, x_1, \ldots, x_n, x_{n+1} = 0$ has exactly two steps $x_{i+1} - x_i$ equal to ± 1 and all other steps equal to 0. The sequence starts and ends at 0, so the two nonzero steps have opposite signs, and the sequence is constant equal to 1 or to -1 on an interval $\{i, i+1, \ldots, j\}$ and 0 outside it. Hence

$$\Phi^+_{A_n} = \{\alpha_i + \alpha_{i+1} + \cdots + \alpha_j \mid 1 \le i \le j \le n\}$$

a set of $\frac{n(n+1)}{2}$ positive roots, and Φ_{A_n} consists of these and their negatives.

The finiteness of W_Γ has a consequence for Coxeter elements, which the proof of theorem 4.1.46 uses to locate the indecomposable representations.

Lemma 4.1.37: Coxeter Element Lemma

Let Γ be a graph with vertex set I whose Tits form is positive definite, and let $c = s_{i_n} \cdots s_{i_1}$ be a Coxeter element of W_Γ (definition 4.1.33). For every nonzero $v \in \mathbb{Z}^I$ there is an integer $m \ge 0$ with $c^m(v) \not\ge 0$.

***Proof*:**

By proposition 4.1.35(4), W_Γ is finite, so c has finite order $h \ge 1$. Suppose, toward a contradiction, that $c^m(v) \ge 0$ for every $m \ge 0$, and set $u = \sum_{m=0}^{h-1} c^m(v) \in \mathbb{Z}^I$. Then $c(u) = \sum_{m=1}^{h} c^m(v) = u$, because $c^h(v) = v$.

Step 1: $u = 0$. Put $u_0 = u$ and $u_k = s_{i_k}(u_{k-1})$ for $1 \le k \le n$, so that $u_n = c(u) = u$. By definition 4.1.33, $u_k - u_{k-1} = -(u_{k-1}, \alpha_{i_k})\alpha_{i_k}$, and summing over k,

$$0 = u_n - u_0 = -\sum_{k=1}^{n} (u_{k-1}, \alpha_{i_k})\alpha_{i_k}$$

The vertices $i_1, \ldots, i_n$ are distinct, so the vectors α_{i_k} are linearly independent, and every coefficient (u_{k-1}, α_{i_k}) vanishes. Then $u_k = u_{k-1}$ for every k, so $u_{k-1} = u$ and $(u, \alpha_{i_k}) = 0$ for every k. Every vertex occurs as some i_k, so $(u, \alpha_i) = 0$ for all $i \in I$, and $(u, u) = \sum_i u_i(u, \alpha_i) = 0$. Positive definiteness gives $u = 0$.

Step 2: the contradiction. The vector $u = 0$ is the sum of the vectors $c^m(v) \ge 0$, $0 \le m \le h-1$, so every coordinate of every $c^m(v)$ is 0. In particular $v = c^0(v) = 0$, against the hypothesis. Hence $c^m(v) \not\ge 0$ for some $m \ge 0$, as claimed.

The roots of a positive definite Tits form are an instance of a structure that first appeared in the classification of the simple Lie algebras over $\mathbb{C}$. Its definition records the properties of Φ_Γ that do not refer to the lattice $\mathbb{Z}^I$.

Definition 4.1.38: Root System

Let E be a finite-dimensional real vector space with an inner product $(\ ,\ )$. For a nonzero $\alpha \in E$, the *reflection* in the hyperplane orthogonal to α is $s_\alpha(v) = v - \frac{2(v,\alpha)}{(\alpha,\alpha)}\alpha$. A *(reduced) root system* in E is a finite set $R \subseteq E$ of nonzero vectors, its *roots*, such that

1. R spans E,
2. $s_\alpha(R) = R$ for every $\alpha \in R$,
3. $\frac{2(\beta,\alpha)}{(\alpha,\alpha)} \in \mathbb{Z}$ for all $\alpha, \beta \in R$,
4. if $\alpha \in R$ and $c\alpha \in R$ for a real number c, then $c = \pm 1$.

A root system is *irreducible* if it is not the union of two nonempty subsets with every root of the first orthogonal to every root of the second, and *simply laced* if all its roots have the same length. Root systems $R \subseteq E$ and $R' \subseteq E'$ are *isomorphic* if a linear isomorphism $\varphi\colon E \to E'$ carries R onto R' and satisfies $\frac{2(\varphi(\beta),\varphi(\alpha))}{(\varphi(\alpha),\varphi(\alpha))} = \frac{2(\beta,\alpha)}{(\alpha,\alpha)}$ for all $\alpha, \beta \in R$.

Definition 4.1.39: Weyl Group And Base Of A Root System

Let R be a root system in E. Since $(s_\alpha(v), \alpha) = -(v, \alpha)$, applying s_α twice returns v, and the *Weyl group* $W(R)$ is the group generated by the reflections s_α, $\alpha \in R$. A *base* of R is a subset $\Pi \subseteq R$ that is a basis of E such that every root is an integer combination of Π whose coefficients are all ≥ 0 or all ≤ 0. The elements of a base are the *simple roots*, and the roots whose coefficients are all ≥ 0 are the *positive roots* relative to Π.

Example 4.1.40: The Root System A_2

Let $E = \left\{v \in \mathbb{R}^3 \;\middle|\; v_1 + v_2 + v_3 = 0\right\}$ with the dot product, let $\varepsilon_1, \varepsilon_2, \varepsilon_3$ be the standard basis of $\mathbb{R}^3$, and let

$$R = \{\pm(\varepsilon_1 - \varepsilon_2), \pm(\varepsilon_2 - \varepsilon_3), \pm(\varepsilon_1 - \varepsilon_3)\}$$

The vectors $\varepsilon_1 - \varepsilon_2$ and $\varepsilon_2 - \varepsilon_3$ are linearly independent and lie in the two-dimensional space E, so R spans E. Every root has $(\alpha, \alpha) = 2$, so $s_\alpha(v) = v - (v, \alpha)\alpha$, and for $\alpha = \pm(\varepsilon_i - \varepsilon_j)$ this exchanges the i-th and j-th coordinates of v. Exchanging two coordinates permutes R, and $W(R)$ is generated by the three transpositions of coordinates. The inner product of two roots lies in $\{\pm 1, \pm 2\}$, so $\frac{2(\beta,\alpha)}{(\alpha,\alpha)} = (\beta, \alpha) \in \mathbb{Z}$, and the only multiples of a root lying in R are the root and its negative. So R is a root system, and $\{\varepsilon_1 - \varepsilon_2, \varepsilon_2 - \varepsilon_3\}$ is a base, the third positive root $\varepsilon_1 - \varepsilon_3$ being their sum.

The linear isomorphism $\mathbb{R}^2 \to E$ sending α_1 to $\varepsilon_1 - \varepsilon_2$ and α_2 to $\varepsilon_2 - \varepsilon_3$ carries the symmetric form of A_2 to the dot product, since $(\alpha_1, \alpha_1) = (\alpha_2, \alpha_2) = 2$ and $(\alpha_1, \alpha_2) = -1$ (proposition 4.1.27) agree with the dot products of the images. By example 4.1.36 it carries $\Phi_{A_2} = \{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2)\}$ onto R.

Definition 4.1.41: Cartan Matrix And Dynkin Diagram

Let R be a root system with base $\Pi = \{\alpha_1, \dots, \alpha_r\}$. The *Cartan matrix* of R relative to Π is the $r \times r$ matrix with entries

$$A_{ij} = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}$$

The *Dynkin diagram* of R has vertex set $\{1, \dots, r\}$, two distinct vertices i, j being joined by $A_{ij}A_{ji}$ edges, and each multiple edge carrying an arrowhead pointing toward the shorter of α_i and α_j.

The integers A_{ij} and A_{ji} are positive multiples of (α_i, α_j), so $A_{ij}A_{ji} \geq 0$. The non-simply-laced case already occurs in rank two.

Example 4.1.42: Cartan Matrix

For each of the four sets R below, the two vectors α_1, α_2 chosen are linearly independent, so R spans the plane containing it. The reflections in the roots act as permutations of the coordinates, possibly combined with changes of sign, and these permute R. The integrality axiom follows from the inner products listed, and no two roots are proportional except a root and its negative, so each R is a root system. To see that $\{\alpha_1, \alpha_2\}$ is a base, it suffices to write each positive root as a nonnegative integer combination of α_1 and α_2, the other roots being their negatives. The diagonal entries of every Cartan matrix are 2.

1. $(A_1 \times A_1)$ In $\mathbb{R}^2$, $R = \{\pm\varepsilon_1, \pm\varepsilon_2\}$, with $\alpha_1 = \varepsilon_1$ and $\alpha_2 = \varepsilon_2$. The reflection s_{α_i} changes the sign of the i-th coordinate, every root has length 1, and inner products of roots lie in $\{0, \pm 1\}$. Since $(\alpha_1, \alpha_2) = 0$,
$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$
and the Dynkin diagram is two vertices with no edge.

2. (A_2) For R as in example 4.1.40, with $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$, the third positive root is $\alpha_1 + \alpha_2$. Here $(\alpha_1, \alpha_1) = (\alpha_2, \alpha_2) = 2$ and $(\alpha_1, \alpha_2) = -1$, so
$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$
and the Dynkin diagram is a single edge.

3. (B_2) In $\mathbb{R}^2$, $R = \{\pm\varepsilon_1, \pm\varepsilon_2, \pm\varepsilon_1 \pm \varepsilon_2\}$, eight roots with independent signs. The reflection in ε_i changes the sign of the i-th coordinate, the reflection in $\varepsilon_1 - \varepsilon_2$ exchanges the coordinates, and the reflection in $\varepsilon_1 + \varepsilon_2$ exchanges them and changes both signs. The inner products of roots lie in $\{0, \pm 1, \pm 2\}$, and $(\alpha, \alpha) \in \{1, 2\}$. With $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2$, the positive roots are α_1, α_2, $\varepsilon_1 = \alpha_1 + \alpha_2$ and $\varepsilon_1 + \varepsilon_2 = \alpha_1 + 2\alpha_2$. Here $(\alpha_1, \alpha_1) = 2$, $(\alpha_2, \alpha_2) = 1$ and $(\alpha_1, \alpha_2) = -1$, so $A_{12} = 2(-1)/2 = -1$ and $A_{21} = 2(-1)/1 = -2$:
$$A = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$$
The Dynkin diagram is a double edge with its arrowhead pointing toward α_2, the shorter root.

4. (G_2) In the plane E of example 4.1.40, let

$$R = \{\pm(\varepsilon_i - \varepsilon_j) \mid i < j\} \cup \{\pm(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3), \pm(2\varepsilon_2 - \varepsilon_1 - \varepsilon_3), \pm(2\varepsilon_3 - \varepsilon_1 - \varepsilon_2)\}$$

twelve roots, six of squared length 2 and six of squared length 6. The reflection in $\varepsilon_i - \varepsilon_j$ exchanges two coordinates, and for $v \in E$ the reflection in $\gamma = 2\varepsilon_i - \varepsilon_j - \varepsilon_l$ is $v \mapsto v - v_i\gamma$, because $(v, \gamma) = 3v_i$. It exchanges the j-th and l-th coordinates and changes all three signs. The inner product of two roots lies in $\{0, \pm1, \pm2\}$ when both are short, in $\{0, \pm3\}$ when exactly one is long, and in $\{\pm3, \pm6\}$ when both are long, so the integrality axiom holds. With $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = -2\varepsilon_1 + \varepsilon_2 + \varepsilon_3$, the positive roots are

$$\alpha_1 \qquad \alpha_2 \qquad \alpha_1 + \alpha_2 \qquad 2\alpha_1 + \alpha_2 \qquad 3\alpha_1 + \alpha_2 \qquad 3\alpha_1 + 2\alpha_2$$

equal to $\varepsilon_1 - \varepsilon_2$, $-2\varepsilon_1 + \varepsilon_2 + \varepsilon_3$, $\varepsilon_3 - \varepsilon_1$, $\varepsilon_3 - \varepsilon_2$, $\varepsilon_1 - 2\varepsilon_2 + \varepsilon_3$ and $2\varepsilon_3 - \varepsilon_1 - \varepsilon_2$. Here $(\alpha_1, \alpha_1) = 2$, $(\alpha_2, \alpha_2) = 6$ and $(\alpha_1, \alpha_2) = -3$, so $A_{12} = 2(-3)/2 = -3$ and $A_{21} = 2(-3)/6 = -1$:

$$A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

The Dynkin diagram is a triple edge with its arrowhead pointing toward α_1, the shorter root.

Definition 4.1.41 normalizes by the row index, so the matrices for B_2 and G_2 are the transposes of those in sources that normalize by the column index, such as [10, §11.1]. In the four cases $A_{12}A_{21}$ takes the values 0, 1, 2, 3.

The two meanings of "Dynkin diagram" in this subsection agree.

Proposition 4.1.43: The Roots Of A Dynkin Diagram Form A Root System

Let Γ be one of the Dynkin diagrams A_n, D_n, E_6, E_7, E_8, with vertex set I, and give $\mathbb{R}^I$ the symmetric form $(\ ,\ )$ of Γ.

1. $(\ ,\ )$ is an inner product on $\mathbb{R}^I$, and Φ_Γ is a root system in $\mathbb{R}^I$.
2. $\{\alpha_i\}_{i \in I}$ is a base of Φ_Γ with positive roots Φ_Γ^+, and the Dynkin diagram of Φ_Γ relative to this base is Γ.
3. Φ_Γ is irreducible and simply laced.
4. Every root is $w(\alpha_i)$ for some $w \in W_\Gamma$ and $i \in I$, and $W(\Phi_\Gamma) = W_\Gamma$.

***Proof*:**
By theorem 4.1.32, the Tits form of Γ is positive definite, so proposition 4.1.35 applies, and Γ has no loops and no multiple edges.

1. The form $(\ ,\ )$ is symmetric and bilinear, and $(x, x) = 2q(x) > 0$ for $x \neq 0$, so it is an inner product. The set Φ_Γ is finite, does not contain 0, and contains every α_i, so it spans $\mathbb{R}^I$ (proposition 4.1.35). Every root has $(\alpha, \alpha) = 2q(\alpha) = 2$, so $s_\alpha(v) = v - (v, \alpha)\alpha$, and $\frac{2(\beta,\alpha)}{(\alpha,\alpha)} = (\beta, \alpha)$ is an integer for $\beta \in \mathbb{Z}^I$ by proposition 4.1.27. As in lemma 4.1.34, s_α maps $\mathbb{Z}^I$ to $\mathbb{Z}^I$, satisfies $(s_\alpha(x), s_\alpha(y)) = (x, y) - 2(x, \alpha)(y, \alpha) + (x, \alpha)(y, \alpha)(\alpha, \alpha) = (x, y)$, and

is its own inverse, so $s_\alpha(\Phi_\Gamma) = \Phi_\Gamma$. If α and $c\alpha$ are roots, then $1 = q(c\alpha) = c^2 q(\alpha) = c^2$ and $c = \pm 1$.

2. The α_i form a basis of $\mathbb{R}^I$, and by proposition 4.1.35(2) every root has coordinates all ≥ 0 or all ≤ 0. So $\{\alpha_i\}$ is a base, whose positive roots are Φ_Γ^+. Its Cartan matrix has $A_{ij} = (\alpha_i, \alpha_j)$, which by proposition 4.1.27 is -1 if i and j are adjacent and 0 if they are distinct and not adjacent. So $A_{ij}A_{ji}$ is 1 for adjacent vertices and 0 for other pairs, the Dynkin diagram has a single edge exactly between adjacent vertices and no arrowheads, and it is Γ.

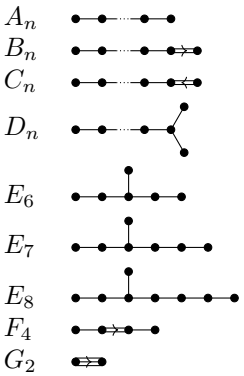
3. Every root has $(\alpha, \alpha) = 2$, so Φ_Γ is simply laced. Suppose $\Phi_\Gamma = R_1 \cup R_2$ with R_1 and R_2 nonempty and orthogonal to each other. No root lies in both, as a root is not orthogonal to itself. If i and j are adjacent, then $(\alpha_i, \alpha_j) = -1 \neq 0$, so α_i and α_j lie in the same R_l, and as Γ is connected, all the α_i lie in one of them, say R_1. A root $\beta \in R_2$ is then orthogonal to every α_i, so $(\beta, \beta) = \sum_i \beta_i(\beta, \alpha_i) = 0$, which is impossible. Hence Φ_Γ is irreducible.

4. First let $x \in \Phi_\Gamma^+$, and argue by induction on $\mathrm{ht}(x) = \sum_i x_i \geq 1$. If $\mathrm{ht}(x) = 1$, then $x = \alpha_i$ for some i. If $\mathrm{ht}(x) \geq 2$, then $2 = (x, x) = \sum_i x_i(x, \alpha_i)$ with every $x_i \geq 0$, so some i has $x_i > 0$ and $(x, \alpha_i) > 0$. As $x \neq \pm\alpha_i$, proposition 4.1.35(3) gives $(x, \alpha_i) = 1$, so $s_i(x) = x - \alpha_i$ is a root (lemma 4.1.34) of height $\mathrm{ht}(x) - 1$. Some x_j with $j \neq i$ is positive, since otherwise $x = x_i\alpha_i$ and $q(x) = x_i^2 = 1$ would force $x = \alpha_i$, and $s_i(x)$ has the same j-th coordinate. So $s_i(x)$ is a positive root (proposition 4.1.35(2)), by induction $s_i(x) = w(\alpha_l)$ with $w \in W_\Gamma$, and $x = s_i w(\alpha_l)$. A negative root x has $-x = w(\alpha_l)$, and then $x = w(-\alpha_l) = ws_l(\alpha_l)$. Next, $s_i = s_{\alpha_i}$, so $W_\Gamma \subseteq W(\Phi_\Gamma)$. Conversely, write a root as $\alpha = w(\alpha_l)$ with $w \in W_\Gamma$. By lemma 4.1.34(2), w preserves $(\ ,\ )$, so $(v, w(\alpha_l)) = (w^{-1}(v), \alpha_l)$ and, for every $v \in \mathbb{R}^I$,

$$s_\alpha(v) = v - (w^{-1}(v), \alpha_l)w(\alpha_l) = w\big(w^{-1}(v) - (w^{-1}(v), \alpha_l)\alpha_l\big) = ws_l w^{-1}(v)$$

So every generator s_α of $W(\Phi_\Gamma)$ lies in W_Γ, and $W(\Phi_\Gamma) = W_\Gamma$, completing the proof.

4.1.44 Remark: The Classification Of Root Systems Every root system is the union of mutually orthogonal irreducible root systems, and an irreducible root system is determined up to isomorphism by its Dynkin diagram, which is one of the connected diagrams



with $n \geq 1$ for A_n, $n \geq 2$ for B_n, $n \geq 3$ for C_n and $n \geq 4$ for D_n, and every diagram on the list occurs [10, §§11.1, 11.3, 11.4, 12.1]. The proof is a case analysis of the possible Cartan matrices, and nothing in this book depends on it. The simply laced diagrams on the list are A_n, D_n, E_6, E_7 and E_8, and by proposition 4.1.43 their root systems are the sets Φ_Γ of the Tits forms. In B_n, C_n, F_4 and G_2 the roots have two different lengths, as in example 4.1.42 for B_2 and G_2. Every root of a graph has $(\alpha, \alpha) = 2$, so these four families never arise as Φ_Γ, and in particular never from the Tits form of a quiver.

4.1.4 Gabriel's Theorem

Theorem 4.1.32 singles out the Dynkin diagrams of type A, D, E among connected graphs by the positivity of their Tits forms, and proposition 4.1.35 attaches to each of them a finite set of positive roots. Gabriel's theorem identifies these positive roots with the indecomposable representations of every quiver whose underlying graph is one of these diagrams. Throughout, $\mathbb{F}$ is a field, representations are over $\mathbb{F}$, and quivers are finite.

Definition 4.1.45: Collection Of Indecomposables

Let Q be a finite quiver. Then $\mathrm{Ind}(Q)$ denotes the set of isomorphism classes of nonzero finite-dimensional indecomposable representations of Q over $\mathbb{F}$, and Q is of *finite type* over $\mathbb{F}$ if $\mathrm{Ind}(Q)$ is finite.

Theorem 4.1.46: Gabriel's Theorem

Let Q be a finite quiver whose underlying graph Γ is connected (definition 4.1.25), and let $\mathbb{F}$ be a field.

1. If Γ is one of the Dynkin diagrams A_n, D_n, E_6, E_7, E_8, then Q is of finite type over $\mathbb{F}$, and the dimension vector
$$\dim\colon \operatorname{Ind}(Q) \longrightarrow \Phi_\Gamma^+ \qquad [V] \longmapsto \dim_Q V$$
is a bijection onto the set of positive roots of Γ.
2. If $\mathbb{F}$ is infinite and Q is of finite type over $\mathbb{F}$, then Γ is one of the Dynkin diagrams A_n, D_n, E_6, E_7, E_8.

By part (1), the number of indecomposable representations of a quiver of type A, D, E depends only on its graph, not on the orientation of its arrows or on the field: a quiver whose underlying graph is A_n has $\frac{n(n+1)}{2}$ of them by example 4.1.36. The proof of part (2) below uses that $\mathbb{F}$ is infinite: over a finite field every representation space is a finite set, so the number of orbits of $G(\underline{d})$ carries no information.

By proposition 4.1.43(4), every positive root is obtained from some α_i by applying simple reflections. The reflection functors of Bernstein, Gelfand and Ponomarev are the corresponding operations on representations. A simple reflection s_i is defined at every vertex without a loop, hence at every vertex of an A, D, E graph, but a reflection functor at i is defined only when i is a sink or a source (definition 4.1.4), and it reverses the arrows at i. For a sink i write s_i^+Q, and for a source i write s_i^-Q, for the quiver iQ of definition 4.1.4, in which every arrow at i is reversed. The sink i of Q is a source of s_i^+Q, and $s_i^-(s_i^+Q) = Q$.

Let Q be $1 \xrightarrow{e} 2$, in which 2 is a sink, and let V be the representation $V(1) \xrightarrow{T_e} V(2)$. The reflection at the sink 2 keeps $V(1)$, replaces $V(2)$ by $\ker(T_e)$, and replaces the arrow by its reversal $2 \to 1$, carrying the inclusion of $\ker(T_e)$ into $V(1)$:

$$\left(\begin{array}{ccc} V(1) & \xrightarrow{T_e} & V(2) \\ \bullet & & \bullet \end{array} \right) \longmapsto \left(\begin{array}{ccc} V(1) & \hookleftarrow & \ker(T_e) \\ \bullet & & \bullet \end{array} \right)$$

If T_e is surjective, the dimension vector (d_1, d_2) becomes $(d_1, d_1 - d_2)$, and since $(d, \alpha_2) = 2d_2 - d_1$ by proposition 4.1.27, this is $s_2(d)$. The reflection at the source 1 replaces $V(1)$ by $\operatorname{coker}(T_e) = V(2)/\operatorname{im}(T_e)$ and the arrow by the quotient map $V(2) \to \operatorname{coker}(T_e)$, and if T_e is injective the dimension vector becomes $(d_2 - d_1, d_2) = s_1(d)$. At an arbitrary sink or source, all the arrows at the vertex are combined into one map. If i is a sink of Q and $V \in \operatorname{Rep}(Q)$, let

$$\Psi_i\colon \bigoplus_{e:j\to i} V(j) \to V(i)$$

be the map whose restriction to the summand indexed by an arrow $e\colon j \to i$ is T_e, the sum having one summand for each arrow into i. If i is a source, let

$$\Phi_i\colon V(i) \to \bigoplus_{e:i\to j} V(j)$$

be the map whose component in the summand indexed by an arrow $e\colon i \to j$ is T_e.

Definition 4.1.47: Reflection Functor

Let Q be a quiver.

1. For a sink i and $V \in \mathrm{Rep}(Q)$, define $S_i^+(V) \in \mathrm{Rep}(s_i^+Q)$ by $S_i^+(V)(k) = V(k)$ for $k \neq i$ and $S_i^+(V)(i) = \ker(\Psi_i)$. The map along an arrow not at i is T_e, and the map along the reversal $i \to k$ of an arrow $e\colon k \to i$ is $\pi_k \circ \iota_i$, where $\iota_i\colon \ker(\Psi_i) \to \bigoplus_{e:j\to i} V(j)$ is the inclusion and π_k is the projection onto the summand $V(k)$ indexed by e (two arrows from k give two summands and two projections).

2. For a source i and $V \in \mathrm{Rep}(Q)$, define $S_i^-(V) \in \mathrm{Rep}(s_i^-Q)$ by $S_i^-(V)(k) = V(k)$ for $k \neq i$ and $S_i^-(V)(i) = \mathrm{coker}(\Phi_i)$. The map along an arrow not at i is T_e, and the map along the reversal $k \to i$ of an arrow $e\colon i \to k$ is $p \circ \iota_k$, where $\iota_k\colon V(k) \to \bigoplus_{e:i\to j} V(j)$ is the inclusion of the summand indexed by e and p is the quotient map onto $\mathrm{coker}(\Phi_i)$.

3. For a morphism $\varphi\colon V \to W$ and a sink i, the morphism $S_i^+(\varphi)$ is $\varphi(k)$ at $k \neq i$ and the restriction of $\bigoplus_{e:j\to i} \varphi(j)$ to $\ker(\Psi_i^V) \to \ker(\Psi_i^W)$ at i. For a source i, the morphism $S_i^-(\varphi)$ is $\varphi(k)$ at $k \neq i$ and the map induced by $\bigoplus_{e:i\to j} \varphi(j)$ on cokernels at i.

The functors $S_i^+\colon \mathrm{Rep}(Q) \to \mathrm{Rep}(s_i^+Q)$ and $S_i^-\colon \mathrm{Rep}(Q) \to \mathrm{Rep}(s_i^-Q)$ are the *Bernstein-Gelfand-Ponomarev reflection functors*, or *reflection functors* for short. That (3) is well defined and gives functors is the first paragraph of the proof of proposition 4.1.49.

Example 4.1.48: Reflection Functor

Let Q be $1 \xrightarrow{e} 2$, so that s_2^+Q is $1 \leftarrow 2$. Three nonzero indecomposable representations of Q are $\mathbb{F} \xrightarrow{1} \mathbb{F}$, $\mathbb{F} \to 0$ and $0 \to \mathbb{F}$ (example 4.1.23), of dimension vectors $\alpha_1 + \alpha_2$, α_1 and α_2. Applying S_2^+:

1. For $\mathbb{F} \xrightarrow{1} \mathbb{F}$, the map Ψ_2 is the identity of $\mathbb{F}$, so $\ker(\Psi_2) = 0$ and S_2^+ gives $\mathbb{F} \leftarrow 0$, of dimension vector $\alpha_1 = s_2(\alpha_1 + \alpha_2)$.

2. For $\mathbb{F} \to 0$, the map Ψ_2 is zero on $\mathbb{F}$, so $\ker(\Psi_2) = \mathbb{F}$ and S_2^+ gives $\mathbb{F} \xleftarrow{1} \mathbb{F}$, of dimension vector $\alpha_1 + \alpha_2 = s_2(\alpha_1)$.

3. For $0 \to \mathbb{F}$, $\ker(\Psi_2) = 0$ and S_2^+ gives 0, while $s_2(\alpha_2) = -\alpha_2$ is negative.

So S_2^+ exchanges the representations of dimension vectors α_1 and $\alpha_1 + \alpha_2$, as s_2 exchanges these positive roots of A_2, and it sends the simple representation at the sink, whose dimension vector s_2 makes negative, to 0.

For a sink i, let $\mathrm{Rep}^{i,-}(Q)$ denote the full subcategory of representations of Q for which the map

$$\Psi_i\colon \bigoplus_{e:j\to i} V(j) \to V(i)$$

is surjective, and for a source i, let $\mathrm{Rep}^{i,+}(Q)$ denote the full subcategory of representations of Q for which the map

$$\Phi_i\colon V(i) \to \bigoplus_{e:i\to j} V(j)$$

is injective. For a vertex i, let $S(i) \in \mathrm{Rep}(Q)$ denote the representation with $S(i)(i) = \mathbb{F}$, with $S(i)(l) = 0$ for $l \neq i$, and with all edge maps zero. A representation is *simple at vertex i* if it is isomorphic to $S(i)$.

Proposition 4.1.49: Properties Of Reflection Functor

Let Q be a quiver. In items (4) to (7), the upper sign refers to a sink i and the lower sign to a source i. For items (6) and (7), assume in addition that Q is finite and that every representation named has finite total dimension $\sum_i \dim V(i)$, and let s_i be the simple reflection at i of the underlying graph of Q (definition 4.1.33), which is defined because a sink or a source carries no loop.

1. If i is a sink, then for any $V \in \mathrm{Rep}(Q)$, $S_i^+(V) \in \mathrm{Rep}^{i,+}(s_i^+(Q))$ (similarly for source)
2. If i is a sink, then for any $V \in \mathrm{Rep}(Q)$, $S_i^- S_i^+(V) \in \mathrm{Rep}^{i,-}(Q)$ (similarly for source)
3. The restrictions of the reflection functors are equivalences

$$S_i^+ : \mathrm{Rep}^{i,-}(Q) \to \mathrm{Rep}^{i,+}(s_i^+ Q) \qquad (i \text{ a sink of } Q)$$
$$S_i^- : \mathrm{Rep}^{i,+}(Q) \to \mathrm{Rep}^{i,-}(s_i^- Q) \qquad (i \text{ a source of } Q)$$

   where the inverse of each restriction is the restriction of the other functor, taken over the reflected quiver
4. If V is simple at vertex i, then $S_i^\pm(V) = 0$.
5. If V is a non-zero indecomposable representation of Q that is not simple at i, then $V \in \mathrm{Rep}^{i,\mp}(Q)$, the representation $S_i^\pm(V)$ of $s_i^\pm(Q)$ is nonzero and indecomposable, and $V \cong S_i^\mp S_i^\pm(V)$.
6. If V is a nonzero indecomposable representation of Q, then $S_i^\pm(V)$ is a nonzero indecomposable representation of $s_i^\pm(Q)$ if and only if $s_i(\dim_Q(V)) \geq 0$
7. If $V \in \mathrm{Rep}^{i,\mp}(Q)$ is a nonzero indecomposable representation, then $\dim_Q(S_i^\pm(V)) = s_i(\dim_Q(V))$

***Proof*:**

For a sink i of a quiver and a representation V, write $\Psi_i^V : \bigoplus_{e:j\to i} V(j) \to V(i)$ for the map whose component at an edge $e : j \to i$ is T_e, and for a source i write $\Phi_i^V : V(i) \to \bigoplus_{e:i\to j} V(j)$ for the map whose component at an edge $e : i \to j$ is T_e. These are the maps in the definition of the subcategories $\mathrm{Rep}^{i,\mp}$ above. On a quiver morphism $\varphi : V \to W$, the functor S_i^+ acts by $S_i^+(\varphi)(k) = \varphi(k)$ for $k \neq i$ and by taking $S_i^+(\varphi)(i)$ to be the restriction of $\bigoplus_{e:j\to i} \varphi(j)$ to a map $\ker(\Psi_i^V) \to \ker(\Psi_i^W)$, as in definition 4.1.47(3). The restriction lands in $\ker(\Psi_i^W)$ because summing the commuting squares of φ over the edges $e : j \to i$ (definition 4.1.9) gives $\Psi_i^W \circ \bigoplus_j \varphi(j) = \varphi(i) \circ \Psi_i^V$. The map $\bigoplus_j \varphi(j)$ acts summand by summand, so $\pi_k \circ \bigoplus_j \varphi(j) = \varphi(k) \circ \pi_k$ for the projection π_k onto the summand indexed by an edge $e\colon k \to i$, and this is the commuting square of $S_i^+(\varphi)$ at the reversal of e. At the edges not at i the squares are those of φ, so $S_i^+(\varphi)$ is a morphism. Dually, summing the squares of φ over the edges $e : i \to j$ gives $\Phi_i^W \circ \varphi(i) = \left(\bigoplus_j \varphi(j)\right) \circ \Phi_i^V$, so $\bigoplus_{e:i\to j} \varphi(j)$ carries $\mathrm{im}(\Phi_i^V)$ into $\mathrm{im}(\Phi_i^W)$ and induces the

map $S_i^-(\varphi)(i)$ on cokernels. It commutes with the inclusions of the summands and with the quotient maps, which gives the squares of $S_i^-(\varphi)$ at the reversed edges. Both constructions send identities to identities, and they respect composition because the restriction of a composite is the composite of the restrictions, and likewise for induced maps on quotients.

For representations V, W and a sink i, the canonical identification

$$\bigoplus_{e:j\to i}(V\oplus W)(j)=\Big(\bigoplus_{e:j\to i}V(j)\Big)\oplus\Big(\bigoplus_{e:j\to i}W(j)\Big)$$

carries $\Psi_i^{V\oplus W}$ to $\Psi_i^V\oplus\Psi_i^W$, and the kernel of a direct sum of linear maps is the direct sum of the kernels, so $S_i^+(V\oplus W)\cong S_i^+(V)\oplus S_i^+(W)$, the edge maps on both sides being assembled summand by summand from the same inclusions and projections. The same identification on cokernels gives $S_i^-(V\oplus W)\cong S_i^-(V)\oplus S_i^-(W)$ at a source.

Item (1). Let i be a sink of Q and $V\in\mathrm{Rep}(Q)$, and write $U=\bigoplus_{e:j\to i}V(j)$ and $K=\ker(\Psi_i^V)\subseteq U$. In $s_i^+(Q)$ the vertex i is a source, and its outgoing edges are exactly the reversals of the edges of Q into i. By definition 4.1.47, the map of $S_i^+(V)$ at the reversal of an edge $e:k\to i$ is $\pi_k\circ\iota_i$, where $\iota_i:K\to U$ is the inclusion and π_k is the projection of U onto the summand $V(k)$ indexed by e. The map $\Phi_i^{S_i^+(V)}$ therefore has components $\pi_k\circ\iota_i$ over all summands of U, and a vector of U is recovered from the family of all its components, so $\Phi_i^{S_i^+(V)}=\iota_i$. An inclusion is injective, so $S_i^+(V)\in\mathrm{Rep}^{i,+}(s_i^+(Q))$. For a source i, the edges of $s_i^-(Q)$ into i are the reversals of the edges of Q out of i, the map of $S_i^-(V)$ at such an edge is $p\circ\iota_k$ with $p:\bigoplus_{e:i\to j}V(j)\to\mathrm{coker}(\Phi_i^V)$ the quotient map, and assembling the components gives $\Psi_i^{S_i^-(V)}=p$. A quotient map is surjective, so $S_i^-(V)\in\mathrm{Rep}^{i,-}(s_i^-(Q))$.

Item (2). Let i be a sink of Q. Reversing the arrows at i twice returns the original quiver, so $s_i^-(s_i^+(Q))=Q$. The vertex i is a source of $s_i^+(Q)$, so the source half of item (1), applied to the quiver $s_i^+(Q)$ and the representation $S_i^+(V)$, gives $S_i^-S_i^+(V)\in\mathrm{Rep}^{i,-}(Q)$. If i is instead a source of Q, the sink half of item (1) applied to $s_i^-(Q)$ gives $S_i^+S_i^-(V)\in\mathrm{Rep}^{i,+}(Q)$.

Item (3). Let i be a sink of Q. By item (1), S_i^+ carries $\mathrm{Rep}^{i,-}(Q)$ into $\mathrm{Rep}^{i,+}(s_i^+(Q))$ and S_i^- carries $\mathrm{Rep}^{i,+}(s_i^+(Q))$ into $\mathrm{Rep}^{i,-}(Q)$, so both restrictions are defined.

Let $V\in\mathrm{Rep}^{i,-}(Q)$ and retain U, K, ι_i from item (1). By item (1), $\Phi_i^{S_i^+(V)}=\iota_i$, so $(S_i^-S_i^+(V))(i)=\mathrm{coker}(\iota_i)=U/K$, and the map of $S_i^-S_i^+(V)$ at an edge $e:j\to i$ of Q is $p\circ\iota_j:V(j)\to U/K$, with $p\colon U\to U/K$ the quotient map. At every other vertex and edge, $S_i^-S_i^+(V)$ agrees with V. Because $V\in\mathrm{Rep}^{i,-}(Q)$ the map Ψ_i^V is surjective, and $\Psi_i^V(u)=\Psi_i^V(u')$ holds exactly when $u-u'\in K$, so

$$\overline{\Psi_i}:U/K\to V(i)\qquad u+K\mapsto\Psi_i^V(u)$$

is a well-defined linear isomorphism. Define $\mu_V:S_i^-S_i^+(V)\to V$ by $\mu_V(i)=\overline{\Psi_i}$ and $\mu_V(k)=\mathrm{id}_{V(k)}$ for $k\neq i$. At an edge $e:j\to i$ the required square commutes because

$$\overline{\Psi_i}(p(\iota_j(x)))=\Psi_i^V(\iota_j(x))=T_e(x)$$

for $x\in V(j)$, and at every other edge both representations carry the same map with identities as the vertical maps. So μ_V is a quiver morphism that is a linear isomorphism at every vertex. Its vertexwise inverse is again a quiver morphism, because from $\mu_V(i')\circ S_e=T_e\circ\mu_V(j')$ at an edge $e:j'\to i'$ one gets $S_e\circ\mu_V(j')^{-1}=\mu_V(i')^{-1}\circ T_e$. Hence μ_V is an isomorphism $S_i^-S_i^+(V)\cong V$.

For naturality, let $\varphi : V \to V'$ be a morphism in $\mathrm{Rep}^{i,-}(Q)$. At a vertex $k \neq i$ both composites $\varphi \circ \mu_V$ and $\mu_{V'} \circ S_i^- S_i^+(\varphi)$ equal $\varphi(k)$. At i, the map $S_i^- S_i^+(\varphi)(i) : U/K \to U'/K'$ is induced by $\bigoplus_{e:j\to i} \varphi(j)$, so both composites send $u + K$ to

$$\varphi(i)(\Psi_i^V(u)) = \Psi_i^{V'}\Big(\bigoplus_j \varphi(j)(u)\Big)$$

the equality being the sum over the edges $e : j \to i$ of the commuting squares of φ.

Conversely, let $W \in \mathrm{Rep}^{i,+}(s_i^+(Q))$, write $U' = \bigoplus_{f:i\to k} W(k)$, summing over the edges of $s_i^+(Q)$ out of i, and write $\Phi = \Phi_i^W : W(i) \to U'$, an injective map because $W \in \mathrm{Rep}^{i,+}$. Then $(S_i^-(W))(i) = \mathrm{coker}(\Phi) = U'/\mathrm{im}(\Phi)$, and by the source half of item (1), $\Psi_i^{S_i^-(W)}$ is the quotient map $p' : U' \to U'/\mathrm{im}(\Phi)$, so

$$(S_i^+ S_i^-(W))(i) = \ker(p') = \mathrm{im}(\Phi)$$

Define $\nu_W : W \to S_i^+ S_i^-(W)$ by $\nu_W(k) = \mathrm{id}_{W(k)}$ for $k \neq i$ and $\nu_W(i) : W(i) \to \mathrm{im}(\Phi)$, $x \mapsto \Phi(x)$, a linear isomorphism because Φ is injective. At an edge $f : i \to k$ of $s_i^+(Q)$, the map of $S_i^+ S_i^-(W)$ is $\pi_k \circ \iota_i$ with $\iota_i : \ker(p') \to U'$ the inclusion, and

$$\pi_k(\iota_i(\Phi(x))) = \pi_k(\Phi(x)) = T_f(x)$$

because the f-component of Φ is T_f. At every edge not at i both representations carry the same map and the vertical maps are identities. As before, the vertexwise linear isomorphism ν_W is an isomorphism of representations. Naturality in W follows as for μ: for $\varphi : W \to W'$, the map $S_i^+ S_i^-(\varphi)(i)$ is the restriction of $\bigoplus_f \varphi(k)$ to the kernels, and $\Phi' \circ \varphi(i) = \big(\bigoplus_f \varphi(k)\big) \circ \Phi$ is the sum over the edges $f : i \to k$ of the commuting squares of φ.

The natural isomorphisms μ and ν exhibit $S_i^- S_i^+ \cong \mathrm{id}$ on $\mathrm{Rep}^{i,-}(Q)$ and $S_i^+ S_i^- \cong \mathrm{id}$ on $\mathrm{Rep}^{i,+}(s_i^+(Q))$, so the two restrictions are mutually inverse equivalences. This proves the first line of the display, together with the statement that its inverse is the restriction of S_i^-. The second line is the case just proved, applied to the quiver $s_i^-(Q)$: if i is a source of Q, then i is a sink of $s_i^-(Q)$ and $s_i^+(s_i^-(Q)) = Q$, so the restrictions

$$S_i^- : \mathrm{Rep}^{i,+}(Q) \to \mathrm{Rep}^{i,-}(s_i^-(Q)) \qquad S_i^+ : \mathrm{Rep}^{i,-}(s_i^-(Q)) \to \mathrm{Rep}^{i,+}(Q)$$

are mutually inverse equivalences.

The unrestricted functors are moreover adjoint. Fix a sink i of Q, and let $A \in \mathrm{Rep}(s_i^+(Q))$ and $B \in \mathrm{Rep}(Q)$. Away from i, the representations $S_i^-(A)$ and A carry the same spaces and maps, as do B and $S_i^+(B)$, so a family $(g(k))_{k\neq i}$ satisfies the square conditions away from i for a morphism $S_i^-(A) \to B$ exactly when it satisfies them for a morphism $A \to S_i^+(B)$. In a morphism $g : S_i^-(A) \to B$, the squares at the edges $e : k \to i$ of Q read $g(i) \circ p \circ \iota_k = T_e^B \circ g(k)$, with p the quotient map onto $\mathrm{coker}(\Phi_i^A)$, so $g(i)$ is the map that $\Psi_i^B \circ \bigoplus_k g(k)$ induces on $\mathrm{coker}(\Phi_i^A)$, and such a map exists, uniquely, exactly when this composite kills $\mathrm{im}(\Phi_i^A)$. In a morphism $h : A \to S_i^+(B)$ carrying the same family $g(k) = h(k)$ away from i, the squares at the edges $f : i \to k$ of $s_i^+(Q)$ read $\iota_i \circ h(i) = \big(\bigoplus_k g(k)\big) \circ \Phi_i^A$, so $h(i)$ exists, uniquely, exactly when the right-hand side lands in $\ker(\Psi_i^B)$. Both conditions state that $\Psi_i^B \circ \big(\bigoplus_k g(k)\big) \circ \Phi_i^A = 0$. Restricting to the components away from i is therefore a bijection

$$\mathrm{Hom}_Q(S_i^-(A), B) \cong \mathrm{Hom}_{s_i^+(Q)}(A, S_i^+(B))$$

natural in A and B, because composition with a morphism acts on the components away from i componentwise.

Item (4). Isomorphic representations have isomorphic images under the functors, so take $V = S(i)$. If i is a sink, every j with an edge $e : j \to i$ satisfies $V(j) = 0$, so $\bigoplus_{e:j\to i} V(j) = 0$ and $\ker(\Psi_i^V) = 0$. The representation $S_i^+(V)$ is then zero at i and equals $V(k) = 0$ at every $k \neq i$, so $S_i^+(V) = 0$. If i is a source, then $\bigoplus_{e:i\to j} V(j) = 0$, so $\operatorname{coker}(\Phi_i^V) = 0$ and $S_i^-(V) = 0$ for the same reason.

Item (5). First a dichotomy: if i is a sink and V is a nonzero indecomposable representation with $V \ncong S(i)$, then Ψ_i^V is surjective, that is, $V \in \operatorname{Rep}^{i,-}(Q)$. Suppose Ψ_i^V is not surjective and choose a subspace $C \subseteq V(i)$ with $V(i) = \operatorname{im}(\Psi_i^V) \oplus C$ and $C \neq 0$. Define $V', V'' \subseteq V$ by

$$V'(i) = \operatorname{im}(\Psi_i^V) \qquad V'(k) = V(k) \text{ for } k \neq i \qquad V''(i) = C \qquad V''(k) = 0 \text{ for } k \neq i$$

Both are subrepresentations (definition 4.1.8): every edge $e : j \to i$ satisfies $T_e(V(j)) \subseteq \operatorname{im}(\Psi_i^V) = V'(i)$ and $T_e(V''(j)) = T_e(0) = 0 \subseteq V''(i)$, the sink i has no outgoing edges, and edges away from i are unaffected. Every edge map is accordingly the direct sum of its restrictions to V' and V'', so $V = V' \oplus V''$ as in definition 4.1.17. Indecomposability of V with $V'' \neq 0$ forces $V' = 0$, so $V(k) = 0$ for all $k \neq i$ and $\operatorname{im}(\Psi_i^V) = 0$. All edge maps of V then vanish, so V is a direct sum of $\dim(C)$ copies of $S(i)$, and indecomposability forces $\dim(C) = 1$ and $V \cong S(i)$, against the hypothesis. The dichotomy follows. For a source the same construction splits off the kernel: if Φ_i^V is not injective, then V'' with $V''(i) = \ker(\Phi_i^V) \neq 0$ and $V''(k) = 0$ for $k \neq i$, together with V' given by a complement of $\ker(\Phi_i^V)$ at i and $V'(k) = V(k)$ for $k \neq i$, are subrepresentations with $V = V' \oplus V''$, and indecomposability forces, by the same argument, $V \cong S(i)$.

Now let V be nonzero indecomposable with $V \ncong S(i)$, and let i be a sink. By the dichotomy, $V \in \operatorname{Rep}^{i,-}(Q)$, which is the first claim of item (5), and item (3) gives $V \cong S_i^- S_i^+(V)$, which is the last. If $S_i^+(V) = 0$, then $V \cong S_i^-(0) = 0$, against $V \neq 0$. So $S_i^+(V) \neq 0$. Suppose $S_i^+(V) = W_1 \oplus W_2$ with both summands nonzero. Item (1) puts $S_i^+(V)$ in $\operatorname{Rep}^{i,+}(s_i^+(Q))$, and $\Phi_i^{W_1\oplus W_2} = \Phi_i^{W_1} \oplus \Phi_i^{W_2}$, so each $\Phi_i^{W_l}$ is injective and $W_l \in \operatorname{Rep}^{i,+}(s_i^+(Q))$. Item (3) gives $W_l \cong S_i^+ S_i^-(W_l)$, and $W_l \neq 0$ forces $S_i^-(W_l) \neq 0$. The additivity of S_i^- recorded at the start of the proof now gives

$$V \cong S_i^-(W_1 \oplus W_2) \cong S_i^-(W_1) \oplus S_i^-(W_2)$$

Let θ be the inverse of the displayed isomorphism, an isomorphism from $S_i^-(W_1) \oplus S_i^-(W_2)$ to V. An isomorphism carries subrepresentations to subrepresentations, so $V_1 = \theta(S_i^-(W_1) \oplus 0)$ and $V_2 = \theta(0 \oplus S_i^-(W_2))$ are nonzero subrepresentations of V with $V = V_1 \oplus V_2$, against definition 4.1.22. Hence $S_i^+(V)$ is indecomposable. For a source the argument runs with S_i^- and S_i^+ exchanged, using the source halves of items (1) and (3) and the additivity of S_i^+.

Item (7). Item (6) uses this item. Let i be a sink and $V \in \operatorname{Rep}^{i,-}(Q)$ of finite total dimension, and write $d = \dim_Q(V)$ (definition 4.1.7). Vertices $k \neq i$ contribute $\dim(S_i^+(V)(k)) = \dim V(k) = d_k$. At i, the map Ψ_i^V is surjective, so rank-nullity gives

$$\dim \ker(\Psi_i^V) = \sum_{e:j\to i} d_j - d_i$$

the sum running over the edges into i, each counted once. On the other side, by definition 4.1.26, $\langle d, \alpha_i\rangle = d_i - \sum_{e:j\to i} d_j$ and $\langle \alpha_i, d\rangle = d_i$, because no edge leaves the sink i. Hence $(d, \alpha_i) =$

$2d_i - \sum_{e:j\to i} d_j$, and $s_i(d) = d - (d,\alpha_i)\alpha_i$ agrees with d away from i and has i-th entry

$$d_i - \Big(2d_i - \sum_{e:j\to i} d_j\Big) = \sum_{e:j\to i} d_j - d_i$$

The two dimension vectors agree, so $\dim_Q(S_i^+(V)) = s_i(\dim_Q(V))$. For a source i and $V \in \mathrm{Rep}^{i,+}(Q)$, the map Φ_i^V is injective, so $\dim \operatorname{coker}(\Phi_i^V) = \sum_{e:i\to j} d_j - d_i$, while $\langle d,\alpha_i\rangle = d_i$ and $\langle \alpha_i, d\rangle = d_i - \sum_{e:i\to j} d_j$, and the same computation gives $\dim_Q(S_i^-(V)) = s_i(\dim_Q(V))$.

Item (6). Let V be nonzero indecomposable and let i be a sink. The source case replaces S_i^+ by S_i^- throughout, using the source halves of items (4), (5), and (7). If $V \cong S(i)$, then $S_i^+(V) = 0$ by item (4), and $\dim_Q(V) = \alpha_i$, so $s_i(\dim_Q(V)) = -\alpha_i$ (lemma 4.1.34), whose i-th entry is -1. Both sides of the equivalence fail. If $V \ncong S(i)$, then item (5) gives that $S_i^+(V)$ is nonzero indecomposable and that $V \in \mathrm{Rep}^{i,-}(Q)$, and item (7) gives $s_i(\dim_Q(V)) = \dim_Q(S_i^+(V))$, a vector of dimensions, hence ≥ 0. Both sides of the equivalence hold. In either case the equivalence is true.

Corollary 4.1.50: Reflection Functors Are Adjoint

Let i be a sink of Q. For $A \in \mathrm{Rep}(s_i^+(Q))$ and $B \in \mathrm{Rep}(Q)$ there is a bijection

$$\mathrm{Hom}_Q(S_i^-(A), B) \cong \mathrm{Hom}_{s_i^+(Q)}(A, S_i^+(B))$$

natural in both variables, so S_i^- is left adjoint to S_i^+.

***Proof*:**
This is the final paragraph of the proof of item (3) of proposition 4.1.49: both morphism spaces biject with the families $(g(k))_{k\neq i}$ satisfying the away-from-i squares together with the single condition $\Psi_i^B \circ \big(\bigoplus_k g(k)\big) \circ \Phi_i^A = 0$, and the bijection is natural in A and B by the uniqueness of the induced components at i.

A reflection functor at i requires i to be a sink or a source, so reflecting at every vertex in turn requires an ordering of the vertices adapted to the arrows.

Lemma 4.1.51: Sink Orderings

Let Q be a finite quiver with n vertices and without oriented cycles (definition 4.1.3). Then the vertices of Q can be listed as $i_1,\dots,i_n$ so that, with $Q_0 = Q$ and $Q_k = s_{i_k}^+ Q_{k-1}$, each i_k is a sink of Q_{k-1}, and then $Q_n = Q$. Moreover, a finite quiver whose underlying graph has no loops, no multiple edges and no cycles has no oriented cycles.

***Proof*:**
Step 1: an ordering in which every arrow points backward. The vertices can be listed so that $t(e)$ comes before $s(e)$ for every arrow e. Induct on n, the case $n = 0$ being empty. For $n \geq 1$, Q has a sink: otherwise, starting at any vertex and repeatedly following an arrow out of the current vertex produces vertices $w_0, w_1, \dots, w_n$ with an arrow from each w_{l-1} to w_l. Two of these $n+1$ vertices coincide, $w_a = w_b$ with $a < b$, and the arrows from w_a to w_b form an oriented cycle. Let

i_1 be a sink. The quiver obtained by deleting i_1 and its arrows has no oriented cycles, so by induction its vertices can be listed as $i_2, \dots, i_n$ with the required property. An arrow of Q at i_1 ends at i_1, which comes first, and every other arrow of Q is an arrow of the smaller quiver.

Step 2: each i_k is a sink of Q_{k-1}. List the vertices as in Step 1. Reflecting at a vertex reverses exactly the arrows at that vertex, and Q has no loops, since a loop is an oriented cycle. So an arrow e of Q appears in Q_{k-1} reversed once for each of its endpoints among $i_1, \dots, i_{k-1}$. Let e be an arrow at i_k. If $t(e) = i_k$, then $s(e)$ comes after i_k, so e has not been reversed and ends at i_k. If $s(e) = i_k$, then $t(e) = i_l$ with $l < k$, so e has been reversed once and, in Q_{k-1}, ends at i_k. Hence i_k is a sink of Q_{k-1}. After all n steps every arrow has been reversed once at each of its two endpoints, so $Q_n = Q$.

Step 3: underlying graphs without cycles. Suppose Q has an oriented cycle, and let $\gamma = e_m \cdots e_1$ be one of minimal length m, with $w_l = s(e_l)$. The vertices $w_1, \dots, w_m$ are distinct: if $w_a = w_b$ with $a < b$, then $e_{b-1} \cdots e_a$ is a shorter oriented cycle. If $m = 1$, then e_1 is a loop. If $m = 2$, then e_1 and e_2 are two arrows between w_1 and w_2, a multiple edge of the underlying graph. If $m \geq 3$, then $(w_1, \dots, w_m)$ is a cycle of the underlying graph, completing the proof.

Definition 4.1.52: Coxeter Functor

Let Q be a finite quiver without oriented cycles, and let $i_1, \dots, i_n$ be an ordering of its vertices as in lemma 4.1.51. The *Coxeter functor* is the composite

$$C^+ = S^+_{i_n} \circ \cdots \circ S^+_{i_1} \colon \mathrm{Rep}(Q) \to \mathrm{Rep}(Q)$$

in which $S^+_{i_k}$ is the reflection functor $\mathrm{Rep}(Q_{k-1}) \to \mathrm{Rep}(Q_k)$ at the sink i_k of Q_{k-1}. The corresponding Coxeter element of the underlying graph is $c = s_{i_n} \cdots s_{i_1}$ (definition 4.1.33).

When the underlying graph is of type A, D or E, proposition 4.1.49(5) and (7) show that, as long as no intermediate representation is simple at the vertex being reflected, C^+ carries a nonzero indecomposable representation to a nonzero indecomposable one and acts on its dimension vector by c. By lemma 4.1.37, the iterates of c eventually produce a vector that is not ≥ 0, so iterating C^+ must reach a representation that is simple at the vertex being reflected.

Lemma 4.1.53: Polynomials Over An Infinite Field

Let $\mathbb{F}$ be an infinite field and $N \geq 0$.

1. Every nonzero $f \in \mathbb{F}[x_1, \dots, x_N]$ satisfies $f(a) \neq 0$ for some $a \in \mathbb{F}^N$.
2. If $X_1, \dots, X_m \subseteq \mathbb{F}^N$ are subsets with $X_1 \cup \cdots \cup X_m = \mathbb{F}^N$, then for some j the only polynomial in $\mathbb{F}[x_1, \dots, x_N]$ vanishing at every point of X_j is 0.

***Proof*:**

1. Induct on N. For $N = 0$, f is a nonzero constant. For $N \geq 1$, write $f = \sum_{l=0}^{m} f_l x_N^l$ with $f_l \in \mathbb{F}[x_1, \dots, x_{N-1}]$ and $f_m \neq 0$. By induction $f_m(a') \neq 0$ for some $a' \in \mathbb{F}^{N-1}$, so $g(x_N) = f(a', x_N) = \sum_l f_l(a') x_N^l$ is a polynomial of degree m in one variable. It has at most m roots by [4, Maximum Number Of Roots], and $\mathbb{F}$ is infinite, so $g(a_N) \neq 0$ for some $a_N \in \mathbb{F}$, and $a = (a', a_N)$ satisfies $f(a) \neq 0$.

2. Otherwise choose, for each j, a nonzero polynomial f_j vanishing on X_j. The ring $\mathbb{F}[x_1,\dots,x_N] = \mathbb{F}[x_1,\dots,x_{N-1}][x_N]$ is an integral domain, by [4, Degree Of Polynomials] applied once for each variable, so $f = f_1 \cdots f_m$ is nonzero. It vanishes on $X_1 \cup \cdots \cup X_m = \mathbb{F}^N$, against (1), completing the proof.

***Proof of theorem 4.1.46*:**
Let I be the vertex set of Q and $n = |I|$.

1. Let Γ be one of A_n, D_n, E_6, E_7, E_8. By theorem 4.1.32 its Tits form is positive definite, so proposition 4.1.35 and lemma 4.1.37 apply to Γ. The graph Γ is a path or a star, so it has no loops, no multiple edges and no cycles, and by lemma 4.1.51 the quiver Q has no oriented cycles and its vertices can be listed as $i_1,\dots,i_n$ with each i_k a sink of Q_{k-1} and $Q_n = Q$. Extend the list periodically: for $t \geq 1$ let $j_t = i_l$, where $1 \leq l \leq n$ and $t \equiv l \pmod n$. Put $Q^{(0)} = Q$ and $Q^{(t)} = s^+_{j_t} Q^{(t-1)}$. Since $Q_n = Q$, the quiver $Q^{(t)}$ is Q_l for $t \equiv l \pmod n$ (with $Q_0 = Q_n$), so j_{t+1} is a sink of $Q^{(t)}$ for every $t \geq 0$, and every $Q^{(t)}$ has underlying graph Γ. For $t = mn$, the product $s_{j_t} \cdots s_{j_1}$ is c^m, where $c = s_{i_n} \cdots s_{i_1}$. A representation U of some $Q^{(t)}$ with $\dim U = \alpha_j$ is isomorphic to $S(j)$: every arrow has a zero space at one of its ends, since Γ has no loops, so all maps of U are zero.

   Step 1: reflecting an indecomposable. Let V be a nonzero finite-dimensional indecomposable representation of Q, and $d = \dim_Q V$. Put $V^{(0)} = V$, and, as long as $V^{(t-1)}$ is not simple at j_t, put $V^{(t)} = S^+_{j_t}(V^{(t-1)})$, a representation of $Q^{(t)}$. Suppose that $V^{(t-1)}$ is not simple at j_t for $1 \leq t \leq N$. Then, for each such t, $V^{(t)}$ is nonzero, indecomposable and finite-dimensional, $V^{(t-1)} \cong S^-_{j_t}(V^{(t)})$, and $\dim V^{(t)} = s_{j_t} \cdots s_{j_1}(d)$. If this holds up to $t-1$, then proposition 4.1.49(5), at the sink j_t of $Q^{(t-1)}$, gives $V^{(t-1)} \in \mathrm{Rep}^{j_t,-}(Q^{(t-1)})$, shows that $V^{(t)}$ is nonzero and indecomposable, and gives $V^{(t-1)} \cong S^-_{j_t} S^+_{j_t}(V^{(t-1)}) = S^-_{j_t}(V^{(t)})$. The space $V^{(t)}(j_t)$ is the kernel of a map out of a finite-dimensional space, and proposition 4.1.49(7) gives $\dim V^{(t)} = s_{j_t}(\dim V^{(t-1)}) = s_{j_t} \cdots s_{j_1}(d)$. By lemma 4.1.37, choose $m \geq 0$ with $c^m(d) \not\geq 0$. The supposition fails for $N = mn$, since $\dim V^{(mn)} = c^m(d)$ would then be a dimension vector. Hence there is a least $T \geq 1$ such that $V^{(T-1)}$ is simple at j_T.

   Step 2: d is a positive root and determines V. By Step 1 with $N = T-1$, and since a functor carries isomorphic representations to isomorphic ones,

$$V \cong S^-_{j_1}(V^{(1)}) \cong S^-_{j_1} S^-_{j_2}(V^{(2)}) \cong \cdots \cong S^-_{j_1} \cdots S^-_{j_{T-1}}(V^{(T-1)}) \cong S^-_{j_1} \cdots S^-_{j_{T-1}}(S(j_T))$$

   where $S(j_T)$ is the simple representation of $Q^{(T-1)}$ at j_T. Also $\alpha_{j_T} = \dim V^{(T-1)} = s_{j_{T-1}} \cdots s_{j_1}(d)$, and each s_j is its own inverse, so $d = s_{j_1} \cdots s_{j_{T-1}}(\alpha_{j_T})$. By proposition 4.1.35(1) and (4), α_{j_T} is a root and W_Γ permutes the roots, so d is a root, and $d \geq 0$, so $d \in \Phi^+_\Gamma$. Thus $\dim\colon \mathrm{Ind}(Q) \to \Phi^+_\Gamma$ is well defined. Moreover T is the least $t \geq 1$ with $s_{j_{t-1}} \cdots s_{j_1}(d) = \alpha_{j_t}$: for $t < T$ this vector is the dimension vector of $V^{(t-1)}$, which is not simple at j_t, and is therefore different from α_{j_t}, while for $t = T$ the equality holds. So T, and with it the representation $S^-_{j_1} \cdots S^-_{j_{T-1}}(S(j_T))$, depends only on d. Two nonzero finite-dimensional indecomposable representations with the same dimension vector are therefore isomorphic, and $\dim$ is injective on $\mathrm{Ind}(Q)$.

   Step 3: every positive root occurs. Let $\alpha \in \Phi^+_\Gamma$. By lemma 4.1.37, $c^m(\alpha) \not\geq 0$ for some m, so there is a least $T \geq 1$ with $s_{j_T} \cdots s_{j_1}(\alpha) \not\geq 0$. The vector $\beta = s_{j_{T-1}} \cdots s_{j_1}(\alpha)$ is

a root (proposition 4.1.35(4)) and $\beta \geq 0$ by minimality. The root $s_{j_T}(\beta)$ is not ≥ 0, so it is negative (proposition 4.1.35(2)). It agrees with β away from j_T (lemma 4.1.34(1)), so β vanishes away from j_T, $\beta = b\alpha_{j_T}$ with $b \geq 0$, and $q(\beta) = b^2 = 1$ gives $\beta = \alpha_{j_T}$. Hence $\alpha = s_{j_1} \cdots s_{j_{T-1}}(\alpha_{j_T})$. For $1 \leq t \leq T$ put $\beta_t = s_{j_t} \cdots s_{j_{T-1}}(\alpha_{j_T})$, so that $\beta_T = \alpha_{j_T}$, $\beta_1 = \alpha$, and $\beta_t = s_{j_{t-1}} \cdots s_{j_1}(\alpha) \geq 0$ by minimality of T. Put $U_T = S(j_T) \in \mathrm{Rep}(Q^{(T-1)})$ and $U_t = S^-_{j_t}(U_{t+1})$ for $t = T-1, \ldots, 1$, which is defined because j_t is a source of $Q^{(t)} = s^+_{j_t} Q^{(t-1)}$ and $s^-_{j_t} Q^{(t)} = Q^{(t-1)}$. We show by downward induction that U_t is nonzero and indecomposable with $\dim U_t = \beta_t$. This holds for $t = T$. If it holds for $t+1 \leq T$, then U_{t+1} is not simple at j_t, since otherwise $\beta_{t+1} = \alpha_{j_t}$ and $\beta_t = s_{j_t}(\alpha_{j_t}) = -\alpha_{j_t}$ would not be ≥ 0. By proposition 4.1.49(5) at the source j_t of $Q^{(t)}$, $U_{t+1} \in \mathrm{Rep}^{j_t,+}(Q^{(t)})$ and U_t is nonzero and indecomposable, and by proposition 4.1.49(7), $\dim U_t = s_{j_t}(\beta_{t+1}) = \beta_t$. So U_1 is a nonzero finite-dimensional indecomposable representation of $Q^{(0)} = Q$ with dimension vector α, and dim is surjective.

Step 4: conclusion. The map $\dim\colon \mathrm{Ind}(Q) \to \Phi^+_\Gamma$ is a bijection, and Φ^+_Γ is finite by proposition 4.1.35(3), so Q is of finite type over $\mathbb{F}$. This proves (1).

2. Let $\mathbb{F}$ be infinite and Q of finite type over $\mathbb{F}$. Let $\underline{d} \in \mathbb{Z}^I$ with $\underline{d} \geq 0$ and $\underline{d} \neq 0$. By theorem 4.1.32, applied to the connected graph Γ, it suffices to show $q(\underline{d}) \geq 1$.

   Step 1: finitely many isomorphism classes of dimension vector $\underline{d}$. Let V be a representation with $\dim_Q V = \underline{d}$. By theorem 4.1.24, V is a direct sum of indecomposable subrepresentations, and discarding the zero summands, $V = V_1 \oplus \cdots \oplus V_r$ with every V_l nonzero, finite-dimensional and indecomposable, so $[V_l] \in \mathrm{Ind}(Q)$. Dimension vectors add, so $\sum_l \dim_Q V_l = \underline{d}$, and each V_l has total dimension at least 1, so $r \leq |\underline{d}| = \sum_i d_i$. A direct sum of isomorphisms is an isomorphism, so the isomorphism class of V is determined by the list $([V_1], \ldots, [V_r])$, and there are at most $\sum_{r=0}^{|\underline{d}|} |\mathrm{Ind}(Q)|^r$ such lists.

   Step 2: an orbit on which no nonzero polynomial vanishes. By theorem 4.1.20, the $G(\underline{d})$-orbits in $\mathrm{Rep}(Q, \underline{d})$ correspond bijectively to these isomorphism classes, so there are finitely many orbits $O_1, \ldots, O_m$, and they cover $\mathrm{Rep}(Q, \underline{d})$. Identify $\mathrm{Rep}(Q, \underline{d})$ with $\mathbb{F}^N$, where $N = \sum_e d_{s(e)} d_{t(e)}$, through the matrix entries, with coordinates $x_1, \ldots, x_N$. By lemma 4.1.53(2), some orbit $O = G(\underline{d}) \cdot A_0$ has the property that the only polynomial in $x_1, \ldots, x_N$ vanishing at every point of O is 0.

   Step 3: the orbit map. Let $M = \sum_i d_i^2$, let $\mathbb{F}[g]$ be the polynomial ring in M variables, one for each entry of each matrix g_i with $d_i \geq 1$, and identify a point of $\mathbb{F}^M$ with a tuple of matrices $a = (a_i)_i$. Let $D = \prod_i \det(g_i) \in \mathbb{F}[g]$, the product over the vertices with $d_i \geq 1$, which is a polynomial by the formula for the determinant ([4, Determinant]) and is nonzero because it takes the value 1 at the identity matrices. The group $G(\underline{d})$ is the set of points a with $D(a) \neq 0$. For such a, $a_i^{-1} = \det(a_i)^{-1} \operatorname{adj}(a_i)$ by [4, Invertibility And Determinants], and the entries of $\operatorname{adj}(a_i)$ are determinants of matrices whose entries are entries of a_i, 0 or 1 ([4, Adjugate Matrix]), hence values of fixed polynomials at a. Multiplying numerator and denominator by $\prod_{i \neq s(e)} \det(a_i)$, each coordinate of $(a \cdot A_0)_e = a_{t(e)} (A_0)_e a_{s(e)}^{-1}$ is $P(a)/D(a)$ for a polynomial $P \in \mathbb{F}[g]$ that does not depend on a. Let $\varphi_1, \ldots, \varphi_N$ be the resulting fractions P/D in the field of fractions $\mathbb{F}(g)$ of $\mathbb{F}[g]$, and let $R = \left\{ P/D^l \;\middle|\; P \in \mathbb{F}[g],\ l \geq 0 \right\} \subseteq \mathbb{F}(g)$. For $a \in G(\underline{d})$, evaluation $P/D^l \mapsto P(a)/D(a)^l$ is a well-defined ring homomorphism $\mathrm{ev}_a\colon R \to \mathbb{F}$. It is well defined, since $P_1/D^{l_1} = P_2/D^{l_2}$ gives $P_1 D^{l_2} = P_2 D^{l_1}$, and evaluating at a gives equal values. It respects sums and products, since evaluation at a respects them on $\mathbb{F}[g]$ and $D(a) \neq 0$, and $\mathrm{ev}_a(\varphi_j) = x_j(a \cdot A_0)$. Substituting φ_j for x_j defines an $\mathbb{F}$-algebra homomorphism $\Lambda\colon \mathbb{F}[x_1, \ldots, x_N] \to R$ ([4, Universal Property of Polynomial

Rings], applied once for each variable). For $a \in G(\underline{d})$, the two $\mathbb{F}$-algebra homomorphisms $f \mapsto \mathrm{ev}_a(\Lambda(f))$ and $f \mapsto f(a \cdot A_0)$ agree on $x_1, \ldots, x_N$, so by the uniqueness in the same universal property, $\mathrm{ev}_a(\Lambda(f)) = f(a \cdot A_0)$ for every polynomial f.

Step 4: one entry normalized. Choose a vertex i_0 with $d_{i_0} \geq 1$, let $z \in \mathbb{F}[g]$ be the variable for the $(1,1)$ entry of g_{i_0}, and let $\mathbb{F}[y]$ be the polynomial ring in the other $M-1$ variables. Substituting 1 for z is an $\mathbb{F}$-algebra homomorphism $\sigma\colon \mathbb{F}[g] \to \mathbb{F}[y]$, and $\sigma(D) \neq 0$ because $\sigma(D)$ takes the value 1 at the entries of the identity matrices. As in Step 3, σ extends to a ring homomorphism $R \to \mathbb{F}(y)$, $P/D^l \mapsto \sigma(P)/\sigma(D)^l$, and $\Theta = \sigma \circ \Lambda\colon \mathbb{F}[x_1, \ldots, x_N] \to \mathbb{F}(y)$ is an $\mathbb{F}$-algebra homomorphism. We show that Θ is injective. Let $\Theta(f) = 0$ and write $\Lambda(f) = P/D^l$, so that $\sigma(P) = 0$. Let $a \in G(\underline{d})$ have $(1,1)$ entry $c = z(a) \neq 0$ in a_{i_0}. Then $c^{-1}a \in G(\underline{d})$ has $(1,1)$ entry 1 in its component at i_0, and $(c^{-1}a) \cdot A_0 = a \cdot A_0$ because the scalars cancel in $a_{t(e)}(A_0)_e a_{s(e)}^{-1}$. At a point where $z = 1$, the value of P is the value of $\sigma(P) = 0$, so

$$f(a \cdot A_0) = f((c^{-1}a) \cdot A_0) = \mathrm{ev}_{c^{-1}a}(\Lambda(f)) = \frac{P(c^{-1}a)}{D(c^{-1}a)^l} = 0$$

Hence $P(a) = D(a)^l \,\mathrm{ev}_a(\Lambda(f)) = D(a)^l f(a \cdot A_0) = 0$ for every $a \in \mathbb{F}^M$ with $D(a)z(a) \neq 0$, so the polynomial PDz vanishes at every point of $\mathbb{F}^M$. By lemma 4.1.53(1) it is 0, and since $\mathbb{F}[g]$ is an integral domain and D and z are nonzero, $P = 0$. Then $\Lambda(f) = 0$, so $f(a \cdot A_0) = \mathrm{ev}_a(\Lambda(f)) = 0$ for every $a \in G(\underline{d})$. Thus f vanishes on O, and $f = 0$ by Step 2.

Step 5: the count. By Step 4, a polynomial f with $f(\Theta(x_1), \ldots, \Theta(x_N)) = \Theta(f) = 0$ is zero, so $\Theta(x_1), \ldots, \Theta(x_N)$ are N distinct elements of $\mathbb{F}(y)$, algebraically independent over $\mathbb{F}$. By [4, Existence of a Transcendence Basis], applied to the extension $\mathbb{F}(y)/\mathbb{F}$ with $S = \mathbb{F}(y)$ and $A = \{\Theta(x_1), \ldots, \Theta(x_N)\}$, they lie in a transcendence basis B of $\mathbb{F}(y)/\mathbb{F}$. The $M-1$ variables y are algebraically independent and generate $\mathbb{F}(y)$, so they form a transcendence basis ([4, Transcendence Basis]), and $|B| = M-1$ by [4, Transcendence Degree Is Well-Defined]. Hence $N \leq M-1$, that is,

$$q(\underline{d}) = \sum_{i \in I} d_i^2 - \sum_{e \in E} d_{s(e)} d_{t(e)} = M - N \geq 1$$

As $\underline{d}$ was an arbitrary nonzero vector with $\underline{d} \geq 0$, condition (3) of theorem 4.1.32 holds, and Γ is one of A_n, D_n, E_6, E_7, E_8, completing the proof.

Example 4.1.54: Indecomposable Representations Of A Linearly Oriented A_n

Let Q be $1 \to 2 \to \cdots \to n$. For $1 \leq i \leq j \leq n$, let $V_{[i,j]}$ be the representation with $V_{[i,j]}(l) = \mathbb{F}$ for $i \leq l \leq j$ and $V_{[i,j]}(l) = 0$ otherwise, the identity along each arrow $l \to l+1$ with $i \leq l < j$, and zero maps elsewhere. Its dimension vector is $\alpha_i + \cdots + \alpha_j$.

Each $V_{[i,j]}$ is indecomposable. Suppose $V_{[i,j]}(l) = U(l) \oplus U'(l)$ at every vertex, for subrepresentations U and U'. For $i \leq l \leq j$ the space $\mathbb{F}$ is one-dimensional, so exactly one of $U(l)$ and $U'(l)$ is $\mathbb{F}$. If $U(l) = \mathbb{F}$ and $l < j$, then $U(l+1)$ contains the image of $U(l)$ under the identity, so $U(l+1) = \mathbb{F}$. If $U(l) = \mathbb{F}$ and $l > i$, then $U'(l-1)$ maps injectively into $U'(l) = 0$, so $U'(l-1) = 0$ and $U(l-1) = \mathbb{F}$. Hence $U(l) = \mathbb{F}$ for all $i \leq l \leq j$ or for none of them, and one of U, U' is zero.

By theorem 4.1.46(1) and example 4.1.36, the dimension vectors of the nonzero finite-dimensional indecomposable representations of Q are exactly the vectors $\alpha_i + \cdots + \alpha_j$, each realized by a

single isomorphism class. So the representations $V_{[i,j]}$, for $1 \leq i \leq j \leq n$, are up to isomorphism all the nonzero finite-dimensional indecomposable representations of Q, over every field, and by theorem 4.1.24 every finite-dimensional representation of Q is isomorphic to a direct sum of them. For $n = 3$ these are the six representations, in which the unlabelled arrows are zero maps,

$$\begin{array}{ll} \mathbb{F} \longrightarrow 0 \longrightarrow 0 & \mathbb{F} \xrightarrow{1} \mathbb{F} \longrightarrow 0 \\ 0 \longrightarrow \mathbb{F} \longrightarrow 0 & 0 \longrightarrow \mathbb{F} \xrightarrow{1} \mathbb{F} \\ 0 \longrightarrow 0 \longrightarrow \mathbb{F} & \mathbb{F} \xrightarrow{1} \mathbb{F} \xrightarrow{1} \mathbb{F} \end{array}$$

with dimension vectors the six positive roots of A_3.

4.1.5 Path Algebra of Quivers

A representation of a quiver assigns a linear map to each arrow, and composing these maps along a path gives a linear map from the space at its source to the space at its target. The path algebra has the paths of Q as a basis and concatenation of paths as multiplication, and theorem 4.1.58 below identifies the representations of Q with the left modules over it. Throughout, $\mathbb{F}$ is a field, arrows are denoted $a, b, \ldots$, and the letter e_i is reserved for the trivial path at the vertex i.

Two paths p and q with $t(q) = s(p)$ have a *concatenation* pq, the path from $s(q)$ to $t(p)$ that traverses the arrows of q and then those of p. If $p = a_k \cdots a_1$ and $q = b_m \cdots b_1$ have positive length, then $pq = a_k \cdots a_1 b_m \cdots b_1$, which is a path because $t(b_m) = t(q) = s(p) = s(a_1)$. A trivial path leaves the other factor unchanged, $pe_{s(p)} = p$ and $e_{t(q)}q = q$. The length of pq is the sum of the lengths of p and q, and concatenation is associative: if also $t(r) = s(q)$, then $(pq)r$ and $p(qr)$ are both the path from $s(r)$ that traverses the arrows of r, then those of q, then those of p.

Definition 4.1.55: Path Algebra

Let $Q = (I, E, s, t)$ be a quiver. The *path algebra* $\mathbb{F}Q$ of Q over $\mathbb{F}$ is the $\mathbb{F}$-vector space with basis the set of all paths in Q (definition 4.1.3), the trivial paths e_i for $i \in I$ included, with the $\mathbb{F}$-bilinear multiplication given on paths p and q by

$$p \cdot q = \begin{cases} pq & \text{if } t(q) = s(p) \\ 0 & \text{otherwise} \end{cases}$$

So $p \cdot q$ traverses q first and then p.

Proposition 4.1.56: Structure Of The Path Algebra

Let $Q = (I, E, s, t)$ be a quiver, and for $n \geq 0$ let $(\mathbb{F}Q)_n \subseteq \mathbb{F}Q$ be the span of the paths of length n.

1. The multiplication of $\mathbb{F}Q$ is associative.
2. The trivial paths are orthogonal idempotents, $e_i \cdot e_j = \delta_{ij} e_i$ for $i, j \in I$. For a path γ and a vertex j, $e_j \cdot \gamma$ is γ if $t(\gamma) = j$ and 0 otherwise, and $\gamma \cdot e_j$ is γ if $s(\gamma) = j$ and 0 otherwise. Consequently $(\mathbb{F}Q)e_i$ has basis the paths with source i, and $e_j(\mathbb{F}Q)e_i$ has basis the paths from i to j.
3. If I is finite, then $1 = \sum_{i \in I} e_i$ is an identity element of $\mathbb{F}Q$, so $\mathbb{F}Q$ is an associative $\mathbb{F}$-algebra with 1.
4. $\mathbb{F}Q = \bigoplus_{n \geq 0} (\mathbb{F}Q)_n$, the trivial paths form a basis of $(\mathbb{F}Q)_0$, and $(\mathbb{F}Q)_m \cdot (\mathbb{F}Q)_n \subseteq (\mathbb{F}Q)_{m+n}$ for all $m, n \geq 0$.
5. $\mathbb{F}Q$ is finite-dimensional if and only if Q is finite and has no oriented cycle.

***Proof*:**

1. Both $(x \cdot y) \cdot z$ and $x \cdot (y \cdot z)$ are trilinear in x, y, z, so it suffices to compare them on paths p, q, r. Recall that pq has source $s(q)$ and target $t(p)$. If $t(r) = s(q)$ and $t(q) = s(p)$, then $(p \cdot q) \cdot r = (pq)r$, because $t(r) = s(q) = s(pq)$, and $p \cdot (q \cdot r) = p(qr)$, because $t(qr) = t(q) = s(p)$, and these agree by the associativity of concatenation. If $t(q) \neq s(p)$, then $p \cdot q = 0$, and $q \cdot r$ is 0 or the path qr with target $t(q) \neq s(p)$, so both sides vanish. If $t(q) = s(p)$ and $t(r) \neq s(q)$, then $q \cdot r = 0$ and $(p \cdot q) \cdot r = (pq) \cdot r = 0$, because $s(pq) = s(q) \neq t(r)$.
2. The product $e_i \cdot e_j$ is nonzero only if $t(e_j) = s(e_i)$, that is, $j = i$, and $e_i e_i = e_i$. The product $e_j \cdot \gamma$ is nonzero only if $t(\gamma) = s(e_j) = j$, and then it is $e_j \gamma = \gamma$. The product $\gamma \cdot e_j$ is nonzero only if $s(\gamma) = t(e_j) = j$, and then it is $\gamma e_j = \gamma$. Hence for $x = \sum_\gamma c_\gamma \gamma \in \mathbb{F}Q$, the element $x \cdot e_i$ is the part of this sum over the paths with source i, and $e_j \cdot x \cdot e_i$ is the part over the paths from i to j. These paths are linearly independent, being part of a basis, which gives the two bases.
3. For a path γ, part (2) gives $(\sum_i e_i) \cdot \gamma = e_{t(\gamma)} \cdot \gamma = \gamma$ and $\gamma \cdot (\sum_i e_i) = \gamma \cdot e_{s(\gamma)} = \gamma$. By bilinearity $\sum_i e_i$ is a two-sided identity element.
4. Every path has exactly one length and the paths form a basis, so $\mathbb{F}Q = \bigoplus_n (\mathbb{F}Q)_n$, and the paths of length 0 are the trivial paths. The product of a path of length m and a path of length n is 0 or their concatenation, which has length $m + n$.
5. Suppose Q is finite with no oriented cycle, and let $\gamma = a_k \cdots a_1$ be a path of positive length. The vertices $v_0 = s(a_1)$ and $v_l = t(a_l)$ for $1 \leq l \leq k$ are pairwise distinct: if $v_p = v_q$ with $p < q$, then $a_q \cdots a_{p+1}$ is a path of positive length from v_p to $v_q = v_p$, an oriented cycle. So $k + 1 \leq |I|$. There are at most $|E|^k$ sequences of k arrows, so Q has at most $|I| + \sum_{k=1}^{|I|-1} |E|^k$ paths, and $\mathbb{F}Q$ is finite-dimensional. Conversely, if I is infinite then the trivial paths are infinitely many basis elements of $\mathbb{F}Q$, and if E is infinite then so are

the paths of length 1. If γ is an oriented cycle, of length $k \geq 1$, then $t(\gamma) = s(\gamma)$, so the products γ, $\gamma \cdot \gamma$, $\gamma \cdot \gamma \cdot \gamma, \ldots$ are paths of the distinct lengths $k, 2k, 3k, \ldots$, and again $\mathbb{F}Q$ is infinite-dimensional.

The five parts are proved, completing the proof.

Example 4.1.57: Path Algebras

1. Let Q be the quiver with one vertex 1 and one loop a. Its paths are e_1 and, for $n \geq 1$, the path a^n that traverses a n times. With $a^0 = e_1$, the product is $a^m \cdot a^n = a^{m+n}$ for all $m, n \geq 0$. The linear map $\mathbb{F}[x] \to \mathbb{F}Q$ with $x^n \mapsto a^n$ carries a basis to a basis and products of basis elements to products, so it is an isomorphism of algebras $\mathbb{F}[x] \cong \mathbb{F}Q$. The loop is an oriented cycle, and $\mathbb{F}Q$ is infinite-dimensional, in accordance with proposition 4.1.56(5).
2. With two loops a and b at one vertex, the paths are e_1 and the words in a and b, multiplied by juxtaposition. The products $a \cdot b = ab$ and $b \cdot a = ba$ are different paths, so $\mathbb{F}Q$ is not commutative.
3. Let K be the Kronecker quiver of example 4.1.28, with two arrows $a, b \colon 1 \to 2$. No arrow has source 2, so there is no path of length 2, and $\mathbb{F}K$ has basis e_1, e_2, a, b. The nonzero products of basis elements are $e_1 \cdot e_1 = e_1$, $e_2 \cdot e_2 = e_2$, $a \cdot e_1 = e_2 \cdot a = a$ and $b \cdot e_1 = e_2 \cdot b = b$.

For a representation V of Q and a path γ from i to j, let $T_\gamma \colon V(i) \to V(j)$ be $\mathrm{id}_{V(i)}$ if $\gamma = e_i$, and the composite $T_{a_k} \circ \cdots \circ T_{a_1}$ if $\gamma = a_k \cdots a_1$, writing T_γ^V when the representation must be named. If $t(q) = s(p)$, then $T_{pq} = T_p \circ T_q$, since both sides apply the maps of the arrows of q and then those of p. A quiver morphism $\varphi \colon V \to W$ satisfies

$$\varphi(j) \circ T_\gamma^V = T_\gamma^W \circ \varphi(i)$$

for every path γ from i to j, where T_a^W is the map written S_a in definition 4.1.9. For $\gamma = e_i$ both sides are $\varphi(i)$. For $\gamma = a_k \cdots a_1$ with $k \geq 1$, write $\gamma = a_k \gamma'$, where γ' is the path $a_{k-1} \cdots a_1$, or e_i if $k = 1$, from i to $l = s(a_k)$. The square of φ at a_k and induction on k give

$$\varphi(j) \circ T_{a_k}^V \circ T_{\gamma'}^V = T_{a_k}^W \circ \varphi(l) \circ T_{\gamma'}^V = T_{a_k}^W \circ T_{\gamma'}^W \circ \varphi(i)$$

and the outer terms are $\varphi(j) \circ T_\gamma^V$ and $T_\gamma^W \circ \varphi(i)$.

When Q has finitely many vertices, $\mathbb{F}Q$ has the identity element $1 = \sum_i e_i$ of proposition 4.1.56(3). Let $\mathbb{F}Q$**-Mod** denote the category of left $\mathbb{F}Q$-modules, on which 1 acts as the identity, with the $\mathbb{F}Q$-linear maps as morphisms. An $\mathbb{F}Q$-module is an $\mathbb{F}$-vector space through the action of the scalars $c = c \cdot 1 \in \mathbb{F}Q$.

Theorem 4.1.58: Representations Are Modules Over The Path Algebra

Let $Q = (I, E, s, t)$ be a quiver with finitely many vertices.

1. For a representation V of Q, let $\mathcal{M}(V) = \bigoplus_{i \in I} V(i)$, with $\iota_j \colon V(j) \to \mathcal{M}(V)$ the inclusion of the summand $V(j)$ and $\pi_j \colon \mathcal{M}(V) \to V(j)$ the projection onto it. Then $\mathcal{M}(V)$ is a left $\mathbb{F}Q$-module in which a path γ from i to j acts by $\iota_j \circ T_\gamma \circ \pi_i$. For a quiver morphism $\varphi \colon V \to W$, the map $\mathcal{M}(\varphi) = \bigoplus_i \varphi(i)$ is $\mathbb{F}Q$-linear, and $\mathcal{M}$ is a functor $\mathrm{Rep}(Q) \to \mathbb{F}Q$**-Mod**.

2. For a left $\mathbb{F}Q$-module M, let $\mathcal{R}(M)$ have the space $e_i M$ at each vertex i and the map $e_i M \to e_j M$, $m \mapsto a \cdot m$, along each arrow $a \colon i \to j$. For an $\mathbb{F}Q$-linear map $f \colon M \to N$, let $\mathcal{R}(f)(i) \colon e_i M \to e_i N$ be the restriction of f. Then $\mathcal{R}(M)$ is a representation of Q, $\mathcal{R}(f)$ is a quiver morphism, and $\mathcal{R}$ is a functor $\mathbb{F}Q$**-Mod** $\to \mathrm{Rep}(Q)$.

3. For every representation V, the maps $\eta_V(i) = \iota_i \colon V(i) \to e_i \mathcal{M}(V)$ form an isomorphism of representations $\eta_V \colon V \to \mathcal{R}(\mathcal{M}(V))$, natural in V.

4. For every left $\mathbb{F}Q$-module M, the map $\epsilon_M \colon \mathcal{M}(\mathcal{R}(M)) = \bigoplus_i e_i M \to M$, $(m_i)_{i \in I} \mapsto \sum_i m_i$, is an isomorphism of $\mathbb{F}Q$-modules, natural in M.

Consequently $\mathcal{M}$ and $\mathcal{R}$ are quasi-inverse equivalences of categories between $\mathrm{Rep}(Q)$ and $\mathbb{F}Q$**-Mod**.

***Proof*:**

1. For a path γ from i to j let $\rho(\gamma) = \iota_j \circ T_\gamma \circ \pi_i$, a linear map from $\mathcal{M}(V)$ to itself, and extend linearly to a linear map ρ from $\mathbb{F}Q$ to the algebra $\mathrm{End}_{\mathbb{F}}(\mathcal{M}(V))$ of linear maps from $\mathcal{M}(V)$ to itself. For paths p and q, the composite $\pi_{s(p)} \circ \iota_{t(q)}$ is the identity of $V(t(q))$ if $t(q) = s(p)$ and 0 otherwise. In the first case

$$\rho(p) \circ \rho(q) = \iota_{t(p)} \circ T_p \circ T_q \circ \pi_{s(q)} = \iota_{t(pq)} \circ T_{pq} \circ \pi_{s(pq)} = \rho(p \cdot q)$$

and in the second $\rho(p) \circ \rho(q) = 0 = \rho(p \cdot q)$. By bilinearity $\rho(x \cdot y) = \rho(x) \circ \rho(y)$ for all $x, y \in \mathbb{F}Q$, and $\rho(1) = \sum_i \iota_i \circ \pi_i$ is the identity. These identities, with the linearity of ρ and of each $\rho(x)$, are the axioms of a left module for $x \cdot v = \rho(x)(v)$, and a scalar $c = c \cdot 1$ acts as multiplication by c. For a quiver morphism $\varphi \colon V \to W$ and a path γ from i to j, the identity $\varphi(j) \circ T_\gamma^V = T_\gamma^W \circ \varphi(i)$ proved before the theorem gives

$$\mathcal{M}(\varphi) \circ \iota_j \circ T_\gamma^V \circ \pi_i = \iota_j \circ \varphi(j) \circ T_\gamma^V \circ \pi_i = \iota_j \circ T_\gamma^W \circ \varphi(i) \circ \pi_i = \iota_j \circ T_\gamma^W \circ \pi_i \circ \mathcal{M}(\varphi)$$

so $\mathcal{M}(\varphi)$ commutes with the action of every path, and by linearity with the action of every element of $\mathbb{F}Q$. The identities $\mathcal{M}(\psi \circ \varphi) = \mathcal{M}(\psi) \circ \mathcal{M}(\varphi)$ and $\mathcal{M}(\mathrm{id}_V) = \mathrm{id}_{\mathcal{M}(V)}$ hold summand by summand.

2. The subset $e_i M$ is an $\mathbb{F}$-subspace of M, since $c \cdot e_i m = e_i \cdot (c \cdot m)$ for $c \in \mathbb{F}$. An arrow $a \colon i \to j$ satisfies $a = e_j \cdot a$ by proposition 4.1.56(2), so $a \cdot m = e_j \cdot (a \cdot m)$ lies in $e_j M$ for every $m \in e_i M$, and $m \mapsto a \cdot m$ is a linear map $e_i M \to e_j M$. So $\mathcal{R}(M)$ is a representation. An $\mathbb{F}Q$-linear map f satisfies $f(e_i m) = e_i f(m)$, so it carries $e_i M$ into $e_i N$, and $f(a \cdot m) = a \cdot f(m)$ is the square of $\mathcal{R}(f)$ at a. Restriction preserves composites and identities.

3. The trivial path e_i acts on $\mathcal{M}(V)$ by $\iota_i \circ \pi_i$, so $e_i\mathcal{M}(V) = \iota_i(V(i))$, and $\iota_i \colon V(i) \to e_i\mathcal{M}(V)$ is a linear bijection. For an arrow $a \colon i \to j$ and $v \in V(i)$, $a \cdot \iota_i(v) = \iota_j(T_a v)$, so $\eta_V(j) \circ T_a$ is the map of $\mathcal{R}(\mathcal{M}(V))$ along a composed with $\eta_V(i)$. Hence η_V is a quiver morphism with bijective components, which is an isomorphism, as noted after definition 4.1.13. For a quiver morphism $\varphi \colon V \to W$, the morphism $\mathcal{R}(\mathcal{M}(\varphi))$ is the restriction of $\bigoplus_l \varphi(l)$, and $(\bigoplus_l \varphi(l)) \circ \iota_i = \iota_i \circ \varphi(i)$, that is, $\mathcal{R}(\mathcal{M}(\varphi)) \circ \eta_V = \eta_W \circ \varphi$, which is the naturality of η.

4. Every $m \in M$ satisfies $m = 1 \cdot m = \sum_i e_i m$, so ϵ_M is surjective. If $\sum_i m_i = 0$ with $m_i = e_i m_i'$, then multiplying by e_j and using $e_j e_i = \delta_{ij} e_i$ gives $0 = \sum_i e_j e_i m_i' = m_j$ for every j, so ϵ_M is injective. Let γ be a path from i to j. Its map in $\mathcal{R}(M)$ is $m \mapsto \gamma \cdot m$: for $\gamma = e_i$ it is the identity of $e_i M$, which is $m \mapsto e_i \cdot m$, and for $\gamma = a_k \cdots a_1$ it is $m \mapsto a_k \cdot (\cdots (a_1 \cdot m)) = \gamma \cdot m$, because the product $a_k \cdot \cdots \cdot a_1$ in $\mathbb{F}Q$ is the path γ. So γ sends $(m_l)_l \in \mathcal{M}(\mathcal{R}(M))$ to $\iota_j(\gamma \cdot m_i)$. Each $m_l \in e_l M$ satisfies $m_l = e_l \cdot m_l$, and $\gamma \cdot e_l = 0$ for $l \neq i$ by proposition 4.1.56(2), so

$$\epsilon_M\left(\gamma \cdot (m_l)_l\right) = \gamma \cdot m_i = \sum_l \gamma \cdot e_l \cdot m_l = \gamma \cdot \epsilon_M\left((m_l)_l\right)$$

   By linearity ϵ_M is $\mathbb{F}Q$-linear, and a bijective $\mathbb{F}Q$-linear map is an isomorphism of modules. For an $\mathbb{F}Q$-linear map $f \colon M \to N$, $\epsilon_N\left((f(m_l))_l\right) = \sum_l f(m_l) = f\left(\epsilon_M((m_l)_l)\right)$, which is the naturality of ϵ.

Items (3) and (4) give natural isomorphisms $\mathrm{id}_{\mathrm{Rep}(Q)} \cong \mathcal{R} \circ \mathcal{M}$ and $\mathcal{M} \circ \mathcal{R} \cong \mathrm{id}_{\mathbb{F}Q\textbf{-Mod}}$ ([4, Natural Transformation and Natural Isomorphism]), which by [4, Equivalence of Categories] make $\mathcal{M}$ and $\mathcal{R}$ quasi-inverse equivalences, completing the proof.

Under this equivalence, the submodules of $\mathcal{M}(V)$ are the subspaces $\bigoplus_i \iota_i(W(i))$ for the subrepresentations $W \subseteq V$. Such a subspace is a submodule because T_γ carries $W(i)$ into $W(j)$ for every path γ from i to j: this holds for trivial paths and for arrows by definition 4.1.8, and for longer paths by induction on the length, as T_γ is the composite of the maps along the arrows of γ. Conversely, a submodule N contains $e_i n$ for each of its elements $n = \sum_i e_i n$, so $N = \bigoplus_i e_i N$ with $e_i N \subseteq \iota_i(V(i))$, and the subspaces $W(i) = \iota_i^{-1}(e_i N)$ form a subrepresentation, because $a \cdot e_i N \subseteq e_j N$ for every arrow $a \colon i \to j$. The two recipes are inverse to each other: starting from W, $e_i\big(\bigoplus_j \iota_j(W(j))\big) = \iota_i(W(i))$, so ι_i^{-1} of it is $W(i)$, and starting from N, $\bigoplus_i \iota_i\big(\iota_i^{-1}(e_i N)\big) = \bigoplus_i e_i N = N$. So V is simple exactly when $\mathcal{M}(V)$ is a simple module, and V is the internal direct sum of subrepresentations $U_1, \ldots, U_n$ exactly when $\mathcal{M}(V)$ is the internal direct sum of the corresponding submodules.

By proposition 4.1.56(2), the left ideal $\mathbb{F}Qe_i$ has basis the paths with source i. The representation with the paths from i to j as a basis at each vertex j is defined for every quiver, and when Q has finitely many vertices it corresponds to $\mathbb{F}Qe_i$ under theorem 4.1.58 (lemma 4.1.59(6) below).

Lemma 4.1.59: The Representation Of Paths From A Vertex

Let $Q = (I, E, s, t)$ be a quiver and $i \in I$. For each vertex j let $P_i(j) = e_j(\mathbb{F}Q)e_i$, the span of the paths from i to j, and for each arrow $a\colon j \to k$ let $P_i(j) \to P_i(k)$ be the map $x \mapsto a \cdot x$.

1. These spaces and maps form a representation P_i of Q, and its map along a path γ from j to k is $x \mapsto \gamma \cdot x$.
2. For every representation V of Q, the map $\mathrm{Hom}_Q(P_i, V) \to V(i)$, $\psi \mapsto \psi(i)(e_i)$, is an isomorphism of vector spaces, natural in V in the sense that $(\varphi \circ \psi)(i)(e_i) = \varphi(i)(\psi(i)(e_i))$ for every quiver morphism $\varphi\colon V \to W$. Its inverse sends $v \in V(i)$ to the quiver morphism ψ^v with $\psi^v(j)(\gamma) = T_\gamma(v)$ for every path γ from i to j.
3. P_i is projective: for every quiver morphism $\sigma\colon V \to W$ with every $\sigma(j)$ surjective and every quiver morphism $f\colon P_i \to W$, there is a quiver morphism $g\colon P_i \to V$ with $\sigma \circ g = f$.
4. The only elements $x \in e_i(\mathbb{F}Q)e_i$ with $x \cdot x = x$ are 0 and e_i, and P_i is nonzero and indecomposable.
5. P_i is simple if and only if i is a sink, and then $P_i \cong S(i)$.
6. If I is finite, then $P_i = \mathcal{R}(\mathbb{F}Qe_i)$ and $\mathcal{M}(P_i) \cong \mathbb{F}Qe_i$ as left $\mathbb{F}Q$-modules, for the functors of theorem 4.1.58.

***Proof*:**

1. For an arrow $a\colon j \to k$, proposition 4.1.56(2) gives $a = e_k \cdot a \cdot e_j$, so left multiplication by a carries $e_j(\mathbb{F}Q)e_i$ into $e_k(\mathbb{F}Q)e_i$, and it is linear. For a path $\gamma = a_l \cdots a_1$ from j to k, the map along γ is $x \mapsto a_l \cdot (\cdots (a_1 \cdot x)) = \gamma \cdot x$, since the product of the arrows $a_l, \ldots, a_1$ in $\mathbb{F}Q$ is γ. For $\gamma = e_j$ it is the identity, which is $x \mapsto e_j \cdot x$ on $e_j(\mathbb{F}Q)e_i$.
2. The map $\psi \mapsto \psi(i)(e_i)$ is linear for the vector space structure of proposition 4.1.10. Given $v \in V(i)$, define $\psi^v(j)\colon P_i(j) \to V(j)$ on the basis of paths from i to j by $\psi^v(j)(\gamma) = T_\gamma(v)$, extended linearly. For an arrow $a\colon j \to k$ and a path γ from i to j,
$$\psi^v(k)(a \cdot \gamma) = T_{a\gamma}(v) = T_a(T_\gamma(v)) = T_a\left(\psi^v(j)(\gamma)\right)$$
so ψ^v is a quiver morphism, and $\psi^v(i)(e_i) = T_{e_i}(v) = v$. Hence $\psi \mapsto \psi(i)(e_i)$ is surjective. Now let $\psi \in \mathrm{Hom}_Q(P_i, V)$ and $v = \psi(i)(e_i)$. By part (1), a path γ from i to j is the image $T_\gamma^{P_i}(e_i)$ of e_i under the map of P_i along γ, so the identity $\psi(j) \circ T_\gamma^{P_i} = T_\gamma^V \circ \psi(i)$ for quiver morphisms gives
$$\psi(j)(\gamma) = \psi(j)\left(T_\gamma^{P_i}(e_i)\right) = T_\gamma^V\left(\psi(i)(e_i)\right) = T_\gamma(v) = \psi^v(j)(\gamma)$$
So $\psi = \psi^v$ is determined by v, the map $\psi \mapsto \psi(i)(e_i)$ is injective, and $v \mapsto \psi^v$ is its inverse. The naturality identity is the definition of the composite, $(\varphi \circ \psi)(i) = \varphi(i) \circ \psi(i)$.
3. Let $w = f(i)(e_i)$, and choose $v \in V(i)$ with $\sigma(i)(v) = w$. Then $g = \psi^v$ satisfies $(\sigma \circ g)(i)(e_i) = \sigma(i)(v) = w = f(i)(e_i)$, and the injectivity in part (2), for the representation W, gives $\sigma \circ g = f$.

4. By proposition 4.1.56(2) and (4), an element $x \in e_i(\mathbb{F}Q)e_i$ is a finite sum $x = \sum_{n\geq 0} x_n$, with x_n in the span of the paths of length n from i to i, and $x_0 = ce_i$ for some $c \in \mathbb{F}$, since e_i is the only path of length 0 from i to i. Suppose $x \cdot x = x$. Every product $x_m \cdot x_n$ with $m+n > 0$ lies in $(\mathbb{F}Q)_{m+n}$, so the component of $x \cdot x$ in $(\mathbb{F}Q)_0$ is $x_0 \cdot x_0 = c^2 e_i$, and $c^2 = c$, that is, c is 0 or 1. If $c = 0$ and $x \neq 0$, let $m \geq 1$ be least with $x_m \neq 0$. Every product $x_p \cdot x_q$ with x_p and x_q nonzero lies in $(\mathbb{F}Q)_{p+q}$ with $p + q \geq 2m > m$, so the component of $x \cdot x$ in $(\mathbb{F}Q)_m$ is 0, while that of x is $x_m \neq 0$, which contradicts $x \cdot x = x$. So $c = 0$ forces $x = 0$. If $c = 1$, then $y = e_i - x$ lies in $e_i(\mathbb{F}Q)e_i$, on which e_i is an identity element by proposition 4.1.56(2), so $y \cdot y = e_i - 2x + x \cdot x = y$, and the component of y in $(\mathbb{F}Q)_0$ is 0. By the case $c = 0$, $y = 0$, that is, $x = e_i$.

   The space $P_i(i)$ contains $e_i \neq 0$, so $P_i \neq 0$. Suppose $\theta\colon V_1 \oplus V_2 \to P_i$ is an isomorphism with V_1 and V_2 nonzero (definition 4.1.22). The maps $(v_1, v_2) \mapsto (v_1, 0)$ at the vertices form a quiver morphism ϖ of $V_1 \oplus V_2$ to itself, and $\kappa = \theta \circ \varpi \circ \theta^{-1}\colon P_i \to P_i$ is a quiver morphism with $\kappa \circ \kappa = \kappa$. It is nonzero, because $V_1 \neq 0$, and it is not id_{P_i}, because it vanishes on the nonzero subrepresentation $\theta(0 \oplus V_2)$. Let $x = \kappa(i)(e_i) \in e_i(\mathbb{F}Q)e_i$. By part (2), $\kappa = \psi^x$, and $\psi^x(j)(\gamma) = T_\gamma^{P_i}(x) = \gamma \cdot x$ by part (1), so $\kappa(j)(y) = y \cdot x$ for every $y \in P_i(j)$. Then
   $$x \cdot x = \kappa(i)(x) = (\kappa \circ \kappa)(i)(e_i) = \kappa(i)(e_i) = x$$
   and x is neither 0 nor e_i, because $\psi^0 = 0$ and $\psi^{e_i} = \mathrm{id}_{P_i}$. This contradicts the first paragraph, so P_i is indecomposable.

5. If i is a sink, then no arrow has source i, so the only path from i is e_i, since a path $a_l \cdots a_1$ of positive length from i has $s(a_1) = i$. Then $P_i(i) = \mathbb{F}e_i$, $P_i(j) = 0$ for $j \neq i$, and every map of P_i is zero, so the maps $e_i \mapsto 1$ at i and 0 elsewhere form an isomorphism $P_i \to S(i)$. As P_i is nonzero and its only nonzero space $P_i(i)$ is one-dimensional, its only subrepresentations are 0 and P_i, so P_i is simple. If i is not a sink, choose an arrow $a\colon i \to k$, and let $U(j) \subseteq P_i(j)$ be the span of the paths of positive length from i to j. Left multiplication by an arrow carries such a path to a path of greater length, so U is a subrepresentation. It is nonzero, since $a \in U(k)$, and it is not P_i, since $e_i \notin U(i)$. So P_i is not simple.

6. The representation $\mathcal{R}(\mathbb{F}Qe_i)$ has the space $e_j(\mathbb{F}Qe_i) = P_i(j)$ at each vertex j, and its map along an arrow a is left multiplication by a, so it is P_i. By theorem 4.1.58(4), $\epsilon_{\mathbb{F}Qe_i}$ is an isomorphism from $\mathcal{M}(P_i) = \mathcal{M}(\mathcal{R}(\mathbb{F}Qe_i))$ onto $\mathbb{F}Qe_i$.

The six parts are proved, completing the proof.

Part (4) holds whether or not Q has oriented cycles, although P_i is infinite-dimensional when an oriented cycle passes through i. Part (3) corresponds to the projectivity of the module $\mathbb{F}Qe_i$: when I is finite, every $x \in \mathbb{F}Q$ is $x \cdot e_i + x \cdot (1 - e_i)$, and $e_i \cdot (1 - e_i) = 0$, so $\mathbb{F}Q = \mathbb{F}Qe_i \oplus \mathbb{F}Q(1 - e_i)$ exhibits $\mathbb{F}Qe_i$ as a direct summand of the free module $\mathbb{F}Q$, which makes it projective by [4, Equivalencies for Projective Modules]. If Q is finite without oriented cycles, then $\mathbb{F}Q$ is finite-dimensional by proposition 4.1.56(5), and by part (4) and theorem 2.3.2 each e_i is a primitive idempotent. So $1 = \sum_i e_i$ is a decomposition into primitive orthogonal idempotents, and the left ideals $\mathbb{F}Qe_i \cong \mathcal{M}(P_i)$ are the indecomposable projective modules Ae_i of definition 2.3.3 for $A = \mathbb{F}Q$.

Example 4.1.60: The Representations P_i

1. For $1 \xrightarrow{a} 2$, the paths from 1 are e_1 and a, so $P_1(1) = \mathbb{F}e_1$ and $P_1(2) = \mathbb{F}a$, with $e_1 \mapsto a$ along a, and $P_1 \cong (\mathbb{F} \xrightarrow{1} \mathbb{F})$. The vertex 2 is a sink, and $P_2 \cong S(2) = (0 \to \mathbb{F})$. So the indecomposable representation $\mathbb{F} \xrightarrow{1} \mathbb{F}$ of example 4.1.23 is projective, and it is not simple, as lemma 4.1.59(5) requires of P_1, the vertex 1 not being a sink.
2. For the loop quiver of example 4.1.57(1), $P_1(1) = \mathbb{F}Q \cong \mathbb{F}[x]$, on which the loop acts by multiplication by x. This representation is infinite-dimensional and indecomposable.
3. For the Kronecker quiver K, $P_1(1) = \mathbb{F}e_1$ and $P_1(2) = \mathbb{F}a \oplus \mathbb{F}b$, with $e_1 \mapsto a$ along a and $e_1 \mapsto b$ along b, and $P_2 \cong S(2)$. By lemma 4.1.59(2), $\mathrm{Hom}_K(P_2, P_1) \cong P_1(2)$ is two-dimensional.

Exercise 4.1.1

1. Let K be the Kronecker quiver of example 4.1.28, with arrows $a, b\colon 1 \to 2$. For $\lambda \in \mathbb{F}$ let V_λ be the representation with $V_\lambda(1) = V_\lambda(2) = \mathbb{F}$, $T_a = 1$ and $T_b = \lambda$. Show that every V_λ is indecomposable and that $V_\lambda \cong V_\mu$ only if $\lambda = \mu$. Conclude that K is not of finite type over an infinite field, as theorem 4.1.46 predicts.
2. Let L be the quiver with one vertex and one loop. For $m \geq 1$ let J_m be the representation $\mathbb{F}^m$ whose loop acts by the nilpotent Jordan block N_m, with $N_m\varepsilon_1 = 0$ and $N_m\varepsilon_l = \varepsilon_{l-1}$ for $l \geq 2$. Show that J_m is indecomposable, and deduce that L is not of finite type over any field, finite or infinite. Which extended Dynkin diagram is the underlying graph of L?
3. Let $\mathbb{F}$ be algebraically closed and let L and N_m be as in exercise 4.1.1 (2). For $\lambda \in \mathbb{F}$ and $m \geq 1$ let $J_m(\lambda)$ be the representation $\mathbb{F}^m$ of L on which the loop acts by $\lambda I_m + N_m$, so that $J_m(0) = J_m$. Show that the simple finite-dimensional representations of L are the representations $J_1(\lambda)$, $\lambda \in \mathbb{F}$, and that these are pairwise non-isomorphic. Using [4, Existence of the Jordan Canonical Form], show that every finite-dimensional representation of L is an internal direct sum of subrepresentations each isomorphic to some $J_m(\lambda)$. Deduce from exercise 4.1.1 (2) that every $J_m(\lambda)$ is indecomposable, and show that $J_m(\lambda) \cong J_{m'}(\lambda')$ only if $m = m'$ and $\lambda = \lambda'$.
4. Show that D_4 has exactly 12 positive roots, and list them. (Write the Tits form as a sum of squares, one for each vertex of degree one and one for the central vertex.)
5. Let $\varepsilon_1, \ldots, \varepsilon_{n+1}$ be the standard basis of $\mathbb{R}^{n+1}$. Show that the linear map $\mathbb{R}^n \to \mathbb{R}^{n+1}$ sending α_i to $\varepsilon_i - \varepsilon_{i+1}$ carries the symmetric form of A_n to the dot product and Φ_{A_n} onto $\{\varepsilon_i - \varepsilon_j \mid i \neq j\}$. Show that, under this map, s_i becomes the exchange of the i-th and $(i+1)$-st coordinates, and deduce that W_{A_n} is isomorphic to the symmetric group on $n+1$ letters.
6. For A_2 and the Coxeter element $c = s_2 s_1$, compute the matrix of c in the basis α_1, α_2, show that $c^3 = \mathrm{id}$, and verify lemma 4.1.37 directly for $v = \alpha_1$ and for $v = \alpha_1 + \alpha_2$.
7. Let Q be $1 \to 2 \leftarrow 3$ and let V be the representation $\mathbb{F} \xrightarrow{1} \mathbb{F} \xleftarrow{1} \mathbb{F}$. Compute $S_2^+(V)$, show that it is indecomposable, and compare its dimension vector with $s_2(\dim_Q V)$.
8. Use theorem 4.1.46 to list, up to isomorphism, the nonzero finite-dimensional indecomposable

representations of $1 \to 2 \leftarrow 3$.

9. For $a, b, c \geq 1$, show that the star $T(a, b, c)$ of definition 4.1.29 has positive definite Tits form if and only if
$$\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} > 1$$

10. Let Q be $1 \to 2 \to \cdots \to n$. Show that for $i \leq j$ there is exactly one path p_{ji} from i to j, and that there is no path from j to i when $i < j$. Show that the linear map sending p_{ji} to the matrix E_{ji}, whose entry in position (j, i) is 1 and whose other entries are 0, is an isomorphism of algebras from $\mathbb{F}Q$ onto the algebra of lower triangular $n \times n$ matrices over $\mathbb{F}$, and deduce that $\dim_{\mathbb{F}} \mathbb{F}Q = \frac{n(n+1)}{2}$.

11. Let Q be a finite quiver without oriented cycles and let V be a simple representation of Q. Show that $V \cong S(i)$ for some vertex i. (Choose a vertex i with $V(i) \neq 0$ such that no arrow with source i has a target j with $V(j) \neq 0$.) Show that for the loop quiver the representation $\mathbb{F}$ on which the loop acts by 1 is simple and not isomorphic to $S(1)$.

4.2 Bound Quiver Algebras

The path algebra of definition 4.1.55 turns a quiver into an associative algebra whose modules are the quiver representations of definition 4.1.5 (theorem 4.1.58). This passage runs in one direction: every quiver produces an algebra. The converse question is the one that completes the picture. Given an abstract finite-dimensional algebra A over a field k, is there a quiver Q whose path algebra is A, or close enough to A that the two have the same representation theory? A path algebra is rarely finite-dimensional (a quiver with an oriented cycle has arbitrarily long paths, proposition 4.1.56(5)), so one cannot hope for equality on the nose. The fix is to impose relations, cutting the path algebra down by an ideal, and the content of this section is that this always suffices: over an algebraically closed field, every finite-dimensional algebra is, up to the coarsening given by Morita equivalence, a path algebra modulo relations. This is the structure theorem of Gabriel, and it makes the theory of quivers the universal combinatorial model for finite-dimensional representation theory.

The route to this statement passes through three reductions. First we cut a path algebra by an *admissible* ideal, which is what makes it finite-dimensional while keeping its arrows intact. Then we isolate the class of algebras for which the reconstruction is cleanest, the *basic* algebras, and show via Morita theory that restricting to them loses nothing. Finally we read a quiver directly off an abstract algebra A (its *Gabriel quiver*) using the Jacobson radical of chapter 2, and prove that A is the path algebra of that quiver modulo an admissible ideal. Throughout, A denotes a finite-dimensional algebra over k, modules are left modules, and $\mathrm{Jac}(A)$ is the Jacobson radical. We write kQ for the path algebra of definition 4.1.55 over k, and recall its multiplication convention: for paths p and q the product $p \cdot q$ is the concatenation "q then p", nonzero exactly when $t(q) = s(p)$. In particular a path γ from i to j satisfies $e_j \gamma e_i = \gamma$, so it lives in $e_j(kQ)e_i$; this ordering of the idempotents matters when counting arrows.

The first task is to identify, inside kQ, the piece built from arrows of positive length. The constant paths e_i span a semisimple part, a product of copies of k, and everything else is generated by the arrows. Collecting the arrow-generated part into an ideal gives the object that measures "length at least one", and its powers measure length.

Definition 4.2.1: Arrow Ideal and Admissible Ideal

Let Q be a finite quiver and kQ its path algebra over k.

1. The *arrow ideal* $R_Q \subseteq kQ$ is the two-sided ideal generated by the arrows (the paths of length 1).
2. A two-sided ideal $I \subseteq kQ$ is *admissible* if there exists an integer $m \geq 2$ with

$$R_Q^m \subseteq I \subseteq R_Q^2$$

Two facts make this definition run, both consequences of the way paths concatenate. First, R_Q^ℓ is spanned by the paths of length at least ℓ: a product of ℓ arrows is a path of length ℓ (when the concatenation is legal) and zero otherwise, and every path of length ℓ factors as such a product. Consequently $kQ = kQ_0 \oplus R_Q$ as vector spaces, where $kQ_0 = \bigoplus_i ke_i$ is the span of the constant paths, and more generally the paths of a fixed length span a complement to $R_Q^{\ell+1}$ inside R_Q^ℓ. Second, the sandwiching $R_Q^m \subseteq I \subseteq R_Q^2$ has a concrete meaning at each end. The lower bound $R_Q^m \subseteq I$ forces I to contain all sufficiently long paths, which is exactly what truncates kQ/I to finite dimension. The upper bound $I \subseteq R_Q^2$ forbids relations of length 0 or 1: no relation may involve a constant path or set a single arrow equal to a combination of other arrows. This is the condition that keeps the arrows of Q visible in the quotient, so that Q can be recovered from the algebra kQ/I rather than being partly erased by the relations.

Definition 4.2.2: Bound Quiver Algebra

Let Q be a finite quiver and $I \subseteq kQ$ an admissible ideal. The quotient

$$kQ/I$$

is called the *bound quiver algebra* (or the path algebra of Q *bound by* the relations I). The pair (Q, I) is called a *bound quiver*.

Because $R_Q^m \subseteq I$, the quotient kQ/I is spanned by the images of the paths of length less than m, of which there are finitely many when Q is finite; so a bound quiver algebra is always finite-dimensional. The constant paths e_i descend to a complete set of orthogonal idempotents summing to 1, and R_Q/I is a nilpotent ideal with $(R_Q/I)^m = 0$. In fact R_Q/I is precisely the Jacobson radical of kQ/I. Being a nilpotent two-sided ideal, it lies in the radical by theorem 2.1.8, since kQ/I is finite-dimensional and hence left Artinian. Conversely, $kQ = kQ_0 \oplus R_Q$ and $I \subseteq R_Q$ give $(kQ/I)/(R_Q/I) \cong kQ_0 = \prod_i k$, and an element $\sum_i c_i e_i + r$ of kQ/I, with $r \in R_Q/I$, acts by c_i on the one-dimensional module on which e_i acts as 1 and every other e_j as 0. The radical annihilates each of these simple modules (theorem 2.1.4), so it lies in R_Q/I. This last observation is the seed of the reconstruction theorem: the arrows of Q will be recovered from $\mathrm{Jac}(kQ/I)/\mathrm{Jac}(kQ/I)^2$.

Example 4.2.3: Loops, Truncation, and the A_2 Quiver

We compute three bound quiver algebras from the ground up.

1. *The dual numbers as a bound one-loop quiver.* Let Q be the quiver with one vertex 1 and

one loop $\alpha\colon 1 \to 1$:



The paths in Q are $e_1, \alpha, \alpha^2, \alpha^3, \ldots$, so as a vector space $kQ = k[\alpha]$ is the polynomial ring in one variable (identifying e_1 with 1 and concatenation of the loop with multiplication). The arrow ideal is $R_Q = (\alpha)$, and $R_Q^m = (\alpha^m)$. Take $I = (\alpha^2) = R_Q^2$, which is admissible with $m = 2$ (indeed $R_Q^2 \subseteq I \subseteq R_Q^2$). Then

$$kQ/I = k[\alpha]/(\alpha^2)$$

is the algebra of *dual numbers*, two-dimensional over k with basis $\{e_1, \alpha\}$ and the single relation $\alpha^2 = 0$. Its radical is $R_Q/I = (\alpha)$, one-dimensional, with square zero, and kQ/I modulo its radical is k: there is exactly one simple module, and it is one-dimensional. This is the smallest example of a bound quiver algebra that is not itself a path algebra, the relation $\alpha^2 = 0$ being unavoidable if one wants a finite-dimensional quotient of $k[\alpha]$.

2. *Truncating a longer loop.* Keeping the same one-loop quiver but taking instead $I = (\alpha^3) = R_Q^3$ gives $kQ/I = k[\alpha]/(\alpha^3)$, three-dimensional with basis $\{e_1, \alpha, \alpha^2\}$. Here $m = 3$; the admissibility bound $I \subseteq R_Q^2$ holds because $\alpha^3 \in R_Q^3 \subseteq R_Q^2$, and the relation still involves no constant path and no bare arrow. More generally $k[\alpha]/(\alpha^n)$ is the one-loop quiver bound by (α^n) for each $n \geq 2$, and these exhaust the local algebras arising from a single loop.

3. *The A_2 quiver, hereditary case.* Let Q be the quiver

$$1 \xrightarrow{\alpha} 2$$

with two vertices and a single arrow $\alpha\colon 1 \to 2$. The only paths are e_1, e_2, and α, so kQ is already three-dimensional and $R_Q^2 = 0$ (there is no legal concatenation of two arrows). The only admissible ideal is therefore $I = 0$: admissibility requires $I \subseteq R_Q^2 = 0$, so nothing may be quotiented, and $R_Q^2 = 0 = I$ gives the lower bound with $m = 2$. Thus $kQ/I = kQ$, a path algebra with no relations. The path algebra of a finite connected quiver is *hereditary*, meaning that every submodule of a projective module is projective (theorem 4.1.58 and [11, Corollary 1.22]), so in particular kQ is hereditary.

   Concretely kQ is isomorphic to the lower-triangular matrix algebra

$$\begin{pmatrix} k & 0 \\ k & k \end{pmatrix}$$

via $e_1 \mapsto E_{11}$, $e_2 \mapsto E_{22}$, and $\alpha \mapsto E_{21}$, where E_{ij} is the elementary matrix with a 1 in position (i, j). Lower- rather than upper-triangular is forced by the multiplication convention: the arrow $\alpha\colon 1 \to 2$ satisfies $e_2 \alpha e_1 = \alpha$, so its image must land in position $(2, 1)$, the strictly lower-triangular slot. The check is direct on the multiplication table: $e_2 \cdot \alpha \cdot e_1 = \alpha$ matches $E_{22}E_{21}E_{11} = E_{21}$, and every other product of basis elements is forced to zero on both sides. (The upper-triangular algebra is the opposite algebra, the A_2 path algebra with the arrow reversed to $2 \to 1$; we recover it from the radical in example 4.2.9.) We return to this identification once we have located the indecomposable projectives.

The dual-numbers example already shows the shape of the reconstruction: a local algebra with

radical square zero came out as a single loop bound by α^2, and the loop is exactly a basis vector of $\mathrm{Jac}/\mathrm{Jac}^2$. Before making this into a theorem, one must decide which algebras to reconstruct. Not every finite-dimensional algebra is a bound quiver algebra on the nose. The matrix algebra $M_2(k)$, for instance, is simple, so its only simple module is two-dimensional, whereas a bound quiver algebra always has all its simples one-dimensional over k when k is algebraically closed (they are the vertex simples S_i). The class for which the reconstruction is exact is singled out by demanding that the semisimple quotient $A/\mathrm{Jac}(A)$ be as small as possible: a product of division rings, with no matrix blocks of size larger than one.

Definition 4.2.4: Basic Algebra

A finite-dimensional k-algebra A is *basic* if the quotient $A/\mathrm{Jac}(A)$ is a product of division rings,

$$A/\mathrm{Jac}(A) \cong D_1 \times \cdots \times D_n$$

with each D_i a division k-algebra. Equivalently, in the Artin–Wedderburn decomposition of $A/\mathrm{Jac}(A)$ (corollary 1.2.18) every matrix block $M_{n_i}(D_i)$ has $n_i = 1$.

When k is algebraically closed the finite-dimensional division k-algebras collapse to k itself (corollary 1.2.20), so a basic algebra over such a k has

$$A/\mathrm{Jac}(A) \cong k \times \cdots \times k$$

a product of n copies of k. Two consequences are worth stating plainly. First, the simple A-modules are the n simple modules of this product, each one-dimensional over k; a basic algebra over an algebraically closed field has *all simple modules one-dimensional*. Second, the number n of simple modules equals the number of factors, and it becomes the number of vertices of the reconstructed quiver. The condition "every matrix block has size one" is precisely the condition that the regular representation of $A/\mathrm{Jac}(A)$ contains each simple exactly once, so that A is, in the language of idempotents, a sum of pairwise non-isomorphic indecomposable projectives.

Example 4.2.5: Basic and Non-Basic Algebras

The bound quiver algebras of example 4.2.3 are all basic: for $k[\alpha]/(\alpha^2)$ the radical quotient is k, and for the A_2 path algebra it is $k \times k$, both products of copies of k. The upper-triangular algebra of 2×2 matrices is basic for the same reason. By contrast $M_2(k)$ is *not* basic: it is already semisimple, so $\mathrm{Jac}(M_2(k)) = 0$ and $M_2(k)/\mathrm{Jac}(M_2(k)) = M_2(k)$ is a single 2×2 block. The group algebra $k[S_3]$ over $k = \mathbb{C}$ is not basic either, since by Maschke's theorem it is semisimple and Artin–Wedderburn gives $\mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$, a product with a 2×2 block coming from the two-dimensional irreducible representation of S_3. These non-basic examples are not pathological; they are the reason the theorem is stated up to Morita equivalence rather than isomorphism.

The point of Morita equivalence is that it does not see the difference between an algebra and its basic model. Two algebras are Morita equivalent when their module categories are equivalent, and Morita's theorem identifies exactly which algebras share a module category with a given one: they are the endomorphism algebras of progenerators. That theory, including generators, progenerators, and the classification of Morita equivalences by bimodules, is developed in *Core Algebra* [4, Morita Theory]. What is used here is the criterion for a corner ring, that Ae is a progenerator exactly when the idempotent e is full, in which case A and eAe are Morita equivalent. The block structure of the radical quotient is this book's own and is settled below.

Proposition 4.2.6: Reduction to the Basic Case

Let A be a finite-dimensional k-algebra. Then there is a basic finite-dimensional k-algebra A^b, the *basic algebra of* A, that is Morita equivalent to A: the categories of left modules A-**Mod** and A^b-**Mod** are equivalent. Up to isomorphism, A^b does not depend on the choices made in its construction, and A and A^b have the same number of simple modules and the same radical quotient block *data* in the sense that $A^b/\mathrm{Jac}(A^b) \cong D_1 \times \cdots \times D_n$ where $M_{n_i}(D_i)$ are the Artin–Wedderburn blocks of $A/\mathrm{Jac}(A)$.

***Proof*:**

Step 1: the idempotent. Write $1 = e_1 + \cdots + e_r$ as a sum of primitive orthogonal idempotents, following theorem 2.3.2, so that $A = \bigoplus_j Ae_j$ expresses the regular module as a direct sum of indecomposable projectives $P_j = Ae_j$. Two of these are isomorphic exactly when the corresponding simples $P_j/\mathrm{rad}(P_j)$ and $P_{j'}/\mathrm{rad}(P_{j'})$ agree, so grouping the e_j by the isomorphism class of P_j and choosing one representative from each class produces an idempotent

$$e = e_{i_1} + \cdots + e_{i_n}$$

with n the number of simple modules. It is idempotent because the e_{i_j} are orthogonal. Set $A^b := eAe$.

Step 2: e is full. Each e_{i_j} satisfies $e_{i_j}e = e_{i_j}$, again by orthogonality, so $Ae_{i_j}A = Ae_{i_j}eA \subseteq AeA$. For an arbitrary l the projective Ae_l is isomorphic to Ae_{i_j} for some j by the choice of representatives. If $\theta\colon Ae_l \to Ae_{i_j}$ is an isomorphism, then $u = \theta(e_l) \in Ae_{i_j}$ satisfies $u = ue_{i_j}$, and the A-linearity of θ^{-1} gives $e_l = \theta^{-1}(ue_{i_j}) = u\,\theta^{-1}(e_{i_j})$, so $e_l \in Ae_{i_j}A$ and $Ae_lA \subseteq Ae_{i_j}A \subseteq AeA$. Summing,

$$1 = \sum_{l=1}^{r} e_l \in \sum_{l=1}^{r} Ae_lA \subseteq AeA$$

so $AeA = A$ and e is full.

Step 3: the equivalence. By [4, the corner ring of a full idempotent], a full idempotent makes Ae a progenerator with $\mathrm{End}_A(Ae)^{\mathrm{op}} \cong eAe$, so A and $A^b = eAe$ are Morita equivalent. The equivalence is $\mathrm{Hom}_A(Ae, -)$, which sends an A-module M to eM.

Step 4: A^b is basic, with the stated radical quotient. The equivalence carries the indecomposable projectives of A to those of A^b and preserves isomorphism, and $Ae = \bigoplus_j Ae_{i_j}$ retains exactly one indecomposable projective from each isomorphism class. So the regular representation of A^b contains each simple exactly once, which is what it means for A^b to be basic, and its radical quotient has all blocks of size one. The division rings are unchanged: the equivalence induces a bijection on simple modules and a ring isomorphism on their endomorphism rings, so the blocks are over the same $D_i = \mathrm{End}_A(S_i)^{\mathrm{op}}$ furnished by Schur's lemma (lemma 1.1.5). Another decomposition of 1 and another choice of representatives give an idempotent e' with $Ae' \cong Ae$, since by theorem 2.2.4 the isomorphism classes of indecomposable summands of A do not depend on the decomposition, and Ae and Ae' are both direct sums of one module from each class. Steps 2 and 3 apply to e' as well, so $e'Ae' \cong \mathrm{End}_A(Ae')^{\mathrm{op}} \cong \mathrm{End}_A(Ae)^{\mathrm{op}} \cong eAe$, and A^b does not depend on the choices up to isomorphism, completing the proof.

The reduction lets us prove the structure theorem only for basic algebras and inherit it for all

finite-dimensional algebras through the equivalence of module categories: whatever we say about the representations of $A^b = kQ_A/I$ transfers verbatim to the representations of A. From now on A is basic, and for the reconstruction assume k is algebraically closed, so that all simple modules are one-dimensional and the radical quotient is a product of copies of k. This is the setting of Gabriel's theorem, and it is where the assumption $k = \overline{k}$ earns its place: over a non-closed field the division algebras D_i reappear and the vertices of the quiver would have to be weighted by them.

The quiver attached to an abstract algebra can now be defined. Its vertices are the simple modules; its arrows count how simples fail to split off one another, measured either homologically by Ext^1 or internally by the radical filtration. That these two counts agree is the bridge between the abstract and combinatorial descriptions.

Definition 4.2.7: Ext-Quiver (Gabriel Quiver)

Let A be a basic finite-dimensional algebra over an algebraically closed field k, with simple modules $S_1, \dots, S_n$ up to isomorphism and a complete set of primitive orthogonal idempotents $e_1, \dots, e_n$ with $S_i = Ae_i/\mathrm{rad}(Ae_i)$. The *Ext-quiver* (or *Gabriel quiver*) Q_A is the quiver with

1. vertex set $\{1, \dots, n\}$, one vertex for each simple module S_i, and
2. exactly
$$\dim_k \mathrm{Ext}^1_A(S_i, S_j) = \dim_k e_j\big(\mathrm{Jac}(A)/\mathrm{Jac}(A)^2\big)e_i$$
arrows from i to j.

The order of the idempotents in the formula is fixed by the multiplication convention of definition 4.1.55 and is worth pausing on, since it is easy to get backwards. In the path algebra a would-be arrow from i to j satisfies $e_j \alpha e_i = \alpha$, so it lives in $e_j(kQ)e_i$; matching this, the arrows from i to j in Q_A are counted by $e_j(\mathrm{Jac}/\mathrm{Jac}^2)e_i$, with e_j on the left and e_i on the right. A source using the opposite (left-to-right) convention for path multiplication would write $e_i(\mathrm{Jac}/\mathrm{Jac}^2)e_j$ instead; the two conventions transpose the quiver, so one must commit to one. Here the commitment is to the book's convention throughout. The equality of the two counts, homological and idempotent-theoretic, is the following lemma, proved here because it is what makes the definition well-posed: the arrow count must not depend on the choice of description.

Lemma 4.2.8: Arrows Count Radical Extensions

Let A be a finite-dimensional k-algebra with primitive orthogonal idempotents $e_1, \dots, e_n$ and simple modules $S_i = Ae_i/\mathrm{rad}(Ae_i)$, and assume k is algebraically closed. Then for all i, j,
$$\mathrm{Ext}^1_A(S_i, S_j) \cong e_j\big(\mathrm{Jac}(A)/\mathrm{Jac}(A)^2\big)e_i$$
as k-vector spaces. In particular the two are equidimensional.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $P_i = Ae_i$, the indecomposable projective with top $S_i = P_i/\mathrm{rad}(P_i)$; recall from definition 2.3.3 that $\mathrm{rad}(P_i) = Je_i$, so that $S_i = Ae_i/Je_i$.

Step 1: A projective presentation of S_i. The surjection $P_i \twoheadrightarrow S_i$ has kernel $\mathrm{rad}(P_i) = Je_i$, giving

the short exact sequence

$$0 \to Je_i \to P_i \to S_i \to 0$$

Applying $\mathrm{Hom}_A(-, S_j)$ produces the exact sequence

$$0 \to \mathrm{Hom}_A(S_i, S_j) \to \mathrm{Hom}_A(P_i, S_j) \to \mathrm{Hom}_A(Je_i, S_j) \to \mathrm{Ext}^1_A(S_i, S_j) \to 0$$

where the last term is $\mathrm{Ext}^1_A(S_i, S_j)$ and not $\mathrm{Ext}^1(P_i, S_j)$ because P_i is projective, so $\mathrm{Ext}^1_A(P_i, S_j) = 0$ and the connecting map surjects onto $\mathrm{Ext}^1(S_i, S_j)$. The definition and basic properties of Ext^1 as the obstruction to lifting homomorphisms along a projective presentation are developed in the companion volume on homological algebra ([5, Computation Of Tor and Ext]). We use only that a projective resolution computes it and that projectives have no higher self-extensions.

Step 2: Evaluating the outer terms. For any simple S_j and any module M, a homomorphism $M \to S_j$ factors through $M/\mathrm{rad}(M)$ because $\mathrm{rad}(M)$ maps into $\mathrm{rad}(S_j) = 0$. Applying this to P_i gives $\mathrm{Hom}_A(P_i, S_j) = \mathrm{Hom}_A(S_i, S_j)$, so the first two terms of the sequence in Step 1 cancel and the map between them is an isomorphism. The sequence therefore collapses to

$$0 \to \mathrm{Hom}_A(Je_i, S_j)/\big(\text{image of } \mathrm{Hom}_A(P_i, S_j)\big) \to \mathrm{Ext}^1_A(S_i, S_j) \to 0$$

but the image of $\mathrm{Hom}_A(P_i, S_j)$ in $\mathrm{Hom}_A(Je_i, S_j)$ is zero: a homomorphism $P_i \to S_j$ vanishes on $\mathrm{rad}(P_i) = Je_i$ by the factoring just noted. Hence the connecting map is an isomorphism

$$\mathrm{Ext}^1_A(S_i, S_j) \cong \mathrm{Hom}_A(Je_i, S_j)$$

Step 3: The Hom into a simple is a radical layer. Again by the factoring through the top, any homomorphism $Je_i \to S_j$ kills $\mathrm{rad}(Je_i) = J^2 e_i$, so

$$\mathrm{Hom}_A(Je_i, S_j) = \mathrm{Hom}_A\big(Je_i/J^2e_i,\ S_j\big)$$

The module Je_i/J^2e_i is semisimple, being a module over the semisimple algebra A/J, and S_j is simple, so by Schur's lemma (lemma 1.1.5) and $\mathrm{End}_A(S_j) = k$ (from $k = \overline{k}$) the space $\mathrm{Hom}_A(Je_i/J^2e_i, S_j)$ has dimension equal to the multiplicity of S_j in Je_i/J^2e_i. That multiplicity is $\dim_k e_j(Je_i/J^2e_i)$. Indeed, for a semisimple module N the map $\varphi \mapsto \varphi(e_j)$ is a linear isomorphism $\mathrm{Hom}_A(Ae_j, N) \to e_jN$, with inverse $n \mapsto (ae_j \mapsto an)$. Every homomorphism $Ae_j \to N$ vanishes on Je_j, because $JN = 0$, so $\mathrm{Hom}_A(Ae_j, N) \cong \mathrm{Hom}_A(S_j, N)$, whose dimension is again the multiplicity of S_j in N by Schur's lemma and $\mathrm{End}_A(S_j) = k$. Therefore

$$\dim_k \mathrm{Ext}^1_A(S_i, S_j) = \dim_k e_j(Je_i/J^2e_i) = \dim_k e_j(J/J^2)e_i$$

the last equality because $e_j(J/J^2)e_i = e_j(Je_i)/e_j(J^2e_i)$ and right-multiplication by e_i identifies Je_i/J^2e_i with the e_i-column of J/J^2. Assembling the three isomorphisms gives $\mathrm{Ext}^1_A(S_i, S_j) \cong e_j(J/J^2)e_i$, as we sought to show.

The lemma says the arrows of Q_A have two readings. Homologically, an arrow $i \to j$ records a one-dimensional worth of non-split extension of S_i by S_j: a module of length two with S_j as a submodule and S_i on top that does not break as $S_i \oplus S_j$. Internally, the same arrow is a basis vector of the (j, i) block of the first radical layer $\mathrm{Jac}/\mathrm{Jac}^2$. The first reading is the reason the quiver is called the Ext-quiver and explains why it is an invariant of the module category, hence of the Morita equivalence class; the second reading is the one to compute with, because $\mathrm{Jac}/\mathrm{Jac}^2$ is a finite matrix of idempotent-sandwiched subspaces. The running examples compute it.

Example 4.2.9: Gabriel Quivers of the Running Algebras
Reading off Q_A for two algebras confirms it recovers the starting quiver.

1. *The dual numbers.* Let $A = k[\alpha]/(\alpha^2)$. This is local: its only maximal ideal is (α), so $\mathrm{Jac}(A) = (\alpha)$ and there is a single simple module $S_1 = A/(\alpha) = k$, giving one vertex. The first radical layer is $\mathrm{Jac}/\mathrm{Jac}^2 = (\alpha)/(\alpha^2)$, one-dimensional with basis the image of α, and with $e_1 = 1$ the block $e_1(\mathrm{Jac}/\mathrm{Jac}^2)e_1$ is all of it. So Q_A has one vertex and one arrow, a loop:

$$\overset{1}{\circlearrowleft}\ \alpha$$

   exactly the quiver we bound in example 4.2.3. The relation to be imposed will be $\alpha^2 = 0$, recovering A.

2. *The upper-triangular algebra.* Let $A = \begin{pmatrix} k & k \\ 0 & k \end{pmatrix}$, the 2×2 upper-triangular matrices. Its radical is the strictly upper-triangular part, $\mathrm{Jac}(A) = kE_{12}$, one-dimensional with square zero. The idempotents $e_1 = E_{11}$ and $e_2 = E_{22}$ are primitive orthogonal with $e_1 + e_2 = 1$, and $A/\mathrm{Jac}(A) \cong k \times k$, so there are two vertices with simples S_1, S_2. Since $\mathrm{Jac}^2 = 0$ we have $\mathrm{Jac}/\mathrm{Jac}^2 = kE_{12}$, and we locate E_{12} by its idempotent sandwich:

$$e_1 E_{12} e_2 = E_{11}E_{12}E_{22} = E_{12}, \qquad e_2 E_{12} e_1 = E_{22}E_{12}E_{11} = 0$$

   so the one basis vector of $\mathrm{Jac}/\mathrm{Jac}^2$ sits in the block $e_1(\mathrm{Jac}/\mathrm{Jac}^2)e_2$. By definition 4.2.7 the block $e_j(\cdots)e_i$ counts arrows $i \to j$, so with $j = 1$, $i = 2$ this is one arrow $2 \to 1$:

$$2 \xrightarrow{\ \alpha\ } 1$$

   Under the identification $\alpha \mapsto E_{12}$ this is the A_2 quiver of example 4.2.3, up to relabelling the vertices, and $\mathrm{Jac}^2 = 0$ forces the admissible ideal to be $I = 0$. The algebra is hereditary, and Q_A carries no relations: the upper-triangular algebra *is* its own path algebra kQ_A.

The two examples describe the same three-dimensional hereditary algebra, once as a bound quiver algebra and once reconstructed from its radical, and the indecomposable projectives match under the identification. Take the bound-quiver presentation of example 4.2.3, the A_2 path algebra with arrow $\alpha\colon 1 \to 2$ realized as the lower-triangular matrices via $e_1 \mapsto E_{11}$, $e_2 \mapsto E_{22}$, $\alpha \mapsto E_{21}$. The indecomposable projective $P_i = kQ \cdot e_i$ is spanned by the paths starting at i: P_1 has basis $\{e_1, \alpha\}$, the trivial path at 1 together with the arrow out of 1, and is two-dimensional with top S_1 and socle S_2, while P_2 has basis $\{e_2\}$ and is the simple S_2 itself. On the matrix side these are the column modules AE_{11} and AE_{22}: the first column of the lower-triangular algebra is two-dimensional, spanned by E_{11} and E_{21} (the entries in rows 1 and 2 of column 1), and the second column is one-dimensional, spanned by E_{22}. So the indecomposable projectives read off from the primitive idempotents of theorem 2.3.2 are exactly the projective quiver representations P_i of the path algebra. The Gabriel reconstruction of example 4.2.9 applied instead the upper-triangular representative, whose radical kE_{12} sits in the opposite corner and so produces the arrow $2 \to 1$; the two matrix algebras are opposite to one another (transpose exchanges upper and lower triangular), which is exactly the transposition of the quiver that the two multiplication conventions differ by. Either way the representation theory of A_2 is the same, and the two constructions coincide.

Everything is now in place for the structure theorem. It asserts that the passage from a basic algebra to its Gabriel quiver and back to a bound quiver algebra is a round trip: A is isomorphic to kQ_A modulo an admissible ideal. The proof builds a surjection $kQ_A \twoheadrightarrow A$ by sending vertices to the primitive idempotents and arrows to a lift of the radical layer, then shows the kernel is admissible using that $\mathrm{Jac}(A)$ is nilpotent.

Theorem 4.2.10: Structure of Basic Algebras (Gabriel)

Let A be a basic finite-dimensional algebra over an algebraically closed field k, with Gabriel quiver Q_A (definition 4.2.7). Then there is an admissible ideal $I \subseteq kQ_A$ and an isomorphism of k-algebras

$$A \cong kQ_A/I$$

Consequently, by the reduction to the basic case (proposition 4.2.6), every finite-dimensional k-algebra over an algebraically closed field is Morita equivalent to a bound quiver algebra kQ/I.

The quiver Q_A is determined by A, but the ideal I is not: different admissible ideals can present isomorphic algebras, so the theorem produces *some* admissible presentation rather than a canonical one. This is the one point where the correspondence between algebras and bound quivers fails to be a clean bijection, and it is the reason bound quiver algebras are studied through their quiver plus a choice of relations rather than through the quiver alone.

***Proof*:**

Write $J = \mathrm{Jac}(A)$, and recall J is nilpotent, say $J^m = 0$ for some $m \geq 2$, because A is finite-dimensional (a nilpotent radical is the Artinian case established in theorem 2.1.8). Let $e_1, \ldots, e_n$ be the complete set of primitive orthogonal idempotents underlying Q_A, with $1 = e_1 + \cdots + e_n$ and $S_i = Ae_i/Je_i$.

Step 1: Send vertices to idempotents and arrows to a radical lift. For each vertex i of Q_A set the constant path $\varepsilon_i \in kQ_A$ to map to $e_i \in A$. For the arrows, fix for each pair (i, j) a k-basis of the block $e_j(J/J^2)e_i$; by definition 4.2.7 and lemma 4.2.8 the arrows of Q_A from i to j are in bijection with this basis. For each arrow $\alpha\colon i \to j$ choose a lift $x_\alpha \in e_jJe_i \subseteq A$ of the corresponding basis vector of $e_j(J/J^2)e_i$; such a lift exists because $e_j(J/J^2)e_i$ is a quotient of e_jJe_i. Note $x_\alpha \in e_jJe_i$ means $e_jx_\alpha e_i = x_\alpha$, which is the algebra image of the path relation $e_j\alpha e_i = \alpha$, so the assignment respects the idempotent bookkeeping.

Step 2: Extend to an algebra homomorphism. Because kQ_A is free on paths, an algebra homomorphism out of it is determined by the images of the vertices and arrows, subject only to the relations $\varepsilon_i\varepsilon_{i'} = \delta_{ii'}\varepsilon_i$, $\sum_i \varepsilon_i = 1$, and the source-target matching for arrows. The chosen images e_i and x_α satisfy all of these: the e_i are orthogonal idempotents summing to 1, and $x_\alpha \in e_jAe_i$ matches the source i and target j of α. Extend multiplicatively, sending a path $\alpha_r \cdots \alpha_1$ to the product $x_{\alpha_r} \cdots x_{\alpha_1} \in A$ (in the algebra's own multiplication, so a path from i to j lands in e_jAe_i). This defines a k-algebra homomorphism

$$\varphi\colon kQ_A \to A$$

Step 3: φ is surjective. The image contains every e_i and every x_α, and the x_α lift a basis of J/J^2. We show $\mathrm{im}\varphi = A$ by descending the filtration $A \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^m = 0$ layer by layer. The images $\varphi(\varepsilon_i) = e_i$ generate $A/J = \prod_i ke_i$ over k, so $\mathrm{im}\varphi + J = A$. Next, the products of

length one, $x_\alpha = \varphi(\alpha)$, form lifts of a basis of J/J^2, so $\operatorname{im}\varphi + J^2 \supseteq J$, hence $\operatorname{im}\varphi + J^2 = A$. Inductively suppose $\operatorname{im}\varphi + J^\ell = A$; then J^ℓ is spanned modulo $J^{\ell+1}$ by products of ℓ elements of J, each of which may be taken from the spanning set $\{x_\alpha\}$ of J modulo J^2, so products $\varphi(\alpha_\ell \cdots \alpha_1)$ of ℓ arrows span J^ℓ modulo $J^{\ell+1}$. Thus $\operatorname{im}\varphi + J^{\ell+1} = A$. Since $J^m = 0$, after m steps $\operatorname{im}\varphi = \operatorname{im}\varphi + J^m = A$, so φ is surjective.

Step 4: The kernel is admissible. Let $I = \ker\varphi$, so $A \cong kQ_A/I$. It remains to check $R_Q^{m'} \subseteq I \subseteq R_Q^2$ for some m', where R_Q is the arrow ideal of kQ_A.

For the lower bound, $\varphi(R_Q^m) \subseteq J^m = 0$: a path of length at least m is a product of at least m arrows, and each arrow maps into J, so a length-$\geq m$ path maps into $J^m = 0$. Hence $R_Q^m \subseteq \ker\varphi = I$, giving the lower bound with $m' = m$.

For the upper bound $I \subseteq R_Q^2$, suppose $u \in I$ and write $u = u_0 + u_1 + u_2$ where $u_0 \in kQ_{A,0}$ is the constant-path part, u_1 is the arrow part (length exactly one), and $u_2 \in R_Q^2$ collects the length-≥ 2 paths. We must show $u_0 = 0$ and $u_1 = 0$. Applying φ and reducing modulo J: $\varphi(u_1) \in J$ and $\varphi(u_2) \in J^2 \subseteq J$, so $0 = \varphi(u) \equiv \varphi(u_0) \pmod J$. But φ restricted to constant paths is the isomorphism $kQ_{A,0} = \prod_i k\varepsilon_i \xrightarrow{\sim} \prod_i ke_i = A/J$ (the e_i are linearly independent modulo J because they are orthogonal idempotents with distinct simple tops), so $\varphi(u_0) \equiv 0 \pmod J$ forces $u_0 = 0$. Now reduce modulo J^2: with $u_0 = 0$ we have $0 = \varphi(u) = \varphi(u_1) + \varphi(u_2)$, and $\varphi(u_2) \in J^2$, so $\varphi(u_1) \equiv 0 \pmod{J^2}$. But $\varphi(u_1)$ is the image in J/J^2 of a linear combination of arrows, and the arrows were chosen to map to a *basis* of J/J^2; a linear combination of basis vectors that is zero in J/J^2 has all coefficients zero, so $u_1 = 0$. Therefore $u = u_2 \in R_Q^2$, proving $I \subseteq R_Q^2$.

Combining the two bounds, $R_Q^m \subseteq I \subseteq R_Q^2$, so I is admissible and $A \cong kQ_A/I$ is a bound quiver algebra. The final clause on general algebras follows by composing with the Morita reduction of proposition 4.2.6: an arbitrary finite-dimensional A is Morita equivalent to its basic algebra A^b, and $A^b \cong kQ_{A^b}/I$ by what was just proved, completing the proof.

Three features of the argument deserve a closing word. The surjectivity in Step 3 is a radical-filtration induction: because $\operatorname{Jac}(A)$ is nilpotent, generating $A/\operatorname{Jac}(A)$ by the idempotents and the first radical layer $\operatorname{Jac}/\operatorname{Jac}^2$ by the arrows suffices to generate all of A, since higher layers are spanned by products. The admissibility of the kernel in Step 4 is the exact counterpart of the two bounds in definition 4.2.1: the lower bound $R_Q^m \subseteq I$ is nilpotence of the radical, and the upper bound $I \subseteq R_Q^2$ is the choice of arrows as a basis of $\operatorname{Jac}/\operatorname{Jac}^2$, which is precisely why an admissible ideal is forbidden to contain constant paths or bare arrows. And the theorem is the structure theorem of Gabriel, to be kept distinct from Gabriel's other theorem in this chapter (theorem 4.1.46), which says that a finite quiver with connected underlying graph is of finite type exactly when its graph is a simply-laced Dynkin diagram A, D, E (the converse direction proved there over an infinite field). The two results are complementary: the present one says every finite-dimensional algebra is a bound quiver algebra, and the Dynkin classification says which of the unbound path algebras have only finitely many indecomposable modules.